

D4.1:

Pilots' cross-domain governance and provisioning activities

Revision: v 1.0



D4.1: PILOT'S CROSS-DOMAIN GOVERNANCE AND PROVISIONING ACTIVITIES

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Abstract	This deliverable presents O-CEI Large Scale Pilots governance strategy. This strategy aims to successfully guide the pilots through the different phases of the project while identifying synergies between the different domains and sectors, thereby promoting the design and dissemination of blueprints based on O-CEI technologies and solutions. The methodology and tools that will enable coordinated progress of the pilots and the achievement of the project milestones are presented. This document also provides an initial overview of the preparatory activities carried out by the 8 pilots and its different scenarios in order to establish a solid foundation on which to move forward in the subsequent phases of development and integration. This deliverable also shows the justified identification of the IT and energy infrastructure of the pilots, as well as the main technologies that will be relevant during their life cycle. In addition, it presents the initial scenario of relationships and interconnections between the pilots and their various sectors as a starting point for the replicability strategy of O-CEI solutions.
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EXECUTIVE SUMMARY

This deliverable **defines how O-CEI will run and coordinate its eight large-scale pilots** through a **pragmatic governance** core that preserves pilot autonomy while enforcing common operating rules **combining a federated model**—each pilot drives its own path under shared guidelines via a Pilots' Governance Committee—a layered structure aligned to WP1 (pilot teams → pilots leader → Steering Committee), a minimal-compliance cadence with standardized templates, clear information channels, bi-weekly leaders' calls, and a **year one roadmap** that consolidates all reporting obligations so work packages do not duplicate requests. Together, these choices create a **"Plan once, use many times"** workflow that frees pilots to execute while still surfacing risks and issues early.

The document also captures a baseline after the **initial phase**—most pilots have already installed sensors, management platforms, and control tools—so the next stage can shift to integration, validation in real environments, and demonstrating the value of the platform around energy flexibility and sustainability. In a nutshell, the following are the main results of each pilot, with further details on their corresponding sections:

- **P1.1 (Ireland):** Engaged households and DSOs, deployed smart metering/heating controls, set up GDPR/consent flows, and mapped data pipelines for community energy flexibility and local trading.
- **P1.2 (France):** Assessed devices across homes and substations, prepared virtualization and data-management stack, and defined control loops for flexibility and substation voltage support.
- **P1.3 (Croatia):** Instrumented buildings for NILM-based analytics, aligned EV/heat-pump coordination logic, and prepared community participation processes for island-grid optimization.
- **P2 (Romania):** Defined SDV edge/AI modules for safety and sustainability, connected vehicle telemetry to decision services, and outlined fleet charging/V2G scheduling workflows.
- **P3 (Austria):** Inventoried chargers and fleet data, established data orchestration for predictive charging, and designed V2G "mobile storage" strategies with operational KPIs.
- **P4 (Malta):** Integrated terminal and vessel energy monitoring, prepared edge control services for peak smoothing, and aligned OPS/terminal priorities for coordinated load management.
- **P5 (Cyprus):** Deployed IoT in livestock/dairy sites, set up data models for factory sharing, and planned energy-mix optimization and product energy-footprint tracking.
- **P6 (Spain):** Collected tractor/field telemetry, prepared route and task-planning prototypes, and defined autonomous recharging strategies based on real-time and seasonal patterns.
- **P7 (Greece/Italy):** Set up reliable m2m communications for EV charging, onboarded charger telemetry to edge analytics, and prepared trust/cybersecurity monitoring for data exchange.
- **P8 (Switzerland):** Rolled out metering and SES components in urban/tourism sites, connected forecasting/LLM data flows, and organized citizen-facing engagement and evaluation.

Cross-domain collaboration is organized around **three reusable pillars** that translate into practical exchange: electromobility and charging strategies (V2G/bidirectional storage, energy-flow optimization, battery-life estimation, multi-sensor fusion, object detection, route planning, fleet-scale V2G), renewable resource management and climate adaptability (resilience/connectivity, energy-mix control, carbon-footprint monitoring, grid balancing), and security/traceability/data exchange to enable trustworthy, interoperable data flows. This exchange sits within a broader standards-aligned ambition—pilots will validate the open platform and blueprints bi- or tri-laterally while referencing initiatives such as GAIA-X, Catena-X, and SAREF4ENER/TIC4.0 to avoid new silos and move toward a European pre-standard for CEI deployments.

To allocate attention where it yields early impact, O-CEI applies a transparent **prioritization** built on seven criteria: reusability/cross-pilot collaboration, social engagement, environmental impact, legacy HW/SW interoperability, business readiness, edge-latency needs, and the salience of key technologies at a given moment. Using this lens, residential/community flexibility pilots (1.1, 1.2, 1.3, 8) and EV-centric pilots (1.3, 2, 3, 7, 8) show the highest synergy potential; Pilot 7 (and Pilot 2's safety scenario) carry the most stringent latency; and Pilots 1, 2, and 3 present flagship business cases. In practice, Pilots 1.2, 2, and 3 were spotlighted at M6 for the technical check, and two pilot pairs—**1.2 with 8, and 3 with 7**—were tasked to **co-lead** the first concrete **blueprints to maximize reuse** by pairing consumer flexibility and social acceptability with high-momentum bidirectional EV charging. Looking ahead, **WP4** will channel pilot effort to **validate adopted tools** with the governance framework and shared templates **acting as the coordination backbone** that **turns diverse pilot work into transferable results across** electricity grids, electromobility, logistics, agrifood, telecom, and urban environments.

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ABBREVIATIONS

AC	Alternating current
AD	Anomaly Detection
ADAS	Advanced Driver assistance systems
AMI	Advanced Metering Infrastructure
B2G	Building/Battery to grid
CAN	Controller Area Network
CHP	Combined Heat and Power
CPU	Central Processing Unit
DC	Direct Current
DER	Decentralized Energy Resource
DPIA	Data Protection Impact Assessment
DSO	Distribution System Operator
ECU	Electronic Control Units
EMS	Energy Management System
EPBD	Energy Performance of Buildings Directive
ESB	Electricity Supply Board
EVSE	Electric Vehicle Supply Equipment
GDPR	General Data Protection Regulation
GPU	Graphic Processing Unit
HMI	Human Machine Interface
HV	High Voltage
IDS	Intrusion Detection Systems
IP	Internet Protocol
IT	Information Technology
KPI	Key Performance Indicator
LAC	Logical Access Control
LoRaWAN	Long Range Wide Area Network
LSP	Large Scale Pilots
LV	Low Voltage
MCU	Micro Controller Unit
MQTT	Message Queuing Telemetry Transport
MV	Medium Voltage
MCU	Micro Controller Unit
NILM	Non-Intrusive Load Monitoring
OCPP	Open Charge Point Protocol
OTA	Over The Air
PLC	Programmable Logic Controller
PME	Power Monitoring Expert
PNire	National Plan for Electric Vehicle Charging Infrastructure
PV	Photo Voltaic
RTU	Remote Terminal Unit
REDIII	Renewable Energy Directive III
RES	Renewable Energy Sources

REST API	Representation State Transfer Application Programming Interface
RL	Reinforcement Learning
Rpi	Raspberry Pi
RTG	Rubber Tyred Gantry
SCADA	Supervision Control and Data Acquisition
SDV	Software Defined Vehicle
SIEM	Security Information and Event Management
STS	Ship to shore
TCP	Transmission Control Protocol
TLS	Transport Layer Security
UI	User Interface
V2B	Vehicle to Building/Battery
V2G	Vehicle to Grid
VFD	Variable Frequency Drives
VPN	Virtual Private Network
VPS	Virtual Private Server
XAI	Explainable AI

....

1 INTRODUCTION

1.1 PURPOSE OF THE DOCUMENT

This deliverable sets out the governance approach for the 8 LSP coordination at a high level until the end of the project at M42. It presents the methodology and tools that will guide the aligned progress of the pilots while facilitating cross domain knowledge and information exchange.

The document provides a first report on the LSP activities that has been carried out during the initial phase of the project with the main setup & procurement activities already accomplished, as well as the main infrastructure that is needed (both IT and Technology related). Moreover, an overview of the implementation activities plan for the complete lifecycle of the pilots is provided.

1.2 STRUCTURE OF THE DOCUMENT

The document is structured to provide both the governance strategy for pilot management and the initial perspective of the activities carried out per pilot.

- **Section 1** offers and introduction and context for the deliverable.
- **Section 2** presents the governance strategy as well as the main tools that will be used for the alignment of the progress.
- **Sections 3–10** present first the governance structure and partners of the pilot, the demonstrator paths and their use cases. Each section reports on the main activities carried out so far, the planning for the upcoming months (M18/M21) the current energy/IT infrastructure and its role in the pilot.
- **Section 11** introduces the cross-domain interactions and relations between the pilots that will build further cross domain data and information exchange.
- **Section 12** reports on the pilots' prioritization, detailing the process, underlying reasoning, and criteria applied in defining the priorities.
- **Section 13** offers some conclusions over the work carried out during this first phase of the project.

1.3 RELATION TO OTHER WORK PACKAGES

D4.1 is the first deliverable of WP4 that is aimed at channeling the execution of the 8 LSP by allowing the verification and validation of the O-CEI technologies and platform. On this regard, a strong relationship with technical work packages is not only vital but mandatory. Specifically, this deliverable is related to:

- **WP2** - It is closely related to the work performed in WP2, as the clear definition of the different scenarios per pilot and their requirements developed there provides the starting point for establishing an implementation program and ensuring that the objectives are achieved.
- **WP3** - The work carried out here on mapping the pilots' objectives serves as a cornerstone input for the design of O-CEI the open platform and blueprints.
- **WP5** - Evaluation and assessment. The different activities carried out in WP4 regarding the implementation of the pilots will be evaluated and assessed in this WP.

- **WP6** - Market up taking, results exploitation and transferability. The different outcomes of each LSP will be analyzed from the market perspective always taking into account the cross-domain impact.
- **WP7** - Outreach and ecosystem enlargement. The implementation of the pilots will be used to expand the impact of the progress made in the project and the different outcomes.

2 GOVERNANCE STRATEGY

This deliverable sets out the governance approach for the O-CEI 8 LSP coordination at a high level until the end of the project at M42. It presents the methodology and tools that will guide the aligned progress of the pilots while facilitating cross domain knowledge.

The main goal of the O-CEI project is to establish the digital backbone that will enable key players in the supply chain of strategic sectors across Europe to find, test, validate and exploit innovative CEI technologies. O-CEI is developing a new framework and methodology to integrate, evaluate and exploit/monetize existing CEI technologies that will be validated in large-scale and multi-dimensional pilots with a common thread focused on energy flexibility.

Moreover, pilots in O-CEI involve SMEs, midcaps, RTOs and Industrial companies that will work together towards Energy Flexibility from different sectors and perspectives. Electricity grids, Electromobility, Logistics, Agrifood, Telecommunications infrastructure and Urban Areas are the 6 main domains in which the pilots, all of them built up on a clear value chain, will showcase and validate the O-CEI platform and Blueprints.

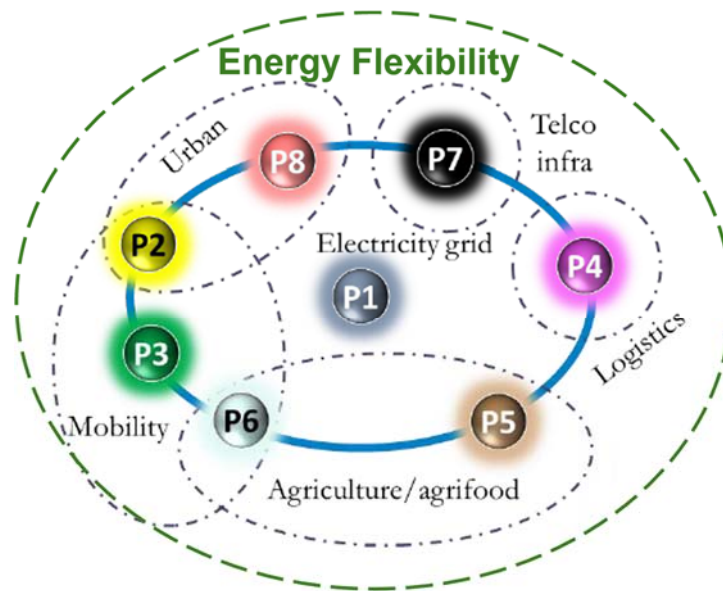


Figure 1. O-CEI Large Scale Pilots and its different sectors

To achieve such an ambitious goal, a clear governance strategy needs to be defined and shared between the pilots but also with all the relevant project partners that will collaborate in the definition and implementation of the O-CEI solutions. This governance strategy has been settled at different levels so as to provide coordination, efficiency and autonomy as well as balance between the information elicitation activities and the time invested by the pilots in collecting and documenting.

In an early stage of the project, and as the initial phase of the governance strategy, a strong effort regarding the design and structuring of pilot templates was made in collaboration with WP2 – Formalization of Pilots and Open Platform requirements elicitation, under Task 2.3 Formalization of pilots' scenarios and use cases including regulatory analysis. This has, together with the coordination activities carried out in WP1 – Project coordination and management, laid the foundations for defining the governance strategy in O-CEI based on different models.

Federated Model

- Each pilot keeps its autonomy working as independent teams with their own implementation paths. Different scenarios under the same pilot with relevant entity may act as sub-pilots with the same level of autonomy.
- There is a Pilots' Governance Committee with representatives from all pilots that participate in the definition of common guidelines.
- Information is shared/requested using standardized templates avoiding ad hoc requests and always with a specific documentation goal aligned between WP.

Layered Model

The governance strategy is aligned with the already existing hierarchical management established in WP1.

- Pilot Teams: They carry out their activities independently, reporting only what is necessary through a single channel. Each pilot team has a spokesperson.
- Pilots' Leader: The leader of the pilots is responsible for coordinating the pilots' activities and representing the pilots during the Steering committee meetings.
- Steering Committee: Defines the overall strategy, aligns with the project objectives and makes key decisions.

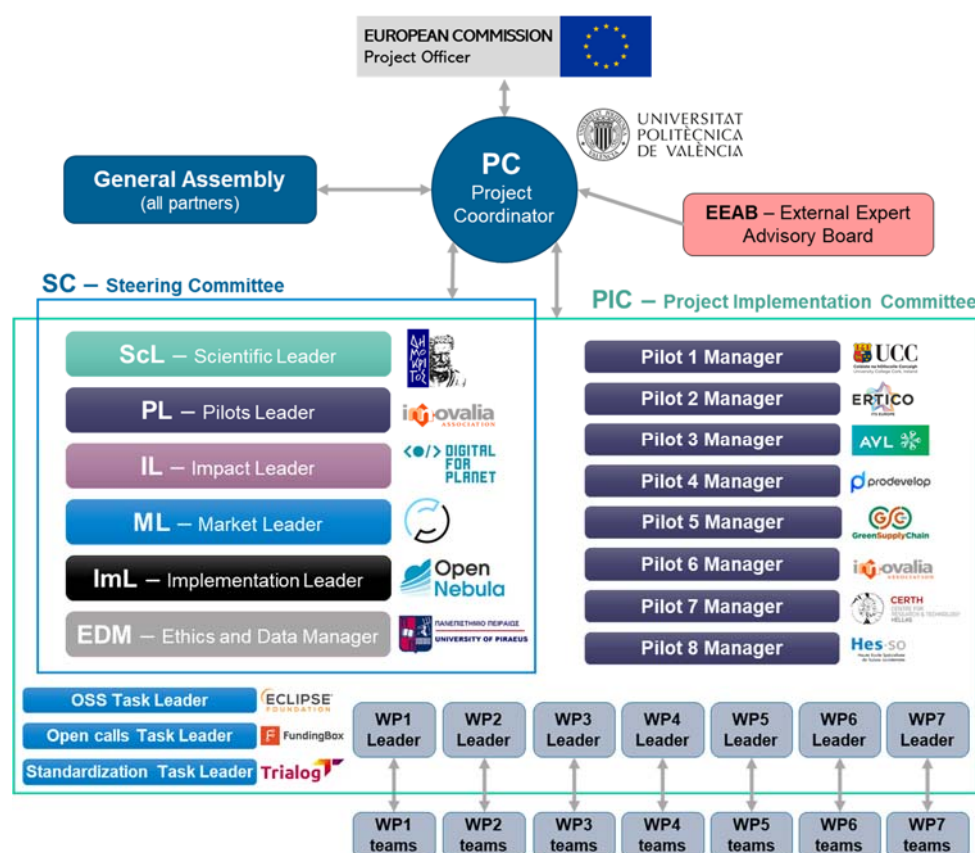


Figure 2. Hierarchical management in O-CEI

Minimal-Compliance Model

- Meetings: bi-weekly teleconferences with all pilot leaders.
- Planned Roadmap for key reporting obligations aligned with the information elicitation required by all WP.
- Overarching goal: maximum focus on execution, not overwhelming on documentation.

At the convergence point of these models, a lightweight yet effective methodology was introduced for monitoring and reporting pilot status at the individual level. Designed to be minimally demanding and as less intrusive as possible, it enables high-level overview of each pilot's progress while allowing for early detection of potential risks or misalignments. Moreover, the bi-weekly teleconferences with all pilot leaders that started in M2 and will take place throughout the project duration, will act as digital collaboration rooms where pilot teams can exchange progress updates, raise issues that may have cross-pilot implications, and receive clarifications from the WP leader on transversal topics - such as deliverable templates, shared reporting practices, workshops, and coordination actions.

These types of front-line actions are the tangible result of the roadmap established in taking into account the main information needed by all the project Work packages from the pilots in terms of key reporting obligations, deliverables and milestones. This roadmap was shared between all the pilot teams to streamline planning and alignment, avoiding duplicating the information request and at the same time maximizing the efficiency of the communications with the pilots, that now have a clear perspective of the workload throughout the project duration.

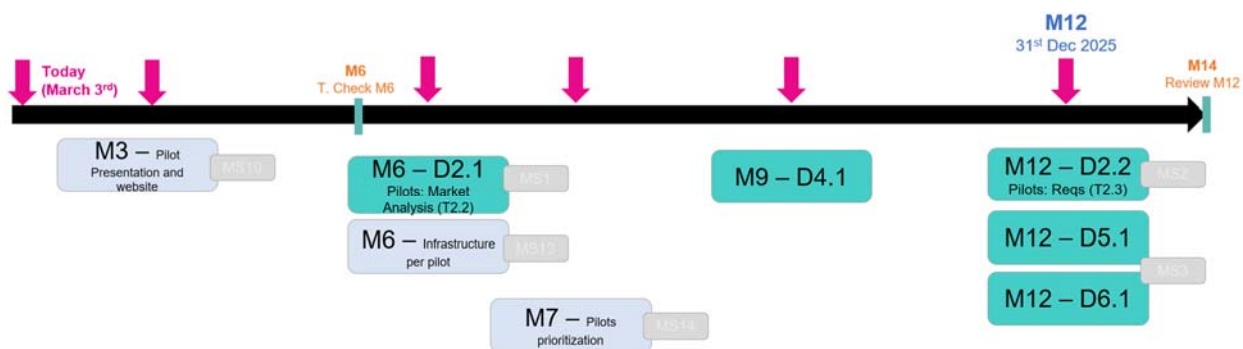


Figure 3. Roadmap for the first year of the O-CEI project

Each of the deliverables and milestones that are related to the pilots have been mapped and they were then analyzed to clearly identify the specific information that should be requested from the pilots so that this call to action comes from a single source. Identifying a single actor as responsible for channeling and organizing the flow of information from pilots to the leaders of the various tasks that require such information avoids duplication of efforts and allows pilots to focus on implementing O-CEI solutions.

Looking ahead, WP4 will play a pivotal role in validating the technologies and tools adopted by the pilots. In this context, T4.1 *Governance of pilots to achieve energy flexibility in cross-domain scenarios* will leverage the outputs of T2.3 *Formalization of pilot scenarios and use cases including regulatory analysis* & T2.4 *Platform requirements elicitation and blueprints design specifications* and channel the effort of the pilots to demonstrate the added value of the O-CEI platform and the future blueprints.

3 PILOT 1 – ELECTRIC GRID PERFORMANCE OPTIMIZATION UPON RES INTEGRATION

3.1 PILOT GOVERNANCE

The next figure depicts the organizational chart of Pilot 1:

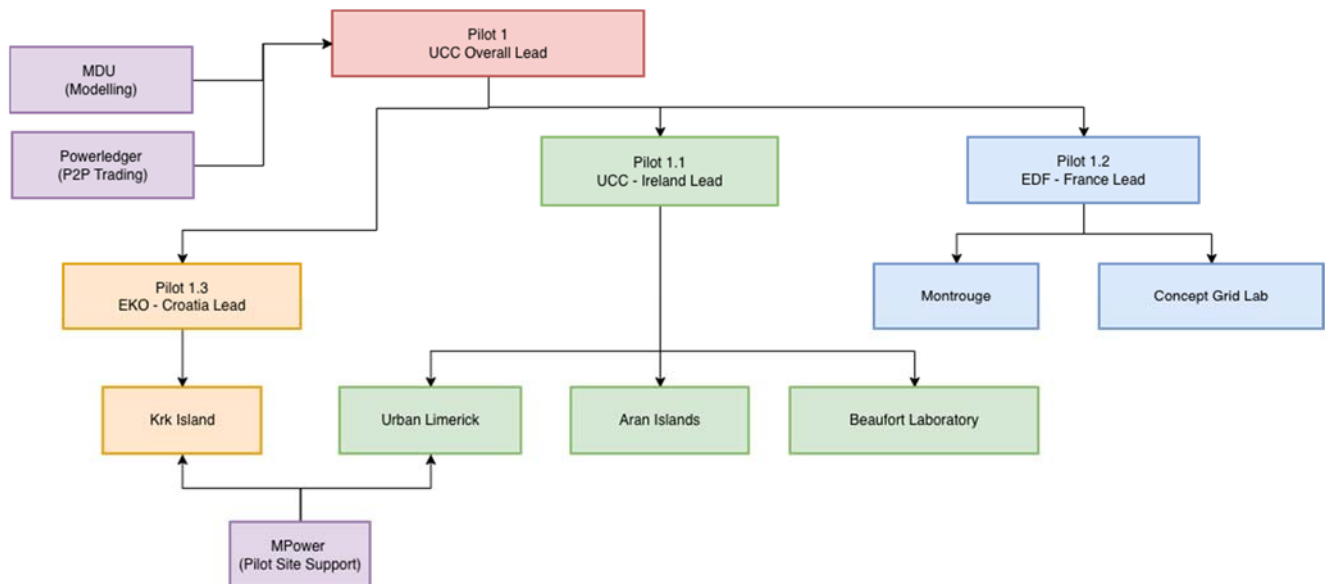


Figure 4. Pilot 1 Governance Diagram

UCC: The University College Cork is the **overall pilot leader**. Oversees all three sub-pilot (P1.1 in Ireland, P1.2 in France and P1.3 in Croatia) ensuring methodological coherence, data interoperability and shared technological backbones. UCC is responsible for cross-pilot integration and coordination of shared components (e.g. NILM, P2P trading, flexibility, orchestration, and system modelling)

EDF: (Électricité de France) **leads** the national implementation in **Pilot 1.2**. Its main focus is on interoperability and grid integration, deploying IoT gateways and programmable logic controllers to enable secure and private energy management near physical assets. EDF ensures that the French pilot aligns with UCC's overall coordination framework, contributing technical expertise in smart thermal load management and proactive scheduling for improved energy efficiency and user comfort.

EKO: Acts as the **pilot leader for Pilot 1.3**, focusing on community-based implementation and energy self-sufficiency on Krk Island. The company coordinates local activities aimed at optimizing renewable energy integration through edge computing and NILM solutions. EKO collaborates closely with UCC for replication and methodological alignment and works with MPOWER for implementation support. Its work emphasizes the adaptability of O-CEI technologies to community energy systems in semi-autonomous environments.

MPOWER is the **implementation partner** supporting **both Ireland and Croatia**. It provides technical assistance for on-site integration, validation, and deployment of CEI technologies. In Limerick, MPOWER supports UCC in the demonstration of smart renewable integration and flexibility management, while in Krk Island, it assists EKO in adapting and implementing orchestration mechanisms. MPOWER's role ensures that technical components and energy management systems operate effectively within real-world grid environments.

MDU contributes as the **modelling and simulation partner** within **the Irish pilot**. Its main task is to provide digital twin and simulation capabilities for UCC, enabling predictive modelling of energy flows and optimization strategies. MDU's analytics help validate system performance in controlled laboratory environments before deployment, enhancing the accuracy and robustness of the pilot's operational models.

PLE (Powerledger, in the process of being introduced as Associated Partner) plays the role of **peer-to-peer (P2P) trading platform provider** for Ireland and Croatia. It supplies the blockchain-based infrastructure that enables secure energy trading between households and prosumers. By connecting with UCC's modelling and orchestration layers, PLE facilitates the creation of local energy markets and enhances the transparency and trustworthiness of renewable energy exchanges. Its technology underpins the market interaction layer for flexibility trading across the pilot sites.

The following table is a summary of the roles, involvements and linkages:

Table 1. Pilot 1 Governance summary

Partner	Role	Pilot(s) Involved	Linkages
UCC	Overall Pilot 1 Leader, Technical Integration, NILM & modelling	P1.1 - Ireland (Lead), France, Croatia	Links to all partners for cross-pilot governance
EDF	Pilot Leader France, interoperability & grid integration	P1.2 - France	Linked to UCC for overall pilot alignment.
EKO	Pilot Leader Croatia, community implementation	P1.3 - Croatia	Linked to UCC for replication & MPower for implementation support
MPOWER	Implementation partner	Ireland, Croatia	Supports Limerick and Krk Island demonstrations
MDU	Modelling and simulation partner	Ireland	Linked to UCC for digital twin/modelling support
Powerledger	P2P trading platform provider	Ireland, Croatia	Linked to UCC for trading infrastructure and market interaction layer

3.2 PILOT 1.1 – CEI-BASED ENERGY MANAGEMENT SERVICES FOR END SERVICES

Pilot 1.1 demonstrates the coupling of electricity and heat in Irish contexts by combining smart thermal assets (heat pumps, combined heat and power, thermal storage), electric vehicle (EV) charging, NILM-based load analytics, and distributed ledger technology for peer-to-peer trading. It is structured around three complementary demonstrators:

- **Urban Limerick:** multi-use buildings at Hartigan's Square and community facilities at Chisel & Oak, where CHP units, smart heat pumps, and EVSE are combined with NILM and blockchain-based flexibility trading.
- **Aran Islands:** a replication site in a rural islanded grid context, testing transferability of NILM models, connectivity resilience, and household flexibility orchestration in a constrained microgrid environment.
- **UCC Beaufort Laboratory:** a controlled hardware-in-the-loop (HIL) environment in Ringaskiddy, enabling systematic validation of PV, battery, PCM, HP, and NILM stacks before field deployment.

The Irish pilot is positioned within national challenges: high renewable penetration in the Irish grid, decarbonization of heat under the Climate Action Plan, and flexibility requirements for ESB Networks and DSOs. The pilot also links to EU-level policy goals (EPBD, REDIII) by showcasing interoperability, trading, and flexibility solutions.

P1-SC1: Community-Driven Renewable Energy Integration

This scenario aims to empower local communities to actively participate in the energy transition by integrating renewable energy generation, enabling P2P energy trading, and improving grid interaction through flexibility services. It focuses on ensuring local renewable generation is efficiently shared, community benefits are maximized, and grid congestion is reduced through distributed ledger and NILM-enabled intelligence.

Limerick Demonstrator (Urban)



Figure 5. Pilot 1 Chisel & Oak Facilities in Limerick City



Figure 6. Pilot 1 Hartigan's Square in Limerick City 1



Figure 7. Pilot 1 Hartigan's Square in Limerick City 2

- **Location:** Hartigan's Square (Picture 2 & 3) and community facilities at Chisel & Oak (Picture 1), Limerick City.
- **Implementation Context:** Mixed-use urban environment with high energy demand diversity.
- **Scenario Fit:** Community-driven renewable integration.

Goals:

- Integrate CHP, smart heat pumps, and EVSE with renewable sources.
- Demonstrate peer-to-peer energy trading via blockchain.
- Leverage NILM (Non-Intrusive Load Monitoring) for intelligent load disaggregation.
- Demonstrate local electricity and heat flexibility services to the DSO.
- Demonstrate how urban communities can manage renewable integration efficiently.

Aran Islands Demonstrator (Rural / Island)



Figure 8. Pilot 1 Inis Mór in Aran Islands (Ireland)

- **Location:** Inis Mór, Aran Islands, Ireland.
- **Implementation Context:** Rural island grid with constrained capacity.
- **Scenario Fit:** Community-driven renewable integration.

Goals:

- Replicate NILM-based flexibility orchestration in a rural microgrid.
- Establish a community data repository to capture all energy related data from the islands.
- Validate renewable integration in a weak grid environment.
- Test P2P energy exchange within an island community.
- Demonstrate connectivity resilience under limited infrastructure.
- Demonstrate local electricity and heat flexibility services to the DSO.
- Explore replicability in other rural or islanded communities across the EU.

P1-SC-2: Smart Thermal Load Management

This scenario focuses on unlocking thermal flexibility as a controllable and dispatchable energy resource. By coordinating smart thermal assets (heat pumps, thermal storage, CHP), the scenario aims to improve demand-side flexibility, enhance grid stability, and decarbonize heat while ensuring comfort and operational reliability.

Urban Limerick Demonstrator (Thermal Layer)

- **Location:** Hartigan's Square (Figure 6 and Figure 7)
- **Scenario Fit:** Smart thermal load management

Goals:

- Optimize CHP units for flexible heat and power generation.
- Implement thermal storage strategies to shift load and balance peaks.
- Test dynamic control strategies under real-time signals.
- Enable multi-vector optimization (heat and electricity) at positive energy district (PED) scale.

UCC Beaufort Laboratory Demonstrator

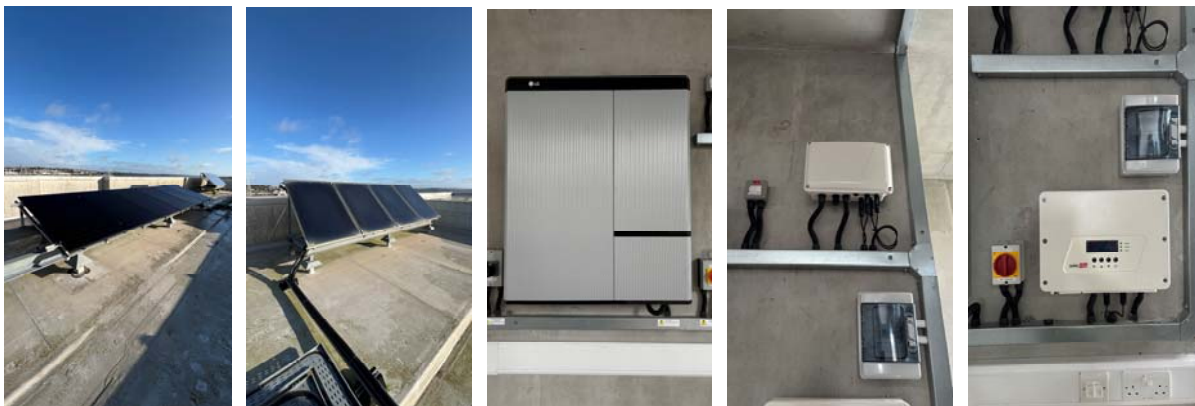


Figure 9. Pilot 1 Ringaskiddy in Cork (UCC Beaufort Laboratory), and details of Solar panels and Energy equipment

- **Location:** Ringaskiddy, Co. Cork (UCC Beaufort Laboratory).
- **Implementation Context:** Controlled HIL (hardware-in-the-loop) environment.
- **Scenario Fit:** Smart thermal load management.

Goals:

- Systematic validation of PV, battery, PCM (Phase Change Material), HP (Heat Pump), and NILM stacks before field rollout.
- Test flexibility control algorithms and interoperability under realistic conditions.

- Provide a replicable testbed for the DSO, and other stakeholders to trial flexibility mechanisms before deployment in live networks.
- Benchmark the impact of thermal flexibility on decarbonization goals and cost optimization.

Strategic Positioning and Impact

- National alignment: Supports Ireland's Climate Action Plan and high renewable penetration targets by showcasing operational solutions for flexibility and decarbonization of heat.
- EU alignment: Demonstrates real-world implementation of EPBD and RED III objectives through interoperability and flexibility.
- Innovation bridge: Integrates cutting-edge NILM analytics, blockchain-based P2P energy trading, and smart thermal management to create replicable blueprints for European communities.

3.2.1 Demonstrator Path

During the phase (M4–M12), Pilot 1.1 concentrates on establishing the infrastructure baseline. Activities included securing ethical/legal frameworks (GDPR, DPIA), procuring smart devices (heat pumps, CHP, EVSE, NILM sensors, edge gateways), commissioning the Beaufort Lab rigs, and preparing the Aran Islands replication package.

The second phase (M10–M18) focuses on rolling out instrumentation and control systems in Limerick buildings, commissioning CHP and thermal storage, and validating connectivity solutions in the Beaufort lab.

From M13 to M33, analytics and orchestration will mature: NILM classifiers will be tuned to detect thermal loads, FlexSim/FLEXUS will simulate co-optimized operation of heat and electricity, and blockchain trading trials will be conducted in Limerick.

Finally, M34–M42 will expand into advanced market participation scenarios, stress tests on the Aran Islands, and packaging of replication guidelines.

3.2.1.1 Setup & procurement activities

P1.1-SA1 – Cybersecurity, GDPR & Data Protection Baseline

A full data protection impact assessment (DPIA) has been developed to ensure lawful processing of household and community energy data. This covers consent mechanisms, data minimization, role assignment (UCC/IERC as controller, technology providers as processors), pseudonymization of NILM outputs, and retention policies. Technical measures include TLS-secured communication between gateways and UCC's AWS back-end, certificate-based device onboarding, and pseudonymization of personal identifiers. This framework guarantees that user privacy is protected while enabling real-time energy flexibility orchestration.

P1.1-SA2 – Instrumentation of Loads and Indoor Environments

Limerick sites are being fitted with a suite of measurement devices: temperature, relative humidity, CO₂, and illuminance sensors for occupancy inference; ultrasonic flow meters for heating circuits; and current transformers for key appliances. This instrumentation captures both comfort parameters and energy flows. The data provides ground truth for NILM model training, supports validation of heat pump COP, and enables demand response strategies that balance comfort and flexibility.

P1.1-SA3 – Smart Heat Pumps and EVSE Procurement

High-efficiency heat pumps with open APIs have been procured, enabling weather-compensated set-point control, curve optimization, and COP logging. These are paired with EVSE units supporting OCPP (Open Charge Point Protocol), allowing orchestration by FLEXUS. Together, the devices

create flexible electrical and thermal loads capable of being shifted in time according to DSO signals, market prices, or community-level agreements.

P1.1-SA4 – CHP and Thermal Storage Integration

At Hartigan's Square, a CHP unit has been coupled with thermal storage tanks. The CHP provides dispatchable electricity while charging thermal stores, which can later release heat without additional electrical demand. This arrangement supports up to 300 kW of aggregated flexibility. Thyristor-controlled charge–discharge systems are being installed to aggregate storage into a controllable asset. This demonstrates sector coupling by linking gas-based CHP operation with electrical demand response and thermal comfort delivery.

P1.1-SA5 – Edge Gateways and NILM Devices

Gateways have been installed to handle MQTT and REST communication with local devices, buffer data during outages, and apply edge-level analytics. NILM devices are being deployed in both Limerick and Beaufort Lab to disaggregate household loads, particularly heat pumps, immersion heaters, and resistive space heating. Their purpose is to extract flexibility without needing intrusive sensors in every circuit, making the solution more scalable.

P1.1-SA6 – Blockchain/DLT Infrastructure for Local Markets

Powerledger nodes are being installed in Limerick to manage peer-to-peer energy trading and flexibility exchange. The system supports transparent settlement, audit trails, renewable energy certificate tagging, and dispute resolution mechanisms. This ensures that households and SMEs can directly benefit from offering their heating and EV flexibility to neighbors or community operators.

P1.1-SA7 – Connectivity Resilience Testing at Beaufort Lab

Before deployment, communication systems are stress-tested in the laboratory using Zigbee and Matter protocols, dual-path Wi-Fi/4G/5G, and store-and-forward mechanisms. Fault injection scenarios include packet loss, latency spikes, and complete outages. The aim is to verify that control loops can gracefully degrade without losing safety or data integrity.

P1.1-SA8 – Aran Islands Replication Package

A replication package has been prepared for deployment in the Aran Islands, including edge caching routers, solar-powered backup communication, offline consent capture tools, and pre-trained NILM models. The kit is designed to operate in rural, intermittently connected environments, allowing testing of resilience and replication potential.

P1.1-SA9 – UCC AWS Cloud Tenant and DevSecOps Pipelines

A dedicated AWS tenant has been set up to host the pilot's data infrastructure. This includes an ingestion pipeline for sensor and NILM data, a data lake, curated marts, and APIs for applications such as FlexSim, FLEXUS, and Powerledger. DevSecOps pipelines have been implemented to automate deployment of firmware updates and NILM models, ensuring security patches are continuously integrated.

P1.1-SA10 – Safety and Commissioning Protocols

Safety is critical in commissioning electrical and thermal systems. Lock-out/tag-out protocols are applied during installation. CHP exhaust safety, thermal store pressure tests, EVSE electrical compliance, and hydraulic balancing are all documented and verified. Checklists are completed and archived for each installation site.

3.2.1.2 Development activities

P1.1-DA11 – FlexSim ↔ FLEXUS Co-Simulation for EV–Heat Integration

FLEXUS and FlexSim are being integrated to perform co-optimization of EV charging and heat pump operation. Multi-period optimization models (using MILP and heuristic algorithms) are applied to minimize peak demand, reduce CO₂, and maintain user comfort. The models can ingest price signals, weather forecasts, and DSO flexibility requests. Scenario libraries are being developed to simulate different levels of EV penetration, occupancy profiles, and renewable availability.

P1.1-DA12 – NILM Classifier Development for Thermal Loads

Thermal loads such as heat pumps and immersion heaters are difficult to classify due to overlapping signatures. NILM classifiers are being trained on ground-truth data from Beaufort Lab rigs and transferred to field sites. Transfer learning is applied to adapt classifiers between building archetypes. Confidence intervals are generated to ensure safe integration with control strategies.

P1.1-DA13 – DLT Smart Contracts for Flexibility Trading

Smart contracts are being developed for community-level trading. These define baseline calculation, settlement procedures, dispute resolution, and renewable certificate allocation. Tariff structures include flat rates, time-of-use pricing, and flexibility premiums. Residents and SMEs will interact with these through a simplified user interface.

P1.1-DA14 – Privacy-Preserving Analytics

To respect privacy, federated learning approaches are being trialed. Edge devices extract features locally and send pseudonymized updates to the cloud. Central models are aggregated without direct access to raw user data. This approach reduces data exposure while maintaining accurate forecasting.

3.2.1.3 Planning & Gantt

The planning of Pilot 1.1 from M4 to M18 focuses on setup and procurement activities, presenting some development activities while, integration and validation are being prepared for D4.2 and D4.3. The following Gantt chart covers therefore the above described activities from SA1 to DA14:

Table 2. Pilot 1.1 Gantt diagram

Pilot 1.1		2025										2026					
Code	Task	M4	M5	M6	M7	M8	M9	M10	M11	M12	M13	M14	M15	M16	M17	M18	
SETUP & PROCUREMENT																	
P1.1-SA1	Cybersecurity, GDPR & Data Protection Baseline																
P1.1-SA2	Instrumentation of Loads and Indoor Environments																
P1.1-SA3	Smart Heat Pumps and EVSE Procurement																
P1.1-SA4	CHP and Thermal Storage Integration																
P1.1-SA5	Edge Gateways and NILM Devices																
P1.1-SA6	Blockchain/DLT Infrastructure for Local Markets																
P1.1-SA7	Connectivity Resilience Testing at Beaufort Lab																
P1.1-SA8	Aran Islands Replication Package																
P1.1-SA9	UCC AWS Cloud Tenant and DevSecOps Pipelines																
P1.1-SA10	Safety and Commissioning Protocols																
DEVELOPMENT																	
P1.1-DA11	FlexSim ↔ FLEXUS Co-Simulation for EV–Heat Integration											→					
P1.1-DA12	NILM Classifier Development for Thermal Loads											→					
P1.1-DA13	DLT Smart Contracts for Flexibility Trading											→					
P1.1-DA14	Privacy-Preserving Analytics											→					

3.2.2 Current Infrastructure

The infrastructure in this sub-pilot includes a full IoT–Edge–Cloud stack supported by services like FLEXUS and Heat Sync. New acquisitions include protocol converters, submeters, environmental sensors, communication routers, and interfaces for PV/Heat Pump systems.

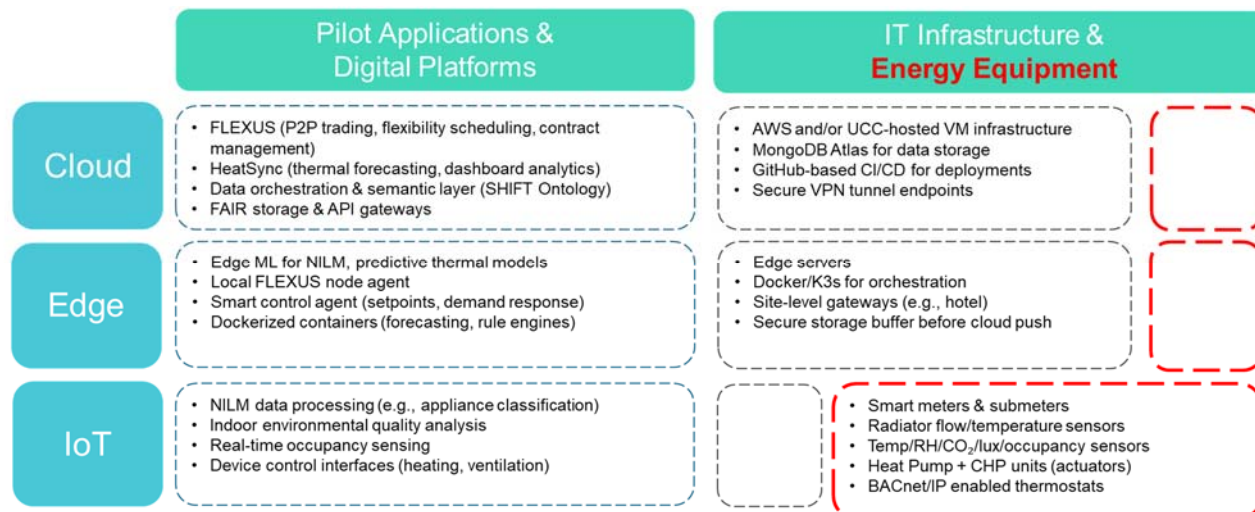


Figure 10. Infrastructure for Pilot 1.1

The figure illustrates a three-layer cyber-physical architecture that connects digital applications with physical energy infrastructure across the Cloud, Edge, and IoT layers. On the left, “Pilot Applications & Digital Platforms” represent the software environment used to manage, analyse, and optimise energy systems, while on the right, “IT Infrastructure & Energy Equipment” captures the physical and computational assets that enable real-world operation. This separation clearly distinguishes where intelligence is hosted versus where energy is physically generated, stored, or consumed.

At the **Cloud layer**, the focus is on centralised intelligence and orchestration. Here, advanced analytics, AI models, and platform services run in a cloud environment (e.g., AWS or private cloud infrastructure), providing a single coordination layer across multiple pilot sites. This layer supports market interfaces, flexibility aggregation, and overall system monitoring but does not directly control energy hardware.

The **Edge layer** introduces decentralised intelligence closer to the physical assets. It enables site-level forecasting, optimisation, and control while maintaining connectivity with the cloud. Edge servers, local gateways, or on-premise controllers buffer data, execute control strategies, and ensure operations continue even if cloud connectivity is interrupted. This is where site-specific optimisation happens, for example, managing thermal loads, DERs, or community-scale flexibility services.

The **IoT layer** forms the foundation of the architecture, directly interfacing with energy assets such as heat pumps, CHP units, EV chargers, PV systems, storage units, and environmental sensors. It enables real-time data acquisition and control through communication protocols such as Modbus, BACnet, and MQTT. This is the physical layer where energy is produced, stored, and consumed, and where NILM devices and sensor networks provide fine-grained visibility of local conditions.

Together, the three layers form a robust and interoperable architecture. Cloud provides coordination and intelligence, Edge ensures local resilience and optimisation, and IoT delivers physical interaction and control. This structure supports scalable, flexible, and secure pilot operation, enabling functions such as P2P energy trading, real-time flexibility management, and multi-site orchestration within the same unified framework.

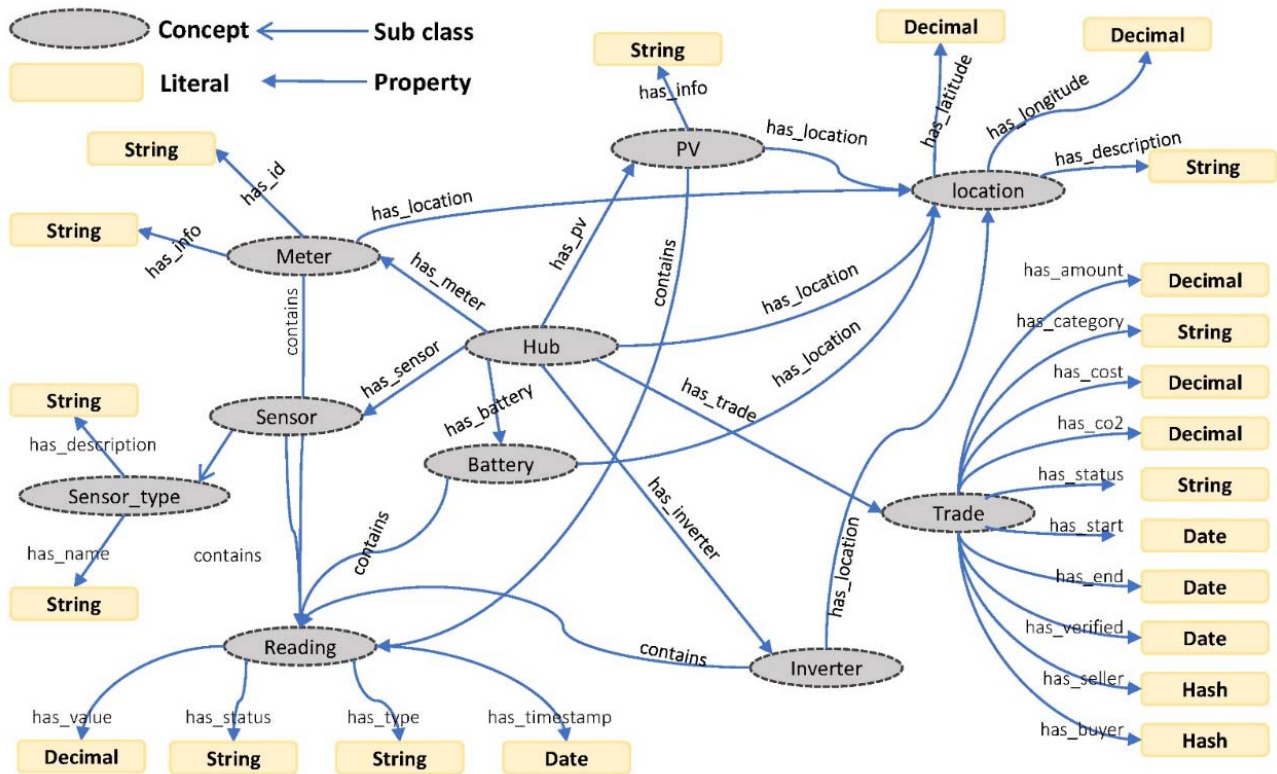


Figure 11. Pilot 1.1 Technical structure

3.2.2.3 Pilot applications and digital platforms

Cloud Layer

- FlexSim/FLEXUS for forecasting and flexibility simulation.
- Blockchain trading platform hosted in Limerick for P2P energy exchange.
- AWS-based hosting of FlexSim/FLEXUS ¹ and trading nodes.
- AWS prepared for other technical solutions from technical work packages to be integrated.
- REST APIs and visualization dashboards for demand/supply forecasting.

Flexus is a modular cloud-based platform designed to enable the integration, orchestration, and management of flexibility services and P2P across multiple domains and actors. Built on a scalable AWS architecture with containerized services, it provides secure APIs, real-time data processing, and semantic interoperability with energy and building ontologies. Flexus acts as the central intelligence layer, connecting edge devices, forecasting and optimization modules, and user-facing applications to support services such as demand response, P2P energy trading, and grid interaction. Its design emphasizes extensibility, allowing new tools, algorithms, and pilot deployments to be seamlessly integrated.

¹ <https://www.sciencedirect.com/science/article/abs/pii/S037722179190131E>

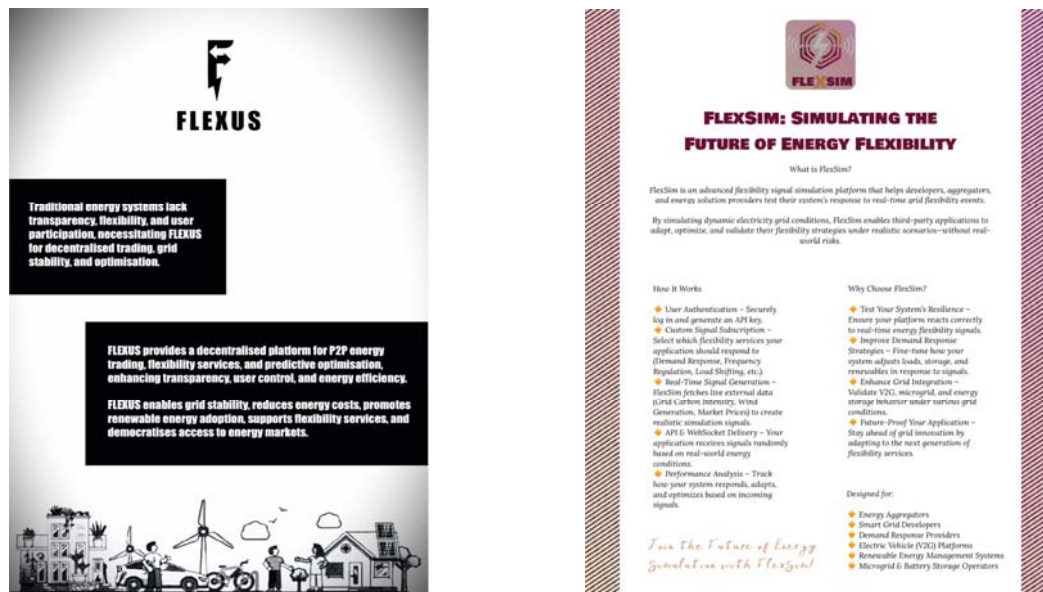


Figure 12. Pilot 1.1 Cloud layer FlexSim/FLEXUS

FlexSim is a simulation environment purpose-built to model, test, and validate flexibility scenarios before deployment on operational platforms like Flexus. It provides deterministic, stochastic, and AI-driven modelling capabilities to replicate building, community, and grid interactions under a range of technical, economic, and behavioral conditions. By simulating multi-agent interactions, energy flows, and market signals, FlexSim supports the evaluation of flexibility strategies, business models, and control mechanisms in a controlled environment. This makes it a critical tool for de-risking deployments, optimizing configurations, and generating synthetic datasets for advanced analytics and machine learning applications.

UCC AWS Architecture

The next figure illustrates a modular and containerized cloud environment designed for secure, scalable, and automated data management and application deployment. The architecture is structured in layers that collectively support continuous integration, real-time operations, and high-availability services across multiple environments (development, testing, and production). It leverages AWS native services, container orchestration, and version control systems to ensure resilience, traceability, and streamlined governance throughout all system operations.

At the core of the setup are compute and orchestration services, primarily built around Amazon EC2 instances running Docker containers. These containers encapsulate applications, APIs, and microservices, enabling isolated, portable, and scalable deployments. Container management is handled through ECS (Elastic Container Service) and ECR (Elastic Container Registry), which store and orchestrate container images pulled directly from version-controlled repositories in GitHub. Continuous integration and deployment pipelines ensure that new commits are automatically built, tested, and pushed to the relevant environment. Traefik, acting as a reverse proxy and load balancer, routes incoming requests to the correct containerized service while managing SSL certificates and internal traffic policies.

The networking and domain management layer is governed by AWS Route 53, which handles DNS routing and directs requests to the appropriate EC2 instances or container clusters. Public and private subnets within the VPC (Virtual Private Cloud) maintain clear separation between internal services and externally facing endpoints. Security Groups and IAM (Identity and Access Management) policies enforce strict access control, ensuring that users, roles, and services have only the permissions necessary for their functions.



Finally, the DevOps and management layer ensures the architecture remains modular, reproducible, and secure. GitHub repositories serve as the single source of truth for all code, configurations, and infrastructure-as-code scripts. Docker Compose and Terraform templates standardize deployments across environments, allowing new instances to be provisioned or scaled automatically. Together, these components create a fully integrated AWS ecosystem that combines automation, governance, resilience, and scalability in a unified architecture suitable for complex, data-driven applications.

This environment was designed to ensure that all required aspects of the pilots in Ireland and Croatia can be met through a secure, scalable, and modular architecture. The layered structure, combining compute, orchestration, networking, and governance, provides a unified framework that supports diverse applications, from data ingestion to analytics and user-facing services. Its containerized design allows rapid deployment of new modules, automated scaling, and seamless integration with external tools and APIs.

Moreover, by adopting version-controlled development workflows, continuous integration pipelines, and standardized security policies, the environment maintains high reliability and transparency across all operational layers. Importantly, the architecture is future-ready, built to accommodate additional tools and services developed under the technical work packages, ensuring long-term adaptability and interoperability as the platform evolves.

3.2.2.4 IT infrastructure and Energy equipment

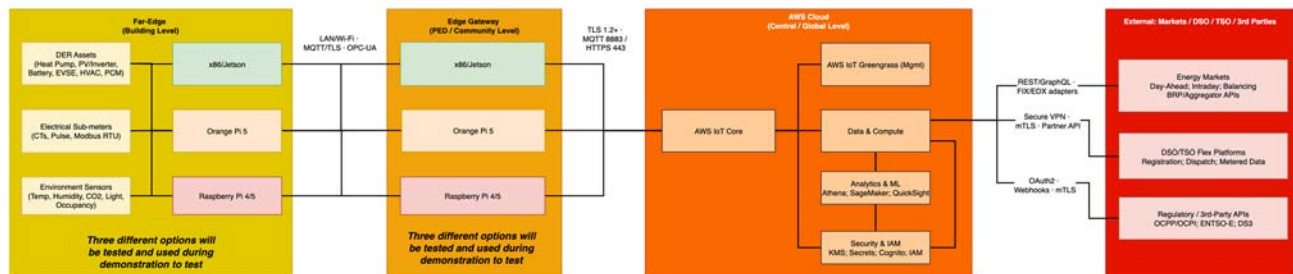


Figure 14. Pilot 1.1 UCC (Limerick and Aran Islands) Architecture

This architecture represents a complete, end-to-end system connecting physical assets within buildings to cloud-based intelligence and external energy systems. It is structured into four integrated layers: Far-Edge (Building Level), Edge Gateway (Community Level), Flexus Platform (Central Level), and External Systems (Markets, DSO/TSO/Partners). Each layer performs a specific role in the overall flow of data, control, and optimization.

At the Far-Edge, the system interfaces directly with the building's physical environment and energy infrastructure. It connects to Distributed Energy Resources (DERs) such as heat pumps, PV inverters, batteries, EV chargers, HVAC systems, and phase-change materials; to electrical sub-meters that measure circuit-level performance; and to environmental sensors monitoring temperature, humidity, CO₂, light, and occupancy. Three hardware configurations can be deployed depending on site requirements: Raspberry Pi 4/5, Orange Pi 5, or a high-performance x86/Jetson platform. These devices handle deterministic control and short-interval sensing, collecting detailed building data and forwarding it securely via LAN or Wi-Fi using encrypted MQTT/TLS or OPC-UA connections.

The Edge Gateway, operating at the district or community level, aggregates and coordinates information from multiple Far-Edge devices. It provides local buffering, rule-based control, and limited predictive functions to maintain autonomy when network connectivity is reduced. Like the Far-Edge, it supports three hardware options. Raspberry Pi 4/5, Orange Pi 5, and x86/Jetson, which can be interchanged and tested depending on performance and cost targets. Each gateway runs AWS IoT Greengrass, containerized services for data handling, and local time-series storage for resilience. It forms the bridge between building-level operations and the Flexus cloud environment, transmitting and receiving data through secure TLS connections.

The Flexus Platform serves as the central intelligence and orchestration layer. Built upon scalable cloud infrastructure, it hosts all major functions for data ingestion, analytics, machine learning, and device management. It collects and processes time-series data from the Edge Gateways, runs predictive and optimization models, and returns updated control schedules and operational parameters. Within Flexus, services equivalent to IoT Core handle device communication and synchronization, while modules based on Kinesis, Timestream, and S3 manage large volumes of telemetry data. Analytics and visualization tools, such as SageMaker, Athena, and QuickSight, enable forecasting, anomaly detection, and performance reporting. Security, identity, and permissions are controlled through a dedicated IAM, KMS, and Cognito framework, ensuring data protection and compliance. Flexus thus represents the operational and analytical "brain" of the architecture, uniting the deterministic, stochastic, and AI layers of the symbiotic building model.

The final layer, External Systems, connects Flexus to wider energy ecosystems, including energy markets, distribution system operators (DSOs), transmission system operators (TSOs), and regulatory or third-party platforms. Through secure APIs and managed interfaces, Flexus exchanges aggregated flexibility data, receives operational or dispatch signals, and provides verification and reporting information. This enables buildings and communities to participate in flexibility programmes, energy trading, and grid-support services while maintaining robust cybersecurity boundaries and privacy controls.

Overall, the architecture ensures seamless interaction from sensor to market. The Far-Edge collects granular, real-world data; the Edge Gateway integrates and contextualizes it locally; the Flexus Platform performs deep analysis and learning; and the External Layer connects the system to the wider energy domain. This four-tier structure creates a resilient, modular, and intelligent digital ecosystem where buildings evolve into active, symbiotic agents capable of optimizing energy performance, enhancing comfort, and contributing to grid stability in real time.

Edge Layer

- **Local Interface & Monitoring:** Each site includes a local HMI dashboard for real-time visualization of household or building-level demand, generation, and environmental data.
- **Gateway Functionality:** Edge Gateways (Raspberry Pi 4/5, Orange Pi 5, or Jetson/x86 mini-PC) aggregate data from multiple Far-Edge devices, execute rule-based control, and perform local analytics.
- **Autonomous Operation:** Gateways host AWS IoT Greengrass components for local buffering, deterministic control, and fallback operation in case of cloud or network interruptions.
- **Data Transmission:** Data and control messages are securely exchanged with the Flexus Platform using TLS-encrypted MQTT channels.

Energy Equipment

- **Smart Building Assets:** Integration of heat pumps, PV inverters, battery storage, EV chargers, HVAC systems, and phase-change materials (PCM) at the building level.
- **Renewable Coupling:** In the Aran Islands, grid-connected PV systems enable local renewable generation and contribute to on-site flexibility.
- **Thermal Storage & Load Shifting:** Domestic hot water tanks and PCM units are monitored and controlled through the Far-Edge devices to support demand response.
- **Local Validation Nodes:** High-performance x86/Jetson units can host distributed ledger or blockchain modules for peer-to-peer energy trading validation and local transaction integrity.

Sensors, Metering and Gateways

At the Far-Edge (Building Level), all physical sensing and sub-metering are integrated through secure, low-latency links (LoRaWAN, Modbus RTU/TCP, BLE, or Wi-Fi). These provide continuous environmental, electrical, and operational data to the Edge Gateway and Flexus Platform.

Table 3. Pilot 1.1 Sensors and Meters

Image	Type	Device / Description
 CONNECTED THINGS	Environmental Monitor (with Display)	Milesight AM319 – Indoor environmental sensor measuring CO ₂ , PM, and O ₃ with on-device display.
 CONNECTED THINGS	Environmental Monitor (without Display)	Enginko LoRaWAN Indoor Sensor – VOC, LUX, and CO ₂ monitoring for distributed rooms and zones.
 CONNECTED THINGS	Electricity Monitor	HotDrop Single Phase Electricity Meter EU868 – sub-metering of circuits and end-use appliances.
 CONNECTED THINGS	Water Meter	Quandify CubicMeter Smart Water Meter & Leak Detector (Copper) – flow and leak detection for domestic hot and cold water lines.

	Window Sensors	Browan Tabs Smart Building Sensors – Door & Window US915 – occupancy and thermal envelope monitoring.
	LoRaWAN Gateway	TEKTELIC Kona Micro LoRaWAN Gateway – Cellular EU868 – connectivity hub for LoRa-enabled sensors and meters.
	Phase Change Materials (Storage)	PCM will be available in the Beaufort Laboratory so we will demonstrate flexible storage of heat
	Raspberry Pi / Orange Pi / x86/Jetson	At the far-edge and edge levels we will test three different options to evaluate their capabilities in providing the necessary operations for flexibility services. As we will be applying a multi-layered modelling approach some operations may prove to be challenging for Raspberry or Orange Pi, so we will see what each is capable of and present our findings throughout the implementation.

3.3 PILOT 1.2 – COMMUNITY-DRIVEN RENEWABLE ENERGY INTEGRATION AND SMART THERMAL LOAD MANAGEMENT

Pilot 1.2, located in France, focuses on enhancing Renewable Energy Sources (RES) integration in smart grids and improving demand-side energy flexibility and management in a context of decarbonization/electrification of energy uses. Pilot 1.2 therefore consists of two distinct scenarios, each with specific setup, procurement and development activities:

P1-SC5 – Residential energy management

This scenario –Cloud-Edge-IoT-based energy management services for end-consumers-- consists in the deployment of a CEI architecture at the participant's homes to enable cloud- and edge-based energy services for energy management and flexibility for the end-consumer. This scenario will be implemented in the Municipality of Mountrouge (France, South of Paris), by deploying CEI-based services to enable real-time energy management and consumption optimization for residential end-users. The solution integrates smart metering, local processing, and AI-based disaggregation algorithms to enhance visibility, efficiency, and privacy.



Figure 15. Pilot 1.2 City of Mountrouge

P1-SC6 – Advanced load scheduling

The second French scenario –Virtualization for smart substations to improve RES integration— is dedicated to the virtualization of voltage control functions at the secondary substation level of a smart grid. The primary objective is to optimize the integration of RES by proposing a more distributed and virtualized voltage management system. This scenario is studied within the EDF Lab Smart Grids and Concept Grid laboratories, which closely replicate the real-world conditions of the network operated by the French Distribution System Operator (Enedis).



Figure 16. Pilot 1.2 Concept grid

The link between the two scenarios is the need for increased flexibility both on the grid and demand side for the resilience of the electric grid and the objectives of CO₂ emissions reduction. The Linky smart meter acts as a pivotal device between the scenarios.

3.3.1 Demonstrator Path

Concerning **P1-SC5**, the deployment of service infrastructures aimed at end-consumers is necessarily constrained by technical requirements and legal frameworks, as well as social and communication obligations towards participants. This leads to several phase in the Pilot implementation. Both technical/legal and social side of P1-SC5 are processed in parallel. The technical/legal side consists in the definition, go/no-go, implementation and deployment of the software and hardware architecture and GDPR processes. The social and communication side consists in the recruitment of participants, sociological characterization, development of the collaboration with the Municipality and continuous communication activities.

The first building phase has concentrated on the definition of the first version of the technical CEI architecture to be deployed in the pilot, and the validation of the proposed architecture, choice of protocols and cybersecurity measures by the internal cybersecurity and GDPR EDF bodies. The validation was obtained in July 2025. The general architecture and protocols of the solution are shown in the next figure. This architecture will serve as the basis for the blueprint definition for P1-SC5.

Several procurement activities were carried out on the hardware side: definition and build of the EDF gateways that will be deployed in the Pilot; laboratory test of the sensors to be integrated. On the software side, software developments consisted in the software integration of the hardware components into the Pilot architecture, and the adaptation of the IoT backend to prepare for O-CEI deployment. Ongoing activities consist in the continuous software development of the solution, and the definition of the data model and database structure that will be used in the Pilot, based on standardized data models and other EU initiatives (e.g., outcomes of the Omega-X project), as well as the continuous development of the mobile application aimed at the end-users, that will act as interface with the energy services.

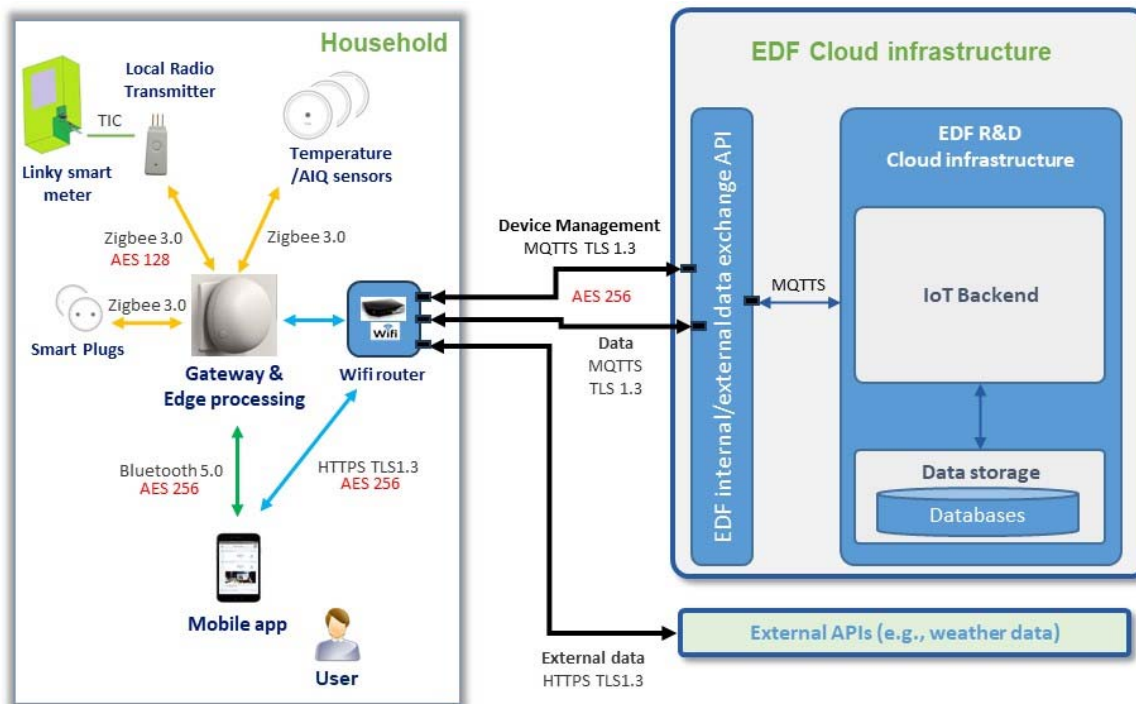


Figure 17. Pilot 1.2 SC5 architecture

P1-SC6 is a scenario deployed in a professional environment, without involvement of the general public. The implementation phases are as follows: use cases definition and problem modeling (investigation of edge- and virtualization-based solutions to resolve grid congestion), choice of software and hardware solutions accessible within the O-CEI consortium, laboratory studies, and deployment of the developed architecture to assess the benefits of decentralization and virtualization.

The first building phase consisted in the definition of the smart grids assets that will be part of the Pilot use case (scope), of the target architecture and its requirements, and of the hardware and software needs (for gateways, application servers, OS and hypervisors). The available options are currently being discussed between technology provider partners in the O-CEI consortium. The architecture of the scenario is shown in the next Figure.

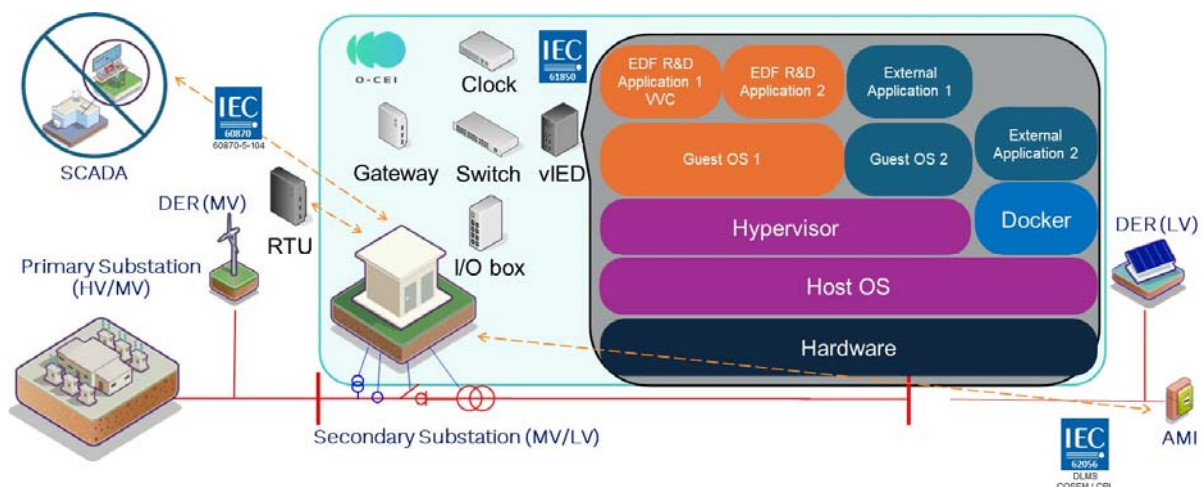


Figure 18. Pilot 1.2 SC-6 architecture

The application site of P1-SC6 is the Concept Grid laboratory, which has the following overall characteristics:

Table 4. Concept Grid laboratory characteristics

ARCHITECTURE	
Networks	7 km LV; 3 km MV; 120 km simulated MV (RLC).
Substations	One primary substation (20 MW transformer); five MV/LV distribution substations (250 kW to 1.5 MW); one H61 transformer (160 kW).
Residential area	5 houses; charging stations (standard and fast); PV panels; micro wind turbines; mobile three-phase resistive loads (600 kW); fixed resistive loads (400 kW).
Storage system (SS)	SS Li-ion battery (50 kW; 106 kWh); SS supercapacitor (70 kW); SS Li-ion battery (160 kWh); hydrogen system.
High-power EV charging test platform	Concrete slab of 160 m ² able to host various ultra-fast EV charging stations; supplied through a 1.5 MW transformer.
STRESS TEST	
MV networks	FAULTS: Overhead or underground; single-phase-to-earth, two-phase, or three-phase; three neutral grounding modes: resistive, compensated, active
LV networks	ROTATING GENERATOR: 50 kVA (source or discharge; voltage disturbance ($\pm 11\%$ around U_n); frequency disturbance (± 3 Hz around 50 Hz) POWER AMPLIFIER: 120 kVA (source) / 60 kVA (load); voltage, current, and frequency disturbances; harmonic generation up to 25 kHz. SHORT-CIRCUIT: single-, two-, or three-phase.
TELECOMMUNICATION	
Single-mode optical fiber	Command and control (IEC 61850); data transfer via IP-MPLS. Supervision, remote control; measurements; advanced network functions.
Power network	Power-line communication (PLC). Linky smart meter architecture; load control.
Wireless	Study of all protocols (including resistance to disturbed electrical environments). Remote terminal unit (RTU) control ; sensors.

3.3.1.1 Setup & procurement activities

From P1.2a-SA1 to P1.2a-SA9 – Setup of P1-SC5 (End consumer side)

For Scenario P1-SC5 Setup & procurement activities on the technical side include:

Hardware improvement of the EDF Gateway that will act as the central device for data retrieval, storage, Edge processing and service deployment, and for sending data to the EDF Cloud. The Gateway is tested in laboratory conditions in the Smart Home Experience Lab (SHEL) at EDF R&D—in which a Linky smart meter and IoT objects are available—, as well as in real test homes for continuous testing and development.



Figure 19. Pilot 1.2 – EDF Gateway

Laboratory test of sensors that will be deployed to measure electrical load curves and energy consumption, as well as air temperature and quality in the homes:

- Onics (formerly Develco Products) **Smart Plugs** (accuracy +/- 5% @ 0,2-16A, resolution 1W, Zigbee 3.0, remote controllable)



Figure 20. Pilot 1.2 – Onics Smart plug

- Lixee Zigbee 3.0 **Local Radio Transmitter** (plugged into the Linky smart meter to retrieve the global electrical load curve)



Figure 21. Pilot 1.2 – Local Radio Transmitter

- Indoor environment **sensors**: temperature (resolution: 0.1°C, accuracy $\pm 0.2^\circ\text{C}$, sample time: 30 s), humidity (resolution: 1% RH, accuracy 2%) & Indoor Air Quality (IAQ) (resolution: 1 - 32 ppb)



Figure 22. Pilot 1.2 – Temperature, humidity, IAQ sensor

This hardware as well as the global architecture has been tested in the Smart Home Experience Lab (SHEL) laboratory on EDF R&D premises. The use of smart cables or relays is being assessed at the SHEL lab.

From P1.2b-SA1 to P1.2b-SA6 – Setup for P1-SC6 (Smart grid side)

Hardware and software requirements are being refined for the layers of the proposed architecture:

- Gateway and switch
 - Gateway mission**: must handle protocol conversion in real-time. In this case, it must interface with Linky smart meter, IEC 61850 client/server and IEC 60870-5-104 Remote Terminal Units (RTU).

- **Switch mission:** must handle the telecommunication between different equipment and virtual applications
- **Target design:** Intel and THDeg's SDN vSwitch in Intel architecture high performing server, capable of SoC partitionning and SR-IOV virtual functions capable of HSR/PRP redundancy.
- **Progress:** still in discussion to assess adequacy with the pilot's need
- Average measure computation & processing application
 - **Mission:** compute and process power measures from Linky smart meter, IEC 61850 client/server and IEC 60870-5-104 RTU, to generate the right input for Dynamic Voltage Regulation application, developed by EDF R&D.
 - **Target design:** data broker, compatible with IEC 61850, processing 1 min average measurements.
 - **Progress:** to be initiated
- Real time application server, Real time OS and hypervisor
 - **Mission:** must be able to host both private applications from EDF R&D and third-party applications
 - **Target design:** undecided yet between E4S architecture, SEAPATH and custom environment, dealing with real-time interactions between equipment external to the secondary substation automation system (Linky, DEIE, ...)
- Local voltage control application
 - **Mission:** ensure the supervision and control of the voltage level output of the transformer to give the right commands to the On Load Tap Changer.
 - **Target design:** it must run in the real time application server, communicating with the other applications through the virtual switch, complying with IEC 61850.
 - **Progress:** discussion ongoing with Schneider Electric

Hardware and software options are in discussion with the consortium's technology providers.

3.3.1.2 Development activities

P1.2a-DA10 – Further development of P1-SC5 IoT backend and mobile application

Software development for adapting the IoT backend architecture to the company's internal IT/API constraints and ensuring operational maintenance. Firmware updates of the Gateway. Ongoing software development of the mobile application to enable data historization and service administration

P1.2b-DA7 – Development of scenario software components for laboratory testing

This activity is scheduled to start on M13 and further details will be provided for consequent deliverables

3.3.1.3 Planning & Gantt

The following Gantt diagram lists all the activities setup and procurement activities, and Development activities since the beginning of the work package on M4 until M18.

Table 5. Pilot 1.2 Gantt diagram

Pilot 1.2 - (End-consumer)		2025										2026					
Code	Task	M4	M5	M6	M7	M8	M9	M10	M11	M12	M13	M14	M15	M16	M17	M18	
SETUP & PROCUREMENT																	
P1.2a-SA1	Cyberscurity and GDPR file for Corporate validation																
P1.2a-SA2	Validation of participation Municipality of Montrouge																
P1.2a-SA3	Preparation oF recruitment material and Participation Agreement																
P1.2a-SA4	Participant recruitment activities																
P1.2a-SA5	Laboratory tests of objets and architecture																
P1.2a-SA6	D+1 load curve retrieving and monitoring																
P1.2a-SA7	Installation of v0 infrastructure (equipment and service platform) at the participant's homes																
P1.2a-SA8	Initial characterisation of recruited participants energy consumption and practices																
P1.2a-SA9	Initial characterisation of recruited participants energy consumption and practices																
DEVELOPMENT																	
P1.2a-DA10	Software development at the Gateway, IoT backend and mobile application level																
Pilot 1.2 – (Smart grids)																	
SETUP & PROCUREMENT																	
P1.2b-SA1	Use case definition and problem modeling																
P1.2b-SA2	Definition of general architecture																
P1.2b-SA3	Definition of target design & O-CEI tech. providers inputs for Gateway and switch																
P1.2b-SA4	Definition of target design & O-CEI tech. providers inputs for measure computation and processing application																
P1.2b-SA5	Definition of target design & O-CEI tech. providers inputs for real time OS and application server																
P1.2b-SA6	Definition of target design & O-CEI tech. providers inputs for control application																
DEVELOPMENT																	
P1.2b-DA7	Development of Scenario software components for laboratory testing																

3.3.2 Current Infrastructure

The infrastructure combines mobile applications, edge AI deployment, and connectivity solutions (e.g., Zigbee) with secure cloud services hosted at EDF R&D premises. New acquisitions include gateways, smart plugs, radio transmitters, and connected temperature sensors.

The following figures show the Pilot Applications & Digital Platforms as well as the IT infrastructure and Energy equipment that have been identified for the correct implementation of the sub pilot:

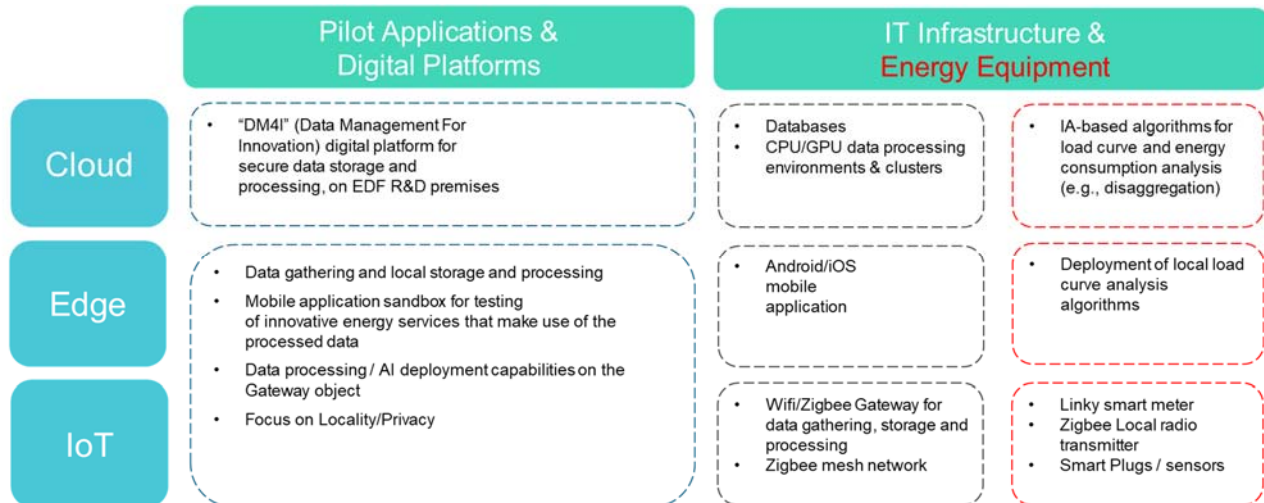


Figure 23. Infrastructure for Pilot 1.2 P1-SC5 - CEI-based energy management services for end-consumers

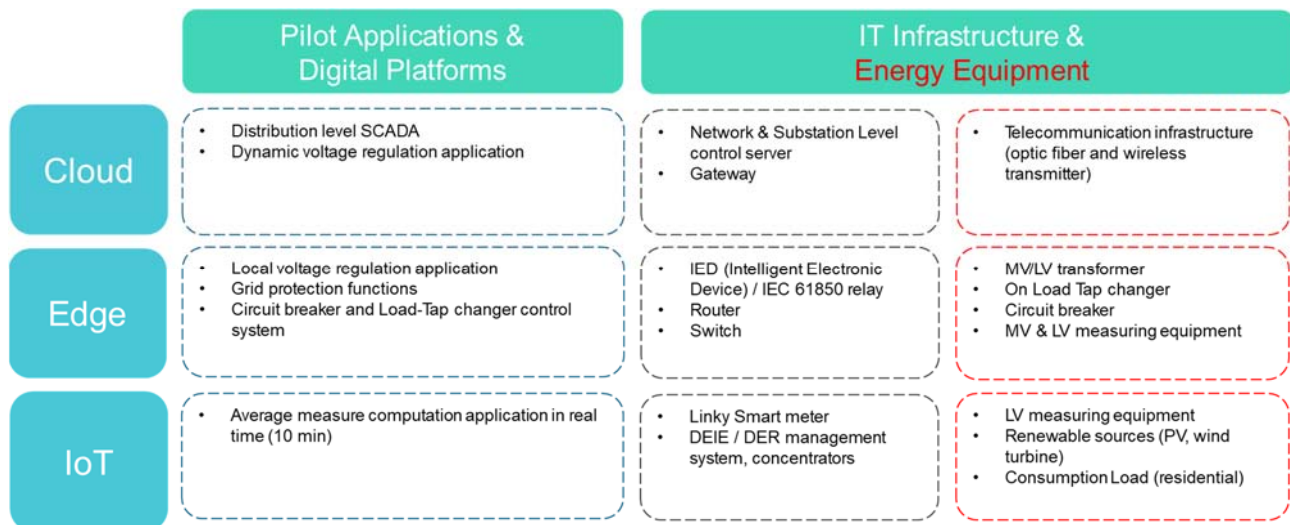


Figure 24. Infrastructure for Pilot 1.2 P1-SC6 - Virtualization for smart substations to improve RES integration

3.3.2.1 Pilot applications and digital platforms

3.3.2.1.1 P1-SC5 – End consumer

The **Cloud layer** is based on the generic EDF and the specific EDF R&D infrastructures:

- The generic EDF infrastructure provides secure connection and routing to/from the outside of the internal EDF network, for specifically approved applications. Data transmission from Gateway to EDF is done using the MQTTS protocol.

- The **IoT backend** of the Pilot architecture is developed and maintained on EDF R&D infrastructures.
- The **DM4I platform** (*"Data Management for Innovation"*), also hosted on EDF R&D infrastructures, supports secure data storage (traditional databases and big data infrastructures) and provides GPU/CPU resources and Linux/python environments for data science applications and developments, such as load curve analysis algorithms (e.g., load curve disaggregation).

Edge & IoT Layers:

- Data centralization, local data aggregation and storage.
- Mobile application sandbox for testing of innovative Edge-based energy services that make use of the processed data.
- Data processing and AI algorithms deployment capabilities on the Gateway object (Rpi Compute Module & linux environment).
- Focus on Locality of data and Privacy thanks to Edge capabilities.

3.3.2.1.2 P1-SC6 – Smart grid virtualization

Cloud Layer

The cloud supports remote monitoring and control of substations using centralized platforms. It collects data from substations, executes analytics for dynamic voltage regulation, and interacts with the distribution-level SCADA system to optimize renewable energy integration and distribution grid stability through dynamic voltage regulation optimization. The goal of this pilot site is to decentralize smart grid function from cloud layer to the edge layer using virtualization technology. Thus, the SCADA system is not needed in this pilot, nor the physical infrastructure running the dynamic voltage regulation, but only the very application to be ported into edge layer infrastructure.

Edge & IoT Layers

- At the secondary substation level, aside from the physical electric infrastructure (like power transformer and voltage transformers), the secondary substation automation system runs in the edge layer real-time applications, especially the local voltage regulation application, grid protection functions and Load-Tap Changer and switchgear control.
- Average measure computation (every 10 minutes). These applications are hosted on local servers directly interfacing with secondary substations and grid devices (DER RTU and client's AMI).

3.3.2.2 IT Infrastructure and Energy equipment

3.3.2.2.1 P1-SC5 – End consumer

Cloud Layer

- **Databases and GPU/CPU clusters** support intensive analytics workloads.
- **AI-based algorithms** for intensive, Cloud-side data processing algorithms such as advanced load curve disaggregation

Edge Layer

- **Gateways** support real-time AI execution and communication with Cloud services.
- **Local storage and processing** on the gateways ensure privacy and latency control.

- **Deployment** on the Gateway of edge-level AI-based algorithms for load curve and energy consumption analysis. This energy analysis encompasses applications such as energy profiling, disaggregation of energy uses, identification of energy use placement to analyse its optimality. The detail of the deployed applications will depend on the co-creation of the energy services with the pilot participants.

IoT Layer

- **Zigbee 3.0 mesh network** for local wireless communication and data retrieval from the sensors. The mesh capabilities of the IoT ecosystem allows for a broader range of radio communication inside the home, allowing to adapt the ecosystem to more complex home geometries.
- **Smart plugs and sensors** for granular consumption measurement of individual appliances. These measurements are used to analyse per device energy consumptions, as well as generate labelled data for load curve disaggregation algorithms.
- **Linky smart meters and local radio transmitters** for household-level energy consumption data acquisition. The Linky smart meter is the central device for the measurement of real-time energy consumptions in the home.
- **Connected temperature sensors** for acquisition of room temperatures in the home. These measurements are necessary for improving heating control and placement applications.

3.3.2.2.2 P1-SC6 – Smart grid virtualization

Cloud Layer

- AWS-based Network and Substation level **control server**
- **Gateway**

Edge Layer

- **Substation-level servers** with embedded control applications for protection, control, and regulation.
- Edge-level compute resources to ensure responsiveness and independence from the cloud in grid-critical tasks.
- **Intelligent Electronic Devices (IEDs) and IEC 61850 relays.**

IoT Layer

- MV/LV transformers, circuit breakers, and Load Tap Changers.
- IEDs, smart meters, concentrators, and full telecom infrastructure to guarantee secure and high-speed data exchange.
- Linky smart meters, DEIE/DER concentrators, and LV/MV measuring equipment. These enable full observability and actuation at the distribution level.
- Monitoring of both renewable generation (PV, wind) and residential loads.

3.4 PILOT 1.3 – WIDE SMART GRID OPTIMIZATION AND PROACTIVE APPLIANCE SYNCHRONIZATION

This pilot demonstrates how citizen-led Renewable Energy Communities (RECs), supported by Cloud-Edge-IoT intelligence, can drive the energy transition in islands and coastal regions. Krk's context is defined by tourism-driven seasonal demand, strong grassroots engagement through EKO Kvarner, and the legal transformation of its cooperative into a Renewable Energy Community. The pilot integrates distributed PV, batteries, and heat pumps into a smart decentralized system, initially targeting 100–300 households, with scaling potential toward 10,000+ homes in Rijeka and the wider coastal area.

P1-SC3: Wide Smart Grid Optimization

This scenario establishes an island-wide intelligent energy orchestration layer designed to optimise distributed renewable generation, storage and consumption in real time. It is based on Krk's strategic goal of achieving energy independence and climate neutrality by 2030 and leverages over 60 MW of planned PV, 30 MW of wind, 10 MW of biomass, and planned MPOWER 500 MWh of advanced battery storage.

The architecture integrates hybrid inverters (Victron MultiPlus-II), smart meters, EV chargers, and heat pumps across 100 – 300 households initially, scaling to thousands in the wider Rijeka coastal region. Data from these devices is collected at 2-second to 15-minute intervals through Zigbee 3.0, Modbus, OPC-UA, MQTT/REST APIs, and IEC 61850 protocols, and processed on HP Enterprise server infrastructure running Linux and MySQL.

Through a Cloud-Edge-IoT orchestration layer, the system enables:

- Real-time forecasting and optimization of generation, consumption and storage capacity.
- Load balancing and congestion management between distributed PV, wind, biomass and battery systems.
- Dynamic demand response using predictive models to flatten peaks and avoid curtailment.
- Integration of peer-to-peer energy trading signals within the Renewable Energy Community (REC).
- Support for islanding mode during grid disturbances to ensure critical services remain powered.

This forms the basis for a Virtual Power Plant (VPP), where distributed citizen-owned assets act in coordination with the local Distribution System Operator (HEP ODS). The wide smart grid optimization scenario demonstrates how a traditionally hierarchical island grid can evolve into a resilient, flexible microgrid ecosystem capable of autonomous operations.

P1-SC4: EV and Heat Pump Synchronization

This scenario targets demand-side flexibility by synchronizing electric vehicle (EV) charging infrastructure and heat pump systems with local renewable generation and grid conditions. On Krk Island, these technologies are key drivers of decarbonization: fossil-fuel boilers in homes and hotels are being replaced with LG Therma V heat pumps, and EV adoption is accelerating as part of transport electrification.

EVs and heat pumps are managed as active flexibility assets through:

- Load shifting and pre-conditioning: EV charging and heat pump operation are aligned with periods of high renewable generation (e.g. midday PV peak or overnight wind).

- Price signal responsiveness: devices respond automatically to time-of-use tariffs and REC trading signals.
- Predictive scheduling: AI-based forecasting tools enable pre-heating/cooling and vehicle charging ahead of demand peaks.
- Vehicle-to-Grid (V2G) integration: bi-directional charging supports grid stability during peak demand.
- Coordinated control via edge intelligence, reducing stress on local substations and avoiding simultaneous high-load events.

The objective is to turn two major end-use sectors—mobility and heating—into controllable, synchronized, and flexible resources. This not only reduces peak load and increases self-consumption of renewables but also supports REC energy trading, improves local grid stability, and enables the VPP to provide ancillary services.

Strategic Impact Across Both Scenarios

- Energy sovereignty: Community-owned assets form the backbone of the island's energy system.
- Grid resilience: Smart orchestration enables the creation of autonomous, microgrid-enabled clusters around key substations.
- Regulatory innovation: Krk's legal transition to a Renewable Energy Community unlocks peer-to-peer trading and collective asset management.
- Replicability: The pilot provides a blueprint for other European islands and coastal regions, particularly those with seasonal demand imbalances.
- Scaling: From 100–300 prosumers to 10 000+ households, enabling regional integration with Rijeka.

Additional Local Support

- Municipalities of Krk, Punat, Baška, Omišalj, Dobrinj, and Vrbnik – policy support and participation.
- Primorje-Gorski Kotar and Istria Counties – replication and regional cooperation.
- HEP ODS – grid operation and connection.
- Local cooperatives and citizens – participation and awareness-raising.

Summary Table

Table 6. Pilot 1.3 Summary table

Scenario	Focus	Core Technologies	Key Functions	Outcome
Wide Smart Grid Optimisation	System-wide orchestration	Hybrid inverters, PV, wind, biomass, 500 MWh storage, smart meters	Real-time forecasting, load balancing, congestion management, islanding, P2P integration	Grid-scale VPP with optimised generation and storage
EV & Heat Pump Synchronisation	Demand-side flexibility	EV chargers, LG Therma V heat pumps, edge controllers	Load shifting, price signal response, predictive scheduling, V2G	Flattened demand curve, renewable maximisation, flexible load control

For delivering these scenarios, a large-scale setup is being put in place, reaching over 300 small-scale solar PV systems, 10 MW solar plant, 30 MW wind, 10 MW biomass facilities, battery storage (500 MWh) and smart grid infrastructure with smart meters and EV charging networks responding to the following generic setup:



Figure 25. Setup in Pilot 1.3

The figures above illustrate the integration of distributed renewable generation (PV 60 MW, wind 30 MW, biomass 10 MW) with 500 MWh community storage and IoT-enabled households. Each node represents a prosumer equipped with a hybrid inverter, local controller, and sensors feeding data to the edge layer.

3.4.1 Demonstrator Path

Since Month 4 (M4), the pilot has made significant progress in both setup/procurement and development activities, laying the groundwork for a fully integrated and secure energy community infrastructure. The deployment of hybrid inverters across households has replaced legacy systems, enabling more advanced battery integration and improved cybersecurity. This has been complemented by the installation of community-level monitoring dashboards, enhancing collective energy awareness and enabling real-time visibility of production, consumption, and storage. Pilot household batteries have been acquired and installed, increasing flexibility, enabling peak shaving, and improving system resilience during critical periods. In parallel, the beta version of the Battery Management System (BMS) has been deployed in real households, with community training sessions ensuring that citizens understand the system's operation and flexibility potential. Furthermore, dedicated servers have been established in collaboration with HP for secure data aggregation, enabling edge-cloud orchestration and supporting future Virtual Power Plant (VPP) integration.

Looking ahead to Month 18 (M18), the focus will shift toward advancing the development activities. This includes the rollout of the community data collection software (beta), which will aggregate smart meter and IoT data to optimise flexibility and energy management at the community level. The full BMS software development will be completed, ensuring robust charge–discharge optimisation, extended battery lifetime, and seamless integration with hybrid inverters and dashboards. In addition, a dashboard and mobile application will be introduced to enhance prosumer engagement, enabling users to monitor energy flows, receive actionable insights, and participate actively in flexibility services. Together, these activities represent a cohesive, stepwise approach to building a secure, flexible, and intelligent energy community that can support VPP participation and interoperability with external energy systems.

3.4.1.1 Setup & procurement activities

3.4.1.1.1 P1.3-SA1 – Deployment of hybrid inverters across households

These devices replace legacy inverters, enabling integration of batteries and advanced control. They are a prerequisite for flexibility services and self-consumption. It is extremely important that with the new (high quality) hybrid inverters we will also increase cybersecurity, which is currently in a critical state.

P1.3-SA2 – Installation of community-level monitoring dashboards

Dashboards allow households and the community to visualize production, consumption, and storage in real time, fostering engagement and collective energy awareness. This optimizes the total production and consumption and prepares the realization of the virtual power plant.

P1.3-SA3 – Acquisition and installation of pilot household batteries

These batteries provide local flexibility, enable peak shaving, and improve resilience by supporting households during high demand or low generation periods. For security reason, batteries are developed using new technology.

P1.3-SA4 – Deployment of BMS (beta) and community training

The Battery Management System (BMS) beta will be tested in real households (including a lot of stress test), with training sessions ensuring citizens understand system behavior and flexibility potential.

P1.3-SA5 – Establishment of servers (via HP collaboration) for data aggregation

Dedicated servers ensure secure storage and processing of household and community data, enabling edge-cloud orchestration and future VPP integration. The server allows for 15-minute data sampling with a tendency to increase to 2 seconds in the future, which is crucial for both network security and the virtual power plant.

3.4.1.2 Development activities

P1.3-DA6 – Development of community data collection software (beta June 2025)

This software aggregates 15-min HEP (National Energy Operator) meter data and IoT device inputs, providing a unified interface for analysis and flexibility optimization. It focuses on the production, storage and consumption in energy community as well as sharing data outside the community.

P1.3-DA7 – Full BMS software development (ready Jan 2026)

The BMS will optimize charge/discharge cycles, extend battery lifetime, and integrate with hybrid inverters and dashboards, creating a foundation for aggregation. It will support all industry standard protocols.

P1.3-DA8 – Dashboard and mobile app for prosumer engagement.

These tools provide intuitive interfaces for households to monitor energy flows, receive recommendations, and participate actively in community flexibility services. It also controls data and energy flow outside community (ex. other energy communities or Stock market

3.4.1.3 Planning & Gantt

The planning of Pilot 1.3 from M4 to M18 focuses on setup and procurement activities, and introduce some development activities. The following Gantt chart covers therefore the above described activities. Technically maps concurrent hardware roll-outs and software developments, ensuring that inverter deployment and BMS testing align with dashboard and data-collection software readiness milestones

Table 7. Pilot 1.3 Gantt diagram

Pilot 1.3		2025										2026					
Code	Task	M4	M5	M6	M7	M8	M9	M10	M11	M12	M13	M14	M15	M16	M17	M18	
SETUP & PROCUREMENT																	
P1.3-SA1	Deployment of hybrid inverters across households																
P1.3-SA2	Installation of community-level monitoring dashboards																
P1.3-SA3	Acquisition and installation of pilot household batteries																
P1.3-SA4	Deployment of BMS and community training																
P1.3-SA5	Establishment of servers (via HP collaboration) for data aggregation																
DEVELOPMENT																	
P1.3-DA6	Development of community data collection software																
P1.3-DA7	Full BMS software development																
P1.3-DA8	Dashboard and mobile app for prosumer engagement																

3.4.2 Current Infrastructure

This sub-pilot supports proactive synchronization of appliances such as EVs and heat pumps, based on energy availability and predicted demand. The infrastructure combines local storage, AI processing, and monitoring systems, and is supported by dedicated IT hardware including servers, databases, and smart metering units. The setup includes a redundant server farm, local Linux/MySQL environments, and advanced prosumer data management systems.

3.4.2.1 Pilot applications and digital platforms

Linux Mint 22.2 was chosen as the operating system, MySQL as the database, and Python and C++ as the programming languages. The use of artificial intelligence is also being considered, but the tools are still being analyzed.

Cloud Layer

- The pilot relies on a **self-managed data center** model using dedicated HP servers.
- Cloud-like services such as **data aggregation**, **smart grid optimization**, and **prosumer dashboards** are hosted locally.
- A **redundant server farm** ensures availability and fault tolerance.

Edge Layer

- **Real-time processing** of data from smart meters and sensors.
- Edge servers manage **grid balancing algorithms**, **load synchronization**, and **prosumer analytics**.
- Hybrid **Inverters**, Local **Controllers**, Battery Management System (**BMS**) **modules**, and Household **gateways**.

IoT Layer

- Local Linux-Mynt-based nodes interface with **prosumer meters**, **PV generation systems**, **storage**, and **consumption tracking devices**.
- **HEP** 15-min data feeds, **Modbus-connected** heat pumps, **Zigbee** smart plugs and sockets, sensors for household monitoring.

3.4.2.2 IT infrastructure and Energy equipment

3.4.2.2.1 P1-SC3 – Wide Smart Grid Optimization

The diagram and description in Scenario P1-SC3 – Wide Smart Grid Optimization present a layered cyber-physical energy system architecture designed to enable real-time orchestration, community energy engagement, and flexibility services for more than 300 households on Krk Island. The architecture combines IoT, Edge, and Cloud layers to deliver an integrated system capable of demand-response, P2P energy exchange, and Virtual Power Plant (VPP) readiness.

Cloud Layer – Central Orchestration and Optimization

At the cloud level, the architecture hosts the community-scale orchestration layer and the federated data platform, which play a crucial role in ensuring coordinated operation across all participating households.

Functions:

- Real-time energy flow optimization between the grid, household batteries, and flexible loads.
- P2P energy exchange settlement, market participation, and VPP dispatch algorithms.
- Central data aggregation, anonymization, and privacy-preserving analytics across 300+ connected households.
- Provision of dashboards and digital interfaces for operators and regulators to monitor community performance.

The cloud ensures scalability and interoperability, hosting high-level algorithms and market interfaces while preserving household privacy through federated data models. It also enables replication and extension of the solution to other islands or cities (e.g., Rijeka).

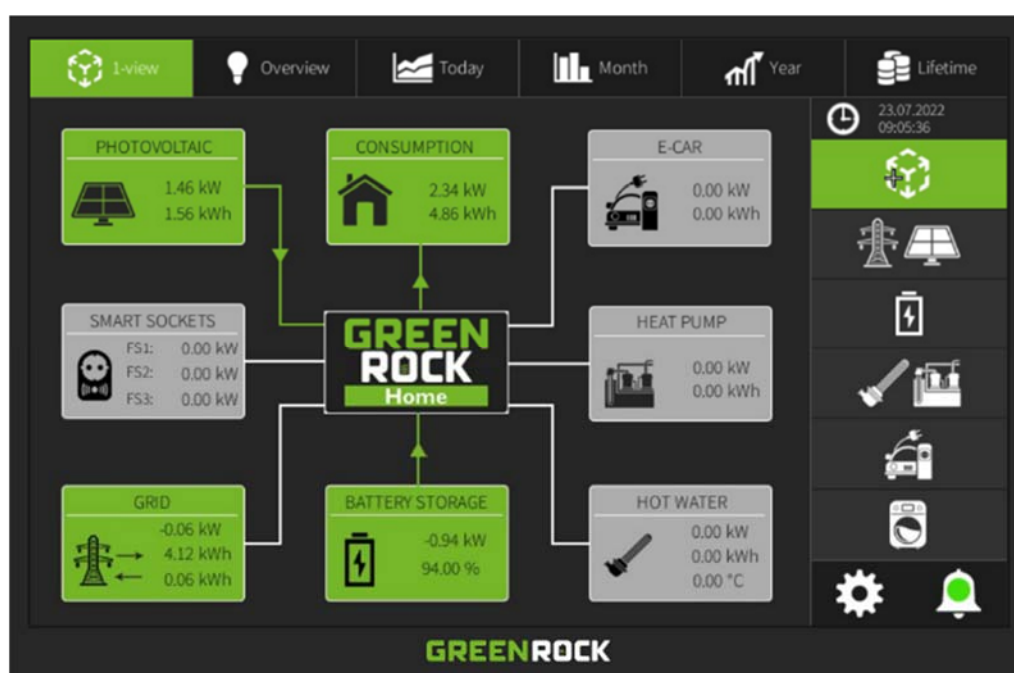


Figure 26. Pilot 1.3 GreenRock dashboard

The above image shows the live community-level monitoring interface hosted on the HP ProLiant server. The dashboard, developed in Python/Flask with Grafana-style visualizations, displays PV generation, battery SOC, consumption, and network performance in real time via MQTT data streams (2–15 s intervals). It forms the operational interface for citizen engagement and community energy awareness.

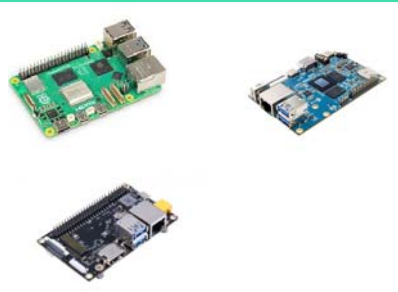
Edge Layer – Local Control and Intelligence

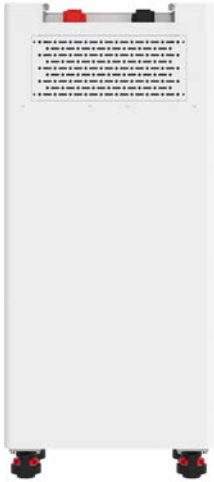
The edge layer is responsible for localized orchestration and site-level decision-making, acting as an intermediary between the IoT assets and the cloud orchestration services.

- Hosting local controllers for batteries, EV chargers, and heat pumps.
- Executing real-time control loops for load balancing and energy flow optimization.
- Ensuring continued operation during network interruptions by locally buffering data and implementing fallback control strategies.
- Enabling secure connectivity through OPC-UA and Modbus protocols for device-to-edge and device-to-cloud communication.

The edge ensures resilience and responsiveness, reducing reliance on continuous cloud connectivity and allowing faster reaction times to local grid fluctuations. It also allows local enforcement of dispatch strategies and peer-to-peer balancing within smaller network segments.

Table 8. Pilot 1.3 IoT devices

Image	Type	Device / Description
	Raspberry Pi / Orange Pi / x86/Jetson	Embedded Linux devices running Dockerized edge services. Each tested for CPU/GPU performance and energy efficiency. Executes local forecasting, load control, and fallback logic when cloud connection is lost. Multi-platform evaluation guarantees best cost-performance ratio for distributed control; essential for low-latency response and autonomy.
	Battery Management System (BMS)	Proprietary hardware (manufacturer TBC) with Modbus/TCP interface; monitors SOC, SOH, temperature. Optimises charge/discharge, ensures safety and data communication with hybrid inverters. Fundamental for battery lifetime extension and secure grid integration; compliance with IEC 62933-5.
	HP Enterprise ProLiant ML350 Gen11 Server	Dual-socket Xeon scalable architecture, 128 GB DDR5 RAM, redundant power and RAID-10 storage array (12 TB NVMe + 24 TB HDD). Core data aggregation and orchestration host at the cloud layer; runs MySQL databases, forecasting algorithms, and VPP orchestration modules. Provides high-performance, secure computing for community-scale optimisation, ensuring scalability, cybersecurity, and compatibility with future regional replication.

	RAID Array / Network Switches / UPS	Managed L3 Gigabit switches with VLAN isolation and redundant APC Smart-UPS 1500 VA. Maintains continuous, low-latency connectivity (< 50 ms) between local edge controllers and the server cluster. Ensures operational resilience, data integrity, and uninterrupted orchestration during short-term outages.
	Victron MultiPlus-II Inverter/Charger	48 V DC hybrid inverter (5 kVA) supporting on/off-grid modes via VE.Direct / Modbus communication. Manages bidirectional power flow between PV, battery, and grid. Replaces legacy inverters; enables advanced control, islanding, and flexibility services.
	Household Battery System (5–15 kWh LiFePO₄)	Lithium-iron-phosphate modules with integrated BMS, Modbus TCP interface. Stores surplus PV energy, enables peak shaving, and provides backup during outages. Improves self-consumption and resilience; key enabler for flexibility participation and VPP aggregation.
	LG Therma V Monoblock R32 Heat Pump	Inverter-driven air-to-water unit (7–16 kW) with Modbus control. Electrified heating/cooling integrated into domestic hot water systems; controllable through edge orchestrator. Replaces fossil boilers, supporting heating decarbonisation and enabling demand-side

		flexibility through predictive control.
	Smart Meters & IoT Sensors (Zigbee 3.0 / Modbus / MQTT)	Electrical, thermal, and environmental sensors gathering data at 2 s – 15 min intervals. Provide granular measurements of consumption, generation, and indoor conditions. Supply real-time inputs for NILM analysis, forecasting, and flexibility estimation.
	Software Stack (Linux Mint 22.2 + MySQL + Python/C++)	Installed across cloud and edge environments; uses open-source libraries for ML, orchestration, and visualisation. Core platform for forecasting, optimisation, and dashboard interfaces. Open-source base ensures transparency, security control, and cost efficiency.

IoT Layer – Field Devices and Real-time Sensing

The IoT layer represents the physical infrastructure deployed at the household and community level. It includes:

- Photovoltaic systems (PV) for renewable generation.
- Battery storage units with BMS software.
- EV chargers, heat pumps, smart plugs, and hot water control systems.
- Sensors and actuators connected via Zigbee 3.0 and other local communication standards.

Functions:

- Real-time sensing of generation, consumption, and flexibility capacity at the appliance level.
- Execution of control commands from edge controllers.
- Enabling prosumer engagement through user interfaces and dashboards at the home level.

The IoT layer enables fine-grained visibility and control, empowering households to actively participate in energy optimization and demand-response. It creates the data foundation upon which edge intelligence and cloud orchestration operate.

3.4.2.2.2 P1-SC4 – EV & Heat pump Synchronization

The **EV & Heat Pump Synchronization** represents a **household-level orchestration architecture** that integrates **PV generation, battery storage, heat pumps, and real-time control** to maximize self-consumption and flexibility provision. The system is structured across the **IoT, Edge, and Cloud** layers to enable efficient interaction between physical assets, local intelligence, and market-level services.

Cloud Layer

At the **cloud level**, the architecture provides the intelligence needed to manage and coordinate multiple households participating in flexibility schemes or Virtual Power Plant (VPP) programmes. The main functions are:

- Aggregation of household data from PV, batteries, and heat pumps.
- Forecasting algorithms for load and generation based on weather and usage patterns.
- Optimization of self-consumption and flexibility provision to the grid.
- Secure API layer (MQTT/REST) for monitoring, control, and integration with utility/market platforms.
- Future integration with IEC 61850 for grid-side communication.

The cloud layer enables scalable coordination, grid-compliant operation, and participation in flexibility markets, ensuring that local household assets contribute to system-wide optimization and decarbonization goals.

Edge Layer

The **edge layer** is the core of the household-level decision system. It enables fast and reliable control of loads and storage without depending on continuous cloud connectivity. The main functions are:

- Real-time orchestration between PV inverter, battery, and heat pump.
- Local control logic embedded in dockerized edge applications.
- Load prioritization and battery charging/discharging decisions based on instantaneous conditions.
- Data buffering and fallback operations in case of cloud disconnection.
- Secure communications with IoT devices through Modbus and local APIs.

The edge layer ensures resilience, low latency control, and energy independence, which are essential for enabling households to participate effectively in demand-response and VPP programmes.

IoT Layer

The IoT layer consists of the actual energy assets and devices installed at the household level. These form the physical backbone of the scenario. The main functions are:

- Photovoltaic (PV) system with hybrid inverter, enabling solar generation and integration with storage.
- Battery Energy Storage System (ESS) to store excess solar energy and provide flexibility.
- Smart heat pump (Therma V) integrated with a Domestic Hot Water (DHW) tank, controlled via Modbus interface.
- Third-party controller for real-time load orchestration and dynamic interaction with the grid.
- Connection to HEP's 15-minute metering service, providing standardized data without requiring proprietary smart meters.

This layer delivers real-time visibility and control of physical energy assets, enabling high levels of self-consumption, peak shaving, and flexible interaction with the grid.

Standards And Interoperability

P1-SC4 scenario applies a multi-protocol communication strategy:

- **Modbus** for inverter–heat pump communication.
- **MQTT/REST APIs** for real-time monitoring and cloud integration.
- **IEC 61850** planned for future VPP and grid-side communication.
- **Dockerized edge apps** enabling modular deployment and updates.

This ensures **interoperability** between multiple device vendors, control systems, and energy market interfaces, allowing the solution to scale and adapt to future needs.

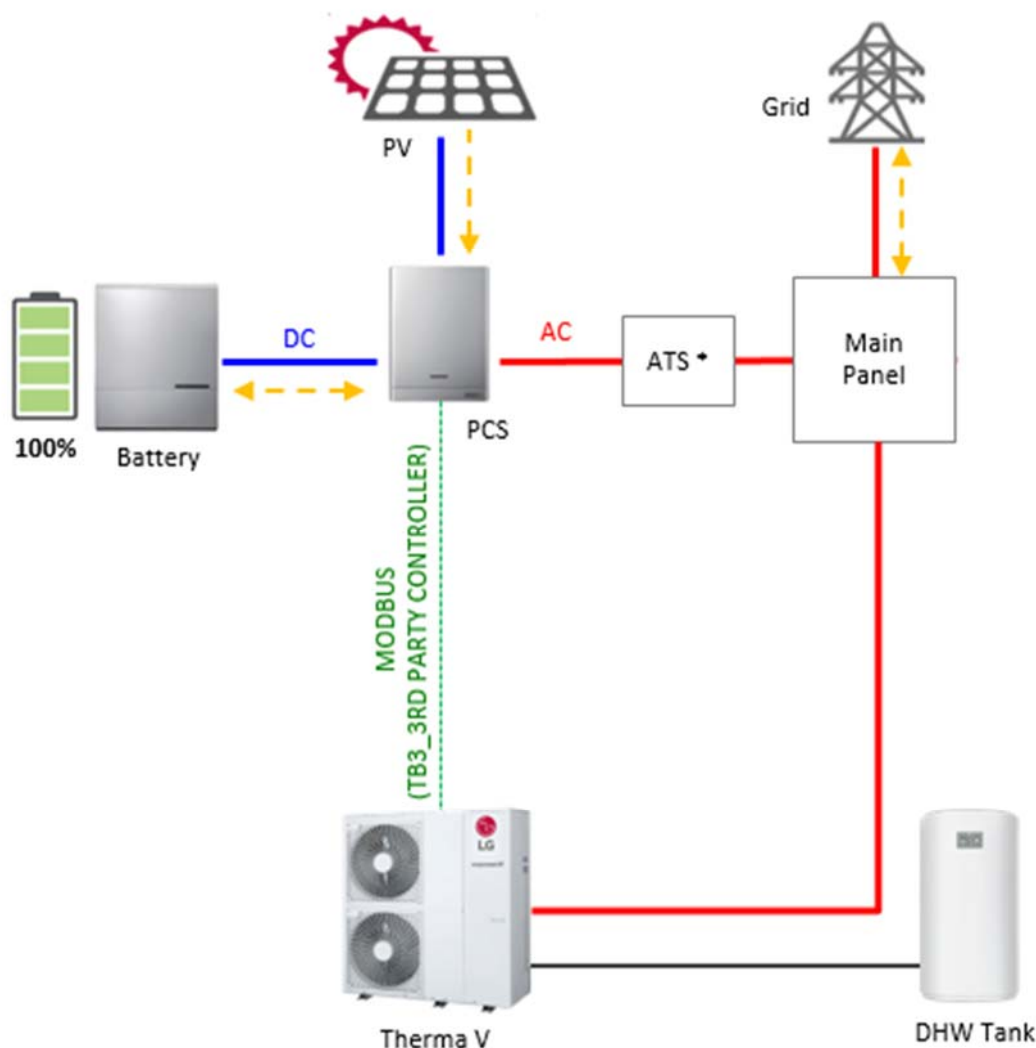


Figure 27. Pilot 1.3 KR-04 Infrastructure

The figure above depicts the three-layer cyber-physical architecture for EV and Heat Pump Synchronization. The IoT layer (PV, ESS, heat pump, EV charger) connects through Modbus and Zigbee to the Edge layer (local controllers running Dockerized applications) and onward to the Cloud layer (forecasting and market interaction). The figure demonstrates how modular control ensures low-latency operation (< 200 ms) even during network interruptions.

4 PILOT 2 – SOFTWARE-DEFINED VEHICLES FOR VAS IN URBAN AREAS

Pilot 2 operationalizes the Vehicle-as-Software (VaS) paradigm by embedding Software-Defined Vehicles (SDVs) into the cloud–edge–IoT continuum for urban mobility in Timișoara, Romania. Vehicles act as intelligent edge nodes that sense, compute, and coordinate with mobility and energy infrastructures via O-CEI-aligned components and European data spaces (Gaia-X/IDSA, Catena-X). Pilot 2 advances safety-critical AI (collision avoidance, lane-keeping assistance, adaptive cruise control), VaS orchestration, and fleet–grid interoperability, with rigorous governance for cybersecurity, GDPR compliance, and standards alignment. The study reports on architecture, methods, KPIs, risk and compliance frameworks, and cross-pilot value exchange to enable replicability across EU cities.

Background and Rationale Urban mobility is undergoing digital transformation driven by electrification, autonomy, and platforms. Fragmented edge–cloud ecosystems, energy-cost pressures, and sustainability mandates hinder scale and interoperability. Pilot 2 leverages the O-CEI approach based on blueprints to:

- **Reduce decision latency** for safety functions via edge inference.
- **Enhance energy flexibility** through V2G and dynamic pricing participation.
- **Improve operability** through interoperable data spaces and semantic alignment.
- **Ensure compliance with EU regulations** for AI, cybersecurity, privacy, and energy.
- **Integrate Vehicle-to-Grid (V2G):** Enable Electric Vehicles (EVs) to function as distributed.

P2-SC1- Vas for Vehicle Safety

Utilize a distributed software architecture to deploy safety-critical AI functions at the vehicle edge. This involves multi-sensor fusion (LiDAR, radar, camera, IMU) for robust environmental perception, enabling real-time object detection, path planning, and control for advanced driver-assistance systems (ADAS) such as collision avoidance, lane-keeping assistance (LKA), and adaptive cruise control (ACC). The implementation leverages O-CEI orchestration for over-the-air (OTA) updates and federated learning to continuously improve model accuracy across the vehicle fleet while ensuring deterministic latency through Time-Sensitive Networking (TSN).

P2-SC2: VaS for Sustainability:

Minimize the environmental impact by implementing AI-driven powertrain optimization and eco-driving strategies. This includes real-time analysis of vehicle state (e.g., State of Charge), proximity to charging infrastructure, and dynamic traffic conditions to calculate energy-efficient torque and regenerative braking limits. Furthermore, the pilot applies the ISO 23795-1 (Physics of Driving) methodology, fusing high-frequency CAN bus data with satellite ground speed from nomadic devices to generate accurate, verifiable per-trip carbon emission estimates. These estimates are published as composite sustainability indicators on the O-CEI marketplace, forming the basis for a credible carbon credit ecosystem.

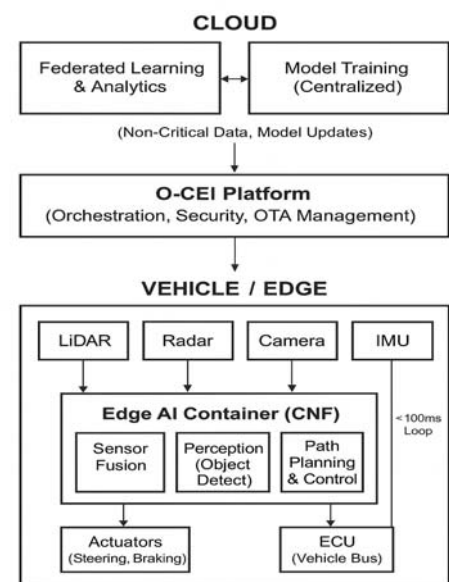


Figure 28. Pilot 2 –VaS Vehicle Safety

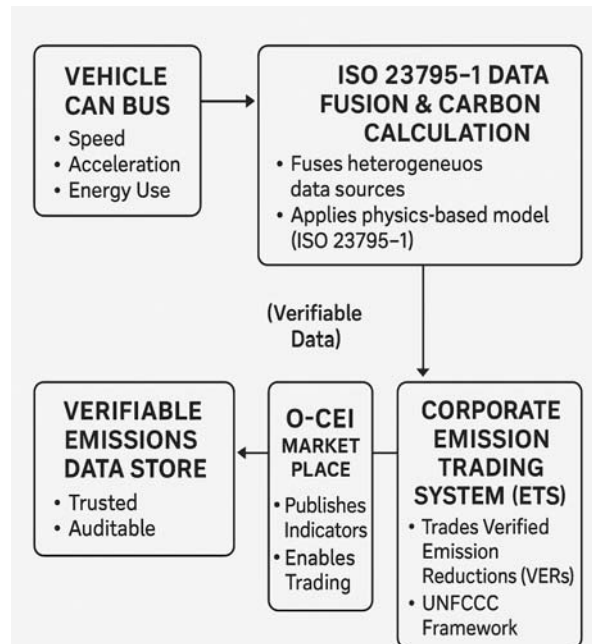


Figure 29. Pilot 2 – VaS for Sustainability

P2-SC3 VaS for fleet V2G integration:

Enable EVs to act as energy storage units, contributing energy back to the grid when needed. Energy resources (DERs) and flexible storage units for the smart grid. This involves developing bidirectional charging capabilities, battery health-aware scheduling algorithms (considering State of Health, Depth of Discharge, and temperature), and integration with grid signals. The O-CEI marketplace facilitates the contractual and policy-based orchestration of these V2G services, allowing fleets to participate in demand response programs and contribute energy back to the grid during peak demand, thus generating revenue streams.

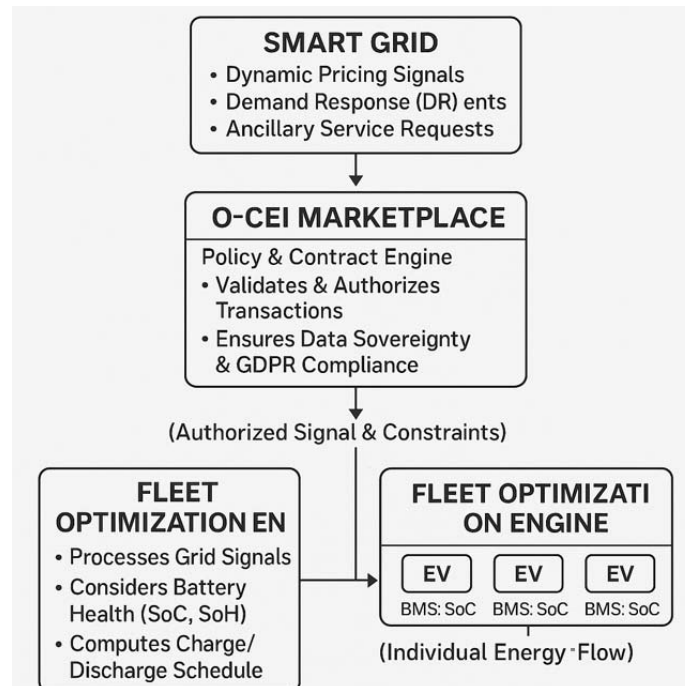


Figure 30. Pilot 2 – VaS for fleet V2G

P2-SC4 VaS for Grid optimization:

Facilitate seamless, standardized interaction between the vehicle fleet and smart grid infrastructure. This is achieved through the adoption of O-CEI semantic models and data space connectors (aligning with Catena-X and Gaia-X/IDSA), ensuring interoperable data exchange for grid service provisioning. The pilot validates the vehicle fleet's ability to provide grid-balancing services such as peak shaving, load shifting, and voltage regulation, thereby optimizing the integration of renewable energy sources and enhancing overall grid stability and efficiency.

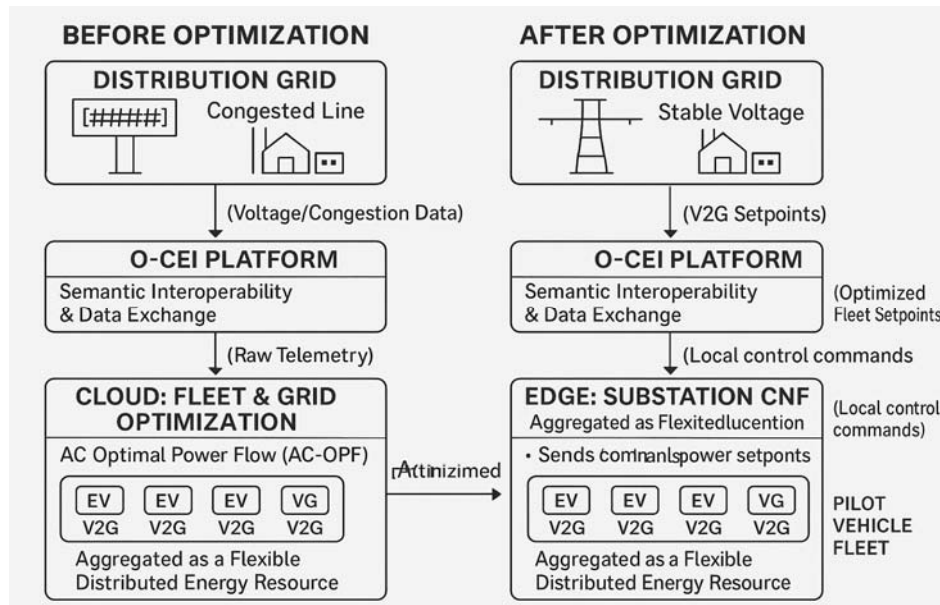


Figure 31. Pilot 2 – VaS for Grid optimization

4.1 PILOT GOVERNANCE

The following section describes the different roles and responsibilities of the assigned partners and how the plan to organize the scenarios of the pilot. ERTICO will coordinate twice per months a pilot site coordination meeting collecting the partner's contribution managing the advance of the pilot activities, previously defined in an activity plan including GANTT chart.

ERTICO serves as the **leader** for Pilot 2, taking responsibility for ecosystem orchestration and overall compliance. As pilot site leader and coordinator, ERTICO bridges the pilot's activities to the broader O-CEI implementation path, ensuring that the outcomes are consistent with the project's architectural and methodological framework. The organization coordinates all partners and scenarios, aligning the work with Gaia-X and IDSA principles, and governs the definition and monitoring of KPIs. ERTICO also manages stakeholder engagement involving cities, OEMs, fleets, and utilities, fostering cross-sector collaboration. Additionally, it oversees the security and compliance posture of the pilot, ensuring conformity with O-CEI standards and marketplace onboarding processes, while promoting the development and reuse of CEI components and blueprints across pilots.

CONTINENTAL plays a central technical role in advancing **Vehicles as Software (VaS)** within the O-CEI continuum, positioning vehicles as active computing nodes in the edge-cloud ecosystem. Building upon prior XRECO innovations, the company leads the implementation of AI-enabled functionalities in Timisoara, including collision avoidance, lane-keeping assistance, adaptive cruise control, sensor fusion, object detection, and path planning. Continental integrates predictive maintenance and real-time decision-making at the edge, validated through on-road testing and over-the-air (OTA) updates. Furthermore, the company enables semantic interoperability for SDV-grid–

fleet exchanges via O-CEI, connecting vehicles to smart grid infrastructures and dynamic pricing systems. Its contribution also includes providing vehicle-to-grid (V2G) scheduling, state-of-charge and state-of-health management, peak-load reduction, and grid-balancing services, while leveraging network slicing for efficient communication and data distribution.

FENIX2.0 focuses on **fleet-grid integration and energy intelligence**, complementing Continental's work by supporting the development of sustainable and interconnected Software-Defined Vehicle (SDV) solutions. The company contributes to the design and implementation of Pilot 2 scenarios, providing expertise in energy data exchange and fleet management optimization. FENIX 2.0 ensures alignment with the European Digital Infrastructure Consortia (EDIC) frameworks on Digital Twins and Mobility, as well as with Gaia-X and IDSA standards, thereby reinforcing interoperability and cross-domain data governance within the pilot's ecosystem.

4.2 DEMONSTRATOR PATH

The pilot progresses from design and integration to deployment and cross-site validation. Foundational work spans vehicle edge enablement, AI frameworks, fleet-grid connectors, and security/compliance controls, culminating in scalable VaS services. The following sections describe the setup and procurements activities, and part of the development already planned.

4.2.1 Setup & procurement activities

P2-SA1 - SDV Edge Enablement

Deploy edge modules and AI accelerators for low-latency inference and control; ensure OTA lifecycle management. The responsible of this activity is Continental.

P2-SA2 – O-CEI Marketplace Onboarding

Register data/services; configure contracts and policies for trusted, sovereign data exchange. The responsible of this activity is Ertico.

P2-SA3 – Security & Data Protection

Implement zero-trust, encryption, provenance, and GDPR-compliant processing with auditable policies. The responsible of this activity is Ertico.

4.2.2 Development activities

P2-DA4 - AI Safety & Comfort

Sensor fusion (LiDAR/Radar/cameras), object detection, path planning for collision avoidance, LKA, ACC. The responsible of this activity is Continental.

P2-DA5 – Predictive maintenance

Health monitoring and early-warning analytics for critical components; federated learning where applicable. The responsible of this activity is Continental.

P2-DA6 – Fleet V2G Enablement

Scheduling charge/discharge, SoC/SOH management, dynamic pricing response. The responsible of this activity is Continental.

4.2.3 Integration Activities

P2-IA7 – Cross-Domain Data Space Integration:

SDV–grid–fleet exchange via O-CEI semantics; align with Catena-X and Gaia-X/IDSA connectors and policies. All pilot partners are responsible of this activity.

P2-IA8 – IoT & Sensor Stack Integration:

ADAS/telematics, LiDAR/Radar data pipelines; network slicing, TSN, CNIs/SDN/service mesh for QoS. The responsible of this activity is Continental

4.2.4 Validation Activities

P2-VA9 – Vehicle & Road Tests:

Validate latency, perception quality, control stability, and comfort in urban scenarios. The responsible of this activity is Continental.

P2-VA10 – V2G & Grid Services:

Validate peak shaving, load shifting, grid balancing alongside utilities. The responsible of this activity is Continental

4.2.5 Planning & Gantt

The following Gantt diagram list all the activities planned since the beginning of the work package on M4 until M18.

Table 9. Pilot 2 Gantt diagram

Pilot 2		2025										2026					
Code	Task	M4	M5	M6	M7	M8	M9	M10	M11	M12	M13	M14	M15	M16	M17	M18	
SETUP & PROCUREMENT																	
P2-SA1	SDV Edge Enablement (CONTI)																
P2-SA2	O-CEI Marketplace Onboarding (ERTICO&FENIX)																
P2-SA3	Security & Data Protection (ERTICO&FENIX)																
DEVELOPMENT																	
P2-DA4	AI Safety & Comfort (CONTI)																
P2-DA5	Predictive Maintenance (CONTI)																
P2-DA6	Fleet V2G Enablement (CONTI)																
INTEGRATION																	
P2-IA1	Cross-Domain Data Space Integration (All)																
P2-IA2	IoT & Sensor Stack Integration (CONTI)																
VALIDATION																	
P2-VA1	Vehicle & Road Tests (CONTI)																
P2-VA2	V2G & Grid Services (Continental, coordinated by ERTICO)																

4.3 CURRENT INFRASTRUCTURE

The infrastructure comprises a dense mix of vehicle-embedded sensors (ADAS, Lidar, Radar), IoT gateways, Edge servers, and cloud-based SDV management platforms. Most components are pre-integrated within the SDV stack, and no major acquisitions are listed, indicating that equipment and computing resources are provided by industrial partners (e.g., CONTI, AVL) as part of their internal platforms. An update with regards to the energy infrastructure is shown in the following Figure.

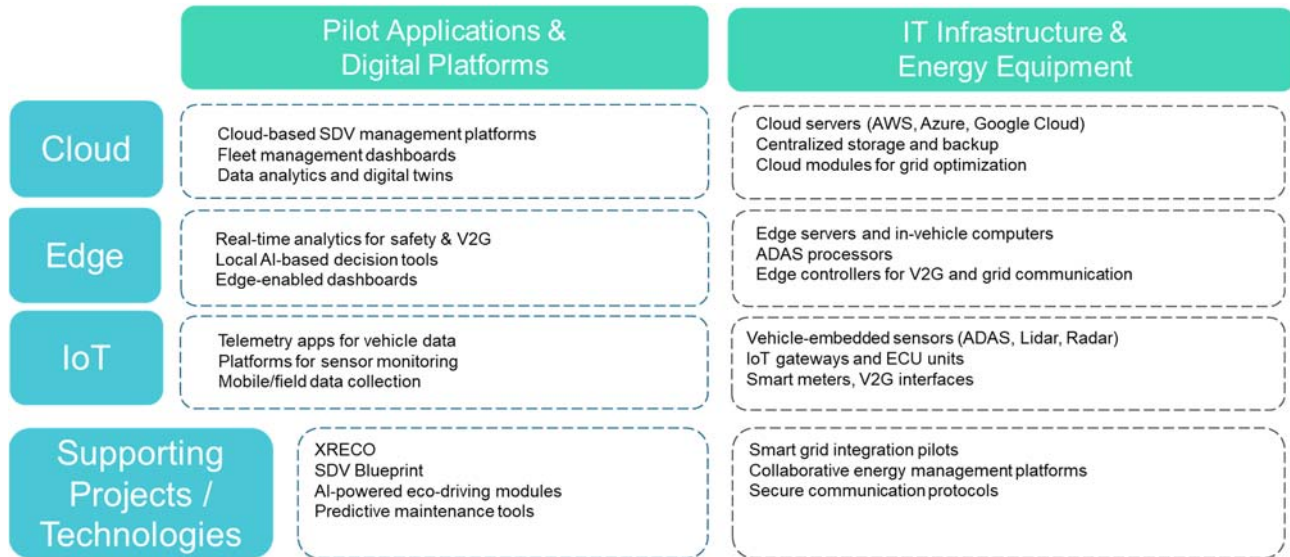


Figure 32. Pilot 2 infrastructure

The figure above gives an overview of the different modules used and to be extended in Pilot 2. This covers the O-CEI layers of Cloud/Edge/IoT and Technology, in particular:

Vehicle & Edge Layer: ADAS sensors, telematics, LiDAR/Radar, ECUs, AI accelerators, and edge modules for real-time inference and control.

- Cloud & Data Space Layer: Mobility platforms, O-CEI marketplace, semantic catalogues (Catena-X/O-CEI), AI orchestration, OTA services, federated learning.
- Security & Governance Layer: Cybersecurity controls, federated identity/access, auditability, GDPR-aligned handling; Gaia-X/IDSA policy enforcement.

In particular for the VAS for sustainability, a system overview is shown in Figure below, reproduced with the permission of the authors. It shows how vehicles equipped with nomadic devices generate speed profiles which can be linked to energy demand (fuel, electricity, etc.) available at the CAN Bus but also to ISO-23795-1 estimated carbon emissions using Physics of Driving calculations. Merging the two information sources will increase the accuracy of carbon footprint calculations significantly and enable eco-routing platforms to trade recorded emissions in corporate marketplace. This will support decarbonization of fleet operators and help regulators to boost Emission Trading Systems for road transport desperately under investigation by climate researchers.



Figure 33. Pilot 2 pipeline linking nomadic-device speed profiles with CAN-Bus energy signals (vehicle sight)

The different parts of the following figure show how the elements work together and represent the current infrastructure used in pilot 2, including the planned development of the O-CEI components to be added by the developer teams. The vehicle view is complemented by the IoT connectivity layer of the edge cloud, either in the MNO base station or by converting the vehicle itself to an edge node as planned to be implemented in O-CEI.

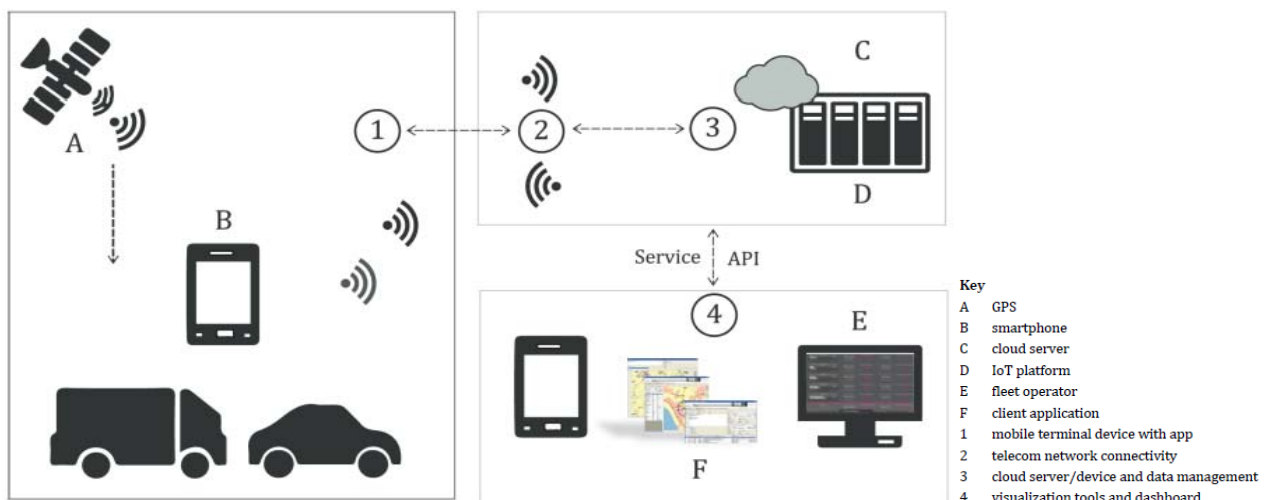


Figure 34. Pilot 2 pipeline linking nomadic-device speed profiles with CAN-Bus energy signals (IoT based). SOURCE: Cavallaro, F., Maino, F., & Morelli, V. (2013), Willenbrock, R., Tischler, J.: Reproduced with the permission of the authors

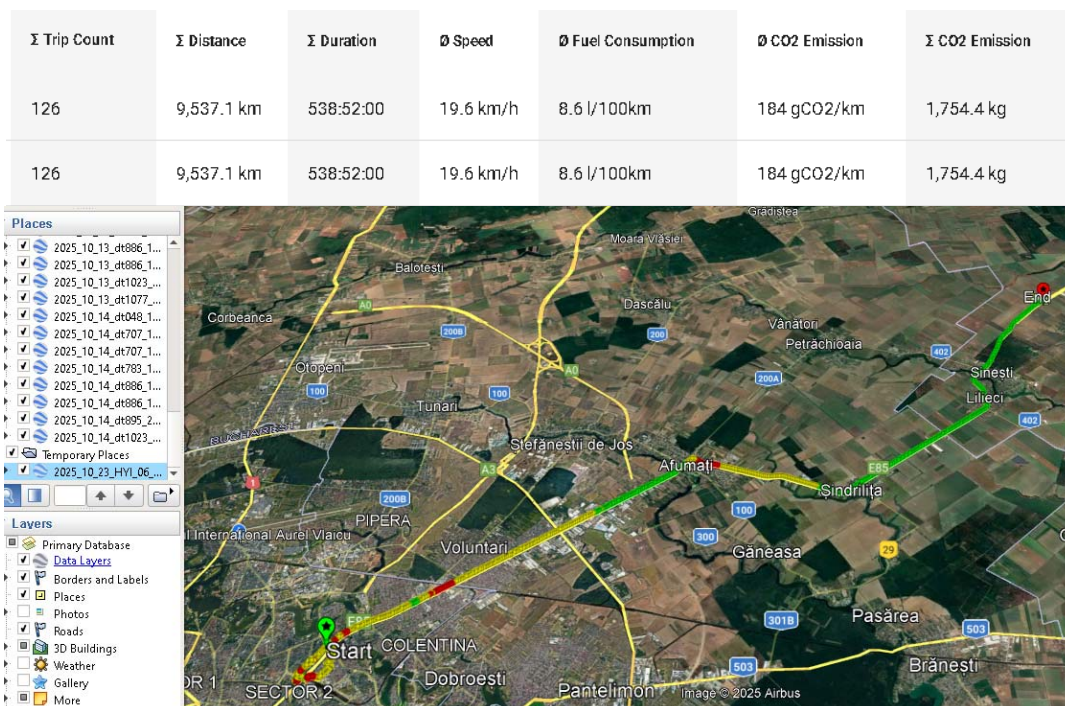


Figure 35. Pilot 2 Map visualization

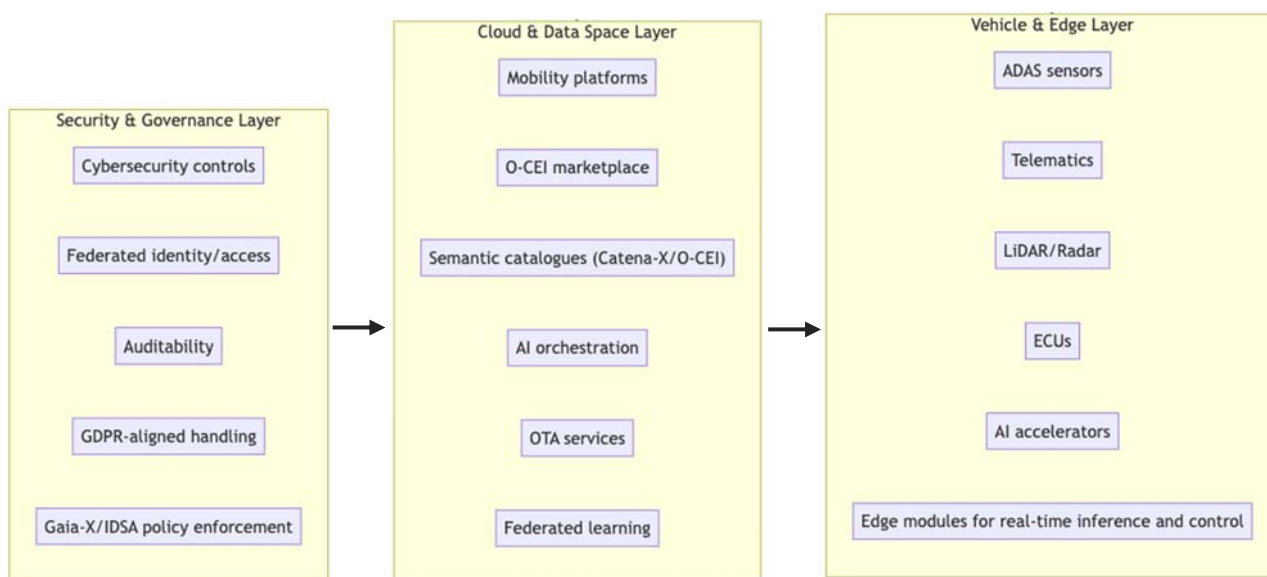


Figure 36. Current infrastructure and set-up

The figure above illustrates the currently planned infrastructure, where all components are available but not yet integrated into the O-CEI environment. The integration process will be carried out over the coming months.

4.3.1 Pilot applications and digital platforms

Cloud Layer

At the cloud level, the pilot utilizes cloud-based SDV management platforms responsible for coordinating vehicles, managing energy exchanges, and optimizing fleet operations. These platforms include:

- **Fleet management dashboards** for centralized supervision.
- **Cloud modules for grid optimization.**
- **Centralized storage and backup** systems.
- **Digital twins** that replicate vehicle behavior for forecasting and optimization purposes. The cloud infrastructure likely runs on commercial providers (e.g., AWS, Azure, or Google Cloud), and interfaces with orchestration tools and data analytics services.

Edge Layer

On the edge, in-vehicle computers and ADAS processors host AI-based decision-making tools and predictive analytics engines that enable real-time vehicle responses. These systems support applications such as collision avoidance, eco-driving assistance, and route optimization. Edge-enabled dashboards provide localized control and monitoring interfaces, while containerized software modules may execute operations such as V2G decision logic or data filtering before cloud transmission.

IoT Layer

The IoT layer comprises vehicle-embedded sensing and communication systems:

- **ADAS sensors, Lidar, and Radar** gather environmental and operational data.
- **Smart meters and V2G interfaces** manage bidirectional energy flow.
- **IoT gateways and Electronic Control Units (ECUs)** handle data acquisition and internal network management. These IoT elements are essential for enabling real-time interactions with the grid and for integrating the vehicle as an active component within urban energy ecosystems.

Supporting Projects and Technologies.

This includes advanced tools and frameworks that enhance the pilot's capabilities:

- **XRECO SDV Blueprint** – A foundational reference for SDV architecture.
- **AI-powered eco-driving modules** – To minimize energy consumption during driving.
- **Predictive maintenance tools** – For real-time fault prediction and servicing optimization.
- **Smart grid integration platforms** – Enabling interaction between vehicles and broader grid infrastructure.
- **Secure communication protocols** – Ensuring trust and resilience in data exchange across the continuum.

These components act as accelerators, ensuring the pilot aligns with state-of-the-art SDV development and energy-aware urban mobility.

4.3.2 IT infrastructure and Energy equipment

Cloud Layer

- **Cloud servers** hosting the SDV management and orchestration tools.
- **Data analytics platforms** supporting grid optimization.
- **Fleet dashboards** and data storage systems, ensuring remote accessibility and secure long-term storage.

Edge Layer

These components allow autonomous decision making and efficient energy use without full dependency on the cloud.

- **Edge servers** and **ADAS processors**.
- **On-board dashboards** for human-machine interaction.
- Embedded AI modules that perform real-time data processing and enable autonomy and flexibility.

IoT Layer

This layer enables full data traceability, vehicle-grid communication, and physical interaction with the energy ecosystem.

- **Sensor systems** integrated into the vehicle (e.g., ADAS, Lidar, Radar).
- **Smart meters** for energy monitoring.
- **V2G interfaces** for power exchange.
- **IoT gateways and ECUs** enabling in-vehicle data networking and communication.

Energy Equipment

These systems enable the vehicle to act not just as a consumer but also as a flexible energy provider to the grid.

- **Electric motors** and **battery systems** for propulsion and V2G services.
- **Energy control units** that manage energy consumption, charging cycles, and interaction with external energy infrastructure.

5 PILOT 3 – ENERGY CONSUMPTION AND EMISSION REDUCTION IN POSTAL SERVICE FLEET OPERATION VIA INTELLIGENT BEV CHARGING STRATEGIES

This pilot aims to reduce energy consumption and emissions in postal service fleet operations by implementing intelligent battery-electric vehicle (BEV) charging and Vehicle-to-Grid (V2G) strategies. The project combines advanced AI-driven optimization tools, bidirectional charging infrastructure, and real-time data exchange between power producers and fleet operators. All three scenarios will be demonstrated in Austria.



Figure 37. Pilot 3 Austrian Postal vehicles

P3-SC1: Optimal storage battery systems

This scenario aims to optimize electrical power usage to lower costs, reduce peak demand, and improve grid stability through ML-driven energy management. The system anticipates demand and controls stationary battery storage in real time, supported by continuous data exchange between power producers and fleet operators. By enabling optimal storage sizing and coordinated charging infrastructure, the approach targets a 10–15% reduction of charging costs. The main objective of this scenario is to optimize electrical power usage to minimize cost, reduce peak demand, and enhance grid stability.

- **Technology:** AI-driven software anticipates demand and controls stationary battery storage in real time, supporting optimal sizing of storage systems.
- **Infrastructure:** Chargers are integrated with continuous data exchange between the power producer and fleet operator.
- **Goals:** Achieve a **10–15 % reduction in total cost of EV ownership and operation**, combined with a robust battery-storage investment plan.
- **Current Gap:** Existing market tools do not yet provide full integration of producer–operator data.

Expected Outcome: An investment plan for optimally sized battery storage systems supported by validated optimization methods.

P3-SC2: Vehicle as Mobile Energy Storage Unit

This scenario focuses on leveraging battery electric vehicles (BEVs) as mobile energy storage assets to enhance fleet flexibility and minimize idle time. Through AI-driven optimization combined with vehicle-to-grid (V2G) and bidirectional charging technologies, intelligent scheduling and routing

are enabled. Real-time data exchange between power producers and fleet operators supports dynamic energy management. The goal is to utilize BEVs as mobile energy storage assets to maximize fleet flexibility and reduce idle time.

- **Technology:** AI-driven optimization combined with V2G and bidirectional charging for intelligent scheduling and routing.
- **Infrastructure:** V2G with real-time data exchange between power producer and fleet operator.
- **Goals:** Achieve a 20–30 % reduction in vehicle idling time without productive activity.
- **Current Gap:** To date, only small-scale pilots and simulations have demonstrated the benefits of bidirectional charging

Expected Outcome: Quantified benefits of mobile energy storage and an investment plan for fleet-wide V2G deployment.

P3-SC3: Smart & Innovative Energy Management with Decentralized Renewables

This scenario aims to create an integrated framework that coordinates fleet operations, stationary storage, renewable energy sources, and market mechanisms to maximize efficiency and cost savings. The AI-based platform combines vehicle-to-grid (V2G) functionality, bidirectional charging, and market-driven energy management while integrating data from EV operations, solar generation, stationary batteries, charging infrastructure, and grid stability inputs. The goal of this is to develop a framework that coordinates all elements - fleet operation, stationary storage, renewables, and market mechanisms - for optimal efficiency and cost savings.

- **Technology:** AI-based platform combining V2G, bidirectional charging, and market-driven energy management.
- **Data Integration:** Incorporates EV operation data, solar generation, stationary battery storage, charging stations, and grid-stability inputs.
- **Goals:** Realize a **10–15 % reduction in total EV ownership and operating costs**, while ensuring regulatory compliance and intelligent billing.
- **Current Gap:** Until now, only simulations and paper studies exist for this level of integration.

Expected Outcome: A next-generation software framework that optimizes the interaction of vehicles, storage, and decentralized renewable resources.

Each of the three scenarios will be **deployed, tested, and validated in the real postal service environment**.

5.1 PILOT GOVERNANCE

AVL serves as the **Pilot Leader** and acts as the **main engineering and technology provider** within the consortium. AVL is responsible for developing the digital agents, the reinforcement learning framework, and the methodology for formal model development. Leveraging its expertise in system engineering and intelligent control, AVL ensures the seamless integration of edge and cloud intelligence into the O-CEI architecture. Its work enables adaptive and predictive decision-making for fleet operation and energy optimization, forming the technical backbone of the pilot's intelligent charging strategies.

ENSTEI (Energie Steiermark) is one of **Austria's leading energy providers**, manages the balance between energy demand and supply, while introducing market mechanisms to maximize efficiency and reduce operational costs for large energy consumers such as fleet operators. Within the pilot, ENSTEI **leads the development** of dynamic pricing models, demand-response coordination, and the integration of renewable sources into the grid. The company ensures that O-CEI technologies align with real-world market conditions and regulatory frameworks, supporting the scalability and sustainability of the energy management solutions demonstrated in the pilot.

Austrian Post, acting as the **end user** and **logistics operator**, provides the real-world demonstration environment for the pilot. As one of Europe's largest postal fleets transitioning toward full electrification, Austrian Post aims to operate around 10,000 battery electric vehicles (BEVs) and 8,000 charging stations by 2030, with an estimated annual electricity consumption of 72,000 MWh by 2025. The Neunkirchen site in Lower Austria, comprising 58 BEVs, 61 charging stations, and an annual consumption of 182 MWh, serves as the main testbed for validating O-CEI's intelligent charging strategies, fleet operation optimization, and interoperability mechanisms.

Since 3rd April 2025 the AVL, ENSTEI and Austrian Post have conducted weekly meetings and dedicated topic-specific F2F workshops on the 17th March (principles of systems engineering, location: Austrian post), 15th March (details on systems engineering, location: AVL) and the 17th of June (behavior models and agents, location: ENSTEI).

5.2 DEMONSTRATOR PATH

Aligned with the four main phases of the project (Build, Demonstrate, Expand, and Grow & Sustain), in the figure below there is a list of actions points for pilot 3. Regarding the conducted procurement activities comprise the setup of the systems engineering framework (P3-SA1, M11-M14), the selection of a pilot demonstration site (P3-SA2, M11), the preparation of basic behavior models (P3-SA3, M11-M15), the implementation of agents for demonstration (v0.1, PIC meeting at AVL, P3-SA4, M11) and the procurement of equipment for the POST base station (P3-SA5, M11-M18). This is also reflected in the Gantt chart in Section 1.2.2.

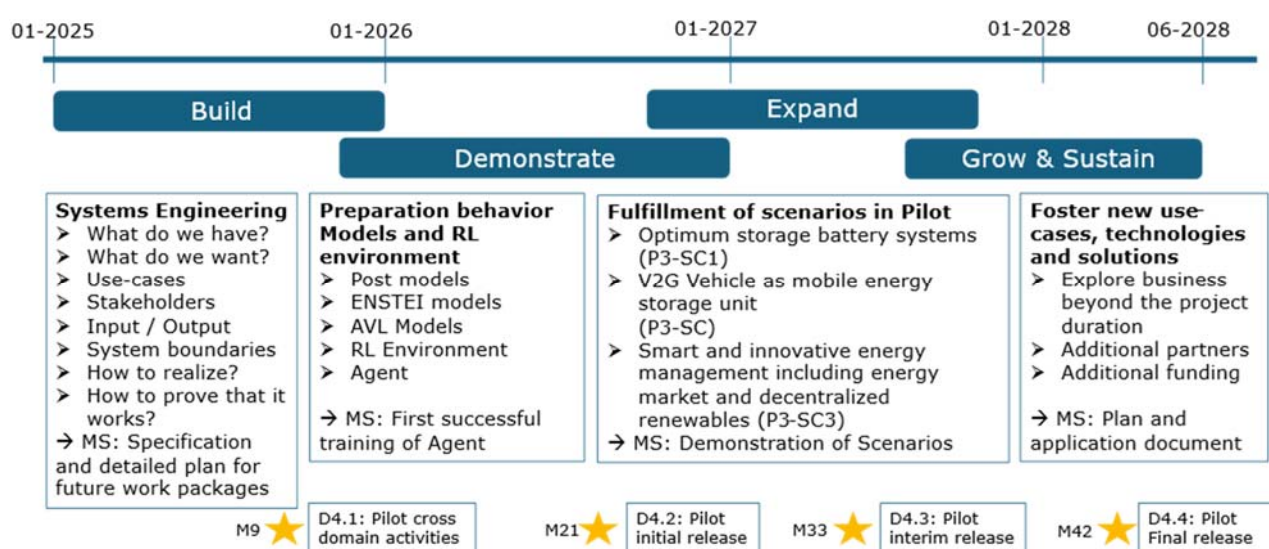


Figure 38. Demonstrator Path for pilot 3

5.2.1 Setup & procurement activities

P3-SA1 – System engineering

Stand up the pilot's systems-engineering backbone: inventory fleet and energy assets (BEVs, chargers, stationary storage, renewables, data interfaces), define the target integrated system (AI-driven charging/V2G, storage sizing, market interaction), key use cases, stakeholders, inputs/outputs, and system boundaries. This establishes the technical baseline for all later work.

First, for the execution of the pilot the **following acquisitions are needed**:

Energy management system: ENSTEI is setting up a state-of-the-art energy management system covering the selected pilot site(s). It will coordinate energy flows, manage demand and storage, and integrate data from various local sources to support optimization strategies.



Figure 39. Pilot 3 Energy management system

Controlled and Sensor equipment: Sensor equipment is needed to provide time-series data for the reinforcement learning agent; the controller is necessary to implement control of the charging process and apply optimization rules at depot level.



Figure 40. Pilot 3 Controlled and Sensor equipment

Wall-boxes: Wall-boxes and vehicle modifications are required to enable more flexible and bidirectional charging. These units are essential for executing V2G strategies and integrating EVs as grid assets.

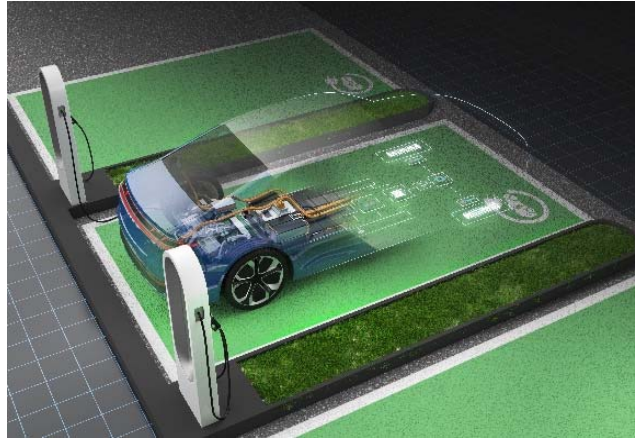


Figure 41. Pilot 3 Wall-boxes

Second, Pilot partners performed a **stakeholder requirements** analysis covering different aspects, including: the development approach, system specifications, behaviour models, agents, cloud usage, knowledge and software re-use, historic data and life data (AVL); data- and energy assets, inputs/outputs of the agents and the potential for cost reduction and revenue maximization (POST); energy management system, energy procurement, self-consumption (Austrian POST), battery capacity simulation and needs w.r.t. to location of the pilot (ENSTEI).

And **third**, detailed use cases analysis were defined and assessed centered around each partner, performing as well an **infrastructure and context analysis**. The second and third steps of the system engineering activity will be documented in D2.2.

P3-SA2 – Choice of Port Bais Location

Confirm Neunkirchen (Lower Austria) as the demonstration site and capture site parameters: ~58 BEVs, 61 charging points, ~182 MWh/yr; frame this as the primary testbed for intelligent charging and fleet optimization.

P3-SA3 – Preparation of behavior models

Draft initial behaviour models for fleet operation, charging strategies, storage control and market-responsive actions; align across partners through the topic-specific workshops (systems engineering and agent behaviour).

P3-SA4 – Preparation of Agent for Demonstration (V0.1)

Implement an initial cloud/edge agent (v0.1) to exercise basic decision loops for charging and storage orchestration; review at the PIC meeting (AVL) and freeze an MVP scope for deployment.

P3-SA5 — Procurement of parts for Post basis

Procure any missing equipment for the site/POST base station necessary to host the agents and data flows (connectivity, metering, edge HW).

5.2.2 Development activities

P3-DA1 — Cloud deployment of Agent

Deploy the agent to the cloud layer of O-CEI, enabling secure data intake (fleet ops, market signals, weather/solar) and remote lifecycle management for updates and experiments.

P3-DA2 — Development of Agent (Version 1.0)

Harden the agent with RL-based decision support and formal modelling methods; add capabilities for dynamic pricing, demand-response coordination and predictive charging policies aligned with ENSTEI's market mechanisms.

5.2.3 Integration activities

P3-IA1 — Integration of cloud-based Agent with Post basis

Integrate agent interfaces with site subsystems (chargers, storage, EMS, data gateways); validate data/command paths using the interface maps and logical decomposition defined for Pilot 3.

P3-IA2 — Live trials to test scenarios (SC1, SC2, SC3)

Execute scenario trials:

- **SC1 – Optimizing Storage Battery Systems:** edge analytics + predictive models to shave peak and coordinate storage.
- **SC2 – V2G as mobile storage:** orchestrate bidirectional charging to support the grid during peaks and improve fleet resilience.
- **SC3 – Smart energy mgmt with decentralized renewables:** combine V2G, storage and market-driven control for maximum efficiency and cost savings.

P3-IA3 — Further development of Agent (Version 2.0)

Iterate the agent with feedback from trials (improved V2G scheduling, SoC/SOH handling, tariff alignment) and tighten interoperability with energy-market processes for scale-out.

5.2.4 Validation activities

P3-VA1 — Live trials for optimum fulfillment of scenarios (SC1, SC2, SC3)

Optimize configurations in situ to reach the Demonstration Path goals (cost reduction, charging efficiency, lower idle/charge time, longer range per charge).

P3-VA2 — Live trials for documentation of KPIs

Measure and document Pilot-3 KPIs: 10–15% TCO reduction, 15–20% charging-process efficiency gain, 15–25% charging-time reduction, 10–15% extra distance/charge, 20–30% idle-time reduction, ~5% CO₂ cut.

P3-VA3 — Final documentation

Consolidate technical results, KPI evidence and an investment case for fleet-wide deployment of mobile energy storage/V2G, feeding the cross-pilot governance and scale-up plans.

5.2.5 Planning & Gantt

The following Gantt diagram list all the activities ongoing and planned since M11 until M21. For the sake of simplicity, the diagram does not contain M4 to M11 because the only performed activity has been the System engineering P3-SA1, where all the analysis and procurements have taken place.

Table 10. Pilot 3 Gantt diagram

Pilot X		2025		2026									
Code	Task	M11	M12	M13	M14	M15	M16	M17	M18	M19	M20	M21	
SETUP & PROCUREMENT													
P3-SA1	System engineering												
P3-SA2	Choise of Post Basis location												
P3-SA3	Preparation of behavior models												
P3-SA4	Preparation of Agent for demonstration (Version 0.1)												
P3-SA5	Procurement of parts for Post basis												
DEVELOPMENT													
P3-DA1	Cloud deployment of Agent												
P3-DA2	Development of Agent (Version 1.0)												
INTEGRATION													
P3-IA1	Integration of cloud based Agent with Post basis												
P3-IA2	Live trails to test scenarios (SC1, SC2 and SC3)												
P3-IA2	Further development of Agent (Version 2.0)												
VALIDATION													
P3-VA1	Live trails for optimum fulfillment of scenarios (SC1, SC2 and SC3)												
P3-VA2	Live trails for documentation of KPIs												
P3-VA3	Final docmention												

5.3 CURRENT INFRASTRUCTURE

The infrastructure in Pilot 3 includes a combination of smart meters, battery energy storage systems, bidirectional chargers, cloud-based optimization platforms, and AI agents. Some key components are being procured as part of the pilot, including an energy management system, controller and sensor equipment, and EV wall-boxes for flexible and bidirectional charging.

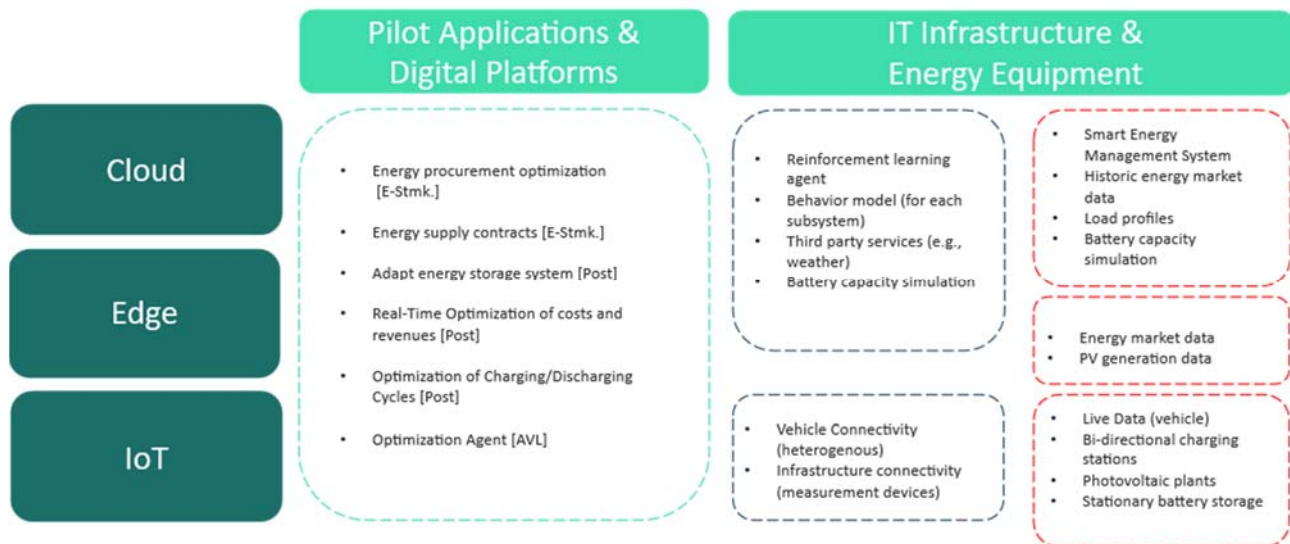


Figure 42. IT Infrastructure and Energy Equipment in Pilot 3

5.3.1 Pilot applications and digital platforms

In this pilot, the Cloud, Edge, and IoT layers are merged into a unified system that spans from data acquisition to real-time control and long-term optimization. The architecture integrates multiple components across this continuum:

- **Optimization agents** based on **Reinforcement Learning (RL)** are used to manage cost-effective charging, discharging, and fleet coordination. These agents run across both local (depot) and cloud environments.
- A **Smart Energy Management System** coordinates the flow of energy between BEVs, depot storage units, charging stations, and the public grid. It incorporates **real-time optimization of cost and revenue**, **battery capacity simulation**, and **forecasting tools**.
- **Behaviour models for each subsystem** are used to simulate operational scenarios and adapt strategies dynamically.
- The system uses data from **third party sources** such as weather services and market platforms to improve accuracy and planning.
- **Charging/discharging cycles** are managed in response to vehicle demand, energy prices, and infrastructure constraints using a combination of edge-based control and cloud orchestration.

5.3.2 IT Infrastructure and Energy Equipment

Cloud Layer:

- Fleet-wide optimization algorithms and behavioural simulations.
- Battery simulation models and energy market data analysis.

- Central dashboards for monitoring energy costs, usage, and operational KPIs. Cloud systems communicate continuously with depot-level edge devices and integrate with third-party forecasting tools.
- Energy Equipment: Though physically located on-premise, battery capacity simulation systems are operated and analysed from the cloud side to support higher-level planning and cost optimization.

Edge Layer:

- Local edge servers and embedded controllers that run optimization agents.
- Control interfaces for initiating or pausing charging operations.
- Data processing units that handle real-time telemetry from vehicles and infrastructure. These devices are essential for low-latency decision-making and continued operation during network outages.
- Energy Equipment: Charging units (wall-boxes) with bidirectional capabilities are deployed to support V2G scenarios, local EMS hardware installed by ENSTEI manages depot-level energy flows, controller and sensor equipment ensure accurate measurement and enforcement of charging rules.

IoT Layer

- **Smart meters and submeters** that record energy consumption at charging points.
- **Onboard vehicle systems** that transmit battery status, availability, and consumption history.
- **Sensors installed on infrastructure components** to monitor ambient conditions and physical energy usage.
- **Energy Equipment: Smart meters** and **energy data loggers** embedded within EV charging units and infrastructure, interfaces with **vehicles' internal energy systems** for seamless integration with EMS and V2G logic.

PV System:

- Type: Trinasolar TSM-NEG9R.28 – 440.
- AC/DC Converter: Huawei SUN2000-115KTL-M2.
- Power: 258,72 kWp.

Vehicles:

- 5x MERCEDES E-SPRINTER (45,2 kWh)
- 38x MAXUS eDeliver 3 L2 (50 kWh)
- 1x MERCEDES E-SPRINTER PRO L 3665 HD 81 kWh (81kWh)
- 12x conventional vehicles (will be exchanged this year --> Type TBD)
- Charging Infrastructure:
- 59x Terra AC wall charging stations
- 2x Terra DC wall charging stations

Energy Storage System:

Currently not available, but it is possible that one us purchased during the project period

In addition, the following list of figures helps to understand the relationships among energy assets and IT assets, and to establish a systemic and logical view among these components:

Logical System Decomposition Overview

The logical decomposition of the overall system is separated into several layers (L0 to L4) and comprises the energy supply system, the energy purchase system, the energy management system, a system for energy trading, and a dedicated system for data management, AC charging stations, DC charging stations as well as mobile charging systems. The vehicle fleet comprises vehicles each of which contains a HV battery and a telemetry data system. There are further energy consumers such as heat pumps and facilities such as PV plants, a PV monitoring system and energy storage facilities. Sourcing contracts complete this high-level system analysis.

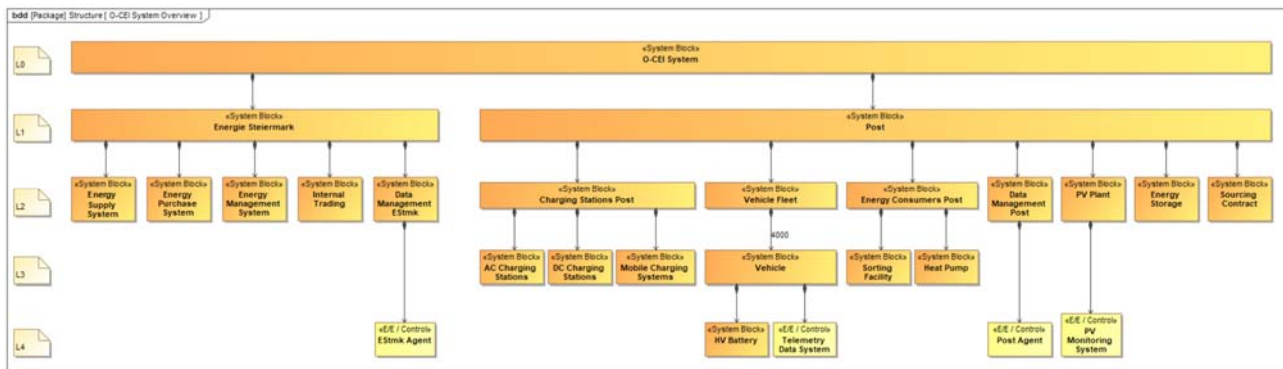


Figure 43. Logical System Decomposition Overview in Pilot 3 Infrastructure

Interfaces of Systems:

For each block we have identified the interfaces to other systems such as interfaces to third party systems. The focus of this analysis is on energy flow and the information flow.

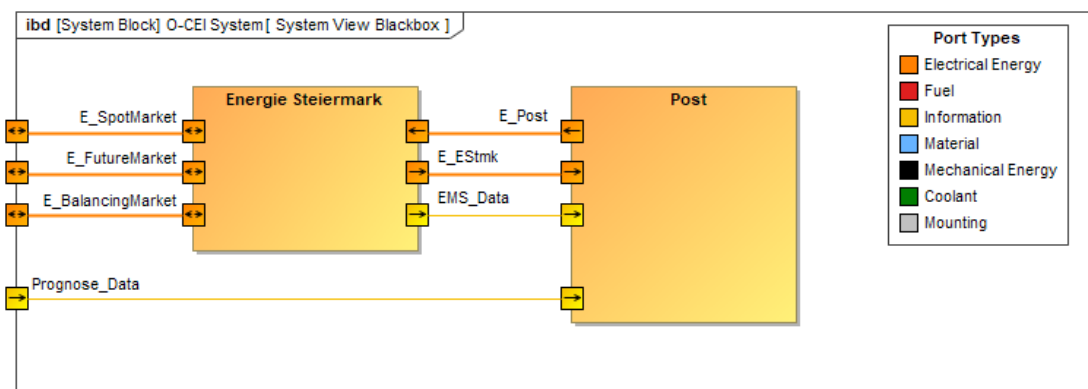


Figure 44. Interfaces of Systems in Pilot 3 Infrastructure

Interfaces of Sub-Systems: ENSTEI

The subsystem analysis for ENSTEI reveals interfaces for components such as Internal Trading, Energy Purchase System, Data Mgmt. System, Energy Supply System and the Energy Management System.

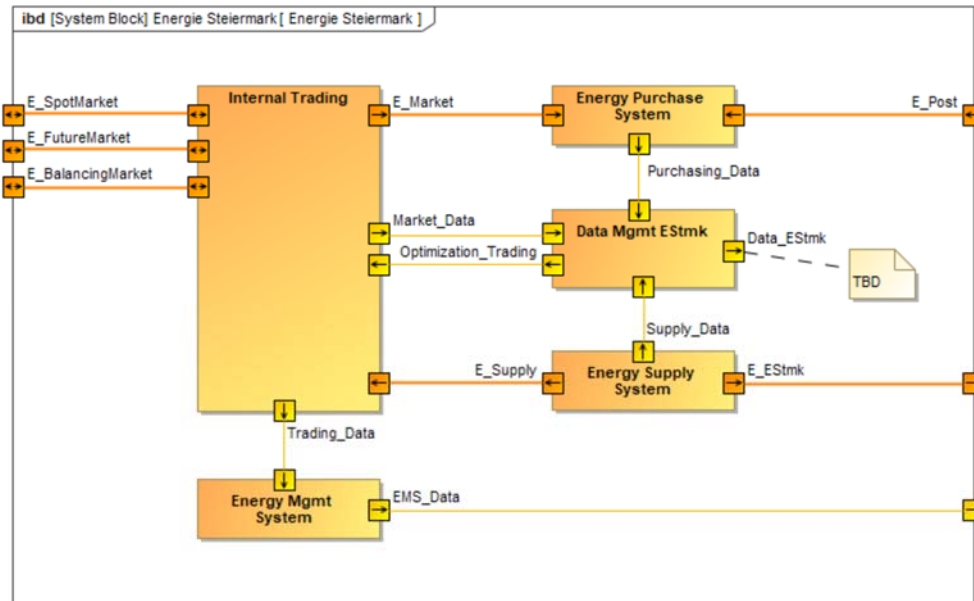


Figure 45. Interfaces of Sub-systems in Pilot 3 Infrastructure for ENSTEI

Interfaces of Subsystems: POST

The subsystem analysis for Austrian Post converts entities such as charging stations, PV plant, vehicle fleet, dedicated components for data management, energy consumers, energy storage facilities and sourcing contracts alongside with the information and energy flows.

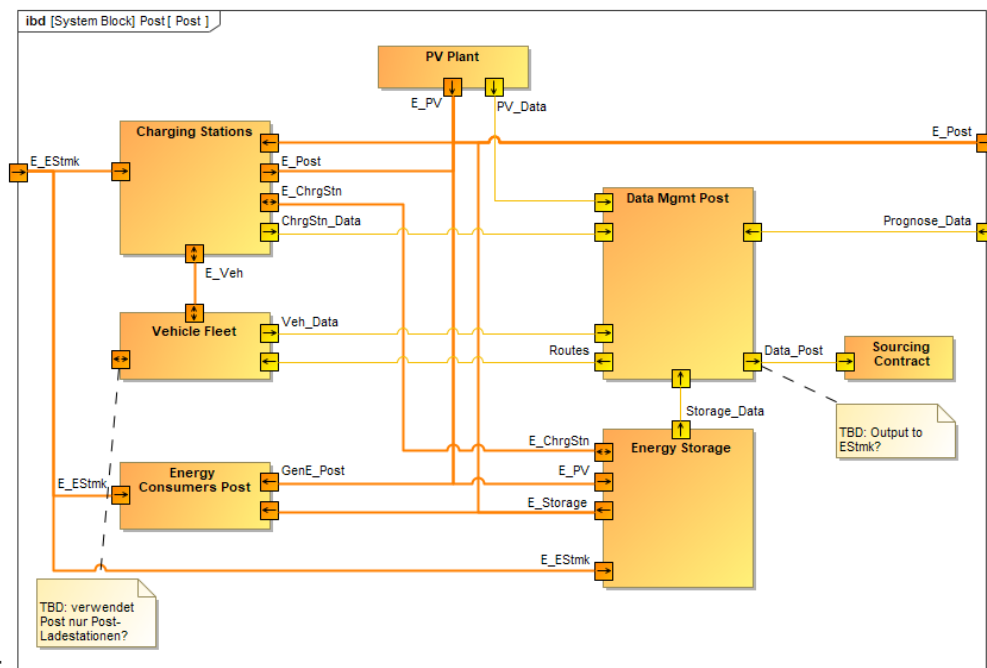


Figure 46. Interfaces of Sub-systems in Pilot 3 Infrastructure for POST

In a further step the functions on the subcomponents discussed above have been formalized. Each function is characterized by the (typed) input and the output. Without going into the details, in the following we provide examples of such functions as a representative example for each partner.

Interfaces of Functions for Energy Management: ENSTEI:

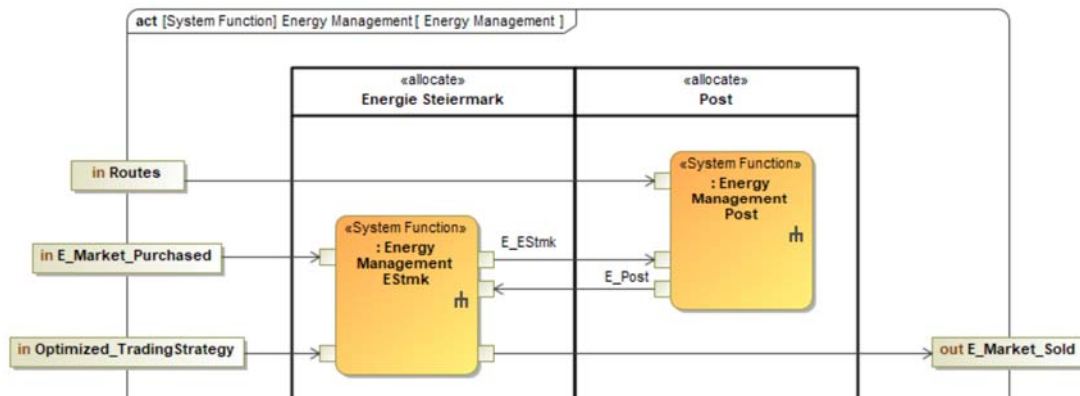


Figure 47. Interfaces of Functions for Energy Management in Pilot 3 Infrastructure for ENSTEI

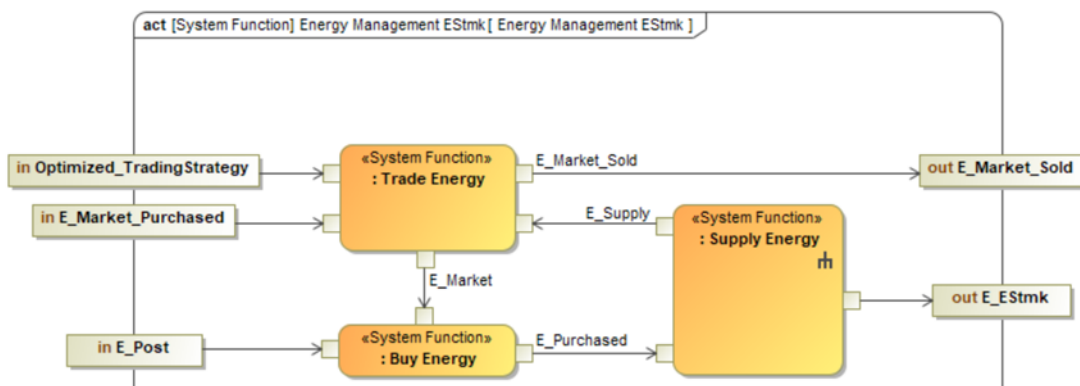


Figure 48. Interfaces of Functions for Energy Management in Pilot 3 infrastructure for ENSTEI II

Interfaces of Functions: POST:

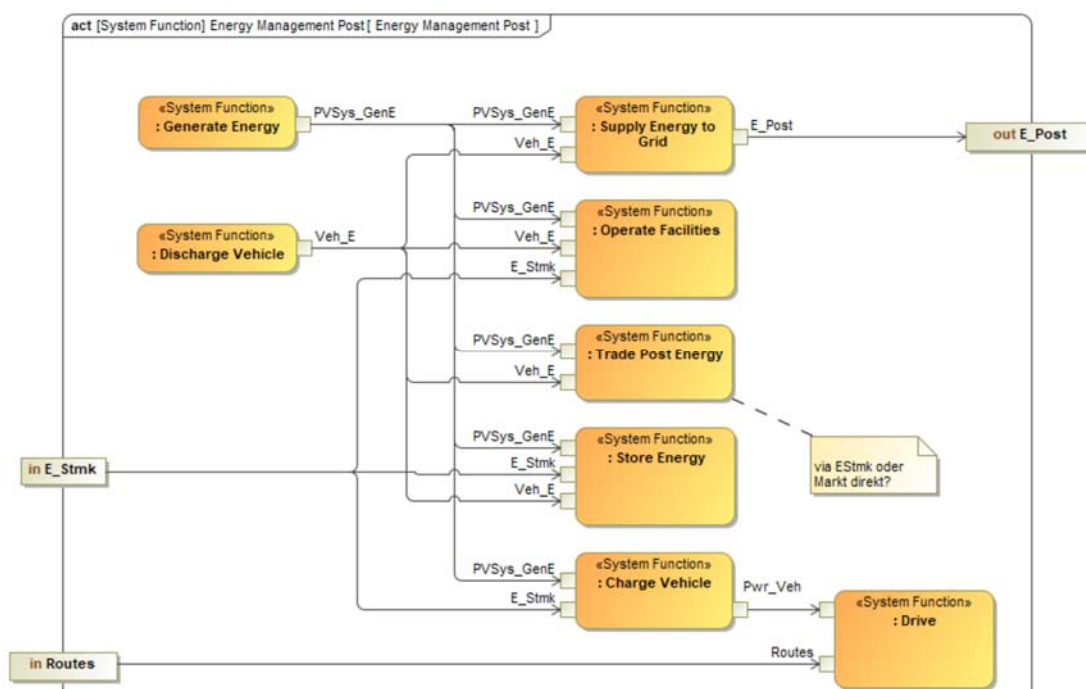


Figure 49. Interfaces of Functions in Pilot 3 Infrastructure for POST

Interfaces of Functions: Data Management, ENSTEI

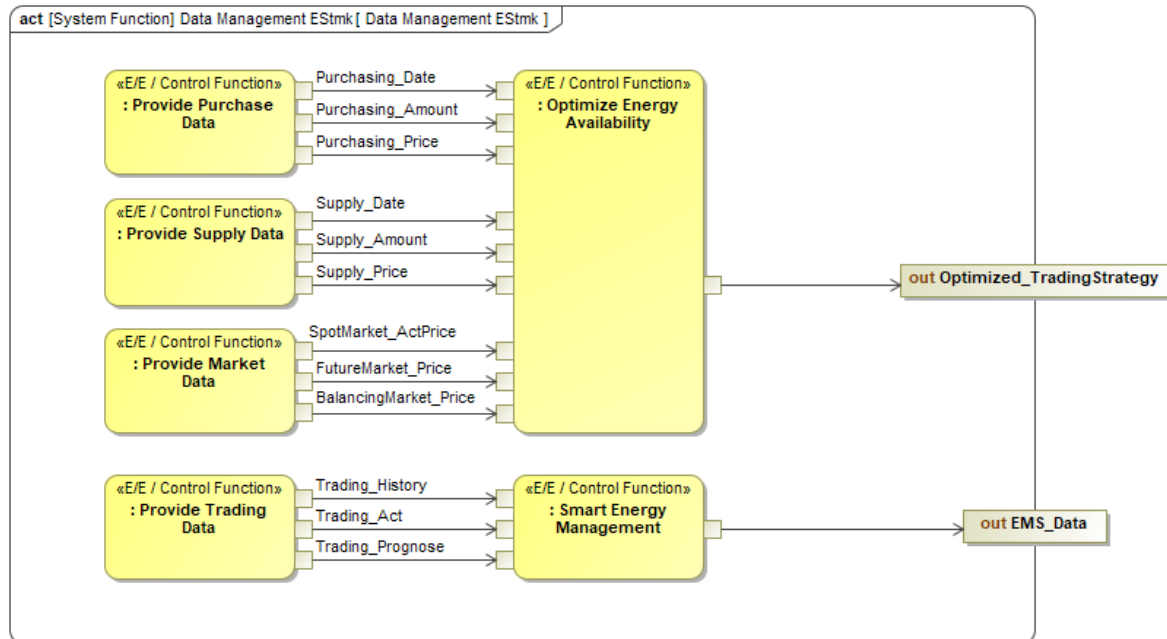


Figure 50. Interfaces of Functions – Data Management in Pilot 3 Infrastructure for ENSTEI

Interfaces of Functions: Data Management, POST:

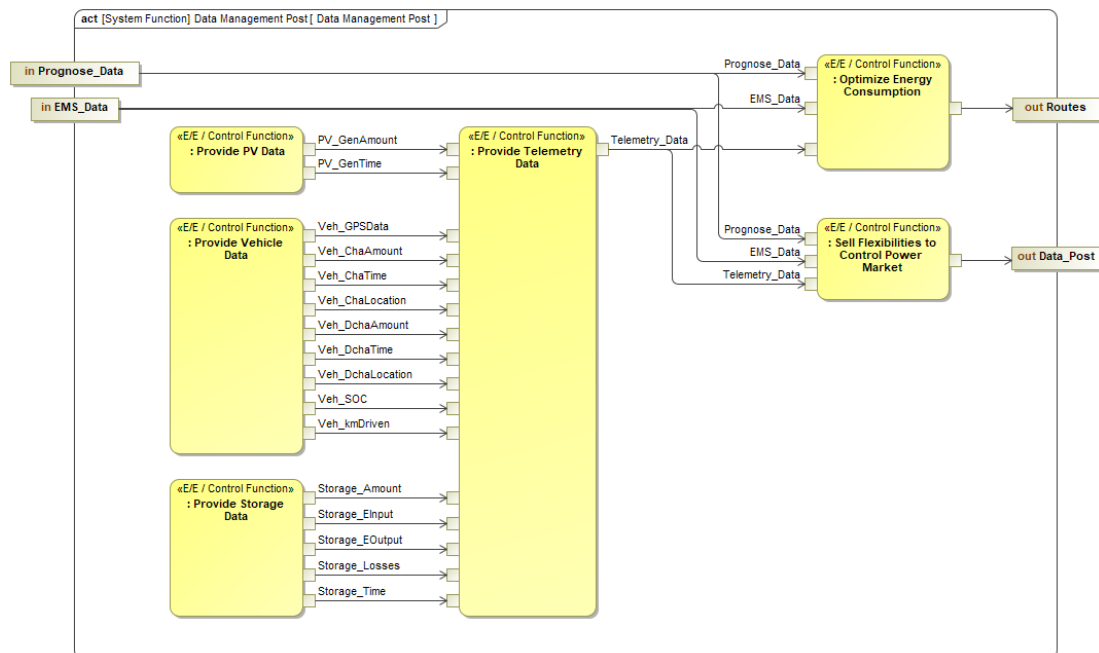


Figure 51. Interfaces of Functions – Data Management in Pilot 3 Infrastructure for POST

6 PILOT 4 – VARIABLE DEMAND IN CHALLENGING MARITIME TERMINAL LANDSCAPE

Pilot 4 shows how ports can manage variable energy demand from cranes, reefers, EV equipment, and berthed vessels, in coordination with the national grid operator Enemalta. It aims to optimize electricity flows at the terminal, reduce fossil fuel dependency by integrating electrified equipment and renewables and test Cloud-Edge-IoT energy flexibility solutions in a real maritime environment.

The pilot combines Malta Freeport's electrified STS cranes, diesel-based RTG cranes, reefer containers platforms, 2xEVs, sensors and SCADA systems with Enemalta's grid control and smart office testbed, creating a real-world environment to monitor, predict and optimize variable port energy demand through IoT, edge and cloud solutions.

P4-SC1 Vessel energy management



Figure 52. CMA-CGM Grace Bay vessel to be monitored during O-CEI Pilot 4 trials

The CMA CGM Grace Bay is a state-of-the-art container ship delivered in May 2024 by China's Jiangnan Shipyard to the French shipping company CMA CGM. This vessel is part of a series of 11 LNG dual-fuel neo-panamax container ships of the 'MARIC 15000 LNG' type, designed to enhance efficiency and reduce environmental impact. Measuring 366 meters in length and 51 meters in width, the Grace Bay boasts a deadweight tonnage of approximately 154,191 tons and a gross tonnage of 154,598. It has a container capacity of 15,254 TEUs, accommodating a significant volume of cargo. The vessel is powered by a WinGD 10X92DF series main engine, delivering around 51,680 kW, enabling speeds up to 22 knots. Additionally, it features five generator sets producing 19,370 kW of electricity to support up to 1,800 refrigerated containers. The CMA CGM Grace Bay operates under the Maltese flag and is deployed on the Mediterranean Club Express service, connecting ports from Valencia, Spain, to Port Klang, Malaysia. Its LNG dual-fuel propulsion system underscores CMA CGM's commitment to sustainable shipping practices, aiming to reduce greenhouse gas emissions and comply with stringent environmental regulations. The O-CEI visualization and decentralized storage utilities will integrate all energy-related flows coming from the vessel. Network connectivity and cybersecurity O-CEI utilities will also be part of the blueprint to be tested in this pilot. O-CEI orchestration will allow the creation of a special control service that will minimize any black-out, i.e., limit the start/stop auxiliary engine of the ship.

P4-SC2: Terminal energy management



Figure 53. Malta Freeport Terminal - location of Pilot 4 Scenario 2

Malta Freeport Terminal commitment to safeguarding environment is demonstrated through the ongoing investments in energy-efficient equipment and renewable energy systems. Significant measures have been implemented to reduce pollution across Quay Cranes and RTGs, equipped with advanced technology to lower energy consumption and improve air quality. In addition, Onshore Power Supply (OPS) system will serve to provide electricity supply to CMA-CGM vessels berthed at Malta Freeport, hence eliminating the necessity of vessels utilizing their own generated power while at berth. This is also expected to bring a considerable reduction in emissions of nitrous oxides, sulphur oxides and particulate matter, as well as a slight reduction in ship-generated noise. Moreover, MFT's total capacity reaches 2,180 reefer units, placing the terminal in an unrivalled position to take on the ever-expanding global reefer container market, as the demand to provide refrigeration for perishable commodities and pharmaceuticals increases. The energy data consumed by the different assets from the container terminal (e.g., CHE data, lighting, buildings, cooling reefers, and EV charging) will be monitored in O-CEI, whose orchestration functionalities will allow to define a priority selection, so that OPS sockets, CHEs or reefer containers with peak power demand can be coordinated, avoiding unnecessary peaks. To do so, every CHE under pilot infrastructure will be equipped with the foreseen control system.

P4-SC3: Home consumers

In the third scenario, the energy capacity adjustment of the island with the vessel and terminal operations will be addressed by replacing the normal electricity meter at homes, by the optimized O-CEI orchestrated control system, able to control the relay protections, providing a basic but O-CEI secure priority connection list of home devices (from critical services, light, water heater, up to home appliances) thanks to the O-CEI data governance schemas. To do so, Enemalta who is the sole Distribution System Operator (DSO) supplying electricity to consumers in the Maltese Islands, will set up an office building, which will be replicated to emulate a whole city. Smart meters and controllers will be installed and will be managed by the optimized O-CEI orchestrated control system. To control the relay protections, a basic O-CEI secure priority connection list of home devices (from critical services, light, water heater, up to home appliances). Finally, thanks to the O-CEI data governance schemas, and the pub/sub models of the project.

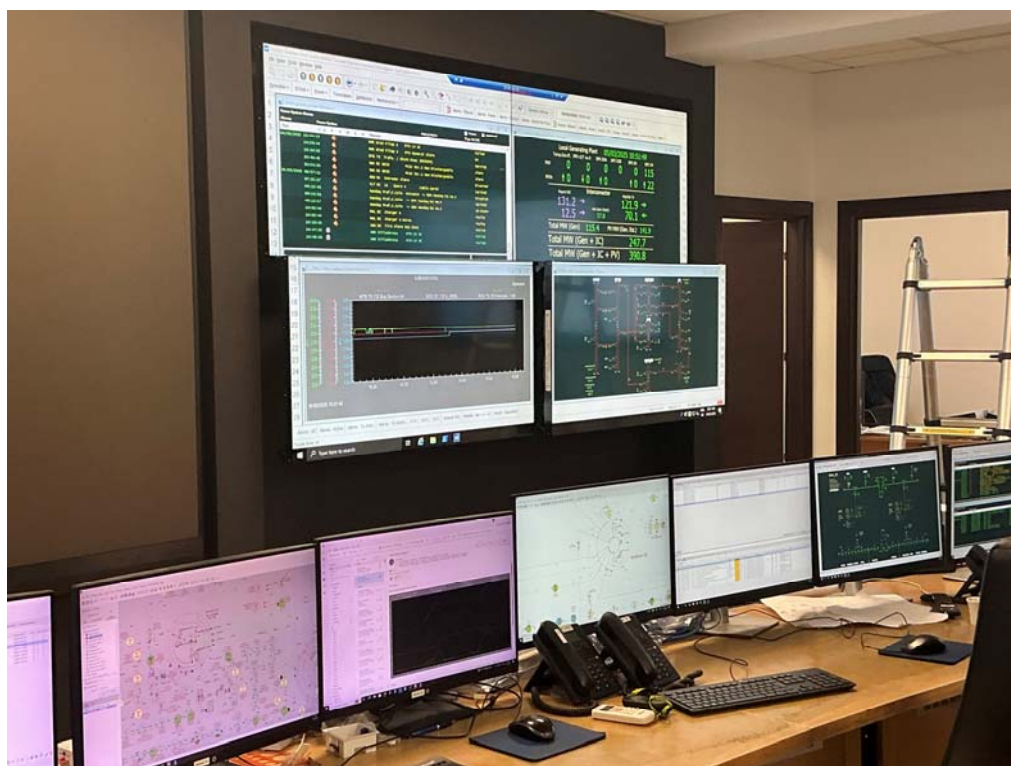


Figure 54. ENEMALTA distribution building - location of Pilot 4 Scenario 3

6.1 PILOT GOVERNANCE

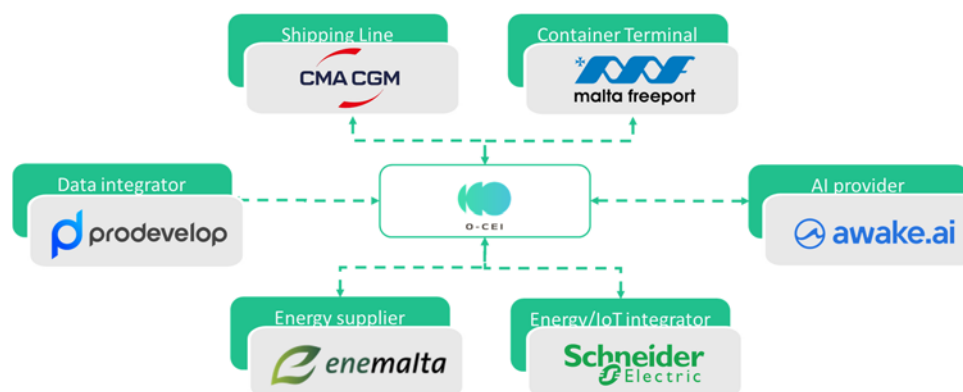


Figure 55. Pilot 4 value chain and governance

Pilot 4 is formed by 6 partners of O-CEI project: CMA-CGM, MFT (Malta Freeport), PRO (Prodevelop), AWAKE, ENEMALTA and SCHN (Schneider Electric). A brief description of their business model, as well as their specific role within the Pilot 4 value chain from a bottom-up approach is presented below.

ENEMALTA: Established in 1977, is the primary energy services provider in the Maltese Islands, responsible for electricity distribution and the development of the national electricity distribution network. The company operates the Delimara Power Station, which includes four electricity generation plants, and manages the Malta-Italy Interconnector, facilitating the import of electricity. ENEMALTA is committed to offering reliable services by continuously investing in modernizing its infrastructure, such as laying over 70 kilometers of new underground cables and upgrading distribution centers, to meet the nation's energy requirements and ensure a stable electricity supply.

SCHN / AVEVA: Schneider Electric is a global leader in energy management and automation, operating in over 100 countries with a workforce exceeding 160,000 employees. The company specializes in electrifying, automating, and digitalizing industries, businesses, and homes to drive efficiency and sustainability. Its comprehensive portfolio includes intelligent devices, software-defined architectures, AI-powered systems, digital services, and expert advisory, enabling buildings, data centers, factories, infrastructure, and grids to function as interconnected ecosystems. SCHN provides engineering for the power data acquisition from the different terminals' typical consumers. Through the different software provided, it gives monitoring and control over selected IO and pushes data to other partners.

PRO is actively involved in the O-CEI project, focusing on enhancing energy efficiency and sustainability in maritime operations. As the leader of the pilot at Malta Freeport, Prodevelop, as main technical leader of the pilot, oversees the implementation of innovative technologies aimed at optimizing energy management flows. Their role as a data integrator involves harmonizing diverse data streams from various sources, ensuring seamless communication between different systems with O-CEI marketplace and orchestration blueprints. Leveraging their expertise in data technology, Prodevelop facilitates the integration of multiple components, including IoT platforms and terminal operating systems, to create a cohesive and efficient operational environment. This comprehensive approach not only streamlines port operations but also contributes significantly to the project's overarching goal of promoting sustainable and energy-efficient practices in the maritime sector.

AWAKE provides cloud-based analytics, planning, and optimization services for maritime logistics. Awake.AI is a research partner and technical provider in the pilot, focusing on data analytics and development of machine learning models and prediction services for forecasting energy requirements in the port. The Awake.AI cloud platform is used in compiling necessary datasets for training ML models, orchestrating data analytics workflows, and building services and APIs to provide prediction functionalities.

CMA CGM / MFT is a leading global shipping company and key shareholder in Malta Freeport Terminals. Both play a pivotal role in the O-CEI project as end-user, providing access to its extensive shipping resources, including CMA-CGM Grace Bay vessel and terminal facilities. This collaboration has led to the introduction of a battery-electric empty container handler developed with Hyster, aiming to reduce greenhouse gas emissions at the port. CMA CGM's governance role and deep understanding of terminal operations are instrumental in aligning the project's objectives with operational needs, thereby promoting cleaner and more efficient maritime activities. By providing access to its extensive network, resources, and industry expertise, CMA CGM enhances the O-CEI project's capacity to implement sustainable and energy-efficient solutions at Malta, thereby contributing to the broader objectives of the initiative.

6.2 DEMONSTRATOR PATH

During the first building phase of the project and since the start of WP4, Pilot 4 has been primarily dedicated to establishing the technical foundation. This phase has centered on setup and procurement activities, ensuring that the essential power monitoring field devices are identified in the architecture for each location to support the pilot's needs of power monitoring data aggregation for the Awake Port Vision (AI model) for visualization and analytics of port activities and energy consumption.

The activities carried out reflect both the technologies identified for fields devices— such as power meter, Scada interfaces, synchronization mechanism—and IT infrastructure identification to run the different historian and connectivity components. From existing power meters at some location to new need requiring brownfield modernization, the pilot's capacity to demonstrate how edge, IoT, and cloud systems can transform port energy management.

This initial work does not only ensure that the power monitoring data are gathered through a secured network architecture, but also ensure connectivity and flexibility for later phases of development, integration, and validation, such as the integration with Prodevelop – MFT big data platform, Schneider Power Monitoring Expert solutions, and AWAKE's Port Vision models for energy consumption prediction.

6.2.1 Setup & procurement activities

P4-SA1 – Existing system at the Grace Bay Vessel

Grace bay CMA-CGM vessel is equipped with reefer distribution boards which are monitored with an energy monitoring system from Schneider that enables precise acquisition of data over time and enables Operator dashboards.



Figure 56. Pilot 4 – Data Gateway (ZigBee)

P4-SA2 – Malta Freeport On-Shore Power Supply

The Government through the Malta Freeport Corporation is in the process of executing the implementation of the High Voltage Shore Connection system at Malta Freeport. The system will serve to provide electricity supply to vessels berthed at Malta Freeport, hence providing the opportunity to berthed vessels to switch off their diesel powered electrical. The system will be available at the Freeport's busiest quays namely North Quay Terminal 1 and North Quay Terminal 2. This will mean a considerable reduction in emissions of nitrous oxides, sulphur oxides and particulate matter as well as a slight reduction in ship-generated noise. The OPS project is being executed in 2 phases:

- **Phase 1:** This phase focuses on equipping the North Quay of Terminal 1 and Terminal 2 with high-voltage shore connections. Construction began in late 2019, with trenching and cable laying nearing completion. Electrical equipment has been tested and is being shipped for installation. Phase 1 is expected to be operational by the end of 2025.
- **Phase 2:** This phase extends the OPS infrastructure to the South Quay of Terminal 2 and the West Quay of Terminal 1. The total investment for this phase is approximately €18.4 million, with the European Union co-financing nearly 70% of the cost through the Connecting Europe Facility (CEF) Transport Call. The tender for Phase 2 was issued online and closed on December 17, 2024, with implementation expected to be completed at the end of 2026.



Figure 57. Malta OPS project² - Trenching and cable laying activities



Figure 58. EV charging stations at MFT

P4-SA3 – EV charging stations

There are two EV charging stations from KEMPOWER used for charging the two Terminal Tractors (TTs), providing up to 800 VDC / 200 KWatts as nominal each (although they have 150 kW each). In principle, KEMPOWER collects the data in their cloud software, but pilot 4 partners will try to get a license for monitoring. The charging stations are connected to a MV/LV substation located 3 meters away.

² <https://maltafreeport.gov.mt/projects/>

P4-SA4 – STS crane switchgear room

MV Switchgear room tag 2E0 contains 10 MV Feeders, 6 to feed STS Cranes, and other feeders for other types of loads. The MV Switchgear is from ABB, model UniSec 12 kV 630 A. The SCADA Supervision and Control System is performed with ABB systems, where the main SCADA system in the control room and local access in the MV Switch room. At least, one of the STS feeders' data will be collected for O-CEI pilot purposes.

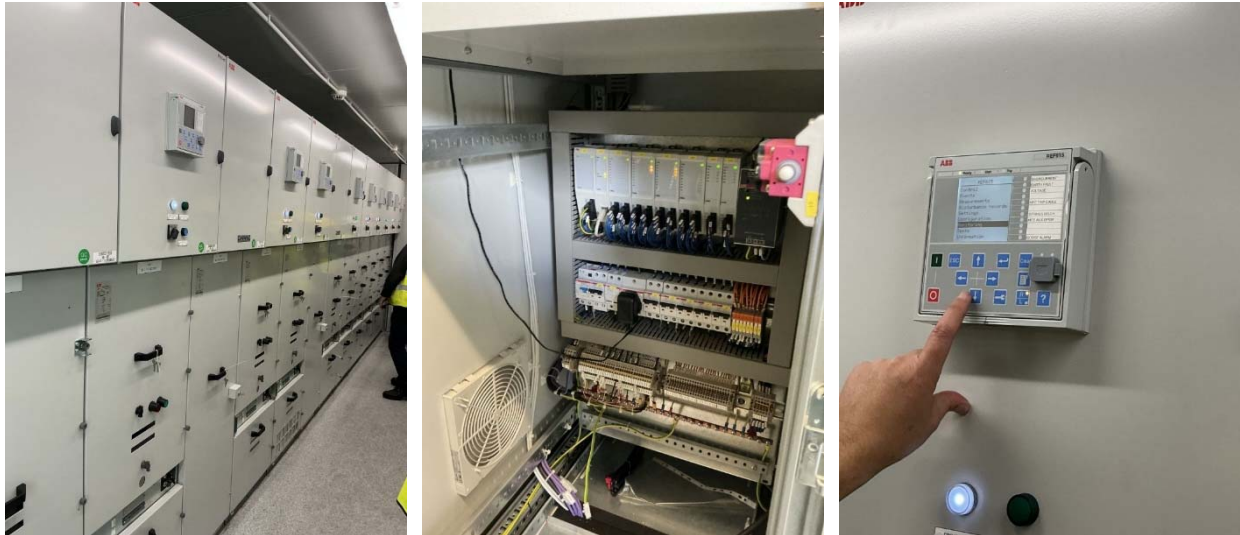


Figure 59. Medium Voltage switchgear for STS electric cranes

P4-SA5 – Reefer station

There are 3 areas of stacking reefers in Malta Freeport Terminal. The monitored stack is able to host up to 30 refrigerated containers (5 levels and 6 rows), that requires 4.5 kilo Watts each as average. Each stack area has a substation of either 500 kVA, 800 kVA or 1 MVA, depending on area size. Although originally, the temperature/humidity/power conditions were monitored by the terminal, they are nowadays in general, monitored by the shipping lines. Nevertheless, new equipment will be installed in the context of O-CEI, including smart meters + disconnected devices + modems to collect the power and control the power supply and additional values to be determined. The goal is to limit energy consumption so that when the monitored temperature is above/below a prefixed value + thresholds (e.g., $-22^{\circ}\text{C} \pm 0.3$), the compressor is started by the system, but when the temperature is within the set threshold, the temperature control system stops the compressor.



Figure 60. Reefer station in MFT premises (I)

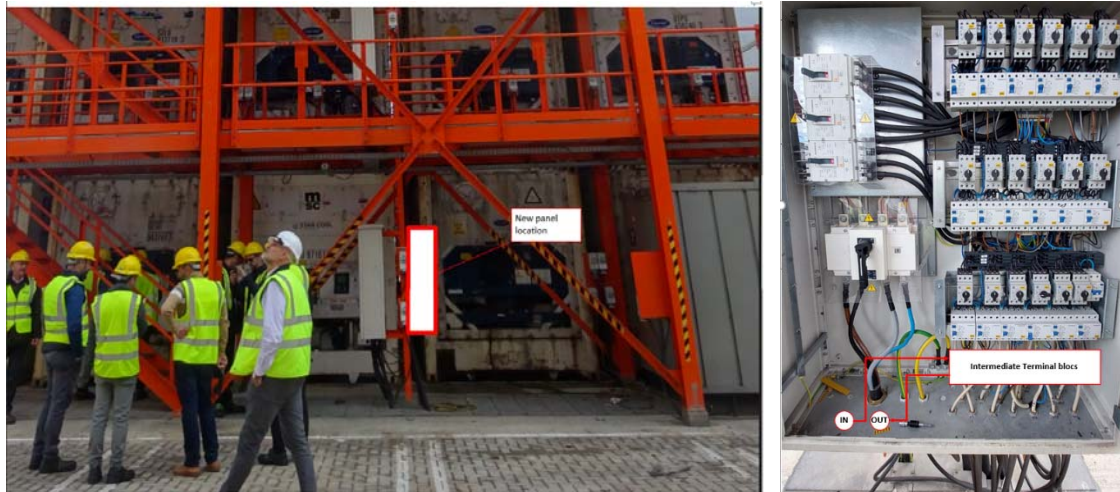


Figure 61. Reefer station in MFT premises (II)

P4-SA6 – Procurement of fields devices: PLC for Crane, Vessel, EV charging, Reefer power station zone

To complement the existing power monitoring sources in the different port areas, several components such as PLC for interfaces are selected for procurement.



Figure 62. Pilot 4 – M340 Interface PLC

P4-SA7 – Creation of Panel for Reefer DB

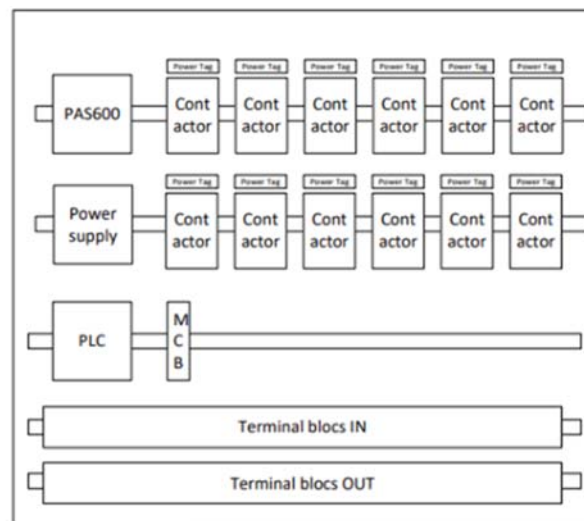


Figure 63. Pilot 4 – Reefer floor DB – Additional panel

Creation of panel specification to enable the monitoring of each reefer individually and provide possibility to un-supply a specific reefer. This includes a PLC controller, a set of connectors, a set of power meters and a network gateway. A small humidity and temperature sensor is also added to get context data associated with reefer consumption. Material inside:



Figure 64. Pilot 4 – Temperature and humidity sensor, connector and power sensor, and IO PLC

P4-SA8 – ENEMALTA Power Plant and Energy distribution system

The ENEMALTA network control room is essential for maintaining the reliable and efficient delivery of electricity along the island, as it serves as the command center for efficient grid operation and management. It continuously monitors the grid from 132kV delivery to 33kV substations and 11kV distribution points in real-time, manages the balance between supply and demand, and detects and responds to faults or outages. The control room coordinates maintenance activities, handles crisis management, and integrates renewable energy sources into the grid. It also plays a key role in data analysis and reporting, ensuring cybersecurity measures are in place to protect the infrastructure. There are several software services from different manufacturers, including those from SIEMENS, ABB and SCHN (2 substations in EPAS/PACIS) for Substation Supervision and Control, furthermore to Distribution Management System by DigPro that helps to monitor main components of the energy system from a generic perspective architecture of the system, as well as from their actual physical placement. Data are provided by local RTU located inside substation and connected to protection relays there. Future interface for energy monitoring could be planned either at substation level by using telecontrol protocol within local RTU or more globally with SCADA.

P4-SA9 – Procurement of fields devices for the building from Scenario 3

It was decided that the fourth floor of the ENEMALTA control building at Marsa will be the smart office of the scenario 3 of pilot 4. Currently, every room has its own AC, which is going to be monitored by connecting to the central controlled system on the roof. The smart meters to be used are described below.

- **AEG Elfa64 C63:** 4-pole miniature circuit breaker designed for industrial applications, offering a rated current of 63A and a breaking capacity of 10kA. It features a C-curve characteristic, suitable for protecting circuits with moderate inrush currents. The device operates at a rated voltage of 230/415V and occupies four EN50022 modules on a DIN rail. It is compatible with electrical auxiliaries and holds CE and IMQ certifications.
- **Shelly i4 Gen3:** A Wi-Fi-operated 4-digital input controller that enables manual activation or deactivation of scenes and synchronization of actions. It supports power supplies of 110-240V AC and features multi-click support for up to 16 possible actions.
- **Schneider PAS800:** Advanced gateway and data logger from the EcoStruxure range, designed to integrate wireless and Modbus RTU devices with control software and cloud applications. It features real-time data collection, email alarm management, and up to three years of data storage. Configuration is user-friendly via embedded web pages or EcoStruxure Power Commission software. The PAS800 includes two Ethernet ports, supports various communication protocols, operates on 110 to 277 V AC/DC, and functions within temperatures from -25°C to +70°C. It holds certifications such as CE, UL/CSA, and IEC61010



Figure 65. Field devices under procurement for Pilot 4 - Scenario 3

6.2.2 Development activities

P4-DA1 – Development Schneider PME configuration (VM1)

The PME (Power Monitoring Expert) server aggregates energy telemetry from sensors and power meters across the terminal. It provides real-time visualization, historical logging, and supports integration with supervisory platforms, contributing to efficient energy management and decision-making. The PME interfaces with the Port field devices through Modbus TCP protocol and provides a dashboard for energy monitoring. It also synchronizes with the PME server onboarded in the Vessel to retrieve onboard Reefer DB power consumption prior to arrival.

P4 DA2 – Development AVEVA System Platform configuration (VM2)

The AVEVA System Platform allows the Operator to perform control over controllable devices. It also allows custom definition of script and alarming. It also has the possibility for many standard OT protocols to interface with that are still under partners' consideration.

P4 DA3 – Development AVEVA PI system (VM3)

PI system is there used as an historian to enable data connectivity for AVEVA Connect. AVEVA connect is the AVEVA platform on Cloud that enable easy data sharing from PI System (On premises: ENEMALTA server) to the AWAKE AI Cloud and the O-CEI Platform.

P4 DA4 – Development PLC configurations

PLC will serve as interface at the different location indicated above. The configuration will be made, and Modbus mapping will be generated for the interface to the 3 VM.

P4 DA5 – TOS data acquisition

Not only energy-related data from the different assets in Malta Freeport Terminal should be collected and aggregated. Operational-related data is also required in order to correlate the energy consumed with the working needs. Therefore, identification of the appropriate SQL tables and views from the TOS of Malta Freeport Terminal are being analyzed and the most suitable ones for the pilot purposes are going to be stored in the Prodevelop big data platform, and securely exposed to other pilot partners, such as Awake.AI port vision activities in P4 DA6.

P4 DA6 – Exploratory data analysis of terminal operations data

Using historical sample datasets from the Malta Freeport Terminal regarding vessel port calls and terminal operations, Awake.AI is performing analysis to evaluate the scope of available data, interconnections between datasets and external data sources, and the potential for using the existing data in machine learning model development. This analysis provides the basis for planning subsequent integrations, continuous data collection, and development of predictive models and simulations.

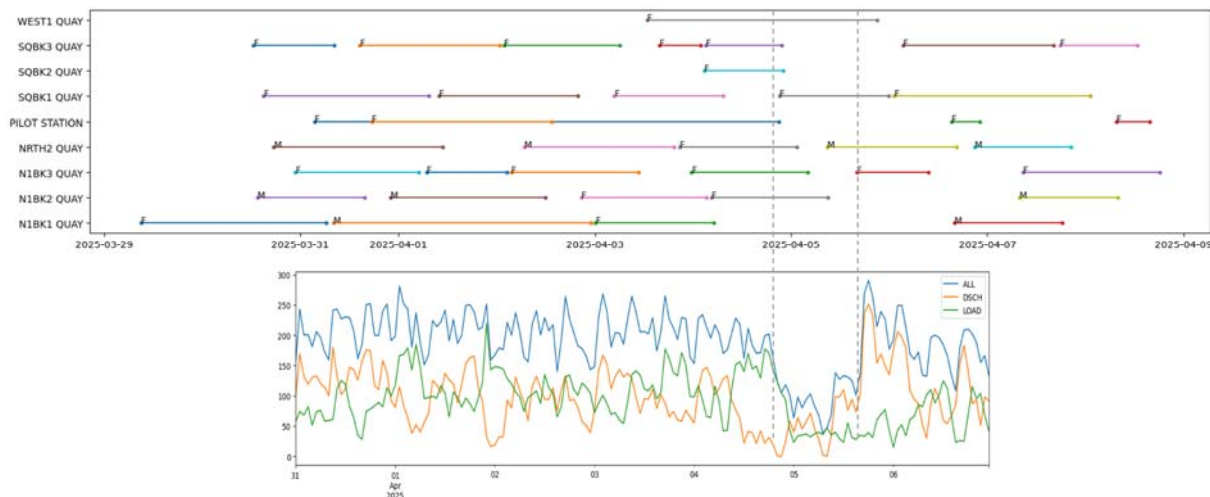


Figure 66. Pilot 4 – Example of exploratory data analysis, illustrating correlation between vessel berth visits (top) and energy usage in cargo handling operations

P4 DA7 – Machine learning model training and validation

Once data on terminal vessel visits, cargo operations, and related energy usage is available in sufficient scale for comprehensive analysis, models will be developed for predicting terminal energy requirements based on current sensor-based and planning data. For this, the available datasets will be analyzed to identify correlations and salient features related to the prediction objectives, and the types of models applicable for the task. For applying machine learning-based models, training workflows will be set up to enable training orchestration, validation, and performance testing using relevant statistical methods such as nested cross validation.

P4 DA8 – Development of prediction services and API interfaces

To provide predictions on future energy requirements based on currently available information on vessel and cargo operations schedules, cloud-based services and infrastructure will be developed for ingesting necessary data, communicating this between cloud microservices, and hosting prediction models. Furthermore, application programming interfaces will be configured to enable access to the predictions from other systems in the pilot.

6.2.3 Integration activities

P4 IA1 – Virtual Machine Integration

Integration of the validated VM1, VM2, VM3 into MFT IT infrastructure (To be determined if they are deployed on premises or on Cloud).

P4 IA2 – Material Integration activity at Malta Freeport

Integration of energy-related material from cranes, reefers power stations and DB floor, EV charging stations, Grace Bay CMA-CGM vessel into the existing network infrastructure in order to establish connectivity with the IT server hosting the above 3 VMs from P4 IA1. There is one specific case still under discussion with the ship owner: the potential need of a VPN from IT infra to Vessel 4G router when sailing in Malta Freeport Terminal proximities.

P4 IA3 – Scada network link with OPS Cold Ironing

Network interface/network infrastructure to be defined.

P4 IA4 – Scada network link with ENEMALTA main power station

Network interface/network infrastructure to be defined.

P4 IA5 – Definition of API infrastructure and data schemas for continuous data collection

To enable development and training of machine learning models, data on vessel port calls and terminal operations needs to be aggregated over long time periods. Based on implemented exploratory data analysis, specification of data to be shared and API interfaces enabling data sharing are planned between PRO and AWAKE. Cloud-based services and APIs are developed for continuous data collection, enabling use of large historical datasets in training machine learning models.

6.2.4 Planning & Gantt

The following Gantt diagram lists all the setup & procurement, development, and integration activities since M11 until M21.

Table 11. Pilot 4 Gantt diagram

Pilot 4		2025		2026										
Code	Task	M11	M12	M13	M14	M15	M16	M17	M18	M19	M20	M21		
SETUP & PROCUREMENT														
P4-SA1	Existing system at the Grace Bay Vessel													
P4-SA2	Malta Freeport On-Shore Power Supply													
P4-SA3	EV charging stations													
P4-SA4	STS crane switchgear room													
P4-SA5	Reefer station													
P4-SA6	Procurement of fields devices for scenario 2													
P4-SA7	Creation of Panel for Reefer DB													
P4-SA8	Enemalata Power Plant and Energy distribution system													
P4-SA9	Procurement of fields devices for the building from Scenario 3													
DEVELOPMENT														
P4-DA1	Development Schenieder PME configuration													
P4-DA2	Development AVEVA System Platform configuration													
P4-DA3	Development AVEVA PI system													
P4-DA4	Development PLC configurations													
P4-DA5	TOS Data acquisition													
P4-DA6	Exploratory data analysis of terminal operations data													
P4-DA7	Machine learning model training and validation													
P4-DA8	Development of prediction services and API interfaces													
INTEGRATION														
P4-IA1	Virtual Machine integration													
P4-IA2	Material integration activity at Malta Freeport													
P4-IA3	Scada network link to OPS cold ironing													
P4-IA4	Scada network link with Enemalta main power station													
P4-IA5	Definition of API infrastructure and data schemas for continuous data collection													

6.3 CURRENT INFRASTRUCTURE

The infrastructure includes vehicle IoT sensors, control ECUs, edge logic for data acquisition, and cloud-based visualization and analysis platforms. Several key acquisitions are required to support the pilot's goals, including a connection to an actual Vessel, Terminal Energy management, programmable telematics gateway, and a weather station for data-driven decision-making.

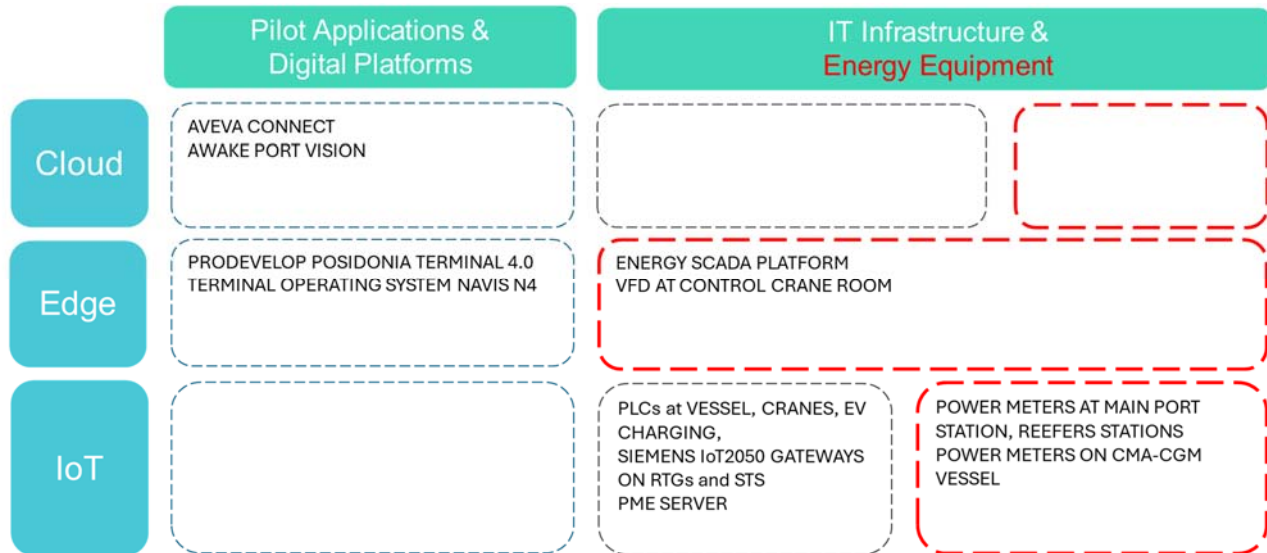


Figure 67. Pilot 4 Infrastructure

6.3.1 Pilot applications and digital platforms

Cloud Layer

- **AVEVA Connect**, a cloud-based industrial software suite for real-time monitoring and control.
- **Awake Port Vision** for visualization and analytics of port activities and energy consumption.

Edge Layer

- **Prodevelop Posidonia terminal 4.0**: a real-time IoT & Big Data platform that leverages CHE telemetry data to measure actual operational data.
- **NAVIS N4**: Malta Freeport Terminal Operating System (TOS), built on a highly scalable fault tolerant architecture. It is based on latest J2EE design concepts, advanced metadata driven model and highly configurable business rules.
- **Schneider PME**: The PME (Power Monitoring Expert) software aggregates energy telemetry from sensors and power meters across the terminal. It provides real-time visualization, historical logging, and supports integration with supervisory platforms, contributing to efficient energy management and decision-making.
- **AVEVA System Platform**: It will enable integration and control of operational data from controllable device and integrate through the OMI the external cloud advisory dashboard.
- **AVEVA PI system**: It will gather information as the main historian and enable the data asset modelling before being shared with MFT's on-prem big data platform managed by PRO.

IoT Layer

- **PLCs** deployed on vessels and cranes.
- **Power meters** installed at reefer stations, vessels, and main port stations.
- Environmental **sensors** for temperature and humidity monitoring of refrigerated containers.

6.3.2 IT Infrastructure & Energy Equipment

Edge Layer

- **Edge Energy Equipment:** energy SCADA platform and VFD at Control Crane room.

IoT Layer IT Infrastructure:

- **PLCs at Vessel, Cranes, and EV Charging:** Programmable Logic Controllers (PLCs) are deployed across vessels, cranes, and EV charging stations to enable real-time control of energy-intensive operations. On vessels, they manage shore power connections; on cranes, they regulate lifting systems and motor behaviour; and at EV chargers, they automate load control and ensure safe charging cycles.
- **Siemens IoT2050 Gateways on RTGs and STS:** Siemens IoT2050 gateways are installed on Rubber-Tyred Gantry (RTG) and Ship-to-Shore (STS) cranes to collect operational data, process it locally, and transmit it securely to edge or cloud systems. They support edge computing functions, reduce latency, and enhance the responsiveness of energy and control applications.
- **Servers** (On premises VM or Cloud VM): The servers will host the VM for PME, Aveva System Platform and AVEVA PI system.

7 PILOT 5 – ENERGY EFFICIENT AND SUSTAINABLE HALLOUMI CHEESE PRODUCTION

Pilot 5 serves as a compelling example of how Europe's agri-food sector can harness cloud–edge IoT technologies to advance sustainability objectives, enhance animal welfare, and optimize energy consumption in the face of challenging environmental conditions.

Implemented in Cyprus, the pilot focuses on the traditional dairy value chain — specifically the production of Halloumi cheese — and aims to demonstrate how data monitoring, integration, and intelligent decision-support for energy-efficient operations can transform a sector confronting both climatic and structural challenges. Its objectives include reducing energy consumption, mitigating livestock heat stress, enabling seamless data exchange across production sites, and achieving tangible sustainability outcomes under increasingly adverse climate conditions.

The next figure provides a high-level overview of the IoT–Edge–Cloud information flow across the entire value chain, from milk production to cheese processing. At each pilot site (Farm A, Farm B, and the factory), an edge mini-server will host the CEI Interoperability and Optimization Enabler — a software component developed specifically to meet the needs of Pilot 5.

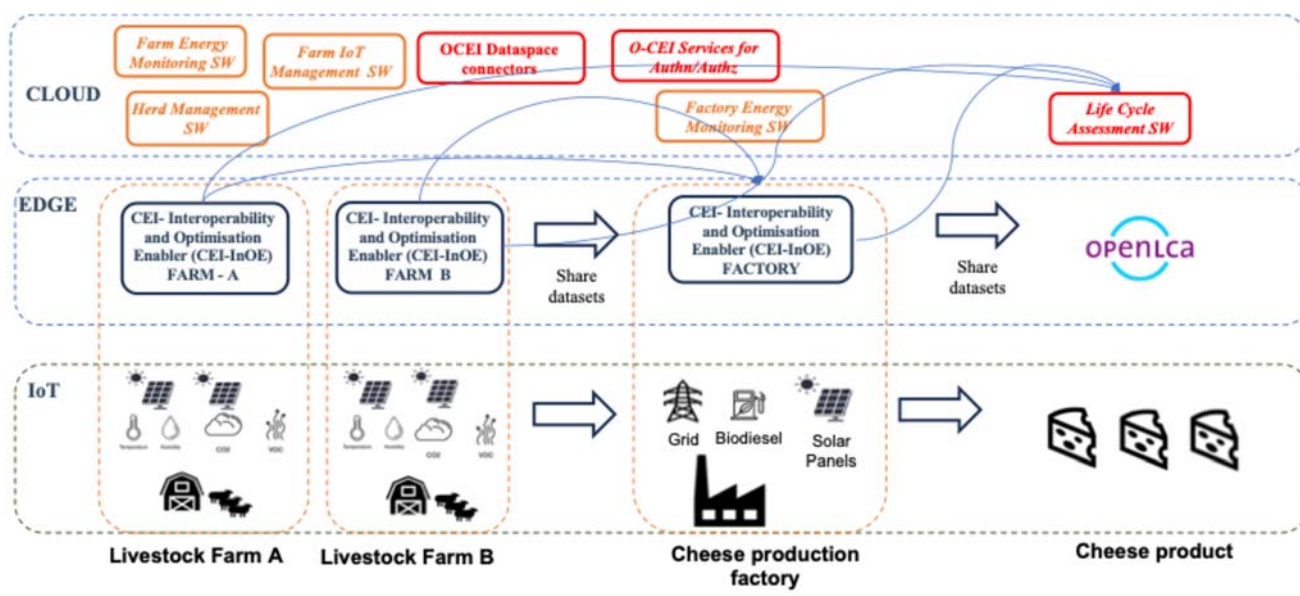


Figure 68. Illustration of the key CEI continuum for the cheese production chain of Pilot 5

This pilot has 4 use-case scenarios that are already under realization:

P5-SC1: IoT-Assisted Livestock Management based on Edge Intelligence and automation

The objective of this scenario is to optimize energy use consumed in stables, while ensuring the animal welfare through the reduction of heat stress. **The main technological focus** is on monitoring and optimizing the energy consumption of various devices used for livestock management and milk production, such as milking stations, cooling fans, freezers, and feed mixing equipment.

Equipment: Milking station, Animal Colling Fans, Milk tank freezers, Food mixing/mill station, energy metering devices, environmental sensors, actuators (for the cooling station), LoRaWan Gateways, PV panels for energy production, batteries for energy storage, mini (edge) server system installed at the farm premises that will host CEI-Interoperability and other edge enablers.

By integrating smart metering devices and environmental sensors, the pilot continuously monitors barn conditions, detects potential heat stress in animals, and generates alerts when critical thresholds are reached. Using advanced modeling and classification techniques, it analyzes animal behavior to distinguish between stressed and non-stressed states and predicts milk production in terms of both quality and quantity. Additionally, the system optimizes energy consumption and costs by dynamically managing HVAC operations and intelligently combining different energy sources, such as the electrical grid and photovoltaic systems.

The main goal is to use smart metering and forecasting of energy needs to optimize energy consumption maintaining optimal conditions for the animals and the workers. **Legacy/previous technologies** include visual inspection of the herd and manual operation of heat mitigation mechanisms. The **expected outcomes** of the first scenario are to ensure that animals (cows) are not stressed, while also ensuring/predicting milk quantity and quality and optimizing energy consumption. The following figures depict the facilities of one of the farms, the installed cooling system, and an example of an already installed energy metering device.



Figure 69. View of one of the cow farms that participate in Pilot 5



Figure 70. View of one of the cow farms that participate in Pilot 5



Figure 71. Cooling System/ Fans installed in one of the farms of Pilot 5



Figure 72. Energy meter devices

Table 12. Energy metering devices already installed in Farms' systems

Energy meter (LK Dairy Farm)	Robotic Milking machine
Energy meter (LK Dairy Farm)	Robotic Milking machine
Energy meter (LK Dairy Farm)	Air compressor for milking machine
Energy meter (LK Dairy Farm)	Fridge
Energy meter (LK Dairy Farm)	Boiler
Energy meter (LK Dairy Farm)	Fans
Energy meter (Blue Farm Dairy)	Fridge
Energy meter (Blue Farm Dairy)	Fridge
Energy meter (Blue Farm Dairy)	Fans of cooling system
Energy meter (Blue Farm Dairy)	Fans of cooling system
Energy meter (Blue Farm Dairy)	Air compressor for milking machine
Energy meter (Blue Farm Dairy)	Air compressor for milking machine
Energy meter (Blue Farm Dairy)	Fans into the farm

P5-SC2: Controlled sharing of data from various stables to the dairy production factory

The 2nd Scenario scene of Pilot 5 focuses on improving the planning and optimization of energy consumption across the dairy production chain through accurate scheduling and forecasting of production cycles. The main technological focus will be given to dataset sharing, semantic interoperability, access control, authorization/authentication and trust.

Selected datasets from stables having a trusted relation with the production factory will be collected and dynamically shared thanks to O-CEI mechanisms and marketplace. O-CEI semantic stack will integrate and expose datasets about milk quantity, quality, properties, feed mix or energy footprint along the supply chain. Standardization trust, security, data protection and data lineage will be applied. The datasets will remain under the control of the owner (e.g. farm administrator) in compliance with existing regulations and best practices. The main goal is to establish secure and trusted generic purpose data sharing mechanisms providing ground truth evidence about the milk production process including parameters about energy consumption and animal/environmental conditions during the milk production. This will allow the planning of the production cycle based on the available quantity and quality of milk.

Equipment: Mini PC systems will be installed in the pilot sites acting as Edge based data interoperability and optimization enablers (figure 6). **The legacy/previously** existing technologies are including data sharing with the use of spreadsheets, paper inventories, heterogeneous formats and no-data sharing. Legacy/previous technologies recording and sharing of milk related information is mainly paper based. **Expected Outcomes:** Establish an interoperable data sharing mechanisms among farm management information systems and the dairy factory Information System.



Figure 73. Mini PC that will host the CEI-INoE software that will ensure interoperability and optimisation

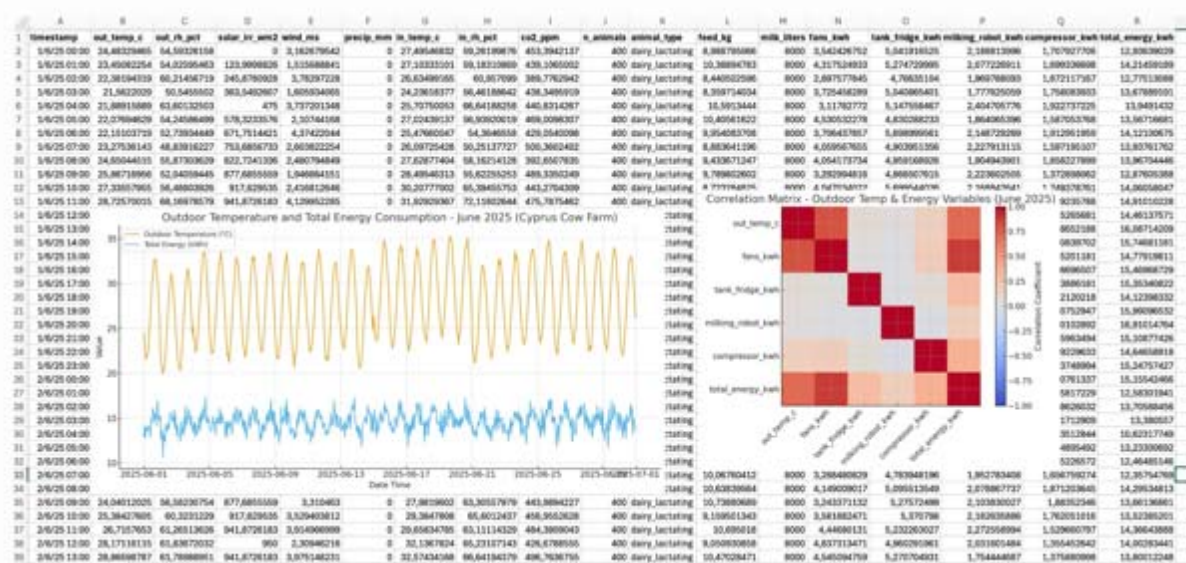


Figure 74. Example of the time-series datasets to be processed towards the energy consumption optimisation objective

P5-SC3: Intelligent recommendations for the selection of energy mix for dairy production



Figure 75. View of the cheese production factory participating in Pilot 5

Cheese production in factory involves several processes that may have varying energy needs depending on aspects such as milk quantity or cheese type. This scenario is oriented to optimise the energy source selection (local grid network, own solar panels, biodiesel engines) according to such needs. The 3rd scenario scene of Pilot 5 is to monitor, analyse, forecast and optimize energy use by balancing different energy sources, such as the local grid, solar panels, and potentially other available energy sources (e.g. biodiesel) to meet the varying energy demands of cheese production efficiently.

The main **technological focus** will be on installing IoT-based energy metering devices to key cheese production machines but also environmental monitoring sensors (e.g. temperature, solar irradiance). In addition, solar panels will be installed on Factory's premises along with energy production metering equipment. Mechanisms for data collection, harmonisation and analytics will be established supporting the forecasting and correlations identification of energy consumption. Finally, recommendation for realising an optimised energy mix combining different energy sources will be provided. The objective of the suggested ideal mix is to reduce costs, environmental impact and footprint while optimizing cheese production operations. The main goals are summarized below:

1. Reduce overall energy consumption costs.
2. Minimize carbon emissions through increased reliance on renewable energy sources.
3. Achieve energy autonomy by dynamically selecting the most efficient and cost-effective energy mix.
4. Improve operational sustainability and reduce the environmental footprint of halloumi cheese production.

Established mechanism allows the recording of energy of the entire production line without being possible to identify the energy needs of individual production subsystems in a more fine-grained manner. In addition, PVs are only partially utilized and not integrated within a holistic decision support approach.

Expected Outcomes: Discover cheese production energy consumption insights and provide optimization recommendations towards reduced environmental impact and cost.

P5-SC4: Calculation of the energy footprint of dairy (halloumi) products

In Scenario 4, the focus is on the aggregated energy footprint per unit of production, which includes energy consumption throughout the production chain.

Main technological focus: Data collection from heterogeneous data sources, an energy assessment algorithm based on the aggregation of the various datasets to calculate the energy footprint. The algorithm will consider, among others, direct electricity measurements (IoT), relevant parameters and trends (AI inference at the edge) and machinery power usage across the livestock and production chain.

The main goal of this scenario is to calculate consumed energy per production unit.

Legacy/previously existing technologies: Energy metering in a disjoint manner. No footprint calculation involving the whole production cycle. **Expected outcomes:** Digital evidence for sustainability labeling and regulatory reporting ([Corporate Sustainability Reporting Directive](#)) will enhance the trustworthiness of the halloumi production.

A key ambition is to establish a scalable and replicable blueprint for digital transformation in agri-food systems, particularly for small and medium-sized farms that lack access to integrated technologies or suffer from poor interoperability between existing systems. While the model may not be suited for large multinational dairy producers, it offers practical, accessible tools for smaller farms across Mediterranean and climate-vulnerable regions. The architectural blueprint will be escorted by actual evidence on the potential environmental and financial gains.

7.1 PILOT GOVERNANCE

Pilot 5 adopts a streamlined governance model that ensures efficient coordination, clear accountability, and seamless integration of activities among all partners.

GSC (GREEN SUPPLY CHAIN DIGITAL INNOVATION HUB) is the Leader of Pilot 5 on “Energetically and environmentally sustainable Halloumi cheese production”. More specifically, it coordinates the pilot design, collection of technical/user/business requirements from all pilot partners, specify the architecture of the technical solutions, and ensures that the developed solutions are aligned with the O-CEI Platform. In addition, GSC contributes to data sharing and data interoperability mechanisms, as well as to DEC related activities.

EMBIO (SK EMBIO DIAGNOSTICS LIMITED) supports the implementation of Pilot 5 by providing IoT cloud/edge technologies and equipment (sensors, actuators, energy monitoring) along with data collection and data processing software. It develops AI-based data analytics and implements the cloud-edge integration needed for smart livestock management and energy optimization. EMBIO also supports deployment and calibration of field devices and the data processing framework.

CHRIS (Charalambides Christis LTD) is the end user of Pilot Program 5 and specializes in dairy processing and product development. It provides the dairy processing unit and the animal stables. At least two medium-sized livestock farms will participate in the O-CEI pilot program (>500 animals in total). CHRIS contributes operational and production data, validates the developed solutions under real conditions, and assesses their impact on energy efficiency, animal welfare, and environmental footprint. It is the largest and longest-running dairy industry in Cyprus, employing more than 600 people. The factory's halloumi production line will be engaged for multiple production cycles, resulting in 100 tons of final product (halloumi).

The intra-pilot governance relies on continuous collaboration and transparent communication among partners. GSC leads biweekly coordination meetings to set short-term and long-term objectives, monitor the advancement of activities, review technical developments, and ensure alignment across

the work plan. In addition, partners maintain daily exchanges through shared online workspaces and internal communication channels, allowing for quick problem resolution, data and document sharing, and synchronized decision-making. Progress reports, technical documentation, and testing results are merged and validated collaboratively under GSC's supervision to maintain a coherent and traceable development process.

7.2 DEMONSTRATOR PATH

Pilot 5 is organized into five clearly defined implementation phases, each marking a critical step in its technical and operational development. The following table details these phases and the specific activities associated with each stage

Table 13. Implementation phases of Pilot 5

Timeframe	Phase	Description of activities
M7 – M12	Phase 1 – Smart Infrastructure Procurement & Installation	Analysis of needs and procurement of infrastructure: Cooling Fans, smart sensors, PV systems, inverters, metering devices for farms and the factory. Installation of the cooling system and fans Installation of metering devices. User and technical requirement analysis. Energy and livestock technology landscape analyses.
M13 – M18	Phase 2 Pilot Infrastructure Integration & Baseline Testing	Installation of PVs at the factory and farm premises. Installation of a PV energy storage system (including inverters and batteries) at one of the farms. First-level system integration tests begin. Verification of baseline system operations. Specification of the CEI-Interoperability and Optimization Enabler (CEI-InOE) architecture that will support data collection, harmonization, and analytics.
M19 – M30	Phase 3 Operational Deployment of IoT & Energy Monitoring	07/2026-06/2027 Full deployment of edge/cloud monitoring systems, THI sensors, automation routines, and energy footprint tracking across the dairy production chain. User dashboards and real-time data collection begin. These will be realized by The CEI-InOE software.
M31 – M36	Phase 4 Scenario Execution & Edge-AI Implementation	07/2027-12/2027 Edge-AI deployment, scenario-based testing, and adaptation. Stakeholder training and interim results evaluation.
M37 – M42	Phase 5 Final Validation and Demonstration	01/2028-06/2028 Real-life demonstrations, validation of KPIs, data storytelling for communication, and documentation of lessons for replication. Engagement of policymakers and agri-cooperatives for scale-up potential.

Until M09 the following activities were carried out:

For P5-SC1: A state-of-the-art review and market research on digital livestock technologies were finalized, along with a detailed assessment of Cypriot farms in terms of digital readiness, data collection practices, and energy usage. Participating farms have been selected, and key livestock management devices for monitoring and optimization (milking stations, cooling systems, food mixers, etc.) have been identified. Procurement for the sub-metering devices has been completed, and the installation is underway to enable real-time monitoring of equipment energy consumption. The next steps include installation and commissioning of sensors, creation of integrated datasets linking energy use with environmental and operational parameters, and development of decision-support insights for optimizing energy consumption and animal welfare.

For P5-SC2: This scenario has advanced into the design and specification phase of data-sharing mechanisms. Current data transfer practices between farms and the factory—mainly manual and paper-based shipment reports—have been documented and analyzed. User and technical requirements for standardized, interoperable, and secure data exchange have been specified, leveraging O-CEI data models and semantic frameworks. Design work has started on a data-sharing information system tailored to the dairy sector, aiming to facilitate trusted and efficient exchange of production and energy data. Next steps include system implementation, integration, and pilot testing with selected farms and the dairy factory.

For P5-SC3: This scenario has reached an advanced stage of technical preparation and partial infrastructure deployment. The cheese-production process has been fully documented, and the energy consumption profile of each stage has been defined. Work for the sub-metering device procurement has been completed while the installation process is ongoing on, as well as the technical analysis for integrating renewable energy sources such as PV panels and biofuel engines. These systems will enable the deployment of a Decision Support System (DSS) for dynamic energy-mix optimization based on real-time data and O-CEI analytics. Next steps involve completing the equipment installation, activating the centralized energy data repository, and correlating consumption data with production outcomes to support data-driven sustainability decisions.

For P5-SC4: Pilot 5 has initiated a comprehensive analysis of the regulatory and sustainability reporting landscape, focusing on the Corporate Sustainability Reporting Directive (CSRD), Protected Designation of Origin (PDO) requirements, and Life Cycle Assessment (LCA) methodologies. Relevant production datasets have been identified along the dairy value chain, and data collection mechanisms are being designed to support automated compliance and traceability. Preparations for the implementation of energy footprint calculations and LCA analysis are in progress. These efforts aim to enable digital evidence-based sustainability certification and to integrate traceability across the entire halloumi production chain, from farm to factory to final product.

Pilot 5 has successfully completed the design, analysis, and preparatory phases, with procurement and partial deployment of energy-monitoring infrastructure in progress.

7.2.1 Setup & procurement activities

During the first implementation phase (M7–M12), the focus is on the needs analysis, procurement and installation of smart infrastructure. This includes a thorough analysis of the existing production process both at the farms and at the factory, analysis of the existing utilized infrastructure focusing on the energy needs of the machinery/devices, the identification of the most energy demanding processes and finally the identification of selected intervention points toward energy consumption optimization. This analysis resulted in acquiring and deploying a dense network of energy metering devices both for the farms and the factory, smart sensors detecting animal presence, cooling fans automatically controlled upon animal presence in a specific spot, PV systems, inverters and batteries. The early installation of cooling systems, fans, and metering devices ensures that everything will be in place to start collecting baseline data for the rest of the phases to be implemented. In parallel, alongside the procurement and installation of the smart infrastructure, data gathering on dairy production and energy consumption has already begun, and the architecture for the use-case scenarios is being designed.

Building on this foundation, Phase 2 will focus on integrating the infrastructure and testing baseline operations to ensure readiness for the pilot activities. In Phase 3, IoT devices, data collection and data processing systems will be fully deployed, enabling real-time data flows, automation, and the first steps in calculating the energy footprint of halloumi products. Phase 4 will then focus on executing the four use-case scenarios under real conditions, enhanced by edge analytics for livestock management, provision of recommendations for intelligent energy mix optimization, and controlled data sharing, paving the way toward the final demonstration phase.

7.2.1.1 System Architecture

The system architecture for Pilot 5 was designed to address two core challenges in the Cypriot dairy sector: high energy consumption and livestock heat stress, both of which impact productivity, animal welfare, and sustainability performance. After technical evaluations, a hybrid cloud-edge IoT approach was selected, integrating energy monitoring, cooling system, and renewable energy assets at both farm and factory levels.

At one of the farms, a smart cooling system (fans and misters designed to lower farm's indoor temperatures) was installed, complemented by LoRa-based energy meters and environmental sensors. This setup enables real-time decision-making for animal comfort and energy profiling. Additionally, a hybrid inverter with PV and battery storage was designed to reduce grid dependency and smooth peak demand.

At the factory, the focus is on integrating a rooftop PV system into the production energy mix, supported by detailed monitoring via energy meters. The PV system will be connected directly to the Halloumi production line, which includes milk storage, processing, and packaging operations. This setup ensures that renewable energy directly offsets the electricity used in the core production stages of the pilot, allowing for improved energy efficiency and a measurable reduction in the carbon footprint per kilogram of Halloumi produced.

In both cases, analytics on historic data collection of energy consumption along with selected contextual information (weather parameters, milk production quantity, milk quality) will facilitate the discovery of correlations and forecasting of energy needs towards recommendations for optimized energy use. This modular, scalable approach was chosen to allow progressive integration, low operational complexity for the end users (farmers and factory operators), and replicability for other Mediterranean agro-industrial sites with similar constraints. These will be realized by a software-based information system planned to be deployed on a mini (edge) server system installed at the farm premises. In figure 4 a high-level functional architecture view of the "CEI-Interoperability and Optimisation Enabler" (CEI-InOE) is presented.

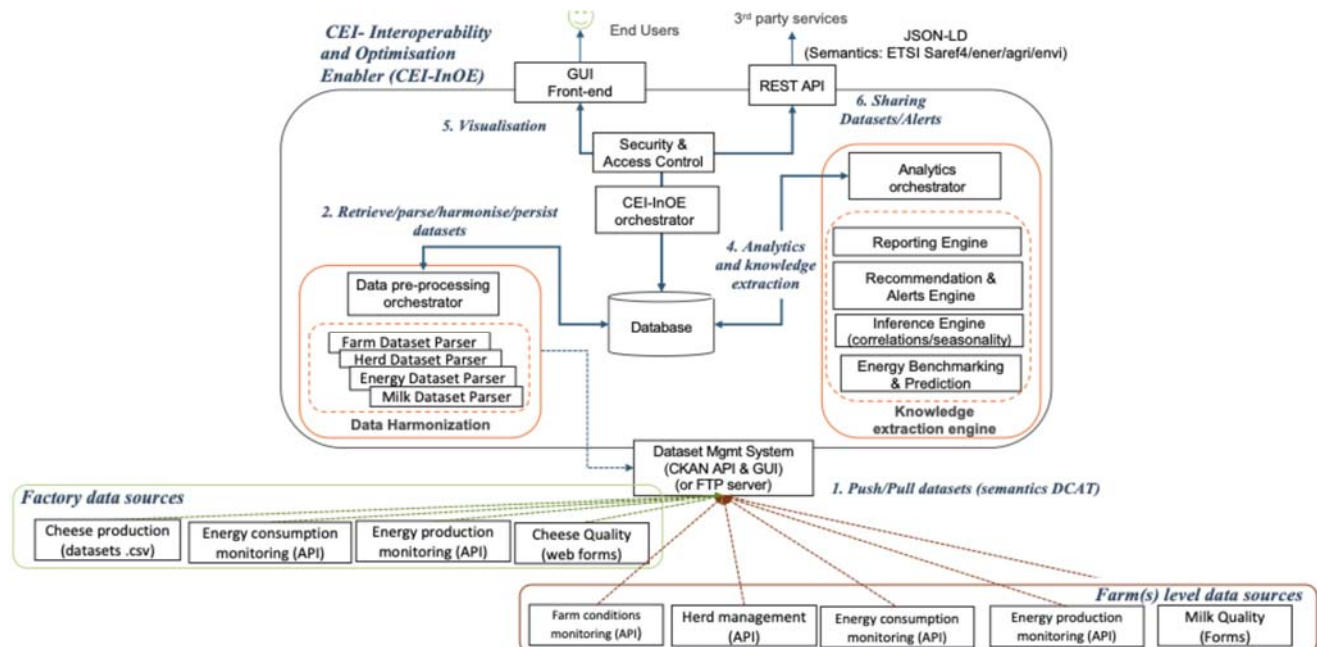


Figure 76. High level functional architecture of the "CEI-Interoperability and Optimisation Enabler (CEI-InOE)"

P5-SA1 – Energy Monitoring via LoRa Electricity Meters

A total of **20 SDM630CT-LoRa series meters** have been purchased and are currently installed across key energy-consuming nodes at both farm and factory locations. These meters will:

- Capture real-time energy consumption data from equipment (e.g., fans, milk cooling tanks, pumps),
- Transmit data over LoRaWAN to edge gateways for energy profiling,
- Enable remote monitoring and insights for performance benchmarking and optimization.



Figure 77. LoRa series meter

This is a key step towards building a **data-driven energy footprint monitoring system**, a foundational aspect of O-CEI Pilot 5's objectives.

At this stage, in farm A, the **electric power submetering devices** have been installed to measure the following systems:

- 2x Robotic Milking machine
- Air compressor for milking machine
- Fridge
- Boiler
- Fans

At this stage, in farm B, the **electric power submetering devices** have been installed to measure the following systems:

- 2x Fridge
- 2x Fans of cooling system
- 2x Air compressor for milking machine
- Fans into the farm

P5-SA2 – PV System + Hybrid Inverter + Batteries at Farm

The site-specific study for renewable energy deployment at the farm has been finalized. Technical configuration includes:

- Azzurro 20kW Hybrid Inverter (10-year warranty),
- Inverter Azzurro 20KW (10-years warranty)
- AIKO 445Wp 90pcs
- 4x 5kW lithium battery modules for energy storage

This system will **maximize solar energy utilization**, reduce peak-time grid dependency, and support resilience under energy-demand fluctuations. Procurement is in the final negotiation stage; installation is expected in the next phase.



Figure 78. Pilot 5 Lithium battery (left) and Azzurro 20kW Hybrid Inverter (right)

P5-SA3 – PV System at Factory

A tailored PV installation is planned at the factory premises. The technical team has:

- Defined system size and configuration,
- Identified the **installation zone**, verified compatibility with factory loads,
- Collected all necessary consumption profiles and structural data to proceed.

The objective is to integrate solar generation into the factory's production energy mix, helping to reduce the carbon footprint per kilogram of Halloumi cheese produced. Final green light is expected shortly.

The PV system at the factory will be connected to the Halloumi production line, which constitutes the core of Pilot 5. This specific line includes all the key energy-intensive processes—from milk storage and pre-treatment to cheese formation, pasteurization, and final packaging. By directly powering this line, the solar system targets the most critical part of the production process, where energy savings will have the highest impact on both cost and sustainability.

This connection ensures that:

- The carbon footprint per kg of Halloumi can be directly monitored and minimized,
- Renewable energy is matched with continuous production loads, maximizing system efficiency,
- Energy performance and traceability can be demonstrated as part of the pilot's KPIs.

P5-SA4 – Installation of Fans and Cooling System for Heat Stress Mitigation

To mitigate heat stress in cattle during the summer period, a comprehensive ventilation and cooling system has been installed at the farm premises. The system is fully operational and designed to **enhance animal welfare**, particularly under extreme climatic conditions during peak summer months. This activity aligns with **sustainability and animal well-being objectives** of the pilot.



Figure 79. Fans and Cooling System for Heat Stress Mitigation (I)



Figure 80. Fans and Cooling System for Heat Stress Mitigation (II)

P5-SA5 - Environmental Sensors for Animal Welfare and PV Efficiency Optimization

As part of the broader data acquisition strategy in Pilot 5, a set of environmental sensors will be installed at the participating farms to continuously monitor ambient temperature, relative humidity, and solar radiation. These parameters are critical not only for assessing the impact of environmental conditions on livestock welfare, but also for optimizing the efficiency and performance of renewable energy systems (e.g., PV panels and cooling systems).

- Temperature and humidity sensors will support the real-time calculation of the Temperature-Humidity Index (THI), a widely accepted indicator of heat stress in dairy cattle.
- Solar irradiance sensors will complement the PV system by providing real-time information on available solar energy. This data will be used to benchmark PV performance, detect anomalies, and better correlate energy generation profiles with consumption patterns at the farm level.

In addition, a weather monitoring system is already installed in one of the farms measuring the following parameters:



Figure 81. Weather Station

Table 14. Weather parameters monitored at the farms

Weather station at Lefkos Farm (LK Dairy Farm)	
Wind speed	m/s
Wind direction	Degrees (0~359)
Humidity	RH
Temperature	°C
Noise	dB
PM2.5	µg/m ³
PM10	µg/m ³
Atmospheric pressure	Kpa
Light intensity	Lux
Solar radiation	W/m ²

The collected data will be processed locally through the pilot's edge infrastructure and integrated into the CEI platform to enable analytics, visualization, and decision support in a harmonized and interoperable manner. This addition enhances the cross-functional value of the pilot, linking animal health and renewable energy under a unified monitoring framework.

7.2.2 Development activities

P5-DA6 – Energy monitoring set up

Following the procurement of LoRa electricity meters, development work will focus on configuring the LoRaWAN-based monitoring framework. Edge gateways are prepared to ingest real-time energy data, enabling energy profiling, benchmarking, and integration with the O-CEI middleware.

P5-DA7- Renewable energy integration logic

For both the farm and factory PV systems, activities in this period focused on the initial design of integration logic. The objective is to ensure that hybrid inverters, batteries, and solar generation assets can later exchange data with the O-CEI platform. Work so far has concentrated on defining the conceptual control flows, identifying relevant data points, and outlining preliminary API requirements for renewable energy reporting. This groundwork will guide the detailed implementation once the PV systems are installed.

P5-DA8- Semantic modelling and data structures

The focus has been on identifying key data categories (e.g., energy consumption, production metrics) and assessing how they can be represented in existing models. This preparatory effort sets the basis for future mapping activities that will ensure uniform data exchange, cross-pilot interoperability, and sustainability reporting.

P5-DA9- Animal welfare automation

The newly installed fan and cooling system for cattle was complemented with **automation logic and monitoring routines**. These developments allow environmental data to be linked with actuation strategies, reducing heat stress while collecting performance data for optimization.

P5-DA10- Data integration, Analytics, and Energy Optimisation

Development of the “CEI-Interoperability and Optimization Enabler” (CEI-InOE) as a software data platform that will: a) collect and harmonize datasets from the various sources of the Farm or the Factory, b) apply analytics on the harmonized datasets to extract knowledge towards the optimization of energy consumption.

7.2.3 Planning & Gantt

The following Gantt diagram list all the activities since the beginning of the work package on M4 until M18. In particular Setup & procurement and Development activities:

Table 15. Pilot 5 Gantt diagram

Pilot 5		2025										2026					
Code	Task	M4	M5	M6	M7	M8	M9	M10	M11	M12	M13	M14	M15	M16	M17	M18	
SETUP & PROCUREMENT																	
P5-SA1	Energy Monitoring via LoRa Electricity Meters																
P5-SA2	PV System + Hybrid Inverter + Batteries at Farm																
P5-SA3	PV System at Factory																
P5-SA4	Installation of Fans and Cooling System for Heat Stress Mitigation																
P5-SA5	Environmental Sensors for Animal Welfare and PV Efficiency Optimization																
DEVELOPMENT																	
P5-DA6	Energy monitoring set up																
P5-DA7	Renewable energy integration logic																
P5-DA8	Semantic modelling and data structures																
P5-DA9	Animal welfare automation																
P5-DA10	Data integration, Analytics, and Energy Optimisation																

7.3 CURRENT INFRASTRUCTURE

Prior to their engagement with the O-CEI project, the digital infrastructure existed in the farms and the factory where limited measuring only partially the required parameters. In addition, all existing systems are acting in silos with limited or no capabilities of integration and data sharing. With regards to IT infrastructure among the core objectives of OCEI's Pilot 5 is twofold:

- Establish a dense metering network that will allow to extract useful analytics with regards to consumed energy
- To break the silos of the existing installed (legacy) information systems and to enable data interoperability. This will improve data richness even with existing systems that will facilitate improved decision making.

The next image provides a high-level overview of the software systems and IT infrastructure operating in Pilot 5. The following subsections provide more details about these systems.

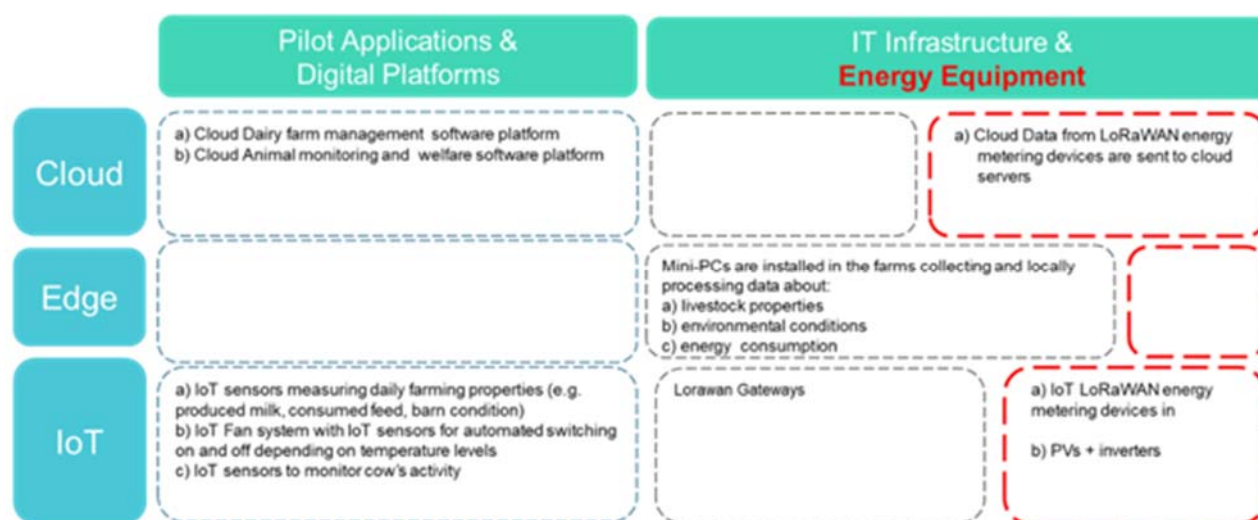


Figure 82. Pilot 5 Infrastructure

7.3.1 Pilot applications and digital platforms

Cloud Layer

Operations at the **farms** are managed with the use of existing commercial software platforms:

- **Dairy Farm Management Platform:** Centralized farm-level herd operation management data including milk production, feed consumption, applied treatments. It should be noted that this platform only supports the creation of reports (pdf format) hence integration of the provided information for energy management purposes is challenging.
- **Animal Monitoring & Welfare Platform:** processes sensor data related to cow behavior, activity, and well-being. Similar with the above platform no API connectivity is available.

At the **Factory** production operations are managed by a combination of cloud-based software platforms that are all using their own proprietary data formats and limited potential for integration. In most of the cases no APIs are available. The following information systems are available:

- PV energy production metering platform for the PV system installed at the factory

- A platform for monitoring real-time cheese production shifts– includes monitoring of the overall equipment effectiveness.
- Milk quantity metering- this is applied on cooling tanks where the milk received by the various farms is stored prior the cheese production process.
- Warehouse management software – this supports the management of the produced cheese products.

IoT Layer

The 2 farms host several types of sensors for different purposes:

- IoT sensors measuring daily farming properties (e.g. produced milk, consumed feed, barn condition).
- IoT sensors to monitor cow's activity IoT sensors measuring daily farming metrics such as milk output, feed usage, barn conditions, and cow activity.
- IoT fan systems with embedded sensors that automatically activate based on temperature thresholds.

7.3.2 IT Infrastructure and Energy equipment

At the heart of the energy monitoring and optimization mechanisms to be implemented for the needs of Pilot 5 is the “CEI-Interoperability and Optimisation Enabler” (CEI-InOE). This software service is designed to be installed on a mini server system deployed at the premises where the data are generated in compliance with the edge computing paradigm.

Edge Layer

Each farm and the factory will be equipped with mini-servers (desktop PCs) for collecting and processing data related to applied processes (e.g. livestock management), environmental metrics, and energy consumption. The “CEI-Interoperability and Optimisation Enabler” (CEI-InOE) will be deployed in these devices, and it will provide:

- Data harmonization on collected datasets from the various sources of the Farm or the Factory
- Analytics on the harmonised datasets or real time decision making
- Optimization for managing energy consumption.

The CEI-InOE will introduce optimization mechanisms without interrupting ongoing dairy production. By processing energy data locally, the system can:

- Enable interoperability even with existing operation (legacy) systems.
- Maintain operational continuity even if the cloud connection is unstable.
- Support predictive maintenance, e.g., detecting abnormal energy spikes in equipment.
- Enable short-term energy flexibility, adjusting consumption patterns in response to renewable energy availability.

Cloud Layer

The following two information systems will provide their functionalities with the use of cloud-based servers:

- **Data from LoRaWAN energy metering devices** are sent to cloud servers. They receive structured energy data from LoRaWAN gateways for centralized access and control. This energy

equipment will be available in the factory and in the farms. This centralized environment enables long-term storage, visualization, and analytics of energy consumption and generation data. The cloud layer acts as the energy intelligence hub, combining:

- Real-time energy profiles from metering devices.
- Historical consumption baselines for benchmarking.
- **Metering of energy generated by PV systems** will be made available through cloud-based server systems.

IoT Layer

- **LoRaWan energy metering devices** to track energy consumed by various devices across the farm and factory environments. Ready for the factory and under installation on the farms.
- Photovoltaic systems (PV) **energy production metering devices**.
- **IoT-controlled fan systems** and environmental sensors support autonomous cooling and livestock comfort.
- **IoT environmental monitoring** systems (e.g. Temperature, Humidity).

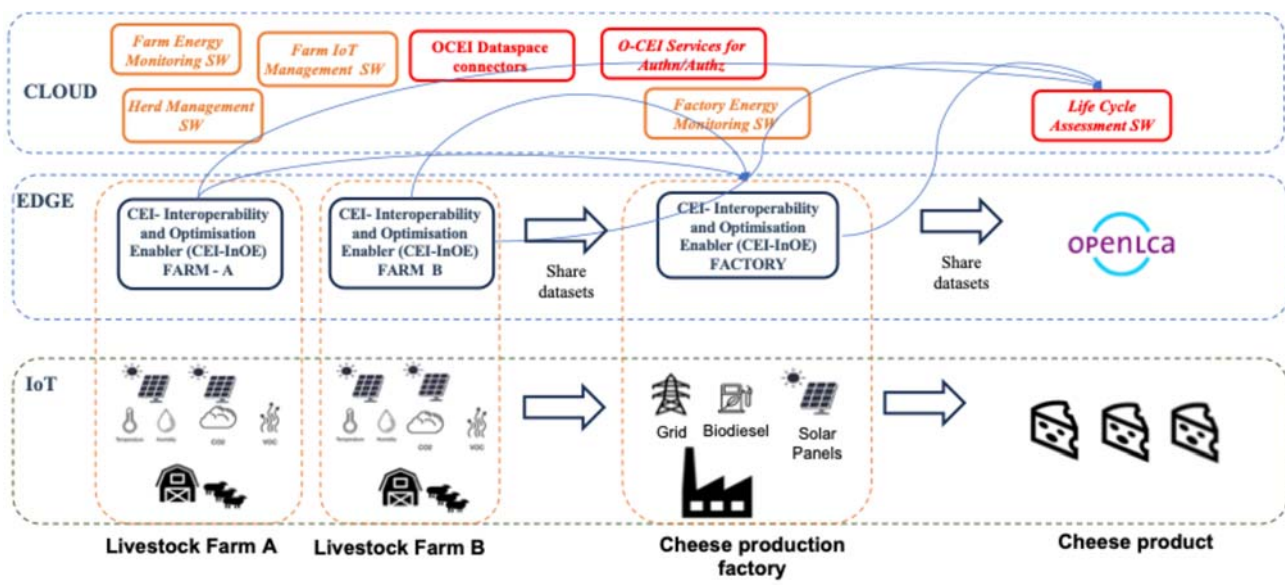


Figure 83. Pilot 5 – Deployment to cloud-edge-IoT and foreseen interactions among the various systems

8 PILOT 6 – SMART AGRICULTURE RECHARGING AND EFFICIENCY OF ROBOT TRACTORS IN AGRICULTURE

Pilot 6 targets the optimization of energy use and charging strategies for electric robot tractors (ZETRABOTs) used in large fruit production fields. The aim is to reduce greenhouse gas emissions, eliminate the use of diesel-based systems, and enable semi-autonomous agricultural operations. The following diagram represents how the fundamental layer is the data management of different sources (e.g., field sensors, tractor signals, climate data, battery data), followed by the trials, business goals, and the final result of higher efficiency and flexibility.

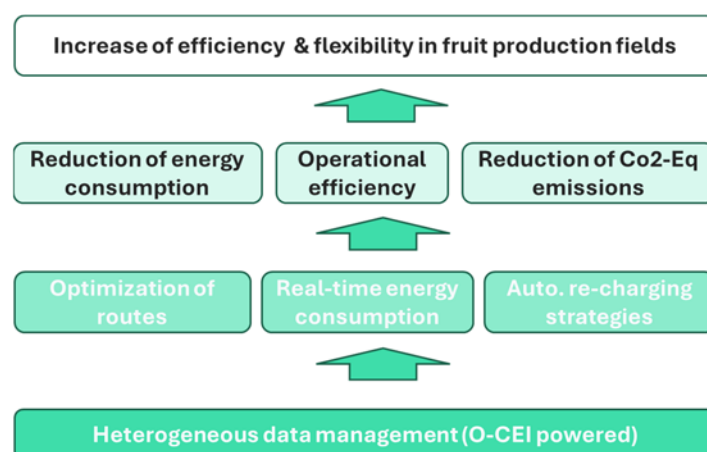


Figure 84. Pilot 6 Summary diagram (From data management to efficiency and flexibility)

The pilot is located in the Basque Country, northern Spain, within an extensive kiwi cultivation area managed by the partner GARAIA. This region is characterized by humid Atlantic conditions, mild temperatures, and fertile soils, making it ideal for fruit production but also demanding in terms of energy management due to frequent rainfall and variable sunlight.

The selected site Figure 85 depicts the setting for testing the ZETRABOT tractors under real agricultural conditions, with sloped terrain, irregular field shapes and high soil moisture that affect traction and energy use. The total available area of the pilot kiwi crops reaches 1.7 hectares (ha) and is located in the following coordinates: 43°22'14.3"N 2°54'56.6"W.

These environmental characteristics make the Basque pilot an excellent testbed for validating O-CEI's edge-enabled optimization and smart-charging strategies before their replication in other European fruit-growing regions.

Pilot 6 has 3 scenarios or business processes that aim to cover the main current and near future challenges faced by agrifood value chain. Energy efficiency, automation, and environmental footprint.

P6-SC1: Real-time energy consumption aid for agricultural decisions

This 1st scenario focuses on real-time energy consumption guidance for farmers. Through the integration of fleet, climate, and location data, the O-CEI platform provides actionable insights to improve task scheduling, route coverage, and overall field productivity. By continuously monitoring each tractor's performance and environmental conditions, the system proposes optimized action plans that balance operational efficiency with energy availability. This ensures longer operating intervals between charges and reduces unnecessary energy expenditure, ultimately enhancing productivity and sustainability across the plantation.



Figure 85. Pilot 6 Site showcasing the Kiwi field and Cottage

P6-SC2: Remote route optimization and task planning

The 2nd scenario builds on this foundation by introducing remote route optimization and task planning. Leveraging the AI-block developed within O-CEI, this process applies state of the art explainable AI models to determine the optimal paths and timing for each agricultural operation. The My ZETRACK application acts as the central hub where data from tractors, sensors, and environmental sources are processed and stored in real time. This enables semi-autonomous management of field operations, where farmers can oversee and adjust routes remotely, improving coordination, reducing idle time, and ensuring that every task is executed with minimal energy waste.

P6-SC3: Autonomous tractor re-charging strategy

Lastly, the 3rd scenario introduces an autonomous electric tractor re-charging strategy via field stations or onboard solar panels. By combining edge analytics and AI-based predictions, the system forecasts energy demand per task and season to manage recharging cycles intelligently. Tractors can autonomously connect to local charging stations or adjust operations based on predicted power needs, while farmers can also control these processes through a dedicated app. This dual approach ensures flexibility between automation and human oversight, extends battery lifetime, and lowers both operational costs and the overall carbon footprint of large-scale fruit production.

8.1 PILOT GOVERNANCE

The governance model of Pilot 6 ensures coordinated collaboration among all participating partners, led by INNOVALIA Association as the Pilot Leader. INNOVALIA handles the technical and organizational management of the pilot, coordinates progress across internal tasks, and ensures alignment with the project architecture and objectives. Partner contributions are collected through regular coordination meetings —typically monthly sessions complemented by informal calls in

between. Progress is tracked through shared documentation, updated planning and milestone reviews. Decision-making is collaborative, with INNOVALIA guiding technical validation, GARAIA ensuring field implementation feasibility, and AGRIMAC and ZETRACK providing input on machinery integration and energy systems.



Figure 86. Pilot 6 Governance model diagram

INNOVALIA acts as the Pilot Leader and System Integrator, managing the overall governance of Pilot 6 and leading tasks in Work Package 4. It coordinates the pilot's roadmap, defines the use cases, ensures the pilot expected progress.

GARAIA serves as the end user and operational pilot site, providing the kiwi plantation in the Basque Country where the electric robot tractors (ZETRABOTs) are tested. As a cooperative specialized in sustainable fruit production, GARAIA leads the field experimentation, monitors KPIs, and is part of the validation of the agronomic and energy performance of the solutions, ensuring their applicability to real world agricultural operations.

AGRIMAC participates as the agricultural equipment provider, contributing its long-standing industrial expertise in tractor design and manufacturing. The company provides the base mechanical platforms and supports their adaptation into electric and intelligent versions compatible with O-CEI technologies. AGRIMAC also assists in evaluating durability, mechanical performance, and energy consumption under field conditions.

ZETRACK operates as the data facilitator and smart machinery developer, responsible for integrating the digital layer that connects the tractors with IoT and edge services. It designs and delivers the energy monitoring systems, autonomous navigation components, and communication interfaces that enable the collection and analysis of real-time operational data across the fleet.

8.2 DEMONSTRATOR PATH

During the first building phase of the project and since the start of WP4 (M4–M12), Pilot 6 has been primarily dedicated to establishing the technical foundation. This phase has centered on setup and procurement activities, ensuring that the essential Agri-equipment, sensors, tools, and parallel infrastructures are in place to support the pilot's scenarios of energy optimization, autonomous navigation, and smart recharging.

During the initial phase of WP4 (M4–M12), Pilot 6 has focused on establishing its operational foundation, deploying the essential agricultural equipment, sensors, and other digital assets needed to support the scenarios on **energy optimization, route planning, and autonomous recharging**. This period has included the integration of the **ZETRABOT prototype, programmable telematics gateway, and battery charging setup**, alongside the initial deployment of **field sensors** and **ZETRACK app first beta version**. These efforts address the technical challenges of operating in complex kiwi fields while advancing O-CEI's sustainability goals by replacing fossil-based practices with renewable, **data-driven** solutions. The work completed so far provides a solid base for the next upcoming stages.

8.2.1 Setup & procurement activities

P6-SA1 – Integration of sensors in the Kiwi field

Initial deployment of proximity sensors in the field, together with supporting cameras, as the guidelines for the electric tractor. The actual field where the pilot takes place has a lot of variability on the terrain and growth of vegetation complicates even more the autonomous driving. The accurate positioning of these sensors is key to enable successful autonomous driving. This activity will be ongoing until all sensors are positioned and calibrated in the field, whose last phase requires the electric tractor to be available for testing while the field conditions allow it.

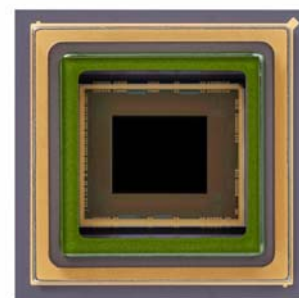


Figure 87. Pilot 6 - Area sensor

P6-SA2 – Integration of agricultural mower into the Electric tractor

Acquisition and installation of an 80 cm inter-row disc mower, designed for mechanical weed control. Its integration with the ZETRABOT is part of the sustainable aspect of the pilot, because partially eliminates the need of herbicide use. Since the tractor is not yet a commercial product, the integration of this tool may require the customization of additional parts. This activity is expected to start during M11 (November 2025) and finish around May 2026, but it may vary depending on the results of the first field test of the ZETRABOT.



Figure 88. Pilot 6 - 80 cm inter-row disc mower

P6-SA3 – Integration of Programmable telematics Gateway (PTG) into the Electric tractor

Currently, in order to update or configure the ZETRABOT systems, a technician needs to be physically near the tractor and plug a laptop for the updates. By integrating the telematic gateway, the mentioned limitation is removed, providing data into and from the tractor. Additionally, it enables remote diagnostics, and cloud connectivity, reducing the need for auxiliary hardware and simplifying integration with the Zetrack platform. The PTG is currently on a testing phase and is expected to be finished by the end of the year (M12), aligned with the start of the first field test in a real environment.



Figure 89. Programmable telematic gateway

P6-SA4 – Setup of the Off-Grid PV system

The optional installation of the photovoltaic system goes beyond simply charging the tractor. By also powering a 3-kW irrigation pump and local sockets, the pilot will demonstrate how on-farm renewable energy can simultaneously support mobility, irrigation, and auxiliary needs. This setup represents a shift toward energy self-sufficiency, ensuring that the ZETRABOT operations are fully decoupled from fossil fuels and resilient against grid fluctuations. In case the PV panels are not deployed, the ZETRABOT will recharge its batteries only from the grid, using special chargers and inside the cottage represented in the following figure. This activity is schedule to start after the field test in December (M12) and its first results.



Figure 90. Pilot 6 - Off-grid PV system

P6-SA5 – Installation of the meteorological station

The weather station is being deployed not only to measure climatic variables, but also to serve as a decision-support enabler. By collecting valuable information such as soil humidity, leaf humidity or evapotranspiration data, it feeds directly into route planning and charging strategies. This integration allows farmers to anticipate how weather conditions—and react to sudden changes—affect tractor performance, optimize irrigation scheduling, and better align field operations with seasonal patterns. The procurement and setup of the meteorological station is aligned with the integration of the field sensors, and both activities should contain a joint validation phase by mid-2026.



Figure 91. Pilot 6 - Meteorological Station

8.2.2 Development activities

P6-DA6 – Development of the ZETRACK App

The ZETRACK platform is evolving from a basic monitoring tool into a comprehensive decision-support environment for farmers. The new web version now includes enhanced dashboards, refined analytics modules, and customized report generation tools that allow users to visualize performance trends and energy indicators at different operational levels. In parallel, a mobile version is under development to provide real-time field insights, enabling farmers to supervise activities directly from handheld devices during daily operations.

Particular attention is given to user experience— the interface simplifies navigation through intuitive icons, color-coded energy indicators, and contextual notifications. The app's five main modules include: manual data ingestion, visualization of key parameters, automatic reporting, alerts, and real-time satellite location tracking. These modules are designed for multi-platform use through a React Native architecture, ensuring accessibility and consistency across devices. The next figures are part of the UI design mockup, the first showcase the live monitoring module, and the second a few of the parameters extracted from the signals of the ZETRABOT.



Figure 92. Pilot 6 Live monitoring UI Mockup

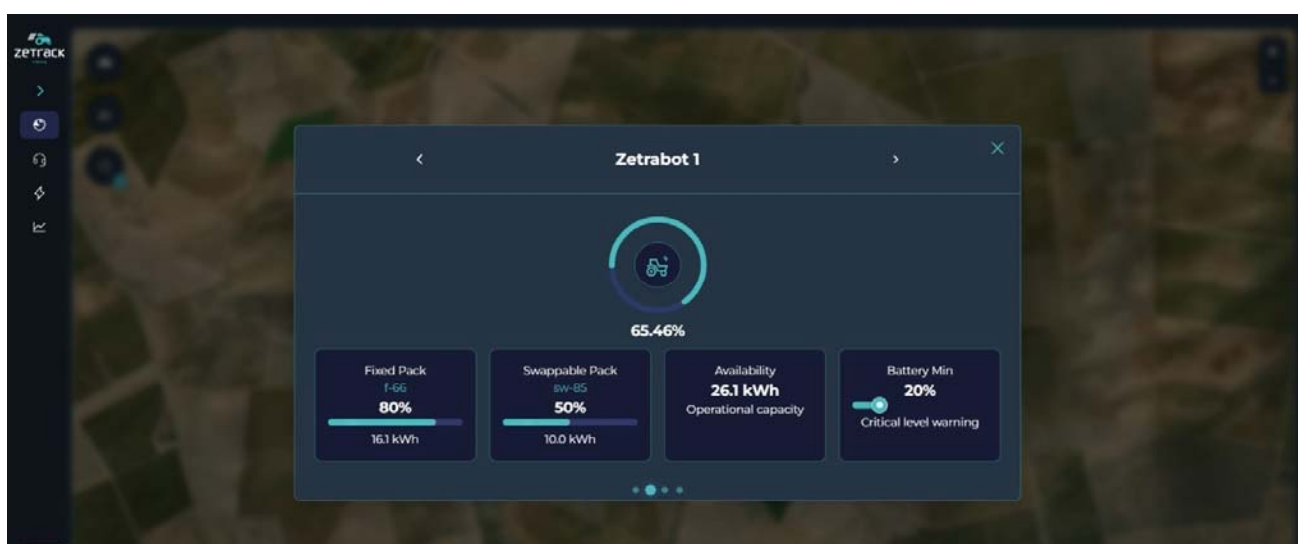


Figure 93. Pilot 6 ZETRABOT Overall parameters UI Mockup

The updated app is connected to the tele-matics gateway (TGC4), which securely transfers more than 100 representative signals from the tractor to the cloud using MQTT and HTTP protocols. This ensures that data captured at the edge —such as battery performance, tool usage, and climatic conditions—is immediately available for cloud-based visualization and AI-driven recommendations. Together, these developments position ZETRACK as the central “Control Room” of the pilot, empowering farmers to monitor the electric tractors, anticipate energy needs, validate test results, and coordinate field activities with higher efficiency and reduced environmental impact.

P6-DA7 – Creation of Battery strategy

The development of the battery strategy is currently ongoing and focused on defining and optimizing the energy management plan for all possible field configuration. Each field section has distinct operational conditions (e.g. connection type, available power, distance to the energy source) requiring tailored charging plans. While the generic strategy is still ongoing, the requirements for the Kiwi field are mostly defined, as summarize in the following table:

Table 16. Pilot 6 Battery strategy of Kiwi field

Parameter	Specification / Description
Total battery capacity	44 kWh (4 battery packs × 11 kWh each)
Chargers to be integrated	2 × ZIVAN SG3 (2.5 kW each)
Total power of simultaneous charging	5 kW
Total charging time	8–9 hours
Required connection	2 industrial plugs – 230 V, 16 A (single-phase)
Power availability	Sufficient for charging, provided it does not coincide with watering pump operation (3 kW)
Charging location	Cottage area with protected outlets and vehicle access

8.2.3 Integration activities

P6-IA8 – Integration of TCG4 Communication Gateway

This activity is about the deployment and configuration of the TGC4 communication gateway, which serves as the secure link between the ZETRABOT onboard systems and the ZETRACK Cloud platform. This gateway ensures reliable and encrypted data transmission through MQTT and HTTP protocols, connecting the tractor's IoT devices with cloud-based analytics and visualization services. The TGC4 gateway is a key enabler for Trial 1 and Trial 3, as it allows real-time monitoring of energy parameters and charging status directly within the ZETRACK environment. This integration ensures that field data collected at the edge can immediately feed the system's dashboards and AI-driven decision-support tools, strengthening the feedback loop between operations and energy optimization.

P6-IA9 – Integration of Agreenculture Navigation Kit and Autonomy system

The installation and calibration of the Agreenculture Navigation and Autonomy Kit equips the ZETRABOT with autonomous driving and task execution capabilities. The system integrates a Central Control Unit, high-precision GNSS antennas, and a network of sensors that allow the tractor to follow preprogrammed routes and perform safe and accurate maneuvers independently. The integration includes connecting the autonomy kit with both the ZETRACK web platform and the mobile application, ensuring synchronized data flow and seamless coordination between route planning and execution. This activity is essential for Trial 2, supporting remote route optimization and task planning scenarios. Once operational, the system will enable semi-autonomous management of field activities, reduce downtime, and improve efficiency through AI-assisted navigation and control.

8.2.4 Validation activities

P6-VA10 – ZETRABOT Laboratory test

The Laboratory Test will be conducted between July (M7) and November 2025 (M11) at the [Masermic](#) facilities. This phase focuses on verifying the correct functioning of the ZETRABOT prototype's electrical, electronic, and control systems under controlled conditions. The test will allow the team to fine-tune parameters, evaluate system stability, and assess the response of critical components when exposed to simulated workloads. In addition, it will serve as the first opportunity to check data transmission and synchronization between the tractor's onboard sensors and the ZETRACK digital platform. The main expected outcomes are the identification of functional errors, the validation of core control logic, and the definition of improvements prior to field deployment.

P6-VA11 – ZETRABOT Asphalt test

The Asphalt Test is planned to be completely executed during November 2025 (M11) and represents the first real-environment evaluation of the ZETRABOT. The objective is to assess the tractor's structural integrity and dynamic performance in outdoor conditions, focusing on the interaction between the chassis, suspension, and electronic control systems during movement. This test will verify mechanical stability, weight balance, and responsiveness, ensuring that all subsystems operate cohesively under real but controlled terrain conditions. The results will be used to adjust the mechanical configuration and confirm readiness for the following real-world agricultural trials.

P6-VA12 – ZETRABOT Field test

The real-environment Kiwi Field test will take place between December 2025 (M12) and Jun 2026 (M18) at the GARAIA kiwi plantation in the Basque Country previously described. This will be the key phase where the ZETRABOT and the ZETRACK application are validated together under authentic farming operations, including mowing, fertilizing, and pruning tasks. The activity will measure parameters such as energy consumption, battery autonomy, and torque efficiency while monitoring tractor behavior in variable soil and weather conditions. Real-time data will be captured and visualized through the ZETRACK app, supporting analysis and performance benchmarking against pilot KPIs. The final goal is to confirm the energy and environmental benefits of the system, optimize recharging strategies, and demonstrate its applicability to large-scale fruit production

8.2.5 Planning & Gantt

The following Gantt diagram list all the activities since the beginning of the work package on M4 until M21. For this pilot, there are all kind of activities –Setup & procurement, Development, Integration and Validation—although more will be added as needed during the next phases.

Table 17. Pilot 6 Gantt diagram

Pilot 6		2025										2026									
Code	Task	M4	M5	M6	M7	M8	M9	M10	M11	M12	M13	M14	M15	M16	M17	M18	M19	M20	M21		
SETUP & PROCUREMENT																					
P6-SA1	Integration of sensors in the Kiwi field																				
P6-SA2	Integration of agricultural mower into the Electric tractor																				
P6-SA3	Integration of Programmable telematics Gateway into the Electric tractor																				
P6-SA4	Setup of the Off-Grid PV system																				
P6-SA5	Installation of the meteorological station																				
DEVELOPMENT																					
P6-DA6	Development of the ZETRACK App																				
P6-DA7	Creation of the Battery strategy																				
INTEGRATION																					
P6-IA8	Integration of TGC4 Communication gateway																				
P6-IA9	Integration of Agreenculture Navigation Kit and Autonomy system																				
VALIDATION																					
P6-VA10	ZETRABOT Laboratory test																				
P6-VA11	ZETRABOT Asphalt test																				
P6-VA12	ZETRABOT Field test																				

8.3 CURRENT INFRASTRUCTURE

The infrastructure includes vehicle IoT sensors, control ECUs, edge logic for data acquisition, and cloud-based visualization and analysis platforms. Several key acquisitions are required to support the pilot's goals, including an agricultural mower, off-grid PV system, programmable telematics gateway, and a weather station for data-driven decision-making. The first subsection –Pilot applications and digital platforms–covers the digital layer (ZETRACK and Data management) while the second subsection –IT infrastructure and energy equipment–covers the physical layer (EVs and energy hardware)

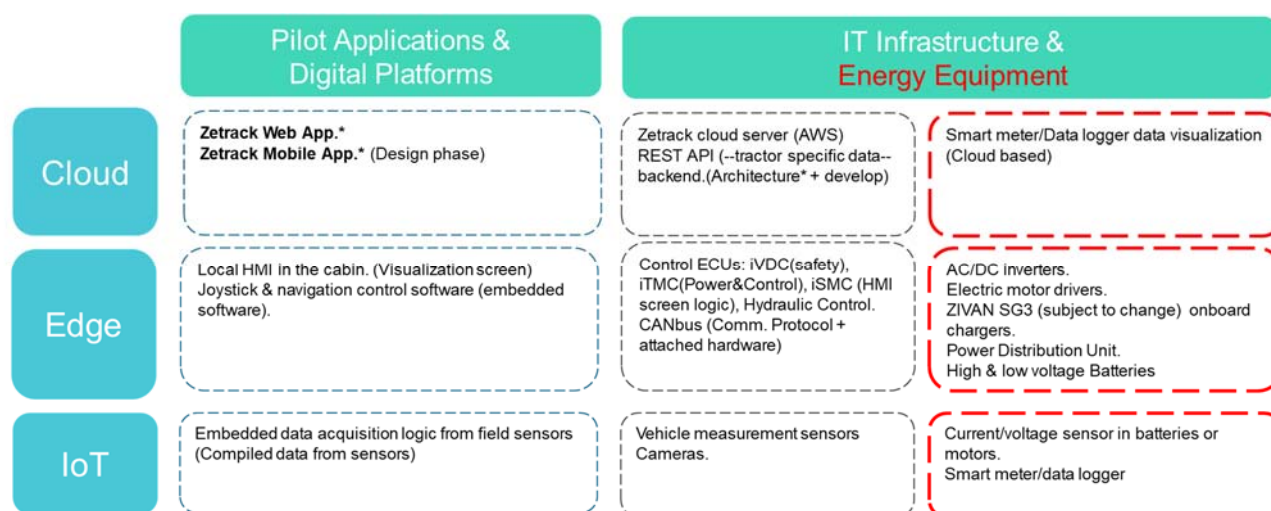


Figure 94. Pilot 6 Infrastructure

8.3.1 Pilot applications and digital platforms

Cloud Layer

- **ZETRACK platform**, hosted on Amazon Web Services (AWS), acts as the main backend and visualization hub of the pilot. It enables remote access to diagnostics, telemetry, and performance dashboards, allowing partners to monitor field activities and battery behavior in real time.
- **Cloud-based data logger** stores and processes data streams from the tractor and field sensors, supporting energy-use analysis and KPI tracking.
- **Secure REST APIs** ensure interoperability with O-CEI components and allow integration with cross-domain AI utilities.
- **ZETRACK mobile version** (under development) mirrors the web environment, providing handheld access to alerts, task monitoring, and quick data visualization for farmers.

Edge Layer

- The edge computing layer within the ZETRABOT will handle **data preprocessing and synchronization** with the cloud through the TGC4 telematics gateway, whose integration is ongoing.

- **Local Human Machine Interface (HMI)** provides real-time visualization of operational data (energy use, battery status, torque, etc.) and supports manual control through the joystick interface.
- **Edge logic modules** perform first-level analytics and ensure data buffering in case of temporary network loss, maintaining system resilience.

IoT Layer

- **IoT network** integrates all field devices and sensors—including environmental sensors, proximity cameras, and smart meters—responsible for **real-time data capture**.
- **Communication** across the system is managed through **CAN bus** and smart metering units, ensuring **synchronization of data** between **tractor, gateway, and cloud**.

This layer provides continuous input for ZETRACK analytics and supports semi-autonomous control of field operations.

8.3.2 IT infrastructure and Energy equipment

Cloud Layer

- **AWS** backend supports the **ZETRACK web platform**, already operational for **data collection, visualization, and telemetry analysis** (Development ongoing)
- Continuous data transfer from the **TGC4 gateway** will ensure that tractor telemetry is safely transmitted to the cloud for ongoing validation.
- The cloud layer also manages **user authentication, data storage, and historical logs**, forming the foundation for later AI-driven recommendations.

Edge Layer

- **ZETRABOT**: First operational prototype of the electric tractor, which is currently under laboratory testing. The prototype is equipped with embedded control software, a local Human–Machine Interface (**HMI**), and a joystick for manual operation. It can already execute basic navigation, energy monitoring, and communication routines, which are being validated through the ongoing test (Lab → Asphalt → Field).



Figure 95. Pilot 6 ZETRABOT first prototype

- **Forklift:** Aside from the ZETRABOT, the demonstration will use an electric forklift a **complementary EV to assist** the ZETRABOT mainly by carrying the products. This vehicle features similar propulsion and control systems of the ZETRABOT and is currently being used to test **energy efficiency and charging behaviour** under different load conditions.



Figure 96. Pilot 6 Electric forklift prototype

- The currently installed energy equipment includes:
 - AC/DC inverters and electric motor drivers
 - ZIVAN SG3 chargers (2.5 kW each)
 - Power distribution unit
 - 44 kWh battery pack (4 × 11 kWh modules)

IoT Layer

- **IoT network** connects all sensors and measurement devices integrated in the ZETRABOT and field setup.
- **Current, voltage, and temperature sensors** are embedded within the tractor's motors and batteries for real-time energy tracking.
- **Cameras and proximity sensors** assist navigation and obstacle detection in the kiwi field.
- **Smart meters and data loggers** monitor energy consumption, charging performance, and power supply metrics at the tractor and charging points.

All IoT elements communicate through the **CAN bus**, synchronizing data between the tractor, the onboard edge system, and the ZETRACK cloud. This layer enables **continuous monitoring of performance parameters** such as battery usage, torque, and energy efficiency, forming the technical baseline for upcoming validation activities.

9 PILOT 7 – PRIVATE 5G DEPLOYMENT FOR ENHANCED EV DISTRIBUTED CHARGING

Pilot 7 focuses on EV distributed charging in one load area equipped with photovoltaic plant and covered by 5G facility supporting a flexible energy management system able to control different types of V2G chargers with respect to different charging requirements. The aim is to demonstrate that **suitable IoT wireless network, edge computing**, and distributed machine learning based pattern recognition and **optimization-based decision support**, allow measurable *cost saving in the EV charging value chain* and, at the same time, enable *data driven ancillary services for grid stability* as well as *security and power quality* during charging operations in presence of unpredictable renewable energy sources.

The pilot includes **two scenarios** based to EV smart charging in a private parking area with AC and DC charging. The following images showcase the location of the Pilot.



Figure 97. Pilot 7 Location of the pilot

P7-SC1: Edge optimization of power flows for smart charging

With the goal of enabling intelligent, edge-optimized EV charging by integrating bidirectional AC/DC chargers with 5G-connected microservices that dynamically manage power flow based on real-time EV data & grid conditions;

P7-SC2: Increasing trustworthiness and security in EV charging data exchange

With the goal of enhancing trust, security & data sovereignty in EV charging operations through continuous monitoring, vulnerability detection & secure, auditable data workflows that ensure the integrity & reliability of all exchanged information.



Figure 98. Benchmark site for Pilot 7

9.1 PILOT GOVERNANCE

CERTH: Leader of the pilot, responsible for coordinating and overseeing all activities to ensure alignment with O-CEI deliverables and objectives. Bi-weekly technical meetings are set to ensure partners are aligned on implementation progress, data collection and validation status. Furthermore, CERTH is responsible for developing, adjusting and deploying the necessary cybersecurity and blockchain tools used in P7-SC2.

E-GAP: Coordinates energy infrastructure deployment and is responsible for selecting, designing, installing and operating the benchmark site for data collection and a physical test bed for scenarios pilot scenarios. The technical execution of activities related to smart grid infrastructure, including the procurement and deployment of battery storage systems, bidirectional AC/DC chargers, and PV generation units. Throughout this process, E-GAP is responsible for ensuring technical and regulatory compliance by managing the coordination between site field operations and technical input for all pilot partners to facilitate the integration of IoT, Edge and Cloud components developed by Ares2t and CERTH. E-GAP ensures that all data, logs and system metrics are shared with partners to guarantee technical feasibility. Beyond the technical setup, E-GAP leverages its industrial know-how and existing partnerships/projects to align the development of the pilot with market expectations and enable a scalable and replicable model across energy and mobility business models. Together with a systematic communication and dissemination strategy aimed at the market through academic institutions, industry events, and social media platforms, these initiatives help the pilot get widespread attention. These programs encourage the commercial adoption of O-CEI technologies for upcoming deployments.

NOVA: As one of the largest telecommunication providers in Greece will assist and provide computing and connectivity infrastructure to merge electromobility benefits, thus aiming allow the implemented cybersecurity measures developed by CERTH, leverage private 5G network to connect

chargers, edge devices, such as cars or busses, and the cloud. Through the O-CEI project, NOVA foresees to improve network resources balancing in EV charging.

ARES2T: serves as the system engineering and integration partner for Pilot 7, leveraging its core expertise in data-driven innovation, Operation Research, and Automation to bridge the gap between energy infrastructure and digital technologies. As specialists in Smart Grid and 5G telecommunication integration, ARES2T is responsible for developing and deploying the IoT, Edge computing, and Cloud infrastructure that enables intelligent energy management across the EV charging ecosystem. Their role focuses on creating the distributed machine learning algorithms and optimization-based decision support systems for P7-SC1, implementing 5G-connected microservices that communicate with bidirectional AC/DC chargers and integrating pattern recognition capabilities for real-time power flow optimization.

9.2 DEMONSTRATOR PATH

Since the beginning of the project, Pilot 7, titled *Private 5G Deployment for Enhanced EV Distributed Charging*, has progressed from the design and definition phase toward the implementation stage, following a coordinated demonstrator path led by **CERTH** with the active participation of **E-GAP**, **Ares2T** and **NOVA**. During the first nine months (M1 to M9), the consortium focused on defining the pilot use cases and scenarios, identifying the test-bed infrastructure, designing the Cloud Edge IoT architecture, and initiating the procurement of compliant energy equipment, including battery storage systems, photovoltaic units, and bidirectional AC/DC chargers under the coordination of E-GAP. In parallel, Ares2t finalised the CEI functional architecture, CERTH defined the cybersecurity and trust framework, and NOVA prepared the 5G integration environment. As the project moves into the next phase from Month 10 to Month 18, activities will focus on completing hardware installation, configuring the 5G-enabled IoT and Edge components, deploying the cybersecurity and optimisation modules, and initiating integration and validation activities. E-GAP will also continue to lead dissemination, industrial engagement and large-scale replication efforts, leveraging its operational know-how to foster business partnerships and support the commercial scalability of the pilot outcomes. This progression ensures that by Month 18, the demonstrator will achieve a fully integrated and operational stage, providing measurable results on energy efficiency, interoperability and data trustworthiness within the O-CEI framework.

Taking into consideration the current state of the art and analyzing technological and business challenges and opportunities, the target load area has been conceived as one smart grid equipped with:

- Bidirectional AC and DC chargers;
- One RES co-generation unit (photovoltaic plant);
- One energy storage unit;
- Mobile energy loads (EVs) supporting AC and/or DC charging.

The reference high level architecture of the target load area is depicted in Figure 99. Implementation of bidirectional energy flow (B2G, V2B, V2G, depicted as yellow lines in Figure 99) allows for **grid stabilization** and **decentralized energy distribution**. The suitable 5G connectivity guarantees **low-latency, real-time and flexible energy management** for optimized EV charging according to time dependent events occurring in the grid (P7-SC1). **Real-time threat detection** and **secure data protocols** ensure cybersecurity for EV charging systems along the value chain (P7-SC2). **Scalable blueprints** are here crucial for future replication, ensuring that project findings and micro-services deployed in the CEI architecture can be adopted across other markets.

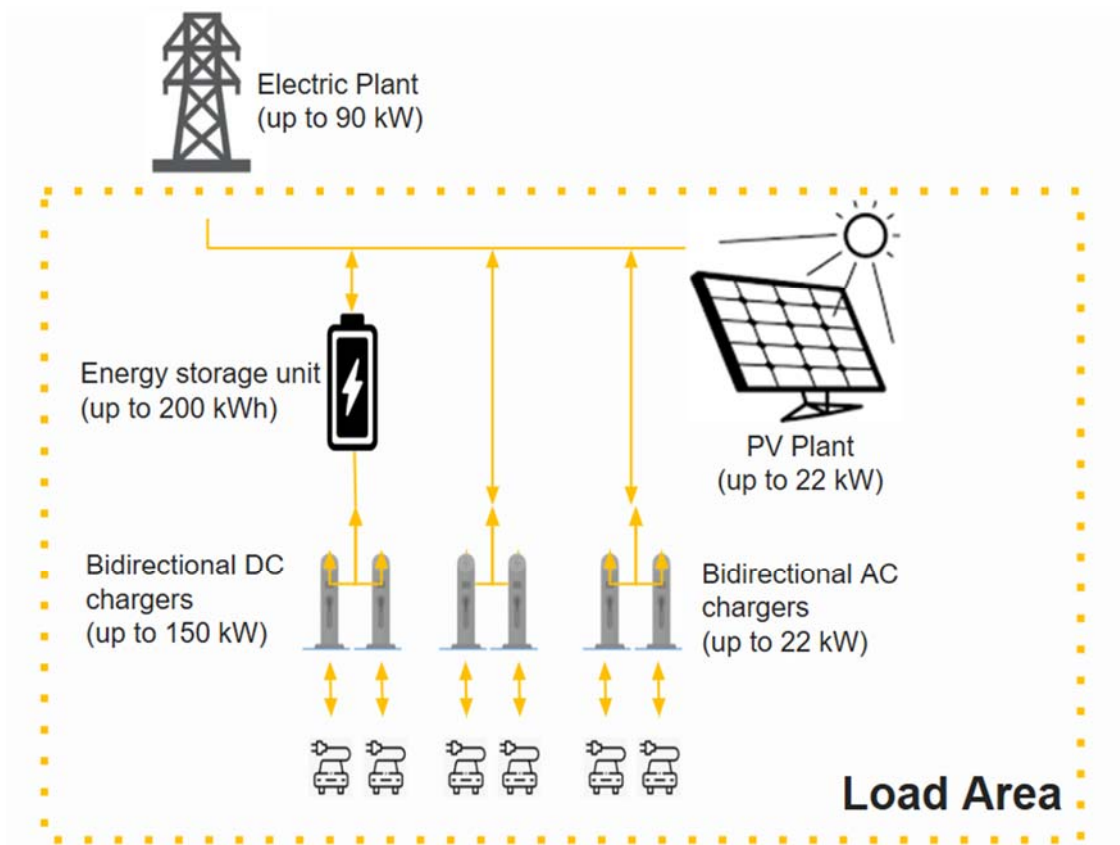


Figure 99. Pilot 7 – Target load area

Once identified the target load area and main building blocks of the energy infrastructure, the implementation of the pilot started from selecting the suitable test-bed and related location to move on with procurement activities as well as with the design and development of the Cloud Edge IoT architecture on top of the energy infrastructure.

The test bed has been selected in a location in Milan consisting of Battery Storage, Bidirectional AC/DC Chargers and PV plant. In some cases, the storage could be integrated within the charging column or a separate unit. The stationary storage will serve as a **buffering and balancing unit**, absorbing excess PV generation during periods of low demand and discharging during peak demand or grid imbalance events.

Flexible charging and discharging modes based on grid requirements will be made possible by the bidirectional chargers' interface with EVs and the stationary battery. Two distinct options for the storage's physical integration are possible:

- **Integrated configuration:** the battery is built directly into the charging column, providing a compact solution suitable for urban environments where space is limited.
- **Decoupled configuration:** the battery is deployed as a separate unit, offering larger storage capacity and more flexibility in scaling the system.

This dual approach allows us to evaluate the benefits and challenges of both, hence allowing us to better evaluate and assess efficiency. The integrated configuration is a compact, ideal for urban areas with limited space for EV charging allowing for simplified installation and localized buffering. This in turn supports urban PNire objectives related to the deployment of compact infrastructure in cities where space is often scarce and aligns with PUN (Piattaforma Unica Nazionale) transparency enabling fewer assets to register/manage. The main challenge here is the storage cannot be scaled independently of chargers limiting scalability.

Meanwhile the decoupled configuration offers higher capacity whereby the storage can be expanded without altering or adding more chargers suited for larger depots in support of high-capacity EU targets (>400-600kW along TEN-T). This configuration requires more complex deployment but results in higher charging capacity and facilitates V2G and ancillary services in compliance with EU energy flexibility ambitions.

The Cloud Edge IoT (CEI) architecture has been designed to support validation and demonstration activities with particular attention to **flexible charging and grid management** (in P7-SC1) and **process security and data trustworthiness** (in P7-SC2).

In P7-SC1, the three layers of the CEI architecture designed by Ares2T and to be realized in Pilot 7 consists of different functional modules, being better described in the following sections.

- IoT Layer for raw energy data collection from local meters installed in the EV chargers and hardware controllers (PLC, PV inverter, microcontrollers, and relevant active components in the target load area) and communication backbone connecting the load area's components through the Edge Layer.
- Edge Layer for raw energy data pre-processing and aggregation from heterogeneous data sources as energy devices, including energy and power meters, inverters measurements and parameters, charging station status, etc.; at this layer, local charging process monitoring and load area control are enabled by pre-processed raw data allowing to visualize current indicators (e.g., instant power absorbed by a charging station, historical power curves, etc.).
- Cloud Layer for load area management by the Charging Point Operator (CPO) or Distribution System Operator (DSO), advanced charging session control and optimization and power forecast via data analytics where the training set include historical data aggregated data potentially communicated by different load areas related to one macro load area operated by one CPO or one DSO.

The three layers CEI architecture guarantees necessary data communication and elaboration to provide:

- **Cost saving in the EV charging value chain** by suitably controlling all active charging sessions operated in the load area by means of the continuous adjustment of power set-points communicated near real time to the EV chargers so that each EV is charged when more convenient by reducing the single cost per charging session and the overall cost for the CPO / DSO
- **Data driven ancillary services for grid stability** as well as **security and power quality** during charging operations in presence of unpredictable renewable energy sources by controlling all active charging sessions in the load area by means of suitable suspension (charging start/stop control or power modulation based control) of the charging process when more suitable based on (i) the grid condition, (ii) the power production forecast from RES and (iii) the power absorption according to the identified charging profiles in the load area.

The current status of the baseline architecture which has been deployed by ARES2T in the site operated by E-GAP is described in section 9.3. This architecture will be enhanced with additional services during the setup, development, integration and validation activities described in the following sections. These activities are strictly needed to reach the target load area as well as the CEI architecture above mentioned.

Furthermore, CERTH is leading the cybersecurity activities described in P7-SC2 and more specifically focuses on the deployment of continuous monitoring and vulnerability detection mechanisms for the infrastructure set up in Pilot 7 (P7), the development of secure and auditable data exchange workflows to ensure data integrity, confidentiality and sovereignty, as well as the

implementation of forensic logging and anomaly detection to enhance trustworthiness and resilience in smart charging operations. Since the beginning of the project, CERTH has prepared the groundwork for the cybersecurity dimension of P7, leveraging its existing EV charging infrastructure for preliminary vulnerability assessment and secure connectivity testing. Following the definition of the P7 use case narrative and especially P7-SC2 architecture and taking advantage of the design of the CEI architecture, cybersecurity-related implementations focus on the below activities:

- **IoT Layer (chargers, PLCs, microcontrollers, communication backbone):** At the IoT level, cybersecurity focuses on securing the interfaces of EV charging stations, PLC controllers and the 5G low-latency communication backbone. This includes implementing **embedded lightweight intrusion detection systems (IDS)** and **Anomaly Detection (AD) mechanisms** directly on smart chargers to detect malicious activity or abnormal charging patterns. Secure multi-protocol data collection ensures that charging session data cannot be tampered with during transmission.
- **Edge Layer (data aggregation, monitoring):** At the edge, the emphasis is on **local threat detection and containment**. Data aggregation services and load area controllers are protected by **lightweight cybersecurity algorithms** that can identify vulnerabilities in (near) real-time. By processing and filtering data closer to the source, the edge reduces attack surfaces and ensures resilience even if cloud connectivity is disrupted. The 5G-enabled capabilities of the infrastructure on which these cybersecurity algorithms are deployed on, provides additional challenges for securing low-latency links, where CERTH is focusing on monitoring for protocol misuse or denial-of-service attempts.
- **Cloud Layer (macro load control, analytics, optimization):** At the cloud level, cybersecurity is centered around **trust, integrity, and sovereignty of data workflows**. This involves deploying **cybersecurity algorithms, trust mechanisms and secure containerized applications** to protect sensitive charging session data and energy forecasts from manipulation or unauthorized access. Continuous monitoring, vulnerability detection and **forensic logging** are implemented to ensure secure orchestration of charging sessions and to provide auditable records for incident investigation.

More specifically, CERTH has made significant progress in developing the cybersecurity framework for Pilot 7 by conducting comprehensive research into intrusion detection systems tailored for EV charging environments. The team explored and reviewed open-source and public datasets relevant to OCPP charging traffic, identifying both application-layer and network-layer data sources suitable for IDS training and validation. Through detailed analysis of existing IDS frameworks, CERTH has designed a dual-path approach that addresses the unique challenges of charging infrastructure security. For network-layer traffic, the team has applied transfer learning, fine-tuning and retraining techniques on existing IDS models using CICFlowMeter-style datasets, while for application-layer OCPP data, identified the need to train new anomaly detection models due to the fundamentally different feature space involving charging session metadata and JSON event logs. CERTH has also reviewed recent academic literature to understand common cyberattacks against OCPP protocols, such as session spoofing and message tampering, which has further helped with the dataset selection and model design strategy. CERTH has evaluated the alignment between various feature extraction tools (including CICFlowMeter and OCPPFlowMeter) and model input requirements, and is preparing to validate their approach using real OCPP traffic from ARES2T once it becomes available, ensuring the models perform effectively under realistic operational conditions.

9.2.1 Setup & procurement activities

P7-SA1 - Full procurement of units for all use cases

Procurement activities are being led within a validation and methodology framework to ensure that all equipment meets the technical, financial, and regulatory compliance required for Pilot 7. The

procurement matrix below has been developed to proceed with an objective evaluation of the vendors. Supplier selection was prioritized from M6 - M9 consisting of:

1. Vendor assessment by contacting a select x5 certified suppliers (CE, CEI, IEC 61851, ISO 15118) to kick-start the procurement plan.
2. The products offered by the vendors via technical and economic quotations were checked for compatibility to ensure that the charges, storage, and PV units support OCPP 1.6 and preferably 2.0.1 to meet interoperability requirements. This is important because OCPP 2.0.1 provides an advantage for V2G bidirectional set-ups; however, OCPP 1.6 remains widely deployed.
3. The delivery and lead time risk were factored into delivery times, which had to be no longer than 10 weeks from the order, including installation and after-sales support.

With specific examples from vendor datasheets, each criterion is weighted based on its strategic value:

Table 18. Pilot 7 Examples from vendor datasheets

Criteria	Description	Weight (%)	Example – Huawei – SUN2000	Example – Nex2 SeedBox LITE 30	Example – DEFA – Power
Standards Compliance	OCPP (1.6 / 2.0.1), ISO 15118, CHAdeMO, V2G readiness	25	No OCPP support, only RS485/ Ethernet	OCPP 1.6J, ISO 15118-20, CHAdeMO 1.2	OCPP 2.0.1, ISO 15118
Grid & Regulatory Certification	Compliance with CEI 0-21, VDE-AR-N, EN50549, IEC, etc.	20	Broad grid code compliance (CEI 0-21, VDE, IEC)	Basic CE/IEC certifications, less grid focus	EU standard grid code compliance for AC units
Interoperability & Connectivity	Communication interfaces (Wi-Fi, LAN, 4G, APIs), roaming capability	15	RS485, optional 4G dongle, no OCPP	Wi-Fi, LAN, 4G, RFID, payment terminal optional	Wi-Fi, 4G, LAN, seamless OCPP 2.0.1
Lifecycle & Warranty	Warranty period, durability, degradation guarantees	15	10–12 years typical inverter warranty	Not specified, typical EVSE 2–5 years	5–10 years standard
Vendor Maturity & Bankability	Track record, financial stability, global presence	15	Huawei = Tier 1, strong bankability	Nex2 = niche EVSE vendor, smaller scale	DEFA = established Nordic supplier
Total Cost of Ownership (TCO)	CAPEX, O&M, software licensing (V2G, OCPP), scalability	10	May require OCPP gateway → extra cost	Software licensing for V2G functions	OCPP 2.0.1 included, fewer add-ons needed

By M9, the procurement process had evolved, and equipment selection and supplier evaluations were completed per the pilot's interoperability and compliance requirements. Delivery schedules have been verified to coincide with the infrastructure installation schedule. E-GAP will supervise the delivery, setup, and functional testing of all acquired assets during the subsequent phase, which runs until Month 18. This will guarantee that chargers, storage units, and PV systems are appropriately set up for integration inside the CEI architecture.

P7-SA2 - Setup of IoT and Edge components

The setup carried out by ARES2T mainly concerns the data communication and exchange between each single component of the energy equipment (depending on the activity P7-SA1) and the Micro Controller Unit (MCU) at the Edge layer through the 5G wireless communication network. The communication relies on standard communication protocols depending on the type of component. As reported in P7-SA1, the reference protocol is OCPP (ranging from 1.6 to the target version 2.1) for EV charges, energy storage, and PV units support but in the activity other protocols for data exchange between electronic devices can be considered if need (e.g., Modbus TCP via Ethernet communication).

In parallel with the backbone communication setup, CERTH focuses on the integration of cybersecurity mechanisms at the IoT and Edge layer. Building on proprietary work already carried out at CERTH's premises with the EV chargers in operation, the activities will extend to Pilot 7 by deploying dedicated edge devices, such as Raspberry Pi 5 boards or NVIDIA Jetson platforms, that will host lightweight cybersecurity algorithms enabling AI capabilities. These devices will be connected to the existing EV chargers and take advantage of the 5G-enabled communication framework, thereby enabling secure, low-latency data flows between chargers, edge controllers and the cloud. The cybersecurity functions to be deployed include Anomaly Detection on charging sessions, embedded IDS for protocol-level monitoring (e.g., OCPP 2.1 and Modbus TCP) and secure logging services for incident traceability. By running these mechanisms directly on the edge, CERTH aims to provide early detection and containment of threats close to the source of data generation, while ensuring that security events can be correlated with cloud-level monitoring for end-to-end protection across Pilot 7 infrastructure.

P7-SA3 – Setup of 5G components

In order to facilitate the whole procedure, a 5G enabled module (hat) could be added on top of CERTH's edge device that will be connected to the EV charger, offering 5G capabilities to the edge device and more specifically to the EV charger, taking advantage of the one of Italy's 5G public network in order to connect to the internet and through a secure VPN tunnel be able to access NOVA's deployed 5G simulation environment in its premises in Greece.

9.2.2 Development activities

P7-DA4 – Development of new features of smart grid

The main objective of our smart grid strategy is to enable bidirectional power flows between car batteries, stationary storage devices, and charging infrastructure. As part of this, batteries at the infrastructure level and EVs are regarded as distributed energy resources (DERs) that both consume and produce power. The goal of the control layer is to maximize power flow management for a variety of assets. This entails adaptive storage resource dispatch, predictive load and generation forecasting, and real-time grid condition monitoring. While EVs offer flexible, distributed capacity that grows with adoption, stationary storage enhances EV batteries by providing more availability and predictable response. Pilot 7 will develop the operating modes of Vehicle-to-grid (V2G), Vehicle-to-Battery (V2B), and Storage-to-grid (S2G), which will be deployed with bidirectional AC/DC chargers and integrated into the grid by coordinating charging and discharging cycles. Key objectives include:

- Charging during off-peak hours or periods of renewable energy surplus.
- Discharging during peak demand or grid imbalance events.
- Providing **ancillary services** such as frequency regulation, voltage stabilization, and spinning reserve.
- Reducing network congestion and deferring costly grid reinforcement.

The goal of the control layer is to maximize power flow management for a variety of assets. This entails adaptive storage resource dispatch, predictive load and generation forecasting, and real-time grid condition monitoring. While EVs offer flexible, distributed capacity that grows with adoption, stationary storage enhances EV batteries by providing more availability and predictable response. The overall development of the smart grid aims to shape the scalability, user access, and grid services to support the EU's regulatory push for interoperability and flexibility, as per EU regulation 2023/194 and Italian Pnre/PUN frameworks (since the Pilot's location is in Italy).

P7-DA5 – Development of new features of the Micro Controller Unit

Development activity P7-DA2 carried out by Ares2T concerns updating the Micro Controller Unit (MCU) to enable data exchange with the PV plant's inverter (voltage, current and power production data, and possibly temperature, humidity and operation condition) and bidirectional EV chargers. Specific data aggregation services are developed in this activity to manage new raw data and complement the target data model enabling higher level (Cloud layer) added value services for optimization and forecasting.

The difference between traditional and bidirectional EV chargers plays a crucial role in Pilot 7 as it enables remote control acting not only on the charging capability but on the discharging capability too, to discharge the EV's battery in favor of other component of the smart grid (the energy storage or other EV charging sessions) through an actual energy flexibility environment.

P7-DA6 - Development of new features of the Cloud Server

Development activities carried out by Ares2T under the umbrella of this activity concern optimization and machine learning based services at Cloud layer to support

- Macro load area and single local load area optimization and control via Mathematical Programming optimization techniques and Model Predictive Control (MPC) control technique for optimal power modulation in one or more than one load areas (enabled by CEI architecture)
- Charging profile identification and load curve estimation via supervised and unsupervised machine learning (clustering algorithms and regression models).

New features strongly rely on data modeling, data gathering distributed services, and large-scale algorithms able to process considerable volumes of data and to return near real time input controlling commands to actuators distributed in the target load area.

P7-DA7- Development of new trust and security features

To further strengthen the cybersecurity posture of Pilot 7, the algorithms under development will go beyond conventional monitoring and protection mechanisms, incorporating advanced and forward-looking features that address both emerging threats and long-term resilience requirements. Given the critical role of EV charging infrastructures in future smart grids, the security framework must not only ensure **data integrity, trust, and privacy**, but also provide **resilience against sophisticated cyber-physical attacks, adaptability to evolving adversarial tactics and compliance with future-proof standards**. The following enhancements extend the trust and security capabilities of P7-SC2, ensuring that the Cloud-Edge-IoT ecosystem of Pilot 7 remains secure, interoperable and trustworthy under real-world operating conditions.

- **Zero-trust enforcement:** Application of zero-trust principles across the Cloud-Edge-IoT stack (continuous authentication, micro-segmentation, and policy-based access) to prevent lateral movement in case of compromise.

- **ML-powered Anomaly Detection:** Continuous monitoring of EV charging patterns and communication flows to identify abnormal behaviour, detect intrusions in real time, and trigger automated security responses.
- **Secure update and patching mechanisms:** Over-the-air (OTA) trusted updates for chargers and edge devices, using cryptographic signing to prevent supply-chain or firmware manipulation.
- **Explainable security analytics:** Explainable AI (XAI) methods applied to anomaly detection models, providing human-understandable explanations for alerts to increase operator trust and facilitate incident investigation.
- **Distributed ledger for secure data provenance:** Ensures traceability and immutability of charging data flows across Cloud, Edge and IoT components, preventing unauthorized manipulation and enabling end-to-end accountability

Related to the next steps in this phase, CERTH will perform model development and testing in Jupyter Notebooks using publicly available datasets. This work will involve loading existing IDS models and applying retraining and fine-tuning techniques for network-layer data based on CICFlowMeter flows, while simultaneously training new models from scratch for application-layer OCPP data due to its incompatible feature space. The team will systematically evaluate model performance on selected public datasets to identify strengths, weaknesses, and implementation gaps before moving to deployment. Once validated, the trained models will be prepared for integration by serializing them in appropriate formats and adapting existing model handler scripts within the IDS system. CERTH will then integrate these trained models into the comprehensive IDS framework, ensuring that inference capabilities, detection mechanisms and alert generation logic are fully preserved and operational. To enhance transparency and trust in the cybersecurity system, CERTH will also integrate explainability libraries such as SHAP and LIME into the machine learning detection pipeline, enabling stakeholders to understand and validate the reasoning behind security alerts and anomaly detections

9.2.3 Integration activities

P7-IA8 Integration of units for all use cases

Both system-level design and real-world field implementation are needed for the integration of EV batteries, stationary storage, and charging infrastructure. The project will be organized around certified design, installation, and commissioning operations in order to accomplish this.

Preliminary analysis will be carried out in the test bed to ensure grid feasibility, including a load-flow analysis to allow for unit sizing. Operational scenarios will be developed based on rigorous charging schedules that simulate electrical fleet needs. After identifying the interconnection standard and compliance requirements, the intelligent grid design will be outlined for integration, which will kick off with electrical professionals and grid engineers commissioning to design, size and install the procured infrastructure following regulatory and safety standards. Once the wiring, protection systems and interconnects are finalized, the system will be tested for compliance. The monitoring system will be configured to allow for real-time management of all operating modes.

P7-IA9 Integration of MCU in the load area

The updated MCU will be integrated in the load area set up by E-Gap (ref. P7-IA8) to guarantee the suitable data exchange with new energy equipment, including the PV plant's inverter and bidirectional EV chargers for AC and DC charging. Once integrated, data aggregation services will be ready for the first round of validation activities (ref. P7-VA11) to check suitable communication of energy data.

9.2.4 Planning & Gantt

The following Gantt diagram lists all the activities of the work package from M4 until M18. In particular Setup & procurement and Development activities, with an introduction of integration activities that continue over M18.

Table 19. Pilot 7 Gantt diagram

Pilot 7		2025										2026				
Code	Task	M4	M5	M6	M7	M8	M9	M10	M11	M12	M13	M14	M15	M16	M17	M18
SETUP & PROCUREMENT																
P7-SA1	Full procurement of units for all use cases															
P7-SA2	Setup of IoT and Edge components															
P7-SA3	Setup of 5G components															
DEVELOPMENT																
P7-DA4	Development of new features of the smart grid															
P7-DA5	Development of new features of the MCU															
P7-DA6	Development of new features of the Cloud Server															
P7-DA7	Development of new trust and security features															
INTEGRATION																
P7-IA8	Integration of units for all use cases															
P7-IA9	Integration of MCU in the load area															

9.3 CURRENT INFRASTRUCTURE

The architecture combines multi-layered data aggregation and control platforms, edge-level optimization, secure communication protocols, and Docker-based containerized services. Multiple new acquisitions are planned, including bidirectional charging infrastructure, a hybrid fast charging unit with integrated battery storage, and solar panels to support local energy generation.

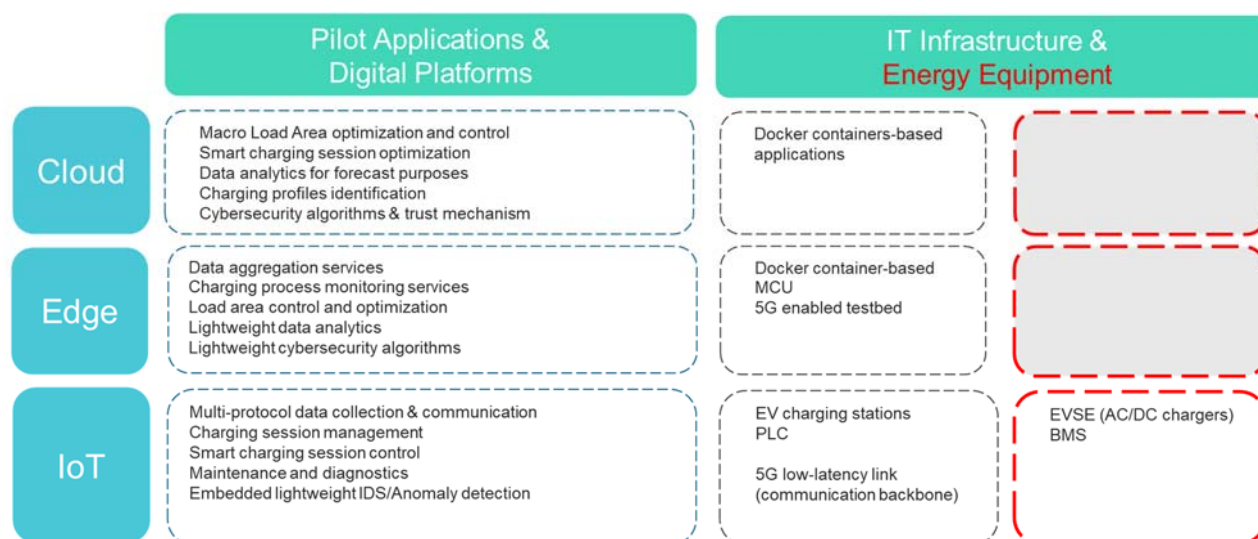


Figure 100. Pilot 7 Infrastructure

9.3.1 Pilot applications and digital platforms

Cloud Layer

Main functionalities supported at the Cloud Layer are:

- Load area management and monitoring
- Smart charging session optimization
- Data analytics for forecast purposes

Raw data from IoT network are collected at the Edge Layer and communicated via 5G infrastructure to the Cloud Layer where the O-CEI platform will publish and make available suitable services to achieve the objectives of P7-SC1.

Optimization algorithm-based services allow for maximizing energy efficiency by controlling near real time the energy loads (EVs) through the charging stations according to power availability in the grid consisting of the power from by the electric plant, the PV plant and the energy storage unit.

Machine learning based data analytics allow for energy and power forecasting to complement power from the PV plant to feed optimization algorithms for actual control.

Edge Layer

- Main functionalities supported at the Edge Layer are:
- Data collection and aggregation services

- Charging process monitoring services (instant power, historical curves, etc.)
- Load area control
- Lightweight data analytics

Data are collected from the EV charge stations on the field by a Micro Controller Unit (**MCU**), which runs a VM and a Linux container through the **Proxmox operating system**. In particular, the data collection is performed by the operating system that runs inside the VM, called **Home Assistant**.

Inside Home Assistant, a **Python implementation of the OCPP protocol** communicates with the AC wallboxes and a **Modbus protocol add-on** (which comes out-of-the-box with the OS) establishes the connection with the DC charging station.

Regardless of the source, the gathered data is represented in Home Assistant as an entity called **sensor**, which works much like a read-only variable.

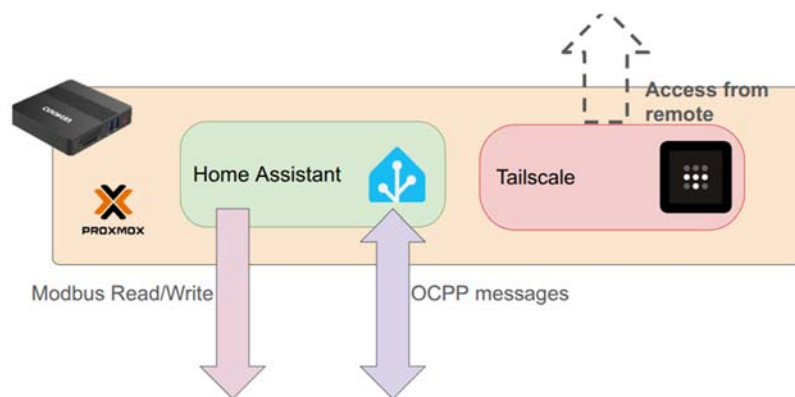


Figure 101. Pilot 7 Edge layer schema

The most relevant sensors are then arranged in user-friendly **views**, which allow the operator to monitor the charging activities on the field as well as the general operating conditions of the devices.

Moreover, the same interface also displays a set of **buttons** which give the operator complete control over the charging sessions performed on the field by each device.

In particular, they allow the operator to 1- **Start** and **stop** a charging session and 2- **Set** and **modify** the **parameters** of a charging session.

Finally, Home Assistant also keeps track of the values every sensor took during its existence. The **history** of the sensors can be fed through graphing and data representation tools (like **Grafana** or Home Assistant's own **History** add-on) to aid the operators during maintenance, debug and inspection activities.



Figure 102. Pilot 7 Representation of sensor values history

Main cybersecurity functionalities supported at the Edge Layer are the following:

- Intrusion and Anomaly Detection for OCPP and Modbus traffic to identify spoofing, injection or abnormal charging patterns.
- Lightweight ML-based analytics for near real-time detection of malicious or abnormal device behavior.
- Secure communication mechanisms (TLS, VPN tunneling) for data aggregation and transfer to the Cloud.
- Forensic logging of charging operations to enable traceability and post-incident investigation.

In this layer, the above-mentioned functionalities will be developed in order to further enhance CERTH's IDS tool to fulfill the requirements of Pilot 7. More specifically, this IDS platform includes several methods and mechanisms, including Deep Learning-based security incident detection, visual analytics for data inspection and cyberattack mitigation suggestions derived from multi-objective optimization algorithms, which work as an all-in-one SIEM solution. CERTH's IDS aims to identify security events from various data sources, such as network traffic and system memory dumps and to provide cyberattack detection, alerts and insights for potential threats while recommending various mitigation suggestions. Its AI models will be further trained with OCPP and Modbus traffic in order to support as many protocols as possible, used in the EV charging domain. The figure below illustrates the UI of the IDS tool.

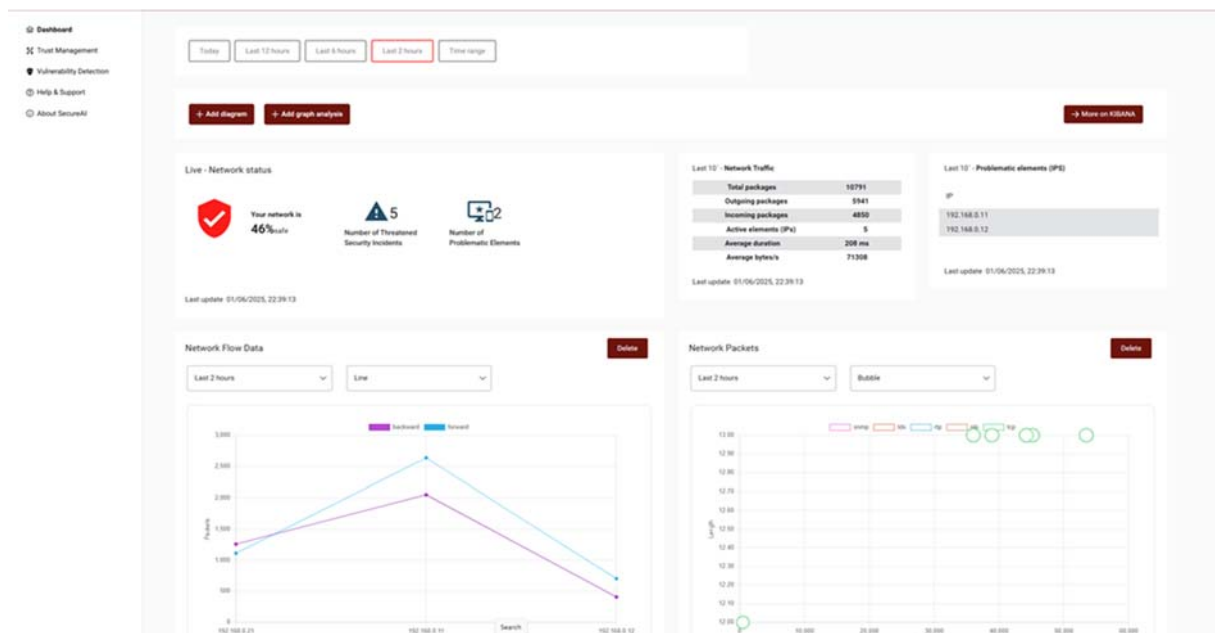


Figure 103. Pilot 7 UI of the IDS Tool

IoT Layer

Main functionalities supported at the IoT Layer are:

- Multi-protocol data collection and communication (OCPP and Modbus TCP).
- Charging session management and control.
- Maintenance and diagnostics.

In this layer, the most important devices are the five EV charge points: **four wallboxes** and **one dual-plug charging station**. Each wallbox is able to output up to 22 kW of AC power, absorbed straight from the grid. The charging station can output over 100 kW of DC power from the **batteries** it's equipped with, which are then recharged at a slower, more manageable rate through the grid.

Two protocols are used to communicate with these five devices: *OCPP* (*Open Charge Point Protocol*), for the AC wallboxes, and *Modbus*, for the DC station. OCPP is based on sequences of messages exchanged between the two actors, while Modbus defines a series of registers which can be read or written.

Beyond just gathering data related to energy equipment and the charging sessions, these protocols enable the upper layers to start, stop and manage them under other aspects as well.

In the case of the four wallboxes, this is performed through different message exchanges between the controller and the devices themselves:

- **RemoteStartTransaction** and **StartTransactionNotification**, that handle the beginning of a charging session;
- **RemoteStopTransaction** and **StopTransactionNotification**, that handle the end of a charging session;
- **MeterValues**, sent periodically by the device while performing a charging session;
- **SetChargingProfile** and **ClearChargingProfile**, to influence the behaviour of the device during a charging session (e.g. how much instant power to deliver).

The format of all these messages is defined by the OCPP protocol. The most relevant piece of data in these exchanges is the **transaction ID**, a unique identifier of each charging session which is generated when the session itself is starting and is then used by all the subsequent message exchanges regarding that session.

With the DC charging station, instead, the management and control of charging sessions is achieved by operating on the registers exposed by the Modbus protocol:

- To **start** and **stop** a charging session, the relevant registers are written and the underlying hardware responds by either starting or stopping the energy delivered;
- To **gather data** about an ongoing session (e.g., power output, energy delivery, duration...) it's sufficient to read the registers related to the desired plug, instead.

Moreover, it is also possible to monitor the status of devices in the load area at any moment. This is possible with the AC wallboxes thanks to the **StatusNotification** OCPP message, which communicates to the upper layers the current conditions of the device: Available or Unavailable, Preparing or Finishing (a charging session), Unavailable or Faulted. This message can either be sent spontaneously by the wallbox or be requested from the outside. Modbus based DC charging station, instead, has **four registers** that can be read, two for the status of the plugs and two for the SoC of the batteries.

9.3.2 IT infrastructure and Energy equipment

Cloud Layer

The IT infrastructure provided by Ares2t to support the cloud-level Pilot 7 energy management services is organized in micro services, implemented using Docker containers and deployed over virtualized infrastructures. Each functional component (database, logic, algorithms, graphical user interface, application-level interfaces, etc.) is hosted in one Docker container and can be dynamically deployed on one or more physical or virtual machines. The reference platform operated and owned by Ares2t, namely the Charge Advisor platform offers the following functional components at the Cloud Layer:

- **Apache2:** Container responsible for running the Apache2 web server. It handles all inbound HTTP requests (GET, POST, ...) directed to the system.
- **PHP:** Container providing the PHP runtime environment. All PHP code execution is performed within this container.
- **MySQL:** Container hosting the backend relational database. It stores all tables and data structures associated with the backend instance.
- **PhpMyAdmin:** Container running the PhpMyAdmin web application, used for database administration and interaction with the MySQL service through a web interface.
- **Workspace:** Development container providing the PHP environment and tools required for project-wide development and testing activities.
- **LAC:** Container running the Python environment responsible for executing the optimization and control algorithm.
- **Central Communication System:** Container managing WebSocket-based communication with the edge-layer field-deployed MCU (Master Control Unit).

In particular, for performance issues and to balance the load the Apache2, PHP, MySQL, PhpMyAdmin, Workspace and Central Communication System run on a VPS (Virtual Private Server) while the LAC runs on another VPS. Both the VPS are based on Ubuntu Server 20.04 LTS 64bit, 4 vCPU, 8 GB RAM and 160GB SSD. The PHP container hosts most of the Logic, Application-level interfaces and Graphic User Interface services which have been developed using the PHP Lavavel v12 framework. While the LAC container hosts most of the forecasting and optimization algorithms which have been developed in Python. Since these two groups of services are mission critical and consumes the majority of the IT resources we decided to deploy them in two separate VPS machines.

Edge Layer

The IT infrastructure provided by Ares2t to support the edge-layer services of pilot 7 has been implemented in the Charge Advisor platform by Docker containers and deployed over virtualized infrastructures. The edge layer is composed of a number of Master Control Units (MCUs) all connected to the Cloud layer and each managing a given load area. The MCU is hosted in an industrial MiniPC (to be installed in potentially harsh environments) equipped with a Proxmox OS. For the purposes of the project Ares2t choose a particularly performant industrial MiniPC with the following characteristics: CPU: Intel Core i5 7267U/7300U, 2 x LAN 1Gbps, 32GB RAM DDR4, 1TB NVME SSD.



Figure 104. MiniPC as infrastructure in the Edge Layer in Pilot 7

Promox OS is a linux based OS dedicated to hosts virtual machines (VMs) and Linux Containers (LXC). In the setup carried out by Ares2t, Promox hosts one LXC and one VM. The LXC hosts a Tailscale client to enable the device remote management via VPN. The VM hosts an instance of the MCU operating system (based on Home Assistant OS) that is in charge of elaborating the localized smart charging logic. The Home Assistant OS is a lightweight, dedicated operating system designed to run the Home Assistant software. It's built specifically for smart building management, providing a stable and reliable platform for controlling IoT devices and related automations. At its core, Home Assistant is primarily written in Python and makes extensive use of Docker containers. Each major component, including the core Home Assistant application, the supervisor, and every single add-on, runs in its own isolated container. In the setup carried out by Ares2t, the following containers were run, among the others:

- **homeassistant:** This is the main brain of your smart home, running the Home Assistant Core software. It's the central hub for all your automations, integrations, and the user interface.
- **hassio_supervisor:** This container is the supervisor for Home Assistant OS. It's responsible for managing all other Home Assistant-related containers, including add-ons, updates, and system health.
- **hassio_dns:** Manages DNS requests for the entire Home Assistant environment, ensuring all containers can connect to external services correctly.
- **hassio_multicast:** Manages multicast networking, which is vital for device discovery and communication with many smart home devices.
- **addon_a0d7b954_ssh:** The SSH & Web Terminal add-on, which allows you to access the Home Assistant command line via SSH from an external device.

The MCU, based on Home Assistant OS, is the core of the edge-layer intelligence in the Charge Advisor platform owned by Ares2t. It leverages a lightweight, Python-based platform that uses Docker containers to house the smart charging logic. This containerized architecture ensures that the system is stable, scalable, and easy to manage in potentially harsh environments, effectively separating the core logic from the host hardware and networking components.

The only IT infrastructure required for deploying CERTH's cybersecurity tool at the edge is lightweight and optimized for AI-enabled IoT devices. The solution can be deployed on compact platforms such as Raspberry Pi 5 with a 5G hat or NVIDIA Jetson devices, which can be directly plugged into EV chargers to provide local monitoring and protection. Each functional module of the tool—including data acquisition, anomaly detection algorithms, logging, and local decision-making—is containerized using Docker, enabling modular deployment and scalability across multiple edge nodes. This setup allows autonomous operation at the edge while maintaining seamless integration with cloud-level energy management services, ensuring real-time detection and response close to the data source.

- ⑩ **5G-enabled testbed** that supports low latency, real-time control and coordination of charging sessions.

IoT Layer

The **IoT Layer in E-GAP's ecosystem** connects electric vehicle charging stations, energy meters, PLCs, and battery systems into a unified smart infrastructure. Ultra-fast AC/DC chargers (up to 150 kW DC with 200 kWh storage) are OCPP 1.6j/2.0.1 and ISO 15118 compliant, enabling interoperability and remote management. While secure LAN connectivity (Ethernet, Wi-Fi, LTE/5G) guarantees smooth data interchange with cloud dashboards for monitoring, reporting, and predictive maintenance, proprietary PLCs with embedded microcontrollers offer real-time control over charging, load balancing, and safety.

At the energy equipment level, **integrated battery management systems (BMS)** oversee advanced LFP/NMC storage modules, combining live thermal monitoring, safety interlocks, and lifecycle optimization. This allows E-GAP's infrastructure to operate grid-parallel or independently while supporting bidirectional functions such as V2G. These IoT-enabled layers create a flexible, scalable, and future-proof architecture.

10 PILOT 8 – HEIGHTENED SOCIAL ENGAGEMENT AND ACCEPTABILITY OF ENERGY FLEXIBILITY IN URBAN AREAS

Pilot 8 demonstrates how smart IoT solutions deployed within energy producer and/or consumer (prosumer) networks can be integrated into the O-CEI platform to foster **social acceptance, citizen engagement, and energy flexibility**. By combining O-CEI orchestration, edge intelligence, and AI-based forecasting, the pilot optimizes energy demand and production in residential areas and tourism sites. It addresses challenges of user trust and adoption, ensuring that energy flexibility models are both socially accepted and technically feasible. The pilot aims to:

- Democratize access to energy flexibility services for prosumers in residential and urban communities;
- Forecast and optimize energy consumption and production using graph-based data models and AI/LLMs;
- Enable self-adaptive energy and computation continuum for tourism areas (villages, ski resorts);
- Assess citizen acceptance of CEI solutions via surveys, user interaction and participatory tools.

The pilot has 3 main scenarios:

P8-SC1 Democratization of Energy Flexibility Services

This scenario focuses on enabling **residential and urban prosumers** to access energy flexibility tools. Through smart IoT devices and the O-CEI orchestration platform, users can monitor and manage their energy consumption, participate in demand-response programs, and contribute to grid stability. The goal is to make energy flexibility **inclusive and accessible**, fostering engagement and trust. The following image showcases the residential apartments of the Polygones cooperative.

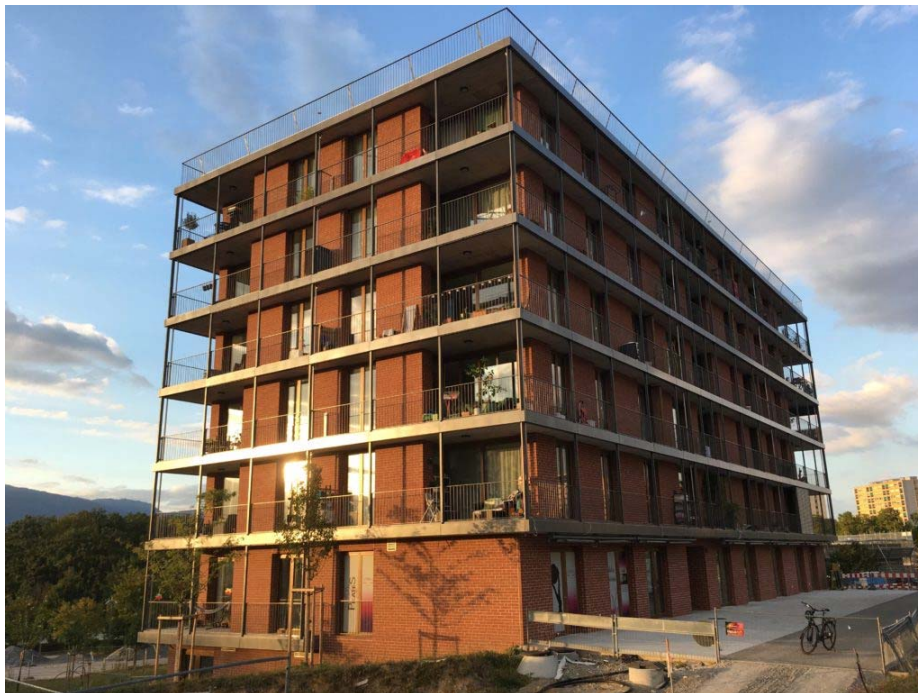


Figure 105. Pilot 8 Polygones cooperative apartments

P8-SC2 Advanced Forecasting with Graph-Based Models and AI/LLMs

Here, the emphasis is on **predictive intelligence**. By leveraging **graph-based data models** and **Large Language Models (LLMs)**, the system can improve the forecast energy consumption and production patterns. Users provide information in natural language. This information is used to refine the predictions provided by ML algorithms.

P8-SC3 Self-Adaptive Energy and Computation Continuum in Tourism Areas

This scenario addresses **distributed and seasonal energy needs** in tourism zones like **villages and ski resorts**. The system adapts to fluctuating demand using edge computing and federated learning, ensuring efficient energy use even in remote or high-variability contexts. It integrates **local infrastructure (hotels, ski lifts, restaurants)** with cloud-based orchestration, creating a seamless continuum of energy and computation. The next images show the tourist attractions of Val d'Anniviers and the hotel Chandolin.



Figure 106. Pilot 8 Ski lifts in Val d'Anniviers



Figure 107. Pilot 8 Hotel Chandolin

10.1 PILOT GOVERNANCE

The pilot is led and managed by HES-SO. Polygones and Val d'Anniviers are the sites that will be equipped with CLEMAP devices and will provide the energy consumption measurements. ***Although another pilot scheme is currently being validated to replace the canton of Geneva.***

HES-SO is the **Pilot Leader** and bears end-to-end accountability for scope, schedule, quality, risk, and reporting. HES-SO also acts as the technical provider for Edge-to-Cloud integration, ensuring that field devices, gateways, data services, and cloud analytics are interoperable and compliant with project standards.

CLEMAP is the **technology provider** and domain expert for energy monitoring and load management. CLEMAP supplies, configures, and maintains CLEMAP Energy Monitor (CEM) and CLEMAP Load Management (CLM) devices, and supports data quality, firmware updates, and troubleshooting in coordination with HES-SO.

Polygones (POLY) and **Val d'Anniviers (VALDA)** are the umbrella organizations for the participating locations. They coordinate on-site activities (installation access, safety, user onboarding) and provide continuous energy consumption measurements via CLEMAP devices. A third pilot scheme (to **replace the canton of Geneva**) is **under validation**; activation follows a formal gate review and change-control update to the site plan. The sites host the end users (prosumers) who interact with the system. These interactions underpin the social-acceptance assessment of the pilot. POLY/VALDA manage user recruitment and local communications; HES-SO provides instruments and protocols (surveys, consent forms, interaction logs); CLEMAP supports device-level UX considerations.

The pilot runs a monthly governance meeting (agenda: status, risks, change requests, KPIs, actions), tracked in WP4 meetings on the project repository. Workstreams (deployment, data/Cloud, social acceptance) may convene ad-hoc technical syncs as needed, with outcomes reported at the monthly forum.

10.2 DEMONSTRATOR PATH

During the first building phase of the project and since the start of WP4 (M4), Pilot 8 has been primarily dedicated to **establishing the technical and social foundation**. This phase has centered on procurement activities, ensuring that the partners that will be providing the energy measurements are on-board. Also, the correct dimensioning of the CLEMAP devices had to be ensured. In-person visits to the installation sites have been carried out. It was necessary to have O-CEI project personnel; owners of the sites and their representative electricians present at the same time so that the process and technical details are understood by all involved. Some sites in Val d'Anniviers still must be visited.

The activities carried out should ensure that the CLEMAPs can be installed correctly and be available for further orchestration of services.

10.2.1 Setup & procurement activities

P8-SA1 – Acquisition and setup of CLEMAP devices

The CLEMAP devices for POLY were ordered in June 2025. They are to be received at HES-SO in September 2025. Then they are to be locally configured with VPN before being sent out to be installed at the sites. Subsequent visit to VALDA and the on-site visits mentioned above lead to a revision of the order. The reason for this is that the window to install devices in VALDA is tight as given below:

- The hotels and restaurants close for the low season. Approximately mid-September to early December.
- Ski lifts have a similar schedule but keep running in the low season for maintenance and similar reasons.
- Electricians that are contracted to various businesses will be very busy in the weeks lead-up to December.

Therefore, it was decided to prioritize the installation to, in order:

- Hotel Chandolin
- Chandolin/St Luc ski lifts
- Polygones
- Other locations in Val d'Anniviers (TBD)

The Polygones is the easiest installation in practical terms. Close to Geneva and the electrician is familiar with the devices already.

10.2.2 Development activities

P8-DA2 – Machine learning

A machine learning pipeline has been developed as a proof of concept. Using standard tools such as dvc and MLFlow. This pipeline is tested on local machines (laptops, local servers).

P8-DA3 – Energy Consumption data

There are some testing-only CLEMAP devices deployed where we do not have problems with data ownership/distribution. The sending of CLEMAP energy consumption data to a local database is tested. In addition to the initial testing of CLEMAP devices, the planning for the deployment of CLEMAP is underway, as is the installation of energy monitors in Val d'Anniviers and Polygones. in line with milestones M4 to M9. A two-day workshop involving key stakeholders from hotels, ski lift operators, restaurants and local governance was held in Val d'Anniviers. The workshop facilitated the coordination of installation logistics and the refinement of deployment strategies for different energy usage profiles. Crucially, the workshop also emphasized the importance of integrating social acceptance by design, recognizing that energy consumption monitoring is a socio-behavioral endeavor as well as a technical one. The pilot aims to capture and understand the behaviors of prosumers within the energy community, ensuring that the orchestration of energy flexibility services is both technically robust and socially embraced.

10.2.3 Planning & Gantt

The following Gantt diagram list all the activities since the beginning of the work package on M4 until M18. In particular Setup & procurement and Development activities.

Table 20. Pilot 8 Gantt diagram

Pilot 8		2025										2026					
Code	Task	M4	M5	M6	M7	M8	M9	M10	M11	M12	M13	M14	M15	M16	M17	M18	
SETUP & PROCUREMENT																	
P8-SA1	CLEMAP order for POLY																
	CLEMAP order for VALDA																
	CLEMAPs received																
	Configure CLEMAP for POLY																
	Configure CLEMAP for VALDA																
	Install CLEMAP to POLY																
	Install CLEMAP to VALDA Hotel Chandolin																
	Install CLEMAP to Ski lifts																
	Install CLEMAP to Ski lift restaurant																
DEVELOPMENT												Continue until M24					
P8-DA2	Machine learning																
P8-DA3	Energy consumption data																

10.3 CURRENT INFRASTRUCTURE

Pilot 8 combines the energy consumption measurements over three distinct regions: Urban apartments; rural touristic; industrial/technical to measure and promote heightened social engagement and acceptability of energy flexibility.

The pilot relies on a layered deployment of CLEMAP energy monitors (CEM) and CLEMAP load management (CLM) devices as Edge devices, and Cloud services to collect and process high-frequency energy data. It combines federated machine learning, local inference, and large language models (LLMs) to create personalized and socially aware interfaces that promote user engagement and behavior change. The current technical setup builds on previous projects.

We use the CLEMAP energy devices as Edge devices as they combine the IoT collection of energy consumption data and Edge computing device (via an onboard Raspberry Pi4) in one housing. This allows the deployment of interesting applications, such as Machine Learning, directly beside the IoT data collection within the Edge layer. Also, CLEMAP device firmware sends energy data to a Cloud-based data store by default, therefore providing a managed data store and backup solution.

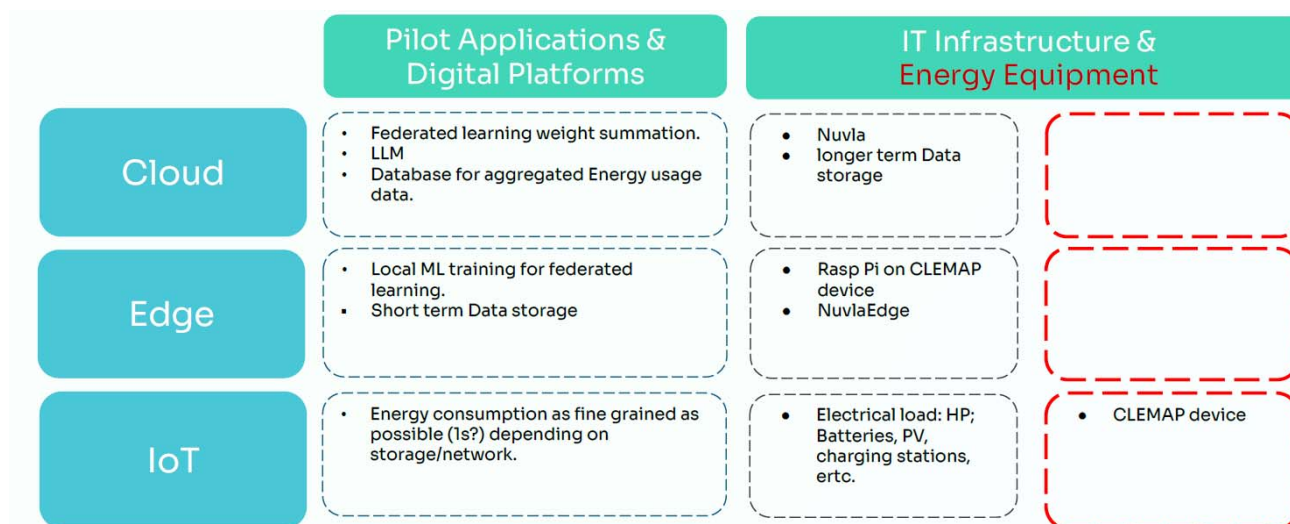


Figure 108. Pilot 8 Infrastructure

Currently, there are CLEMAP energy monitors (CEM) collecting data at two locations in Geneva: 5 energy monitors in a set of townhouse buildings; 7 energy monitors in a school. These energy monitors provide a real-time set of energy consumption data that is useful for testing the Machine Learning routines and pipelines.

The deployments planned so far, combine a selection of each site's different usage profiles. In the case of Polygones, one of each size of apartments plus selected technical areas (laundry rooms and bike charging station) aim to capture the overall energy usage profile of the building. For the Val d'Anniviers village, the overall energy usage of visitors is captured via a selection of energy monitoring sites. These include hotels, restaurants and ski lifts, the typical infrastructure that tourists use when visiting such sites.

The CLEMAP devices at each monitoring point function as the IoT devices (current measuring clamps) and Edge capacity (the onboard Raspberry Pi computing power). The Cloud computing and storage capacity will be bought from a (preferably) Swiss Cloud provider so that we can be sure that the energy monitoring data of individuals and businesses is handled according to the regulations.

Table 21. Pilot 8 list of Infrastructure elements

Polygones cooperative apartments	Val d'Anniviers ski station
- Each CLEMAP of 80A capacity - One apartment each of 3,4,5,6,7 rooms. - Bicycle charging station. - Two laundry rooms. - Overall current to the whole building (to be confirmed).	- Boutique hotel Chandolin - Each CLEMAP of 80A capacity - Tesla EV chargers - Guest Rooms - Kitchen - Laundry - Spa/Sauna - Ski lifts - Funicular railway (200A) - TS Tsape (800A) - TS Foret (200A) - TS Rotse (630A) - Restaurant Tignousa (400A)

The figure below provides a simplified view of the On-Prem/Cloud, Edge and IoT layers in Pilot 8:

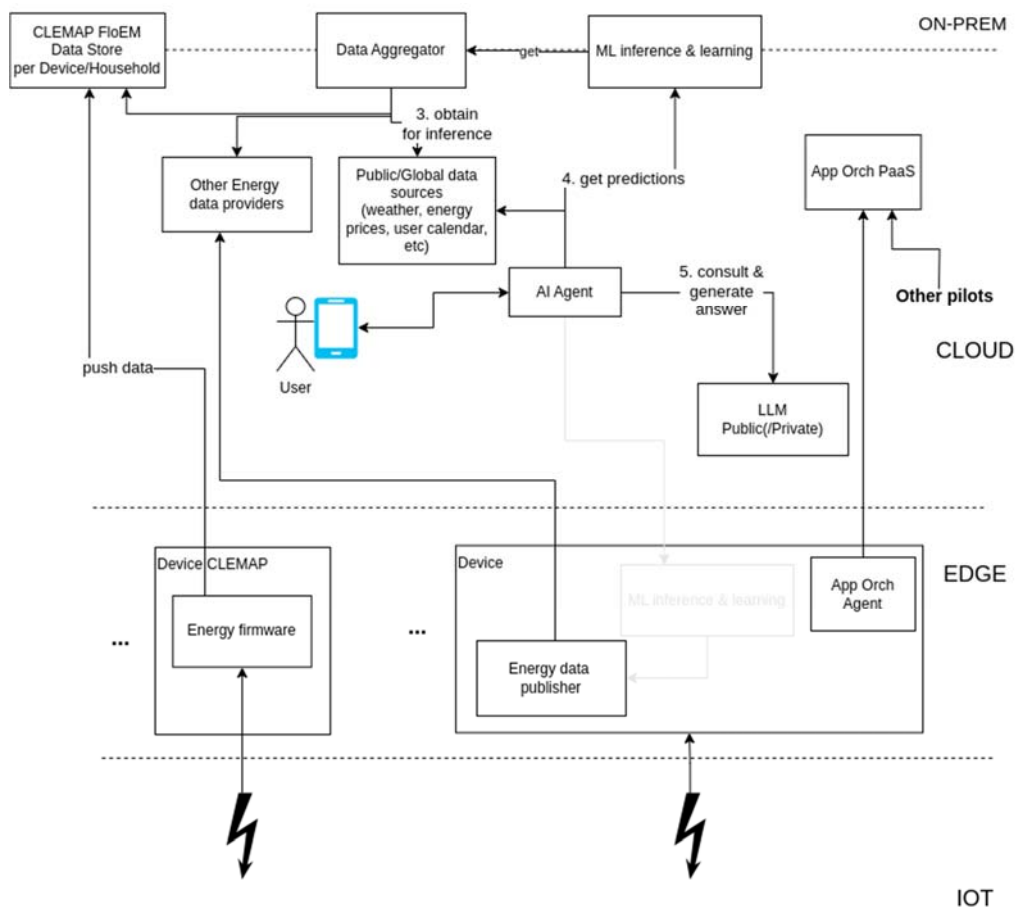


Figure 109. Pilot 8 Integration layers' schema

It shows how the energy providers, CEMs and the edge/cloud environment are connected through a mix of well-documented industrial standards and vendor-provided APIs. The figure is assumed to support two architectures, 1- Edge-to-Cloud (to support ML on the cloud) and, in the future, 2- Edge-to-edge (to implement Federated Machine Learning at the level of the (CLEMAP) edge devices). The CLEMAP devices have a native data collection from the Edge devices to a CLEMAP-managed data storage (FLOEM). The applications that run on the Edge devices are deployed and maintained by an Orchestrator that runs in the Cloud and is available to other pilots to use. This high-level overview helps applicants identify where their solutions can plug in — whether at the energy provider, Edge or Cloud level — and how to ensure smooth integration with existing systems across pilots.

A prosumer in one of the pilot locations (POLY or VALDA) can interact with the system via an AI Agent that could run in their Edge device (CLEMAP) or in the Cloud. Their interface is a user equipment (UE) device (e.g. CLEMAP or other energy monitoring/Edge device) or webpage where they can answer pre-canned questions or questions of their own. The questions/responses can be used to measure the Social Acceptability. The UE device interacts with an AI agent that retrieves the relevant global data (e.g. weather or personal calendar) and triggers a request for an energy consumption inference (prediction). These data are given to an LLM that generates the answer/suggestion.

10.3.1 Pilot applications and digital platforms

Cloud Layer

- Federated learning weight summation.
- Large Language Model (LLM).
- Database for aggregated energy usage data.

Edge Layer

- **Local ML training** for federated learning.
- **Short-term data storage.**

IoT Layer

- **Energy consumption monitoring at high frequency** (e.g., 1-second granularity), depending on network/storage.

10.3.2 IT Infrastructure & Energy Equipment

Cloud Layer:

- Nuvla for orchestrating infrastructure.
- Long-term data storage.

Edge Layer:

- Raspberry Pi on CLEMAP device.
- Nuvla Edge. Agent on Edge device.

IoT Layer:

- **Electrical loads:** HP (heat pump), batteries, PV panels, charging stations, etc.
- **CLEMAP voltage measurements** from power supply input and current from current clamps.

11 CROSS-DOMAIN INTERACTIONS & GOVERNANCE

11.1 INTERACTIONS AND GOVERNANCE IN A CROSS-DOMAIN CONTEXT

O-CEI project is aimed at addressing key challenges in the European supply chain by creating an open and cross-sector CEI continuum platform with a strong emphasis on interoperability, security, reliability, and suitability for hyper distributed applications, supported by a solid, well-defined and efficient data cooperation approach, easing data sharing and processing.

In this regard, in O-CEI LSPs' cross-domain concept will extend beyond data sharing. Relevant artefacts such as AI models will also be available on the marketplace to be used across different pilots. Focused on energy flexibility, **pilots' main concerns** such as energy demand prediction, EV charging recommendations, virtual network analysis, and monitoring techniques **will be addressed collectively** too. Key activities such as federated identity management, security scanning or accountability and traceability will also be infused with the cross-domain concept among all the pilots in O-CEI. As a result, being also one of the main goals in O-CEI, resources within the continuum will be shared, fostering cooperative learning and enhancing computing efficiency and transparency in key strategic sectors like electricity grids, Electromobility, Logistics, Agrifood, Telecommunications infrastructure and Urban Area,

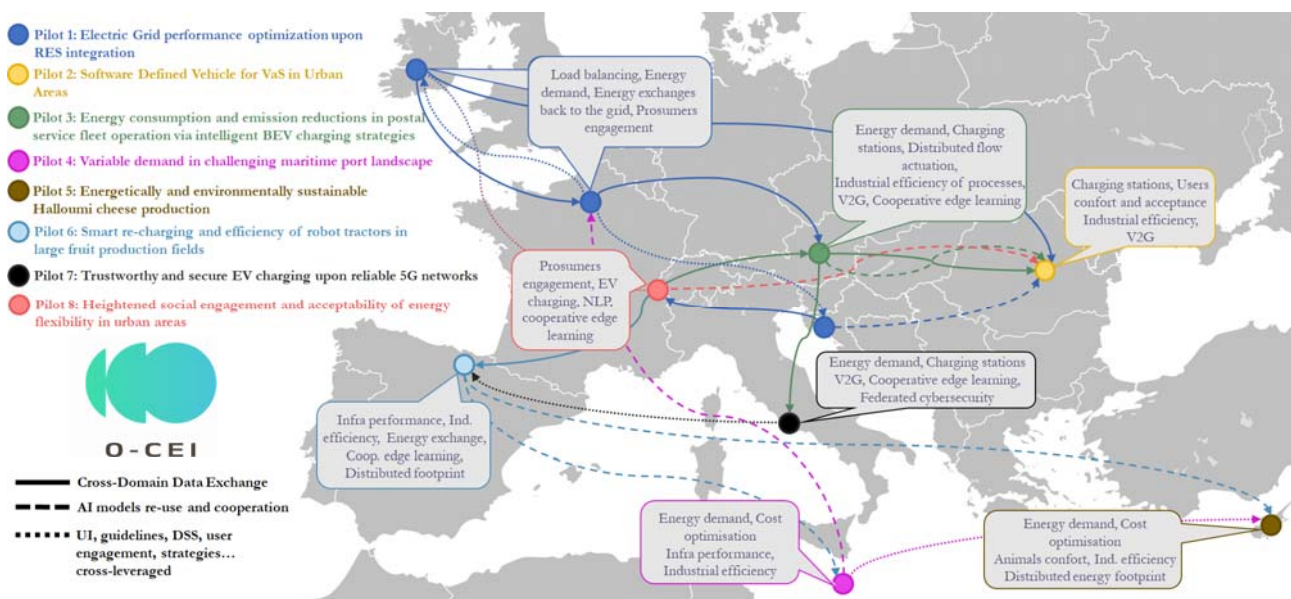


Figure 110. O-CEI Pilots Cross Domain interactions

LSP in O-CEI will cooperate bi-laterally or tri-laterally to validate the O-CEI open platform and solutions effectively, using key standards and initiatives such as GAIA-X, Catena-X, SAREF4ENER TIC4.0 and others, while contributing to the European Standard for CEI deployments (ambition 6, 1.1.2.6).

Some of the main challenges that not only the energy sector but also the main industrial sectors represented in the O-CEI project, is the proliferation of heterogeneous data sources that has resulted into data silos scattered throughout the entities. As a result, the value of data is locked within them, hindering data consumers from finding, accessing, understanding, and even combining these data with others. There are different types of data management mechanisms that can be combined to remove these barriers, from traditional data storage schemas (e.g., databases, data warehouses,

data lakes) and data routing protocols (e.g., MQTT, AMQP, Zenoh); to data fabrics, which facilitate the integration of those silos and expose their data in a uniform way. In these architectures, the knowledge graph is getting traction as candidate solution for implementing a data fabric architecture which can integrate any data linking them with their semantics; data spaces, which provide trustworthy and secure ecosystems to pool, access, share, process and use data of a specific domain, open for the participation of all organizations and individuals. Convergence and standardization effort is ongoing to foster interoperability and avoid the creation of new silos and data marketplaces/hubs, which facilitate the buying, selling or sharing of data. However, implementing effective data governance, sovereignty and trustworthy mechanisms can be challenging, especially in cross-company data sharing networks and ecosystems. From that situation, O-CEI is working to materialize an open platform and set of blueprints that will facilitate the exchange of data within and across computing continuum ecosystems, paving the path for real cross domain interactions between the 8 LSP. Moreover, Pilots utilizing the O-CEI platform to collaborate bilaterally or trilaterally, will facilitate coherent and complementary execution of cross-domain use cases, showcasing the value of openness, interoperability, and standardization at a European level.

11.2 O-CEI PILOTS CROSS-DOMAIN INTERACTIONS

During the first period of the project, the 8 LSP have been working together on the identification of data, assets, strategies or any relevant input that can be shared between the pilots. This section offers the first iteration on this identification exercise both from the perspective of what the pilot can share with others, and what the pilot can reuse from others.



Figure 111. Value to other pilots

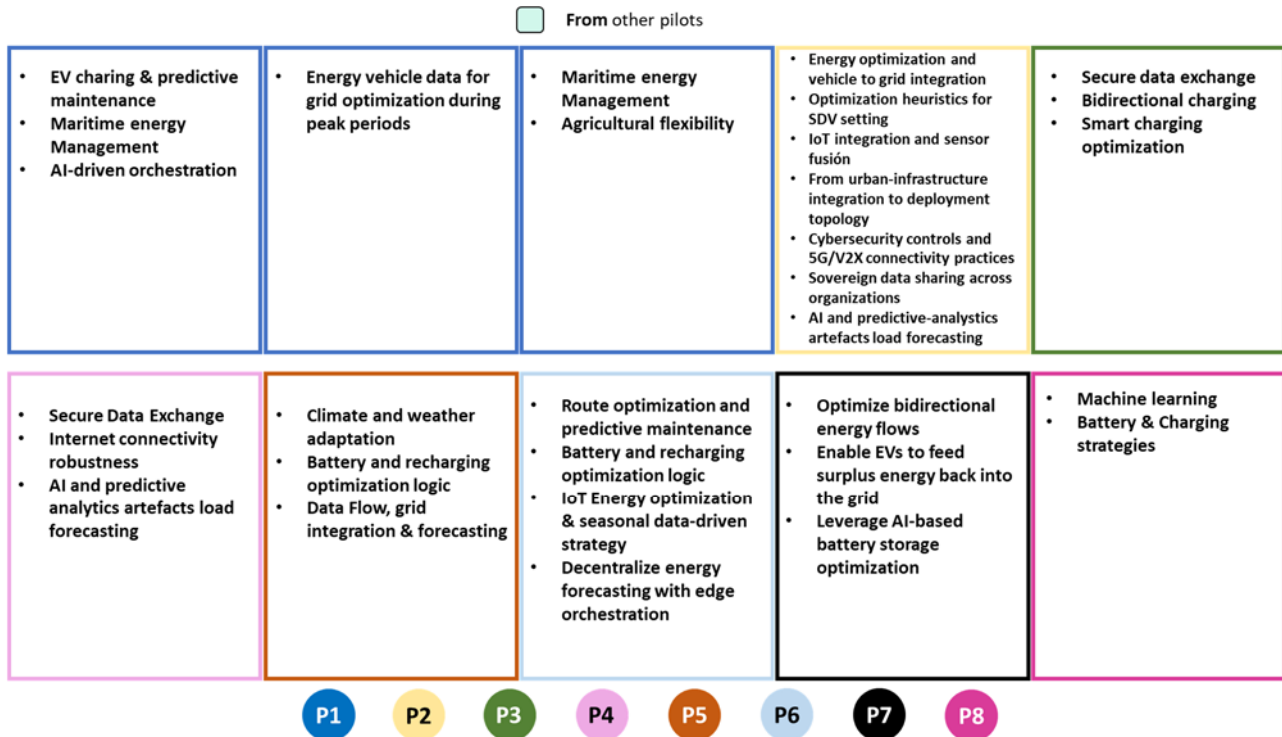


Figure 112. Value from other pilots

In the next section, a more detailed perspective of the cross-domain interactions per pilot is shown:

11.2.1 Pilot 1

Pilot 1.1

Value from other pilots

- **EV charging & predictive maintenance** (from Pilot 2) → P2's Software-Defined Vehicle fleet and predictive EV charging models can inform P1's integration of electric vehicles with heating systems, ensuring grid stability while coordinating heat pump and EV loads
- **Battery & Charging Strategies** (from Pilot 3) → P3's intelligent BEV charging and storage strategies can be adapted to P1 for synchronizing residential and district heating loads with EV charging in Positive Energy Districts.
- **AI-driven orchestration** (from Pilot 8) → P8's advanced AI models for decentralised demand forecasting can be reused in P1 to improve short- and long-term forecasting of electricity and heating demands.

Value to other pilots

- **Grid balancing and flexibility services** → P1's real-time grid demand modelling (Ireland, Croatia, France) provides validated flexibility services that can be exported to P6 (agriculture) and P4 (ports).
- **Community-driven energy trading** → P1's blockchain/DLT-enabled local trading in Limerick can be a template for P6 (farm energy communities) and P7 (urban energy collectives).

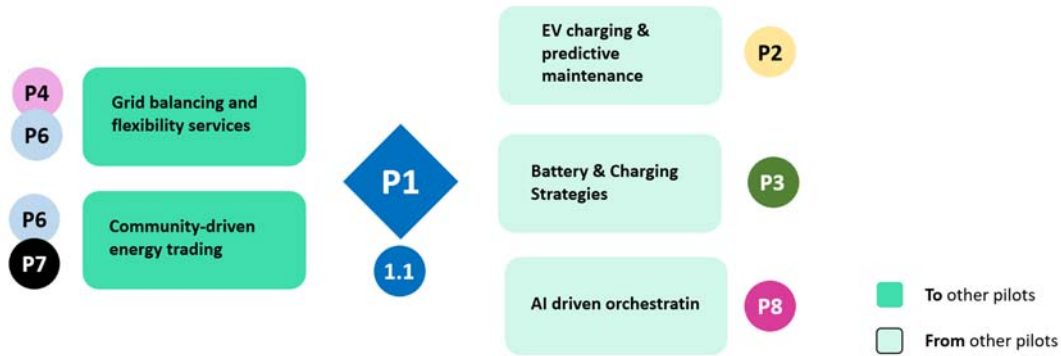


Figure 113. Pilot 1.1 Cross-domain interactions

Pilot 1.2

Value from other pilots:

- **Electric vehicle data for grid optimization during peak periods** (Pilot 2, 3 and 7). Depending on the presence of EVs in the home of P1-SC5 participants, knowledge on electric vehicle data, models, charging needs, etc., may be valuable for optimal placement of this energy use when optimizing the overall energy consumption of a consumer.

Value to other pilots:

- **Processes and algorithms for pilots with personal energy consumption data** (Pilot 4, 8): Setting up a pilot site with real-world end-consumers implies dealing with identifying data (personal data and energy data). Experience in setting up GDPR and data protection processes throughout the whole pilot lifecycle can be valuable to other pilots. Methodologies and algorithms used for energy consumption measurements, analysis or disaggregation can be valuable for other pilots, particularly if they include a “end-consumer” aspect. IA-based algorithms might be applicable to other contexts.
- **Green condition data (e.g. peak loads) for optimal EV charging strategies** (Pilot 2): The identification of optimal moments for end-consumer energy consumption (outside of peak periods, or during high-RES production periods) may be valuable to optimize EV charging strategies.
- **Energy flexibility trough the definition of load shifting needs and set points** (Pilot 2, 3, 4, 7, 8): P1-SC6 will define flexibility needs for the grid depending on RES share for maintaining grid voltage in an acceptable range. These flexibility needs could be used to define equipment set points sent to flexi-consumers who would be ready to participate in voltage-related grid services

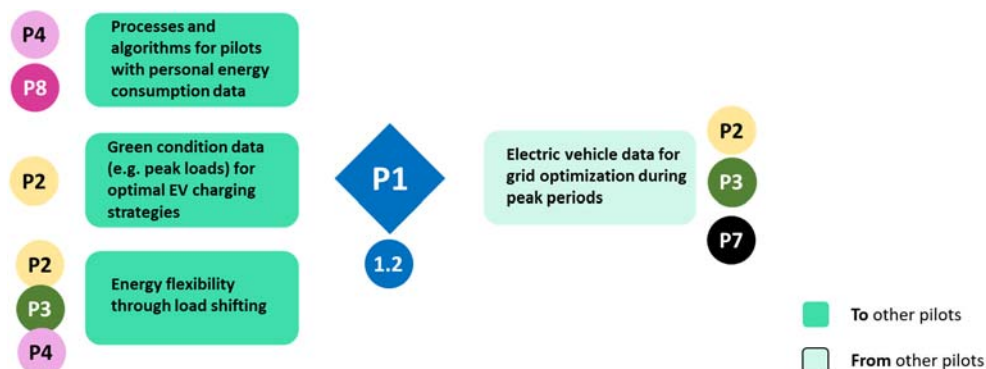


Figure 114. Pilot 1.2 Cross-domain interactions

Pilot 1.3

Value from other pilots

- **Maritime energy management** (from Pilot 4) → Lessons from P4's port energy balancing under variable demand can be reused in P1's Krk Island scenario, where fluctuating renewables mirror port demand variability.
- **Agricultural flexibility** (from Pilot 6) → P6's edge-based forecasting and renewable energy production methods (e.g., solar, biomass) can support rural parts of P1, particularly in Croatia, for balancing heating and mobility loads.

Value to other pilots

- **Energy & Heat coupling** → P1's scenarios on synchronising EVs and heat pumps (Krk and Limerick) provide transferable models for P2 (SDVs) and P3 (postal fleets), where thermal loads and charging events need coordinated scheduling.

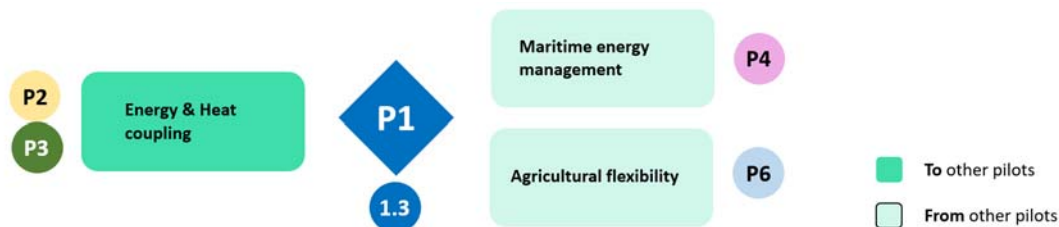


Figure 115. Pilot 1.3 Cross-domain interactions

11.2.2 Pilot 2

Value from other pilots

- The pilot benefits from advances in **energy optimization and V2G integration** developed in sister initiatives. Demand-response strategies, dynamic pricing participation, and renewable-aware charging policies inform the design of UC3 (Fleet V2G Integration) and UC4 (Grid Optimization). These imports sharpen our control logic for when and how EVs should charge or discharge, and they translate directly into operational indicators—kilowatt-hours returned to the grid, the proportion of off-peak charging events, improvements in forecast error (e.g., MAE/MAPE), and grid stability indices observed during stress conditions. (Pilots 3 and 8)
- The corpus of fleet-management and logistics research contributes **optimization heuristics—multi-depot routing, dynamic dispatch, and constraint-aware scheduling**—that we adapt to the SDV setting. In UC2 (Sustainability) and UC3, these heuristics become levers for reducing energy per kilometer and CO₂ per trip while improving schedule adherence and cost per kilometer across heterogeneous vehicle classes. (Pilot 3)
- Progress on **IoT integration and sensor fusion** elsewhere strengthens UC1 (Vehicle Safety). Mature pipelines for multi-sensor perception—combining LiDAR, radar, and camera streams—improve object detection fidelity and robustness to environmental variance. The quality of perception (precision/recall) and decision latency constitute our principal signals that such transfers have been successfully absorbed into the Pilot 2 stack. (Pilots 3 and 6)
- **Urban-infrastructure integration** practices—roadside units, signal-phase and timing interfaces, curb-management policies, and quality-of-service mechanisms such as Time-Sensitive Networking—guide our **deployment topology** for UC1 and UC2. The objective is not merely connectivity but predictable service levels along designated corridors, measured through latency-budget adherence, coverage of priority routes, and user comfort assessments. (Pilots 3)

- Fifth, **AI and predictive-analytics artefacts—load forecasting**, eco-routing, and predictive-maintenance models—are adapted to the urban SDV milieu. Here the emphasis is on reproducibility and governance: models are versioned, monitored for drift, and benchmarked with domain-appropriate error metrics. Downtime reduction and adoption of eco-routes provide pragmatic evidence of impact. (Pilots 7 and 8)
- Sixth, cross-domain collaboration frameworks originating in other pilots—data-space connectors, semantic catalogues (e.g., Catena-X/O-CEI), and Gaia-X/IDSA policy patterns—enable **sovereign data sharing across organisations**. Their assimilation raises data availability and quality while increasing the reuse of CEI components across UC1–UC4. (Pilots 3, 7 and 8)
- Finally, **cybersecurity controls and 5G/V2X connectivity practices**—ranging from zero-trust authentication to network slicing—underpin safety-critical exchanges in UC1 and energy-critical flows in UC3–UC4. We evaluate these imports through reductions in vulnerability exposure and end-to-end latency/jitter on representative traffic and grid-interaction paths. (Pilots 3 and 7)

Value to other pilots:

- Symmetrically, Pilot 2 contributes mature artefacts back to the programme. The most immediate export comprises **fleet-scale V2G strategies** that are explicitly State-of-Charge/State-of-Health-aware and optimised for dynamic tariffs. These strategies offer portable decision policies for energy pilots and logistics fleets seeking demand-side flexibility. (Pilot 3 and 7)
- In the safety domain, our **methodologies for multi-sensor fusion, object detection, and path planning**—shaped by edge-centric constraints—form a reusable toolkit. Pilots in mobility, logistics, and public safety can adopt these methods, benefiting from documented latency envelopes and accuracy benchmarks gathered on public roads. (Pilots 3 and 6)
- Urban-mobility insights constitute a further contribution. We provide operational recipes for **interfacing SDVs with municipal infrastructure**—traffic coordination, kerbside operations, and priority signaling—together with governance templates that clarify roles and data-sharing responsibilities between cities, OEMs, and operators. (Pilot 3)
- **Sustainability practices**—eco-routing, eco-driving feedback loops, and powertrain optimisation—are **bundled as service** components and KPI ‘recipes’. These allow other pilots to reproduce carbon-reduction outcomes using consistent measurement protocols and dashboards. (Pilot 3 and 6)
- At fleet level, we codify **orchestration patterns for dynamic charging schedules and predictive maintenance** into playbooks that scale from small municipal fleets to multi-regional operators. Their value lies in the combination of algorithmic guidance and operational checklists, which shortens the path from prototype to routine practice. (Pilots 3, 6, 7 and 8)
- Critically, Pilot2 demonstrates cross-domain interoperability through Catena-X-aligned data spaces and the O-CEI marketplace. The associated **semantic assets and access-control policies** are published as reusable components, reducing integration friction for pilots that straddle mobility, energy, and city platforms. (Pilots 3, 7 and 8)
- Finally, our **cybersecurity and edge-computing patterns**—secure data exchange, provenance tracing, and real-time inference on the vehicle—serve as reference implementations for edge/AI and critical-infrastructure pilots. Each is accompanied by configuration profiles that balance performance with safety and regulatory requirements. (Pilots 3,7 and 8)

In sum, the value exchange is not a loose catalogue of ‘nice-to-have’ ideas but a disciplined circulation of tested patterns. By importing rigorously evaluated methods and exporting standardised blueprints, Pilot 2 accelerates collective learning while preserving the rigor necessary for safety-critical and energy-critical operations. This reciprocity is the mechanism by which the O-CEI programme achieves replicability at scale.

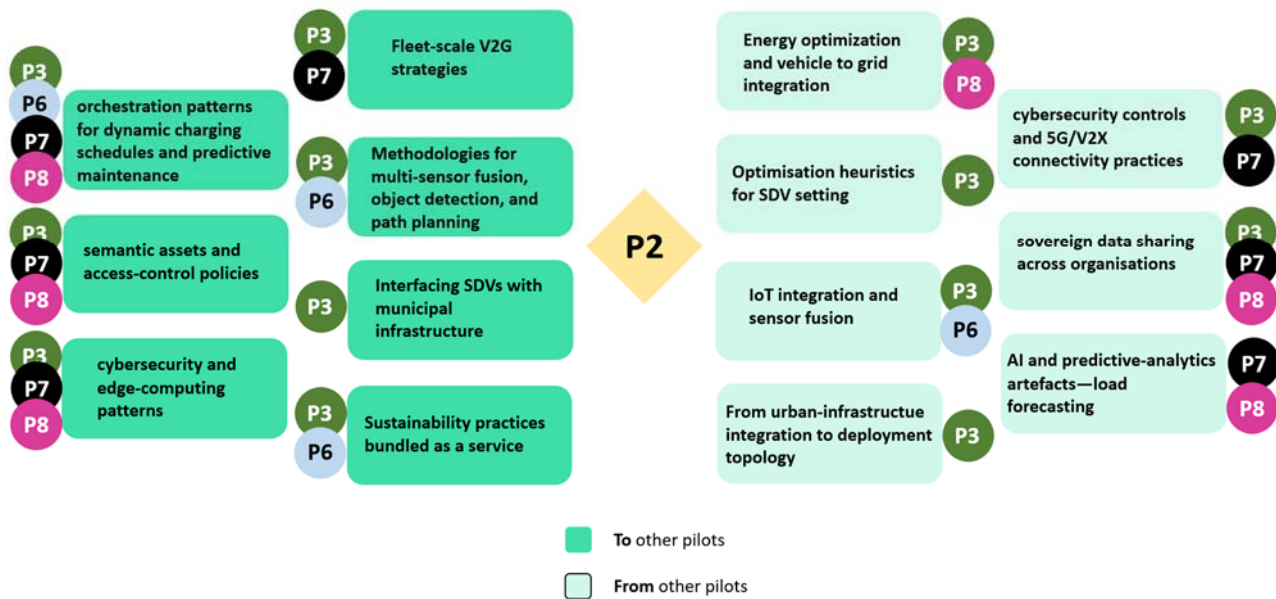


Figure 116. Pilot 2 Cross-Domain Interactions

11.2.3 Pilot 3

Value from other pilots

- **Secure Data Exchange** (Pilot 7): Methods and software frameworks to securely exchange time series data while protecting privacy.
- **Bidirectional Charging** (Pilot 7): Standards on bidirectional charging, concrete charging infrastructure for bi-directional charging.
- **Smart Charging Optimization** (Pilot 7): Complementary methods to reinforcement learning to optimize the charging schedule of a vehicle fleet potentially addressing other criteria.

Value to other pilots

- **Enable EVs to feed surplus energy back into the grid** (Pilot 2): Quantitative assessment on achievable surplus by exploiting the volatility of prices on the spot market.
- **EV battery lifespan estimation** (Pilots 2, 6 and 7): Accurate estimation of the state of health of the battery including quantification of the expected end of life.
- **Overall management of charging station** (Pilots 2 and 7): The charging optimization from Pilot 3 results in an operational charging plan that can be used as a template for similar vehicle fleets.
- **Scalable Reinforcement Learning Framework** (Pilots 2 and 7): The framework for reinforcement learning and the methodology for refinement of optimizations can be applied within other pilots dealing with time-series processing.

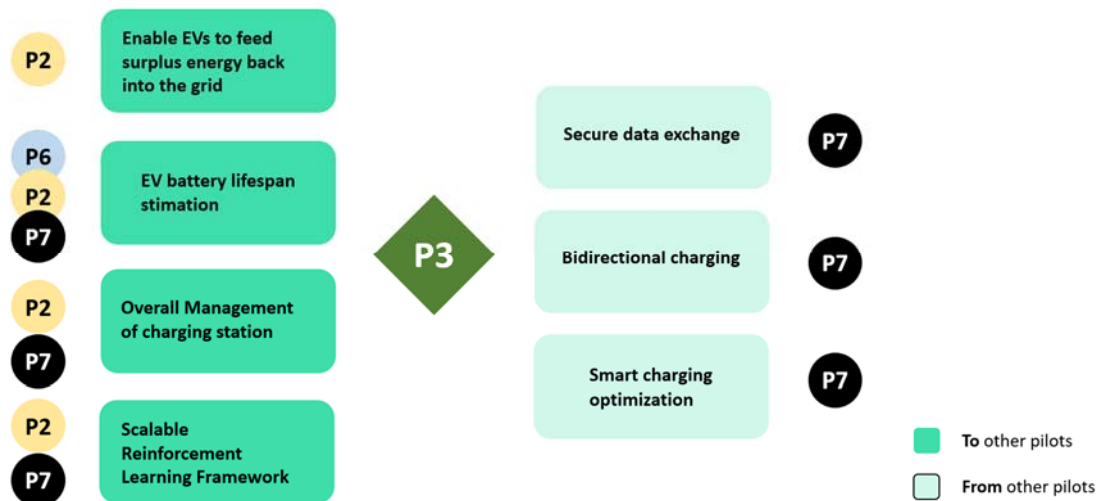


Figure 117. Pilot 3 Cross-domain interactions

11.2.4 Pilot 4

Value from other pilots

- **Secure Data Exchange (from Pilot 7):** Pilot 7 focuses on trustworthy and secure EV charging upon reliable 5G networks. The security measures and data sovereignty protocols to be established in this pilot may enhance the integrity and confidentiality of data exchanges in Pilot 4, ensuring secure communication between maritime terminal systems.
- **Internet Connectivity Robustness (from Pilots 5 and 6):** Pilots 5 and 6 address energy efficiency in agriculture through smart recharging and monitoring systems. The robust internet connectivity solutions developed by these pilots are adapted to Pilot 4, ensuring reliable communication and data transfer within the maritime terminal's complex operational environment.
- **AI and Predictive Analytics Artifacts – Load Forecasting (from Pilots 2, 7, and 8):** Pilots 2, 7, and 8 implement AI-driven predictive analytics for various applications. The load forecasting models developed in these pilots may be applied in Pilot 4 to further enhance the predicted energy demand fluctuations, in case the models developed by Pilot 4 partners are not sufficiently accurate to enable proactive energy management and operational planning in the maritime terminal.

Value to other pilots

- **Energy Grid Balancing and Renewable Energy Production (to Pilot 1):** The energy management strategies and renewable energy integration techniques developed in Pilot 4 can help on informing Pilot 1's efforts to optimize electric grid performance upon renewable energy source integration.
- **Energy Flexibility and Demand Response Capabilities (to Pilots 1 and 8):** Pilot 4's advancements in energy flexibility and demand response mechanisms can be used to provide a framework for Pilots 1 and 8 to enhance their energy management and social engagement strategies.
- **Real-Time Energy Consumption and Flexibility Data (to Pilots 3, 5, and 8):** The real-time energy monitoring and data analytics systems to be implemented in Pilot 4, might offer valuable insights and methodologies for Pilots 3, 5, and 8 to improve their energy consumption tracking and flexibility assessments.

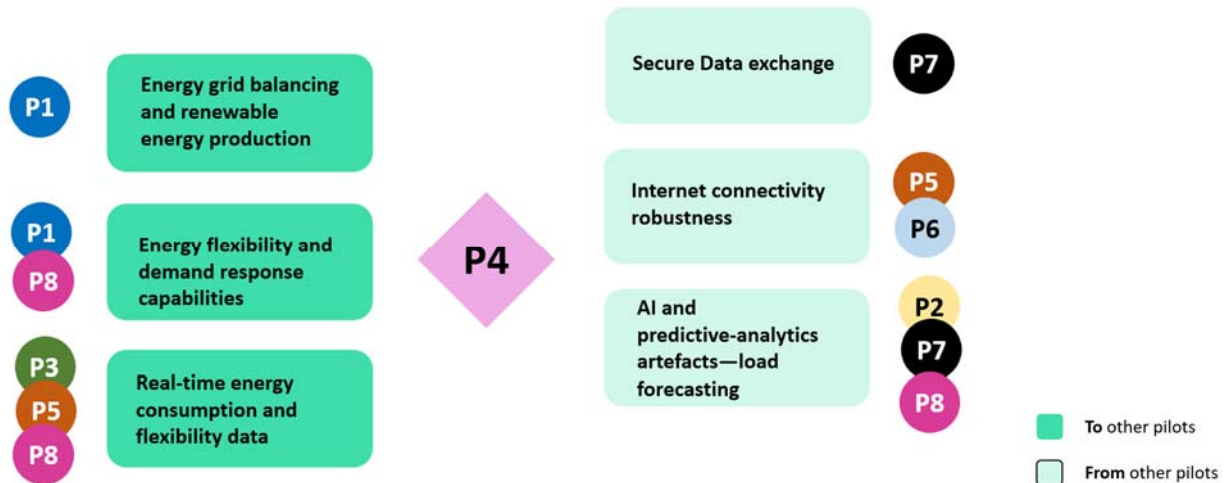


Figure 118. Pilot 4 Cross-domain Interactions

11.2.5 Pilot 5

Value from other pilots

- **Battery & charging optimization logic** (from Pilot 3) -> Pilot 3's integration of metering into BEV charging and V2G/grid management can support Pilot 5 in optimising machinery charging aligned with renewable supply.
- **Climate and weather adaptation** (from Pilot 6) -> Pilot 6 long-term strategies for climate change and short-term measures for sudden weather events can help Pilot 5 exploit real-time data, historical patterns, and seasonal demand inference to strengthen resilience and rapid response.
- **Data flow, grid integration & forecasting** (from Pilots 4, 7 & 8) -> Real-time monitoring from Pilot 4, smart meters for demand-response from Pilot 7, and forecasting/orchestration methods from Pilot 8 can enhance Pilot 5 controlled farm&factory data exchange and predictive energy management.

Value to other pilots

- **Energy mix selection & footprint monitoring** -> Pilot 5's IoT-enabled strategies for seasonal energy optimisation and calculation of the dairy energy footprint can be transferred to other pilots (e.g., Pilots 3, 6, and 7) to support renewable integration, sustainability reporting, and efficient energy source selection.
- **Cross-domain data interoperability** -> Pilot 5's methods for controlled sharing of farm–factory data, combining real-time monitoring with historical energy use, can be applied in logistics (Pilot 4) and urban energy flexibility (Pilot 7) to improve traceability and coordination.
- **Edge-AI & sustainable agrifood strategies** -> Pilot 5's deployment of Edge-AI for livestock/process optimisation and its renewable-powered agrifood strategies can inspire decentralised intelligence in transport (Pilot 2), robotics (Pilot 6), and smart grids (Pilot 8), supporting adaptive and sustainable operations.

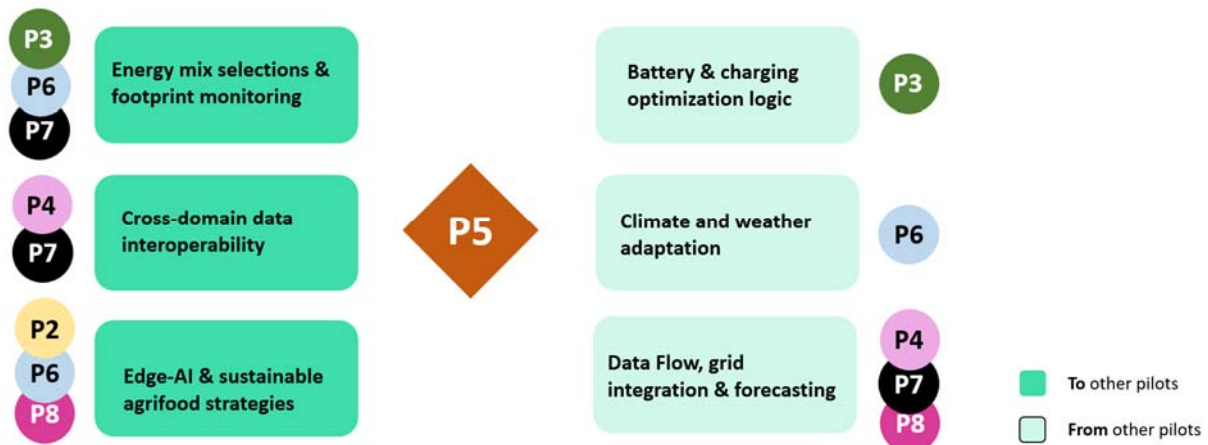


Figure 119. Pilot 5 Cross-domain Interactions

11.2.6 Pilot 6

Value from other pilots

- **Route optimization and predictive maintenance from Pilot 2** -> Pilot 2 will be focused in some of the scenarios on optimizing EV charging (including Battery electric vehicles) while fostering a data-driven ecosystem for efficient transportation. The specific effort on route optimization and data analyzing for monitoring could be reused in Pilot 6.
- **Battery & Charging Strategies from Pilot 3** -> It focuses on intelligent BEV charging strategies for a commercial fleet. Its approaches for battery scheduling, V2G or grid-load management could be directly leveraged in our pilot 6. Specially how to coordinate multiple vehicles with limited charging points.
- **Energy mix selection from Pilot 5** -> Pilot 5 IoT energy optimization & seasonal data-driven energy strategy used for the selection of the most efficient energy source at any given moment can help our pilot 6 tractor batteries to choose between straight from the grid or renewable stored energy (e.g. solar panels).
- **Decentralize energy forecasting with edge orchestration from pilot 8** -> Pilot 8 will focus in Sc 1 on the energy demand forecasting for EV charging. The AI models used could be reused in Pilot 6 for establishing the strategy for recharging the batteries of the tractors.

Value to other pilots

- **Energy grid balancing and renewable energy production** -> In remote agricultural locations as the one where the Pilot 6 is located, renewable energy (solar panels, e.g.) is always an option but it has a relevant impact on the balancing of the grid. Facing this challenge could contribute to Pilot 1 and Pilot 8.
- **Semi-autonomous route planning** -> While mowing the kiwi plantation with the semi-autonomous tractors, the route planning will be the key as the field has complicated elements that change overtime that will need to be mapped. The AI models and logic could be exported to other mobility pilots like pilot 3 with their EV fleet and Pilot 2 with the advanced safety systems for SDVs.
- **Internet Connectivity Robustness** -> Kiwi plantations may have network limitations due to connectivity issues, and the robustness required could be extrapolated to other pilots with similar circumstances, like maritime terminal in Pilot 4 or the farms in Pilot 5.

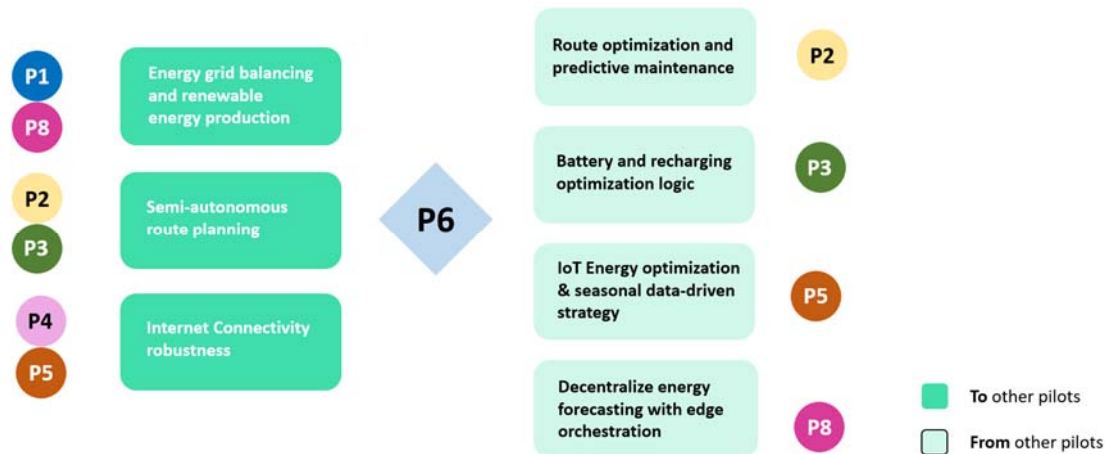


Figure 120. Pilot 6 Cross-domain Interactions

11.2.7 Pilot 7

Value from other pilots:

- **Optimize bidirectional energy flows (V2G & B2G)** by using fleet operation data to enable smarter energy dispatch between EVs, charging stations, and the grid. From Pilot 3.
- **Enable EVs to feed surplus energy back into the grid** by improving both charging station efficiency and overall grid stability. From Pilot 3
- Leverage AI-based battery storage optimization. From Pilot 3
 - Reduce strain on charging infrastructure.
 - Extend EV battery lifespan.
 - Improve overall energy management at charging stations.

Value to other pilots:

- **Secure data exchange:** Supports “controlled data sharing” in Pilot 4 through secure EV charging communication. To Pilot 4
- **Bidirectional charging:** V2B/V2G solutions allow electric farm vehicles to supply surplus energy back to operations, enhancing sustainability and reducing reliance on external power sources. To Pilot 5
- **Smart charging optimization:** AI-driven analytics from O-CEI optimize charging based on tractor usage, terrain, and weather - reducing idle time, preventing energy shortages, and improving efficiency in agriculture. To Pilot 6

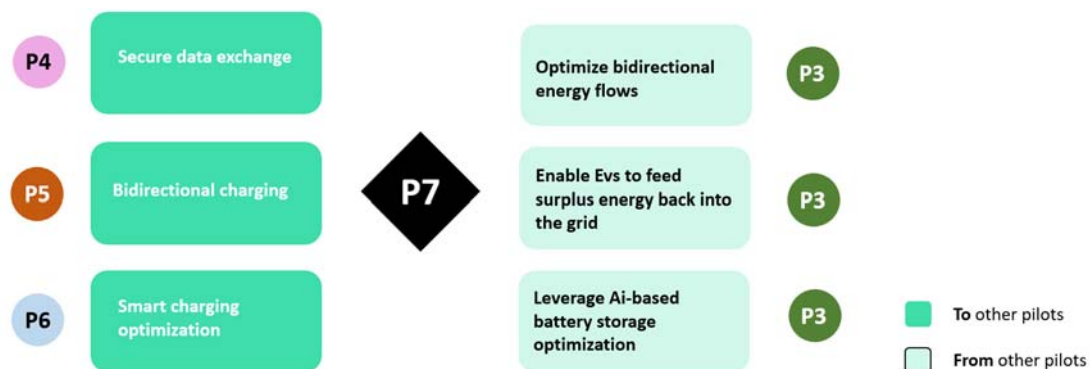


Figure 121. Pilot 7 Cross-domain Interactions

11.2.8 Pilot 8

Value from other pilots:

- **Machine learning from Pilot 1:** Pilot 1 is already making energy predictions from French smart meters. Probably have some pipeline/frameworks that could be interesting to Pilot 8.
- **Battery & Charging Strategies from Pilot 3:** It focuses on intelligent BEV charging strategies for a commercial fleet. Its approaches for battery scheduling, V2G or grid-load management could be directly leveraged in our pilot. Specially how to coordinate multiple vehicles with limited charging points.

Value to other pilots:

- **Social Acceptability:** For other pilots like pilot 2 and 3, the techniques to gather data for social acceptability could be useful.
- **Energy meters** The CLEMAP meters that can function as an Edge computing resource could be interesting to other pilots such as pilot 3, 4 and 7.
- **Decentralized energy forecasting** with edge orchestration Using the Nuvla Orchestrator to deploy decentralized energy forecasting ML could be reused in Pilot 6 for establishing the strategy for recharging the batteries of the tractors.

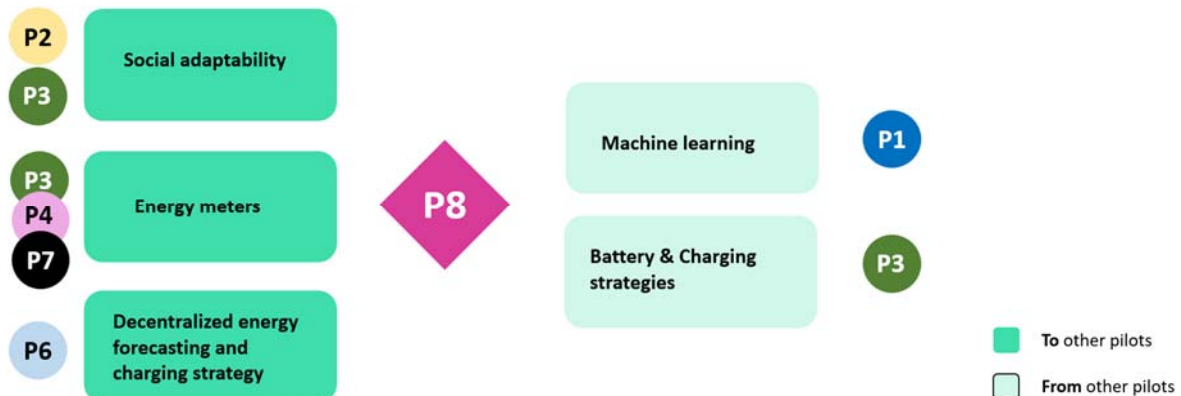


Figure 122. Pilot 8 Cross-domain Interactions

12 PILOT PRIORITIZATION

As reported in July for the Milestone 14, we had to establish a list of prioritized pilots **to take the lead at specific times**, ensuring early tangible results, better use of resources, and stronger demonstrations for stakeholders.

In order to select the most suitable ones, several criteria were defined and analyzed for each pilot, ending with the following results:

- A **set of seven criteria** were defined for prioritization: cross-pilot reusability, social engagement, environmental sustainability, interoperability with legacy systems, business readiness, edge intelligence needs, and alignment with key technologies (like V2G, DLT, decentralized identity).
- All **8 pilots were assessed** against these criteria.
 - **Strong cross-pilot collaboration potential:** community energy flexibility pilots (1.1, 1.2, 1.3, 8) and EV-related pilots (2, 3, 7, 1.3).
 - **Highest social engagement:** residential/community pilots (1.1, 1.2, 1.3, 8).
 - **Most demanding edge latency:** Pilot 7 (EV charging with 5G) and Pilot 2 (vehicle safety).
 - **Business models closer to maturity:** Pilots 1, 2, and 3.
- At month 6, the project **prioritized Pilots 1.2, 2, and 3** for the technical check, as they best represented flagship use cases in energy flexibility, EV bidirectional charging, and electromobility.
- Following evaluator recommendations, **two pilot groups (1.2 & 8, and 3 & 7)** were chosen to lead the **first concrete blueprint designs**, maximizing cross-pilot collaboration and reusability.
- The prioritization is described as a **dynamic and strategic tool**, not a ranking of “better/worse” pilots. It will be revisited regularly to adapt to project needs.

More details about the criteria, selection and results are available in the ANNEX.

13 CONCLUSIONS

This deliverable establishes the **basis for the governance and management of the eight Large-Scale Pilots** (LSPs) of O-CEI project, providing a common framework for coordinating efforts and facilitating interoperability between different domains and sectors. To this end, coordination strategies, information collection and standardization processes have been defined, resulting in this **initial characterization of the Pilots**, the presentation of the **setup and procurement work** carried out to date, and the infrastructure put in place in each of the pilots and their scenarios as preparatory actions for their development and implementation.

At the governance level, the importance of a federated model that allows each pilot to operate autonomously, while maintaining overall consistency through shared governance mechanisms, has been demonstrated. This model is based on the synergies already identified between the key sectors of energy, transport, logistics, agrifood and telecommunications, in order to move forward in an aligned and coordinated manner **towards replicable and scalable solutions** across Europe.

During the first months of the project, significant progress has been made in the preparation and initial deployment phase of the LSPs. During this period, efforts have focused mainly on defining the ethical and regulatory frameworks, **acquiring equipment**, and establishing the necessary **technological infrastructure**, both in terms of hardware and software, to enable the **following development and validation of the proposed solutions**. The work carried out in this first phase provides a solid foundation for the next stage of the project, where the technologies developed will be validated and their integration into real environments will be promoted. This process will be essential to demonstrate the value of the O-CEI platform and lay the foundations for a digital ecosystem geared towards energy flexibility and sustainability. An overall analysis of the status of the pilots shows that, although with different degrees, **all have achieved key milestones** related to the installation of sensors, management platforms and control tools. This initial progress has **laid the groundwork for the next phases**, which will address integration, testing and optimization activities, ensuring that the solutions are robust and transferable to other EU environments.

Cross-domain collaboration relies on sharing proven methods to speed up collective learning and ensure reliability in energy flexibility. **Potential synergies have been mapped out**, both the value each pilot can contribute and the assets they can adopt from others.

In summary, the key pillars of cross-domain interaction are: **(1) Electromobility and charging strategies**, including V2G and storage approaches for bidirectional charging, energy flow optimization, battery life estimation, multi-sensor fusion methodologies, object detection, route planning, and fleet-scale V2G strategies; **(2) Renewable resource management and climate adaptability**, covering resilience and connectivity, energy mix management, carbon footprint monitoring, and grid balancing; and **(3) Security, traceability, and data exchange**, with a primary focus on secure, trustworthy data exchange and cross-domain interoperability.

Pilots' prioritization is introduced as a strategic and dynamic tool to assign leadership or effort to specific activities, optimizing resources while addressing near-term needs. The report proposes a four-phase approach—criteria selection, activity listing, pilot assessment, and prioritization—with the last 3 phases updated over time. The followed criteria align with EU guidance (Competitiveness Compass) and cover synergies, social engagement, environmental impact, HW/SW constraints, business maturity, edge-latency, and technological relevance.

It should be highlighted that the initial phase of the project has not only succeeded in establishing the technical and organizational foundations for the pilots, but also in laying the groundwork for the collaboration needed to maximize the impact of the O-CEI project.

14 ANNEX

PILOT PRIORITIZATION FRAMEWORK

The proposed framework consists of 4 main phases: (i) the selection of the criteria used to support the prioritization exercise, (ii) the listing of key activities where prioritization applies, (iii) the analysis of pilots based on such criteria, and (iv) the current assessment of pilots based on the exposed criteria on M6 of the project. This last exercise ensures the dynamicity of the prioritization, tailoring it to the specific needs of the target activity.

SELECTION OF CRITERIA

The selection of prioritization factors was also made in alignment with strategic guidance from the European Commission, particularly the Competitiveness Compass, which emphasizes reusability, sustainability, trust, and technological sovereignty as core pillars for Europe's digital transformation. This ensures that the framework is not only internally consistent but also resonates with the EU's broader digital and green objectives, providing a foundation for strategic planning across technical, dissemination, and exploitation Work Packages.

To ensure a consistent and transparent approach to pilot prioritization, O-CEI defined a set of seven criteria that reflect current project's ambitions and large-scale demonstration interests, societal objectives, and strategic relevance within the European digital and green transition (reflected as per M6 in Section 3). These criteria have been selected through internal discussions of the Steering Committee (SC), informed by the interaction with the EC, the assessment of relevant EU's priorities (e.g., Green Deal, Fit for 55...), regulatory considerations, and lessons from previous CEI initiatives. The chosen dimensions aim to capture a holistic view – encompassing (i) reusability and knowledge sharing, (ii) citizen implication, (iii) environmental impact, (iv) HW/SW constraints, (v) business readiness, (vi) edge latency requirements, and (vii) relevance of key technologies according to the project momentum. The seven criteria used in this framework are:

1. Reusability / Cross-Pilot Collaboration

Definition: The extent to which results (components, methodologies, data flows, lessons) can be transferred or reused in other pilots.

Rationale: The project fosters cross-domain collaborations, which shall occur inherently within the pilots as having stakeholders from different sectors involved. Apart from intra-pilot, an inter-pilot collaboration fosters collective value creation and accelerates blueprint definition and implementation. Cross-pilot collaboration is foreseen in all pilots, yet their number and scale may vary. Prioritizing reusable results supports coherence across domains and strengthens the federated architecture of O-CEI.

2. Social Engagement

Definition: Degree of citizen or stakeholder implication in pilot scenarios.

Rationale: Enhances acceptability, visibility, and potential adoption of the O-CEI outcomes. Pilots that actively engage users or communities are better positioned to influence behavior, policy, and long-term societal outcomes. Besides, dedicated material for the general public may need to be prepared in advance to engage citizens/stakeholders effectively.

3. Environmental Sustainability

Definition: The main angle from which the pilot is contributing to the sustainability goals.

Rationale: Aligns with the EU Green Deal and there are related KPI in all pilots. All of them are expected to contribute measurably to sustainability goals, which are key to long-term energy resilience and competitiveness. The subcategories include residential/community energy flexibility, EV-grid optimization, grid stability (which the two previous also contributes to), energy optimization and emissions reduction.

4. Relevance of the Expected Interoperability with Legacy Hardware and Software

Definition: Relevance of the interoperability required with existing hardware equipment and software platforms.

Rationale: All pilots come with inherent business, standards, regulatory and legal constraints influencing the project. Yet, regarding the former, there might be some scenarios in which the relevance of the expected interoperability with legacy infrastructure and software will be higher or lower. This varying degree is positive: more constraint scenarios foster more conscious designs, whereas less constraint ones facilitate the validation of larger number of CEI tools.

The O-CEI outcomes should be applicable in the pilots' ecosystems (e.g., considering the applicable standards, key relevant software), therefore their constraints should not be considered as an afterthought. Also, legal readiness is essential for real-world deployment and trust. Pilots handling sensitive data or operating in regulated domains must demonstrate early attention to compliance to avoid future blockers, and thus they are key to define valuable and realistic solutions.

5. Business Models Comprehension

Definition: Level of the establishment of the business models of the pilots' scenarios, or comprehensible path for their implementation.

Rationale: Pilots which, according to the market analysis performed and reported in D2.1, present more actionable short-term business models, with clearer identified constraints. While the potential is present in all of them, pilots with less uncertainties can better serve as example to communicate the features brought by CEI technologies. Also, those acting on greenfields may offer wider room for innovation.

6. Edge Intelligence Relevance

Definition: Critical use of edge computing for decision-making and real-time operations.

Rationale: A foundational element of O-CEI. While all pilots are powered by edge computing, pilots that showcase the need for advanced edge capabilities that help validate the project outcomes in latency-sensitive environments are prioritized according to this criterion (with respect to other motivations such as local aggregation, traffic reduction, etc.).

7. Relevance of Key Technologies According to the Project Momentum

Definition: Presence and relevance of a given technology in the framework of a given technology.

Rationale: The necessity of novel technologies for the continuum is expected in all pilots, contributing to "closing the innovation gap" in the competitive compass. Yet some technologies may be prioritized according to the project momentum or within the scope of a given activity (related to the EUCEI building blocks, or explicitly manifested by the O-CEI community including the EC representatives, e.g., EV Bidirectional Charging – V2G/V2V, DLT, IAA (Identity, Authentication and Authorization) Decentralization (VC and DID), etc.

SELECTION OF ACTIVITIES REQUIRING PRIORITIZATION

A relevant set of activities are now listed, some already executed, some ongoing, and others planned. This list will be enlarged along the way, once new activities requiring selection, prioritization or examples for guidance are identified by the Project Implementation Committee (PIC) – in virtual or physical meetings:

- Workshop on “Decentralized ID, Data Sovereignty and DLT for Energy in Cloud-Edge-IoT” (done – yet methodology was not in place).
- Pilots presented and discussed in the technical check which took place during June in Brussels. (done).
- Initial definition of blueprints aligned with the agile and structured approach towards MS12 (ongoing).
- Deepened elaboration of the available infrastructure, focusing on distributed energy resources – to fulfil MS13 (done).
- Mapping of technological needs and tools identified in D2.1 (ongoing).
- Preference for a prominent role in V2GLeaders event in November 2025 (done).
- Presentations for Open call webinars and the Advisory Board (planned).
- Pilots showcasing in the RP1 review (planned)
- Complete deployment proved via MVP of O-CEI Platform (planned).

PILOTS ASSESSMENT

A first pilots' assessment exercise based on the aforementioned criteria has been performed at M6 of the project, reported in Section 3. It is presented in a short, narrative manner, in a way that reflects either the nature of the pilot (immutable characteristics) or its status in a specific time period. This short analysis is expected to be revised periodically before the execution of a physical PIC meeting (taking place every 6 months), led by the Pilot managers together with WP6 (market) and WP7 (impact) leaders.

PILOTS PRIORITIZATION EXERCISE

The Consortium reserves the right to continuously re-assess the prioritization of pilots. Typically, physical PIC meetings or virtual encounters (happening every month) are used to discuss this matter based on the exposed criteria.

It is worth noting that not every single activity within WP4 is prone to apply this prioritization, but as a rule of thumb, those enumerated in Section 2.2 set illustrative examples of qualified occasions. The exercise of prioritization involves a mapping first between the criteria and the target activity, and once the selected criteria and the pilots (assessing at this moment the Pilot alignment criterion). An example is as follows:

For the Technical check, the Consortium deemed appropriate to bring only a subset of the pilots to devote enough time to introduce and discuss the main concepts and approaches of O-CEI and avoid delving into the particularities of 8 heterogeneous pilots, thus a prioritization was required. At that moment, the pilots were still at definition phase (use cases, requirements, procurement activities...), therefore the main criteria revolved around the stickiness of the pilots to main narrative line of O-CEI (energy flexibility) as well as the capacity of showing impact in relevant fora (such as SDOs) and their potential for future cross-pilot synergies (data, AI models, practices or Cloud-Edge-IoT technologies contributing towards environmental sustainability). Also, these pilots are well-aligned

with current technological priorities of the European ecosystem, namely related to the energy grid, autonomous driving and bidirectional EV charging.

Table 22. Main criteria for pilot prioritization

Cross-pilot	Social Engagement	Environmental Sustainability	Magnitude of expected interoperability	Business readiness	Critical edge intelligence	Relevance of key technology
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By month M6 – facing the technical check – a strategic consideration was made based on these criteria. The result derived in deciding that Pilots 1.2, 2 and 3 should be highlighted by then. These convey leadership and exemplification in the energy flexibility, electromobility and bidirectional EV charging realm; which due to their nature, could be used as examples for cross-piloting activities and for potential CEI innovation in their businesses. The rest of criteria were not neglected (e.g., social engagement aspects are present within Pilot 1.2, significant magnitude of interoperability constraints can be seen in all them...) but were less weighted for the selection at this stage.

It is worth highlighting again that this process will be repeated over time, with potentially diverse results. Other pilots will excel from different angles, and the criteria for selection might be later tilted towards other compasses such as intangibles, particular concern/technology alignment or demonstration potential.

ASSESSMENT OF O-CEI PILOTS

To ensure that the O-CEI outcomes are designed, developed, validated, scaled and communicated in alignment with real-world priorities and technological readiness, the pilot prioritization framework was applied across all pilots – individually and consistently. Each pilot has been assessed across the seven evaluation criteria described in Section 2. This multi-tiered approach allowed the Consortium to identify which pilots are best positioned to drive early impact, e.g., for blueprint design and other activities (see Section 2.2), and will be re-assessed periodically in the physical meetings of the PIC. A summary is provided in Section 3.9.

PILOT 1

Pilot 1 is split into 3 sub-pilots, where involved partners will collaborate to achieve the expected outcomes and impacts in all the included scenarios. They have some analogous use cases around energy flexibility in residential/urban areas, and the social engagement aspect is strong in all of them. Therefore, the legal and regulatory framework as well as the interoperability challenges barriers and challenges share several commonalities, therefore opening the room to deep cross-pilot collaboration, knowledge sharing and best practices application.

Pilot 1.1

Pilot 1.1 main sustainability goal is focused on increasing electric grid flexibility and performance in Limerick through two key use cases: (i) community-driven renewable energy integration using smart heat pumps and DLT-enabled energy trading, and (ii) smart thermal load management based on edge AI that adjusts heating controls using occupancy and weather predictions. Ireland's pressing challenge in decarbonizing residential heat makes this pilot socially and environmentally impactful.

Among employed criteria, this pilot has a strong social engagement aspect. So far, the pilot has been focusing on GDPR and formal agreements (e.g., Consent forms) to securely manage participant and pilot data. Since an active engagement of a household community of prosumers is expected, a cross-pilot interaction is foreseen primarily with Pilot 8, Pilot 1.2 & Pilot 1.3, to share knowledge, approaches and success stories related to the involvement of citizens and the creation of awareness.

From the technological perspective, this pilot integrates different sensors, historic databases, local metering and actuation solutions. Besides, the applicability of DLT for energy trading is one of the main technical characteristics of this pilot, which insights could be extrapolated to other pilots if deemed relevant. Despite having a strong edge nature, its latency characteristics are limited, as for now the pilot team has formalized as requirement the need of logging data in intervals starting from 1 second.

This pilot focuses on residential energy flexibility, one of the most commented business applicability. While having significant presence in relevant fora, there are relevant challenges that limit its applicability as identified in D2.1, being upfront costs and low consumer engagement, some of its main barriers. This pilot also highlights the need for interoperable solutions, given the number of devices, formats and systems already in place to work with, as presented in MS13 (later on to be further refined in D4.1). This was expected due to the fragmentation of solutions as well as the number of players involved in this value chain (DSOs, utility & tech providers, municipalities, citizens).

Pilot 1.2

Pilot 1.2 also focuses on energy grid flexibility and stability, in this case, in the French region of Île-de-France, where increasing electrification and the decentralization of energy sources demand enhanced local control. This pilot considers two scenarios: (i) the first one related to energy management services for end-consumers and (ii) a second related to the virtualization of substations for RES integration. The cross-pilot interaction with Pilot 1.1 & Pilot 8 is foreseen, especially within the scope of the first scenario.

Apart from the motivation given for the previous pilot in the side of social engagement (now for residents rather than a community), the fact of having different devices and systems involved claims for devising blueprints that have the same (or a very similar) target, yet they are technologically realized differently. Overall, the pilot introduces a broad CEI architecture spanning from houses to substations. Technologically, virtualization (following Cloud Native paradigm) and data management/processing are key enablers to implement the pilot. While the latency requirements of the first scenario are similar to those from Pilot 1.1 given their analogous nature, the voltage control at the substation scenario require of real-time processing and action.

With regard to business aspect, the challenges for the first scenario are equivalent to those previously presented. Notably, the second scenario may have a clearer realization path, yet with a more limited scope (mainly involving the DSO). As with Pilot 1.1, constraints in terms of interoperability with existing devices, systems and standards (notably UEC 61850) are significant – adding to those restrictions coming from EU (like GDPR) those related to EDF internal policies, which solutions must comply with.

Pilot 1.3

Pilot 1.3 is deployed on Krk Island, Croatia – a location where energy self-sufficiency is critical due to limited connectivity with the mainland grid. In this case, the energy sustainability aspect is also related to energy grid flexibility and future independence. This pilot explores two advanced use cases: (i) wide smart grid optimization using edge-based analytics and NILM (Non-Intrusive Load Monitoring), and (ii) EVs and heat pump synchronization for proactive load balancing. These capabilities enable predictive energy management in both residential and commercial buildings. Overall, this pilot follows up with the citizen involvement (thus with clear cross-pilot interaction points with Pilots 1.1, 1.2 and 8), here considering not just residents but also commercial engagement. Since it explores EVs also as part of the environment, some synergies could be found with Pilots 2, 3 and 7.

This pilot will test how decentralized coordination of generation, consumption, and storage – combined with community participation – can deliver a replicable model for smart, self-sufficient energy systems in island and remote regions; focusing primarily thus on CEI solutions for data processing and orchestration plus best practices and strategies. In this case, the latency requirements are comparable to those in Pilot 1.1.

Finally, the business aspect faces similar interoperability (GDPR, standards) and mass adoption challenges as the previous sub-pilots: the upfront costs are high, there is a large fragmentation of existing solutions and creating awareness on end-users in complex, as highlighted in D2.1. All in all, business cases are yet immature especially on the part of creating revenue (at least, until being socially-demanded), although this could rapidly change if this is encouraged and fostered by policy-makers.

PILOT 2

Pilot 2 delves into how Software-Defined Vehicles (SDVs) can become intelligent agents in the energy ecosystem and urban environment. Deployed in Timisoara, Romania, this pilot features four scenarios that focus on: (i) Vehicle safety, specifically on monitoring driver behaviour collisions avoidance, automatic energy braking and notification to emergency services; (ii) Sustainability, optimizing routes, recommending behaviour, monitoring emissions and sharing reports; (iii) Fleet V2G integration, working on scheduling charges/discharges, monitoring of battery status and fleet integration strategy; and (iv) Grid optimization, enhancing the previous scheduling by analysing and predicting its performance. Therefore, apart from the inherent sustainability of leveraging electric vehicles, the sustainability aspect comes from the optimization of V2G strategies and consumption monitoring and awareness.

While the pilot has strong urban relevance, citizen implication is less pivotal compared to other community-driven pilots. Cross-pilot interaction is primarily envisioned with P3 and P7, within the use cases related to EV charging. While this pilot can contribute with its vast knowledge in the automotive field, the others can offer the expert view of EV charger provider.

The pilot will integrate AI-driven modules, edge processing, containerized decision support, and predictive analytics in urban mobility contexts. Indeed, the latency requirements of the pilot are of high relevance, especially for the safety scenario (despite not being an autonomous driving scenario, where the impact would be even higher). From the technical perspective, scalability is the main aspect to focus, applied to the serving of AI models, edge processing and monitoring modules (applied for the aforementioned purposes). Also, data spaces will be interfaced to gain access to relevant data.

Moving to business, it is one of the most socially-visible pilots due to its direct link with urban infrastructure and EV users. While the business models face uncertainties in terms of long-term return of investments, they are being fostered and they have been at least partially implemented. Still, these scenarios show different technical, administrative, ethical and (changing) regulatory constraints to face with. In particular, emissions, privacy and data ownership aspects are of high relevance, yet still, there is room for innovation with respect to the aforementioned scalable strategies and data interoperability solutions to address pilot challenges.

PILOT 3

Pilot 3, deployed in Austria, focuses on reducing energy consumption and CO₂ emissions from the national postal fleet (10,000 BEVs and 3,000 charging stations). It includes three scenarios, namely (i) battery storage system optimization, (ii) vehicle as mobile grid supporter, and (iii) market-integrated energy management. In this case, a cross-pilot connection can be found primarily with Pilots 7 and 2, both with EV charging traits (Pilot 7 more from the perspective of an energy charger provider, Pilot 2 from the fleet management and actual charging strategies). The social implications

in this pilot are not as relevant as those mentioned for Pilot 1, since it considers mostly B2B scenarios.

From the technical standpoint, technological momentum can be observed in the field of bidirectional EV charging, which the pilot is delving into, yet scalability aspects related to data orchestration and management (including integration with data spaces) are also relevant, as in Pilot 2. While this pilot includes a large number of edge enablers, real-time requirements have been mentioned for the defined energy assets, yet the latency has not been specified in the current analysis.

Considering a mass market perspective, business models are defined with strong path to market despite the numerous challenges present in the ecosystem, as highlighted in D2.1 – which create difficulties in establishing end-to-end value chains. From the technical perspective, challenges appear first from the infrastructure perspective (lack of two-way energy systems for bidirectional flows) as well as large data heterogeneity. The changing legal and ethical concerns are also complex, related to liability concerns and slow standardization (shared with Pilot 2). Still, these challenges open the room for advanced CEI innovations that can help address the scalability and interoperability issues. In any case, solutions will have to consider the strict compliance and security rules from the sector and thus encouraged by the pilot partners.

PILOT 4

Pilot 4 addresses energy flexibility at Malta Freeport, focusing on energy optimization aspects aligned with decarbonization policies, spanning from (i) vessel electrification, to (ii) port terminal load optimization, and (iii) the inclusion of home consumers in energy balancing. These actions are critical given the number of limitations in the ecosystem stemming from the fact of being located in a small island, thus not having enough space to deploy wind or hydro renewable sources – just relying on photovoltaic plants. Being in an isolated vertical domain, less cross-pilot interaction is expected yet knowledge is expected to be shared with Pilot 1.3 due to the similarity of the ecosystem. Some synergies could be found with Pilot 1.1 & 3 with respect to predictive energy balancing. It should be mentioned that even in cases where pilots might seem more isolated, technological approaches and best practices could be also shared (with logical adaptations), for instance, to server AI models, capturing metrics from energy equipment, effectively sharing data, orchestrating services, etc. The social engagement will not be an angle in this pilot, since validation activities will be performed in an office managed by the pilot stakeholders.

This pilot will foster the use of service orchestration, connection and monitoring of critical services in an Industrial environment to cope with the aforementioned optimization goals. Regarding edge aspects, the pilot requests sub-second latency requirements, thus more demanding than the previous pilots (with the exception of the safety scenario from Pilot 2).

Moving to business aspects, this is a case fostered by policy-makers. While the business cases are logical and indeed relevant for the island, the ROI is uncertain and the market still needs to cope with the needs. Apart from interoperability, there are challenges related to market fragmentation and lack of standards (e.g., in OPS), one critical aspect which is difficult to address without serious investment is related to the limits imposed by the available infrastructure: the equipment is expensive and complicated to replace.

PILOT 5

Pilot 5 tackles energy efficiency and sustainability in the dairy sector, focusing on Halloumi production in Cyprus. It integrates IoT sensors, edge intelligence, and DLT to (i) monitor and optimize the energy used for livestock, (ii) enable inter-factory data sharing, (iii) select energy mix for dairy production, and (iv) calculate the energy footprint of dairy products. The overall goal of the pilot is to reduce energy usage and the emissions without decreasing the product quality. Cross-pilot interaction is expected with the agriculture pilot, in areas such as best practices for energy

optimization related to climate change or sudden weather changes. While not citizen-facing, it benefits rural communities and therefore further farmer implication strategies should be considered.

The pilot will technologically focus on the metering methodology, data models and semantics, trusted data sharing (through DLT), analytics, preparation and serving of AI models. In contrast to previously-presented pilots, the number of existing legacy infrastructure and digital platforms is lower, as depicted in MS13. While Pilot 5 showcases the need for edge processing technologies, the latency aspect is not critical.

Regarding to market adoption, this vertical can hugely benefit from the digitalization features brought by CEI technologies, yet presents some particular traits that should be properly addressed – such as farmers' skepticism about sustainability and digital skills gap. Additionally, funding structures are less clear, and there is a weak coordination between farmers, technological providers and policy makers in this field. Besides, regulation and legacy HW/SW is less complex in contrast to previous verticals, which gives implementation freedom for early PoC validation – despite sustainability certifications can get complex if actors involvement is not properly managed. Still, solutions should foster replicability and adaptability to foster the future adoption of related blueprints.

PILOT 6

Pilot 6 focuses on improving the energy efficiency and sustainability of kiwi production through electric robot tractors. It considers 3 scenarios: (i) real-time consumption aid, (ii) remote tractors' route optimization, and (iii) autonomous recharging strategy. Although implemented in a different sector, this project shares several commonalities with Pilot 5, which enables some of the aforementioned interactions. Additionally, because of involving autonomous tractors, cross-synergies can be found with Pilot 3 (battery charging) as well as SDV principles from Pilot 2. No citizen engagement is foreseen, since the pilot is mostly B2B and automated.

From the technical perspective, the focus is similar to that of Pilot 5 – yet applied to a different sector. In this case, the pilot already considers some existing hardware and software to interoperate with, yet there is also need of integrating additional sensors. The pilot does not present exhaustive latency requirements, except for the autonomous, timely mowing service. Regarding business readiness, the analysis is similar to that of Pilot 5 (see Section 3.5), with one particular note: the introduction of a robotic element introduces a certain level of trust issues (related to both liability and mowing effectiveness).

PILOT 7

Pilot 7 explores how 5G connectivity can enhance the performance, trust, and orchestration of electric vehicle (EV) charging infrastructures. It includes scenarios for (i) energy flow optimization in presence of bidirectional EV charging, and (ii) trustworthiness and security in EV charging data exchange. Through low-latency data exchange and edge analytics, this pilot aims to improve grid stability, reduce peak loads, and ensure real-time responsiveness to user and system demands. It can benefit from cross-pilot collaboration, mainly with Pilot 3 in the field of optimizing bidirectional energy flows. Additional synergies can be found with other pilots within the scope of the application of Blockchain/DLT (with Pilot 1.1 and 5), controlled data sharing approaches (Pilot 4) and EV charging (Pilot 6). No citizen engagement is foreseen in this pilot.

From its technological positioning, this pilot integrates some interesting technological enablers SUCH as 5G, bidirectional EV charging, and Blockchain. The orchestration and processing of data will be also a central element of the pilot, and cybersecurity monitoring has been also identified as a target for the pilot. This pilot is demanding significant low-latency requirements, below 10 milliseconds.

As highlighted in D2.1, the scenarios of this pilot need to address a plethora of technical, administrative, legal and mass-market challenges, therefore the path to (mass) market is still unclear. Apart from the inherent complexities and interoperability needs highlighted for Pilot 3, adding the telco in the mix and fostering such low latency further deepens the issues. Regarding the second scenario, there is still a lack of regulation which both facilitates the execution of PoCs although complicates their scaling.

PILOT 8

Pilot 8 aims to democratize energy management in Swiss urban and semi-urban environments through Smart Energy Services (SES). It promotes citizen participation (prosumers), decentralized energy production, and smart coordination of electric vehicles using V2G and V2V strategies. Scenarios include (i) EV charging coordination, (ii) graph/LLM-based energy forecasting, and (iii) adaptive energy and computing orchestration in tourism-heavy areas. It is the most socially-centered pilot as it takes social acceptability aspects by design, focusing on engagement, transparency, and flexibility. Cross-pilot collaboration with Pilots 1.1, 1.2, 1.3 is thus relevant, yet further synergies can be found in the EV charging field with Pilots 2, 3 & 7.

Pilot 8 will include several technologies, from energy and continuum orchestration ones to EV charging, LLMs, MLOps and self-sovereign identity. For the sake of social acceptance, cybersecurity enablers as well as compliance with privacy regulations are also relevant. No specific real-time requirements have been described, apart from ensuring smooth interaction with users when information is presented to them.

Finally, market readiness could be depicted as a combination of Pilots 1 and 3 (depending on the scenario). Yet the LLM-related scenario may gain huge relevance given the momentum of analogous technologies. Still, several challenges and barriers were identified and reported in D2.1, which among others include the gaps on available data and LLM penetration in this field. Finally, the pilot presents a set of HW equipment and SW platforms to interoperate with, yet the familiarity with them plus the fact of already having data to work with them facilitates the implementation of the blueprints.

PILOTS ANALYSIS SUMMARY

This subsection summarizes the pilots' assessment performed, framed within the selected criteria:

- Cross-pilot collaboration: while all pilots exhibit potential synergies and areas for collaboration, the residential/community energy flexibility (Pilots 1.1, 1.2, 1.3, 8) and the EV fields (Pilots 3, 7, 2, 8, 1.3) show higher possibilities.
- Social engagement: those pilots with residential/community scenarios also include plans to implicate and create awareness in participants (Pilots 1.1, 1.2, 1.3, 8).
- Environmental sustainability: while identified as criteria, all pilots show traits of it, from different angles: residential/community energy flexibility (Pilots 1.1, 1.2, 1.3, 8), grid stability (Pilots 1.2, 7, 4), EV-grid optimization (Pilots 3, 7, 2, 8, 1.3), energy optimization and emissions reduction (Pilots 2, 5, 6).
- Relevance of expected interoperability: the degree of the business, standards, regulatory and legal constraints (including the existing of legacy hardware and software) are complex to value, yet as a tentative exercise Pilots 2 & 3 seem the most constrained ones whereas Pilots 5 & 6 seem the less ones.
- Business readiness: Business models are not ready, otherwise they would be mature enough to be already in the market. Yet those from Pilots 1, 2 & 3 present flagship use cases from the energy grid, SDV and EV ecosystems that can be fostered as initial examples of O-CEI applicability.

- Critical edge intelligence: Focusing solely on latency requirements (as all exhibit strong edge computing needs), Pilot 7 is the one with higher requirements, followed by Pilot 2 (safety scenario). Grid flexibility scenarios from the rest of the pilots will also require of low latencies, yet not at the level of the previous.
- Relevance of key technology: All pilots are data-driven and all are expected to implement monitoring, data sharing and AI models – apart from other CEI enablers. At M6, some technologies in which more attention has been paid are the bidirectional EV charging (Pilots 3, 7) and DLT/Blockchain (Pilots 1.1, 7).

PRIORITIZATION OF ONGOING ACTIVITIES

An example for the prioritized pilots for the technical check was presented in Section 2.4. As mentioned, pilots with flagship, clearer business models were prioritized, with showcasing potential in relevant fora. Energy grid, autonomous driving and bidirectional EV charging have big momentum in the current European landscape, and thus were deemed as good representatives.

Table 23. Main criteria for pilot prioritization before technical review

Cross-pilot	Social Engagement	Environmental Sustainability	Relevance of expected interoperability	Business readiness	Critical edge intelligence	Relevance of key technology
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The first recommendation received from the technical check evaluation was the need of a concrete blueprint definition: “there is a lack of concrete, pilot-specific technical overviews and visualizations showing how key components are operationalized. Thus, to enhance clarity and facilitate stakeholder engagement ... it is strongly recommended to provide at least one concrete, detailed blueprint from at least one of the selected pilots within the next 3 months. This should be quickly followed up by respective blueprints for all pilots wherever possible”.

Thus, the activity “Initial definition of Pilots Blueprints (ongoing)”, already identified by the Consortium yet now even more relevant, was ignited. One decision made was the preparation not 1 but 2 examples of blueprints, led by the pilots. This was agreed to refine the application of the methodology based on blueprints, reducing the potential bias of having a single pilot co-leading the activity.

In this case, the main criterion was the cross-pilot potential, partially fulfilling the recommendations received. The plan is to design 2 blueprints that could be implemented in scenarios of different pilots, aiming at reducing the bias on its design, maximize its reusability potential and share knowledge and ideas in their preparation. Other factors were considered: having one of them around social engagement, so it could help the project to have a social trait in one of the designs; and keep diving into the technological aspects shown in the technical check (i.e., EV bidirectional charging and flexibility of the energy grid). Besides, although not initially considered as criteria, selecting pilots with significant interoperability constraints can help to assess how the presence of relevant equipment, software and standards, regulations can affect to the blueprint design.

Table 24. Main criteria for pilot prioritization after the technical review

Cross-pilot	Social Engagement	Environmental Sustainability	Relevance of expected interoperability	Business readiness	Critical edge intelligence	Relevance of key technology
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The selected pilots have been **Pilots 1.2 & 8 + 3 & 7** to construct these two initial blueprints, focusing on their potential cross-pilot collaboration with respect to consumer energy flexibility and bidirectional EV charging (presented in the technical check).